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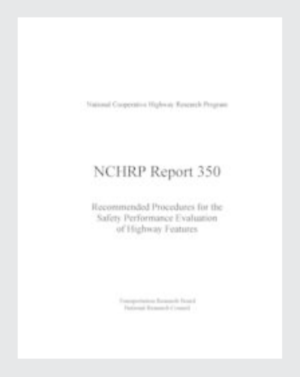

National Cooperative Highway Research Program

NCHRP Report 350

Recommended Procedures for the
Safety Performance Evaluation
of Highway Features

Transportation Research Board
National Research Council

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Recommended Procedures for the Safety Performance Evaluation of Highway Features

Transportation Research Board
National Research Council

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National Cooperative Highway Research Program

Report 350

Recommended Procedures for the Safety Performance Evaluation of Highway Features

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TRANSPORTATION RESEARCH BOARD
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NATIONAL COOPERATIVE HIGHWAY RESEARCH PROGRAM

Systematic, well-designed research provides the most effective approach to the solution of many problems facing highway administrators and engineers. Often, highway problems are of local interest and can best be studied by highway departments individually or in cooperation with their state universities and others. However, the accelerating growth of highway transportation develops increasingly complex problems of wide interest to highway authorities. These problems are best studied through a coordinated program of cooperative research.

In recognition of these needs, the highway administrators of the American Association of State Highway and Transportation Officials initiated in 1962 an objective national highway research program employing modern scientific techniques. This program is supported on a continuing basis by funds from participating member states of the Association and it receives the full cooperation and support of the Federal Highway Administration, United States Department of Transportation.

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The needs for highway research are many, and the National Cooperative Highway Research Program can make significant contributions to the solution of highway transportation problems of mutual concern to many responsible groups. The program, however, is intended to complement rather than to substitute for or duplicate other highway research programs.

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Each report is reviewed and accepted for publication by the technical committee according to procedures established and monitored by the Transportation Research Board Executive Committee and the Governing Board of the National Research Council.

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FOREWORD

By Staff
Transportation Research Board

This report is recommended to highway design engineers, bridge engineers, safety engineers, maintenance engineers, researchers, hardware developers, and others concerned with safety features used in the highway environment. It contains recommended procedures for evaluating the safety performance of various highway safety features. These procedures are based on a comprehensive literature review, an analysis of the state of the art for performance evaluation (including procedures adopted by foreign agencies), and the advice of a selected group of acknowledged experts. It is believed that this report will contribute toward providing safer highways.

Effective traffic barrier systems, end treatments, crash cushions, breakaway devices, truck-mounted attenuators, and other hardware must be used to achieve the highest levels of highway safety. Many devices have been developed by state agencies, universities, and private firms to address particular roadside safety concerns. New systems are continually emerging to address safety problems, and traditional devices and practices for their use are being improved in response to an increased understanding of safety performance, a changing vehicle fleet, the emergence of new materials, and other factors. Full-scale impact testing has been and will continue to be the most common method of evaluating the safety performance of guardrails, median barriers, bridge railings, crash cushions, breakaway supports, truck-mounted attenuators, work zone traffic control devices, and other hardware. Because a number of agencies in the United States conduct such tests, there is a need for uniformity in the procedures and criteria used to evaluate traffic barriers and other roadside safety features. There are also important precedents for promoting international harmonization of procedures to take advantage of relevant experience and research findings in other countries.

Procedures for full-scale vehicle crash testing of guardrails were first published in *Highway Research Correlation Services Circular 482* in 1962. This one-page document specified vehicle mass, impact speed, and approach angle for the crash tests. Although *Circular 482* did bring a measure of uniformity to traffic barrier research then being performed at several research agencies, a number of questions arose that were not addressed.

NCHRP Project 22-2 was initiated at Southwest Research Institute in 1973 to address the questions that were not covered in *Circular 482.* The final report was published as "Recommended Procedures for Vehicle Crash Testing of Highway Appurtenances," *NCHRP Report 153.* This 16-page document was based on technical input from more than 70 individuals and agencies as well as extensive deliberations by a special ad hoc panel. Several parts of the document were known to be based on inadequate information, but coverage of these areas was included to provide a more complete set of testing procedures. These procedures gained wide acceptance following their publication in 1974, but it was recognized at that time that periodic updating would be needed; and, in January 1976, TRB Committee A2A04 accepted the responsibility for reviewing the efficacy of the procedures. Questionnaires were submitted to committee members to identify areas of the document that needed revision. The responses generally fell into two categories: (1) minor changes requiring modified treatment of particular problem areas; and (2) major changes that would require broadening the scope to include, for example, testing with trucks and buses, reevaluating the criteria for impact severity, and treating special highway appurtenances such as construction barriers. The committee addressed the minor changes through special committee action; and *Transportation Research Circular 191,* published in 1978, was the product of this effort.

NCHRP Project 22-2(4) was initiated in 1979 by Southwest Research Institute to address the major changes. Its objective was to review, revise, and expand the scope of *Circular 191* to reflect current technology. The final report of **NCHRP** Project 22-2(4) was published as *NCHRP Report 230,* "Recommended Procedures for the Safety Performance Evaluation of Highway Safety Appurtenances," in 1980. This 36-page document incorporated new procedures, updated the evaluation criteria, and brought the procedures up to date with available technology and practices. This document served as the primary reference for full-scale crash testing of highway safety appurtenances in the U.S. and in many other parts of the world.

In 1987, American Association of State Highway and Transportation Officials (AASHTO) recognized that the evolution of roadside safety concepts, technology, and practices necessitated an update to *Report 230.* The reasons included significant changes in the vehicle fleet, the emergence of many new barrier designs, increased interest in matching safety performance to levels of roadway utilization, new policies requiring the use of safety belts, and advances in computer simulation and other evaluation methods. **NCHRP** Project 22-7 was approved to consider these factors and to prepare the update to *Report 230.*

The research team of Principal Investigator Dr. Hayes Ross, Jr., Texas A & M University, and Co-Principal Investigator Mr. Jarvis Michie, Dynatech Engineering, began this effort in 1989 with a series of white papers on the subjects of the test matrix, assessment procedures and criteria, specification and control of test parameters, in-service evaluation, use of surrogate test vehicles, computer simulations and other evaluation procedures, instrumentation of crash testing, the purpose of the document, and conversion to SI units. (Unedited drafts of these documents can be obtained on request from the **NCHRP**.) The panel reviewed each white paper and provided comments on the issues raised in them. A meeting of the panel was held to discuss the various issues, debate the implications for an updated set of procedures, and develop a consensus on the procedures that would be incorporated in the update. The research team prepared the first draft of the updated procedures after this meeting. This draft was thoroughly reviewed by the panel and a second meeting was held to discuss the report and establish panel consensus on the updated procedures.

The second draft of this document was mailed to some 100 individuals, and comments were received from about 65. The project panel met a third time to consider the review comments and determine where changes were needed in the updated procedures. Although the report originated with the research agency, each recommendation has the consensus endorsement of the NCHRP Project Panel. Where recommendations are founded on less than clear-cut evidence, the judgment of the project panel prevailed.

NCHRP Report 350 represents a comprehensive update of the procedures for safety performance evaluation. *Report 350* differs from *Report 230* in the following ways:

1. It is presented as an all-metric document in anticipation of the U.S. conversion to SI units of measurement. Hard conversions were made in the update process, which will

alter the mass, speeds, and tolerances used in testing. For example, tests previously specified for 60 mph (97 km/h) are now specified at 100 km/h. The impacts of such increases of speed on the kinetic energy of the tests were recognized and appropriate criteria adjustments made.

2. It provides a wider range of test procedures to permit safety performance evaluations for a wider range of barriers, terminals, crash cushions, breakaway support structures and utility poles, truck-mounted attenuators, and work zone traffic control devices.

3. It uses a $^3/_4$-ton pickup truck as the standard test vehicle in place of the 4500-lb passenger car to reflect the fact that almost one-quarter of the passenger vehicles on U.S. roads are in the "light truck" category. This change was made recognizing the differences in wheel bases, bumper heights, body stiffness and structure, front overhang, and other vehicular design factors.

4. It defines other supplemental test vehicles including a mini-compact passenger car (700 kg), single-unit cargo trucks (8000 kg), and tractor-trailer vehicles (36,000 kg) to provide the basis for optional testing to meet higher performance levels.

5. It includes a broader range of tests for each category of safety feature to provide a uniform basis for establishing warrants for the application of roadside safety hardware that consider the levels of use of the roadway facility. Six basic test levels are defined for the various classes of roadside safety features, and a number of optional test levels are defined to provide the basis for safety evaluations to support more or less stringent performance criteria.

6. The report includes guidelines for the selection of the critical impact point for crash tests on redirecting-type safety hardware.

7. It provides information related to enhanced measurement techniques related to occupant risk and it incorporates guidelines for device installation and test instrumentation.

8. The three basic evaluation criteria categories remain the same. The occupant risk criteria retains the use of the flail space model, but defines preferred and maximum levels of occupant impact velocity and acceleration. The lateral occupant impact velocity limits were altered to be equivalent to longitudinal limits to reflect recent research findings. The redirection criteria were altered to incorporate a limiting 12 m/s vehicular velocity change requirement in the longitudinal direction.

9. It reflects a critical review of methods and technologies for safety-performance evaluation, such as surrogate test vehicles and computer simulations, and incorporates state-of-the-art methods in the procedures.

10. It provides optional criteria, established by others, for side impact testing.

The evolution of the knowledge of roadside safety and performance evaluations is reflected in this document. Inevitably, parts of this document will need to be revised in the future, but it is the consensus opinion of the project panel and the many reviewers of these procedures that this document will effectively meet the needs for uniform safety performance evaluation procedures into the 21st century.

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RECOMMENDED PROCEDURES FOR THE SAFETY PERFORMANCE EVALUATION OF HIGHWAY FEATURES

SUMMARY

Procedures are presented for conducting vehicle crash tests and in-service evaluation of roadside safety features or appurtenances. The features covered by these procedures include (1) longitudinal barriers such as bridge rails, guardrails, median barriers, transitions, and terminals; (2) crash cushions; (3) breakaway or yielding supports for signs and luminaries; (4) breakaway utility poles; (5) truck-mounted attenuators; and (6) work zone traffic control devices. The purpose of the procedures is to promote the uniform testing and inservice evaluation of roadside safety features so that highway engineers may confidently compare the safety performance of designs that are tested and evaluated by different agencies. These procedures are presented in the form of guidelines that describe how a feature should be tested and evaluated. The identification of specific new, existing, or modified features for testing and evaluation; the selection of a level of service that the feature is to meet; and the establishment of acceptable performance criteria for an agency are policy decisions that are beyond the purview of this document. The procedures are directed at the safety performance of roadside safety features; other service requirements such as economics and aesthetics are not considered here.

These procedures are devised to subject roadside safety features to severe vehicle impact conditions rather than to typical or average highway situations. Although the innumerable highway-site and safety-feature application conditions that exist are recognized, it is impractical or impossible to duplicate these in a limited number of standardized tests. Hence, the approach has been to normalize test conditions: straight longitudinal barriers are tested although curved installations exist; flat grade is recommended even though installations are sometimes situated on sloped shoulders and behind curbs; idealized soils are specified although roadside safety hardware are often founded in poor soil or frozen ground. These normalized test conditions have a significant effect on a feature's performance but are of secondary importance when comparing results of two or more systems.

For vehicle crash testing, specific impact conditions are presented for vehicle mass, speed, approach angle, and point on the safety feature to be hit. Standard test vehicle types are defined for mini-compact and subcompact passenger cars, standard ¾-ton pickup trucks, single-unit trucks, and tractor-trailer cargo trucks. Impact speeds range from 35 to 100 km/h (approximately 20 to 60 mph), and approach angles vary from 0 to 25 degrees. Three primary appraisal factors are presented for evaluating the crash test performance:

structural adequacy, occupant risk, and after-collision vehicle trajectory. Depending on the safety feature's function, it should contain, redirect, permit controlled penetration of the impacting vehicle, or permit a controlled stop in a predictable manner to satisfy structural adequacy requirements. Occupant risk relates to the degree of hazard to which occupants in the impacting vehicle would be subjected. It is measured in terms of the velocity at which a hypothetical unrestrained occupant strikes some part of the vehicle interior such as the instrument panel, window, or door, and the subsequent occupant ridedown accelerations. The after-collision vehicle trajectory is assessed based on the probable involvement of other traffic due to the path or final position of the impacting car and on the chances of undesirable post-impact vehicle behaviors such as pocketing or snagging. It is recognized that vehicle crash tests are complex experiments and are difficult to replicate because of the imprecise control of test conditions and the sometimes random and unstable behavior of dynamic crush and fracture mechanisms. Accordingly, care should be exercised in interpreting the results.

In-service evaluation is used in the final stage of develoment of new or extensively modified roadside safety features and has the purpose of appraising actual performance during a broad range of collision, environmental, operational, and maintenance situations for typical site and traffic conditions. This report updates the guidelines for in-service evaluation first provided in *NCHRP Report 230,* recognizing the complex nature of vehicular accidents and the limited resources of agencies responsible for monitoring the performance of new or modified safety features.

Appendices to this report provide (a) a commentary on the basis for the procedures; (b) soil specifications for the installation of safety features; (c) specifications for the instrumentation of the tests; (d) a summary on the analytical and experimental tools that can be used in the research and development phase for safety features; (e) a methodology for quantifying the damage to the occupant compartment of a test vehicle; (f) a description of calculation procedures for alternate measures of occupant risk (e.g., THIV, PHD, and ASI); and (g) a methodology for conducting and evaluating side impact tests.

CHAPTER 1
INTRODUCTION

1.1 PURPOSE AND SCOPE

The purpose of this report is to present uniform guidelines for the crash testing of both permanent and temporary highway safety features and recommended evaluation criteria to assess test results. Guidelines are also presented for the in-service evaluation of safety features. These guidelines and criteria, which have evolved over the past 30 years, incorporate current technology and the collective judgment and expertise of professionals in the field of roadside safety design. They provide (1) a basis on which researchers and user agencies can compare the impact performance merits of candidate safety features, (2) guidance for developers of new safety features, and (3) a basis on which user agencies can formulate performance specification for safety features.

Although these guidelines are representative and applicable to an array of highway features and traffic conditions, they should not be viewed as all-inclusive. Experience has shown that as new designs are developed, current test procedures may not properly evaluate critical conditions for these designs. Experience has also shown that evaluation and testing of features not addressed by the current guidelines will be made. Therefore, specific features and site conditions may arise that require special tests and evaluation criteria. Deviations from the guidelines are warranted when other tests or evaluation criteria are more appropriate and representative of site or design conditions.

With the guidelines, a given feature may be tested to one of six "test levels." A test level is defined by impact conditions (speed and angle of approach) and the type of test vehicle (ranging in size from a small car to a fully loaded tractor-trailer truck). A feature designed and tested for a low test level would generally be used on a low service level roadway; for example, a rural collector or local road, an urban street, or perhaps in a restricted work zone. A feature designed and tested for a high test level would typically be used on a high service level roadway; for example, a freeway.

It must also be noted that features that meet a given test level will generally have different performance characteristics. A concrete barrier and a cable barrier can be designed to satisfy a given test level, but it is obvious that the cable barrier will deflect much more, impact forces will be less for the cable barrier, and vehicle trajectory will certainly not be the same for both barriers. A terminal can be designed to have full redirection capabilities along its entire length (termed a *nongating terminal),* or it can be designed to allow controlled penetration along a portion of its length (termed a *gating terminal).* Both designs can be made to satisfy a given test level. A crash cushion can be designed to redirect a vehicle impacting the side of the cushion (termed a *redirective crash cushion),* or it can be designed to decelerate the vehicle to a stop when impacted on the side (termed a *nonredirective crash cushion).* While the guidelines were formulated purposely to offer the user considerable latitude in the design and testing of a feature, it is not the purpose nor is it within the purview of this document to determine where a feature, satisfying a given test level and having specific performance characteristics, would have application. That determination rests with the appropriate transportation agency responsible for the implementation of the safety feature.

These guidelines supersede those contained in *NCHRP Report 230 (1).* Major revisions incorporated herein relative to *Report 230* include (a) changes to the test vehicles, (b) changes to the number and impact conditions of the test matrices, (c) adoption of the concept of "test levels" as opposed to "service levels," (d) changes to the evaluation criteria, (e) inclusion of test guidelines for additional features, and (e) adoption of the International System of Units (SI). The **SI** was adopted in concert with national efforts to convert the English system of units to the internationally accepted SI.

Procedures presented herein involve vehicular tests to evaluate the impact performance of permanent and temporary highway safety features. Performance is evaluated in terms of the degree of hazard to which occupants of the impacting vehicle would be exposed, the structural adequacy of the safety feature, the hazard to workers and pedestrians that may be behind a barrier or in the path of debris resulting from impact with a safety feature, and the post-impact behavior of the test vehicle. Other factors that should be evaluated in the design of a safety feature, such as aesthetics, costs (initial and maintenance), and durability (ability to withstand environmental conditions such as freezing and thawing, wind-induced fatigue loading, effects of moisture, ultraviolet radiation, etc.) are not addressed.

1.2 DEVELOPMENT OF A SAFETY FEATURE

Development of a safety feature from its inception to the time it becomes operational is often a long and arduous process. Depending on its developmental phase, a feature is generally placed in one of three categories: (1) research and development, (2) experimental, and (3) operational. During the research and development phase, the design evolves and is eventually subjected to a set of crash tests. Results of the crash tests are assessed in accordance with a set of evaluation criteria. If the feature satisfies the evaluation criteria, it is then placed in the experimental phase. Subject to interest by user agencies, it may then be installed, often on a limited basis. The purpose of the experimental phase is to closely monitor the in-service performance of the feature. If it performs satisfactorily in service, it may then be placed in the operational phase; but its performance should continue to be monitored.

Shown in Figure 1.1 is a flow chart illustrating the process through which a new safety feature is typically subjected. Steps 1 through 4 involve the research and development phase. Factors that should be considered and design techniques that may be used during this phase are indicated. Note that during step 3 it may be beneficial to use analytical and experimental tools described in Appendix D. Step 4 is addressed in Chapters 2 through 6. If a device meets the recommended impact performance criteria, it is then classified as experimental and steps 6 through 9 may then be performed. If acceptable to a user agency, standard plans are prepared (step 8) and the feature is installed on a limited basis. Step 9 involves in-service evaluation, and recommended procedures for conducting this evaluation are given in Chapter 7. This step is perhaps the most important part of the development of a feature. If its in-service is acceptable, it may then be classified as operational (step 11) and fully implemented. However, monitoring of the in-service performance of the device should continue in this phase (step 12).

To be noted is that, while there are specific guidelines (as presented herein) for determining if a device is acceptable as an experimental device, there are no widely accepted, specific criteria to assess its in-service performance. Consequently, considerable judgment must be exercised in determining when a device has met the in-service requirements and thus becomes operational. Some agencies may choose to skip steps 9 and 10, declaring the feature operational on the basis of expert opinion. However, it is strongly recommended that step 12 never be omitted.

1.3 PERFORMANCE GOALS

A goal of a highway safety feature is to provide a forgiving roadway and roadside for an errant motorist. The safety goal is met when the feature either contains and redirects the vehicle away from a hazardous area, decelerates the vehicle to a stop over a relatively short distance, readily breaks away or fractures or yields, allows a controlled penetration, or is traversable, without causing serious injuries to the vehicle's occupants or to other motorists, pedestrians, or work zone personnel.

Ideally, the roadside would be clear of all obstructions, including unnecessary roadside hardware, and be traversable so that an errant motorist could recover control of the vehicle and stop or return to the pavement. However, there are numerous roadside areas that cannot practically be cleared of all fixed objects or made traversable. At these sites, the use of an appropriate safety feature or safety treatment is intended to reduce the consequences of a run-off-the-road incident.

As discussed in Section 1.2, the safety performance of a highway feature cannot be measured by a series of crash tests only. Testing must be viewed as a necessary but not sufficient condition to qualify a feature for operational status. Tests results are judged on the basis of evaluation criteria presented in Chapter 5.

1.4 PERFORMANCE LIMITATIONS

Even the most carefully researched device has performance limits dictated by physical laws, crashworthiness of vehicles, and limitation of resources. For example, at some sites, sufficient space is lacking to safely decelerate a vehicle, regardless of the crash cushion design. Irrespective of the breakaway feature, certain timber utility poles may be so massive that the impacting vehicle is abruptly decelerated, thus limiting achievable safety performance without a change in support technology. There is no assurance that a feature meeting the test recommendations herein for a tracking vehicle will perform satisfactorily if impacted by a vehicle sliding sideways. Some vehicle types may lack sufficient size or mass or necessary crashworthiness features such as interface strength, stiffness, controlled crush properties, and stability to provide occupants with an acceptable level of protection. Note that no provisions are made herein for the design and testing of safety features for two-wheeled vehicles-powered or unpowered. Longitudinal barriers that will contain and redirect the smaller passenger cars and yet have strength capability to redirect a tractor-trailer or intercity bus are relatively expensive. Seemingly insignificant site conditions such as curbs, slopes, and soft soil conditions can cause or contribute to the unsuccessful performance of a safety feature.

For these reasons, safety features are generally developed and tested for selected idealized situations that are intended to encompass a large majority, but not all, of the possible in-service collisions. Even so, it is essential that test results be evaluated and interpreted by competent researchers and that the evaluation be guided by sound engineering judgment.

While it is to be expected that certain features, meeting all test and evaluation criteria recommended herein, will have untested "windows of vulnerability" in service, the corollary of this is also to be expected. That is, it is to be expected that certain features not meeting all test and evaluation criteria recommended herein will still be cost-effective alternatives for selected in-service applications.

1.5 ORGANIZATION OF REPORT

Chapter 2 describes the test parameters, including those associated with the test facility, the test article, and the test and its simulated occupants. Chapter 3 gives the test conditions recommended for each of the respective features. It also presents recommended tolerances on impact conditions and a procedure for identifying the critical impact point for certain features. Chapter 4 describes the data acquisition systems recommended and identifies parameters that should be measured before, during, and after the test. The evaluation criteria used to assess test results are contained in Chapter 5. Chapter 6 recommends the manner in which a given test and its results are to be documented. Chapter 7 contains guidelines on how the in-service performance of a feature should be evaluated. Appendix A is a commentary on those items contained in Chapters 2 through 6 and presents further elaboration and discussion. Appendix B contains soil specifications, reproduced with permission from the American Association of Highway and Transportation Officials, and Appendix C contains instrumentation specifications, reproduced with permission from the Society of Automotive Engineers, Inc. Appendix D summarizes analytical and experimental tools that can be used in the research and development phase; Appendix E contains a methodology for quantifying the damage to the occupant compartment; Appendix F describes calculation procedures for the Theoretical Head Impact Velocity (**THIV**), Post-Impact

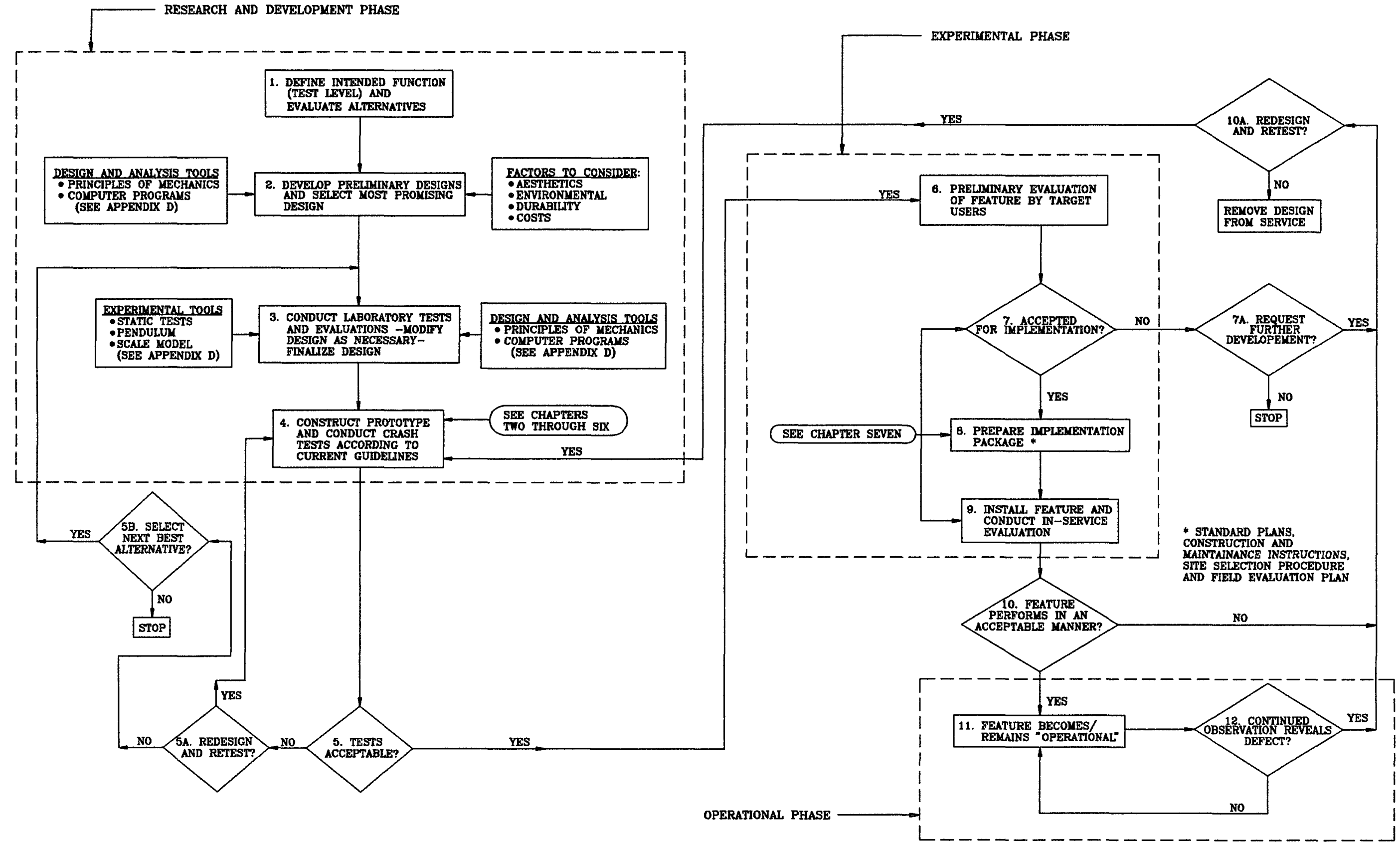

RESEARCH AND DEVELOPMENT PHASE
1. DEFINE INTENDED FUNCTION (TEST LEVEL) AND EVALUATE ALTERNATIVES
DESIGN AND ANALYSIS TOOLS
• PRINCIPLES OF MECHANICS
• COMPUTER PROGRAMS (SEE APPENDIX D)
2. DEVELOP PRELIMINARY DESIGNS AND SELECT MOST PROMISING DESIGN
FACTORS TO CONSIDER:
• AESTHETICS
• ENVIRONMENTAL
• DURABILITY
• COSTS
EXPERIMENTAL TOOLS
• STATIC TESTS
• PENDULUM
• SCALE MODEL (SEE APPENDIX D)
3. CONDUCT LABORATORY TESTS AND EVALUATIONS –MODIFY DESIGN AS NECESSARY– FINALIZE DESIGN
DESIGN AND ANALYSIS TOOLS
• PRINCIPLES OF MECHANICS
• COMPUTER PROGRAMS (SEE APPENDIX D)
4. CONSTRUCT PROTOTYPE AND CONDUCT CRASH TESTS ACCORDING TO CURRENT GUIDELINES
SEE CHAPTERS TWO THROUGH SIX
YES
5B. SELECT NEXT BEST ALTERNATIVE?
YES
NO
STOP
YES
NO
5A. REDESIGN AND RETEST?
NO
5. TESTS ACCEPTABLE?
YES
EXPERIMENTAL PHASE
YES
10A. REDESIGN AND RETEST?
NO
REMOVE DESIGN FROM SERVICE
YES
6. PRELIMINARY EVALUATION OF FEATURE BY TARGET USERS
7. ACCEPTED FOR IMPLEMENTATION?
NO
7A. REQUEST FURTHER DEVELOPEMENT?
YES
NO
STOP
YES
SEE CHAPTER SEVEN
8. PREPARE IMPLEMENTATION PACKAGE *
9. INSTALL FEATURE AND CONDUCT IN–SERVICE EVALUATION
* STANDARD PLANS, CONSTRUCTION AND MAINTAINANCE INSTRUCTIONS, SITE SELECTION PROCEDURE AND FIELD EVALUATION PLAN
10. FEATURE PERFORMS IN AN ACCEPTABLE MANNER?
NO
YES
11. FEATURE BECOMES/ REMAINS "OPERATIONAL"
12. CONTINUED OBSERVATION REVEALS DEFECT?
YES
NO
OPERATIONAL PHASE

Figure 1.1. Flow chart for development of a safety feature.

Head Deceleration (PHD), and the Acceleration Severity Index (ASI); and Appendix G contains a methodology for conducting and evaluating side impacts. Appendix H contains the references and bibliography, and Appendix I is a glossary of terms. Sample SI conversion factors are presented in Appendix J.

1.6 INTERNATIONAL HARMONIZATION OF PERFORMANCE STANDARDS

Concurrent with the preparation of this report, the European Committee for Standardization (CEN) was preparing a similar document (77). Discussions were held to explore ways impact performance standards between the U.S. and CEN could possibly be harmonized. However, scheduling deadlines precluded the development of substantive agreements. Nonetheless, based on expressed interests by U.S. government agencies and CEN agencies, it is expected that common standards will be adopted in future reports, at least for certain elements of the test and evaluation process. Toward this end, it was decided to incorporate the SI system of units herein. Furthermore, it is suggested that certain evaluation criteria adopted by CEN be calculated and reported for tests in the U.S. for purposes of comparison with comparable U.S. criteria (see Section 5.3).

CHAPTER 2

TEST PARAMETERS

2.1 GENERAL

Highway safety features are evaluated for a particular test level through a series of vehicular crash tests for a limited range of impact conditions, as presented in Chapter 3. Many important test parameters have been standardized to arrive at practical stratification of tests (test matrices) and to enhance the degree of test replication. Care must be exercised in the interpretation of test results and in the projection of the results to in-service performance. Good performance under ideal test conditions does not ensure comparable performance under in-service conditions. As discussed in Chapter 1, the evaluation process should not stop with successful completion of tests recommended herein. In-service evaluation of the feature is perhaps more important than crash test evaluation and should be pursued as recommended in Chapter 7.

2.2 TESTING FACILITY

A flat surface, preferably paved, should be used when accelerating the test vehicle to the desired speed and to provide for unrestricted trajectory of the vehicle following impact. The surface should be free of curbs, swales, ditches, or other irregularities that could influence impact or post-impact behavior of the vehicle except when test conditions require such features. If necessary, a flat compacted soil or sod surface should adjoin the paved approach area to replicate conditions for safety features normally surrounded by an unpaved surface, so that post-impact vehicular behavior can be properly assessed.

2.2.1 Soil

Impact performance of some features depends on dynamic soil-structure interaction. Longitudinal barriers with soil embedded posts and soil-embedded support structures for signs and luminaires are such features. When feasible, these features should be tested with soil conditions that replicate typical inservice conditions. Unfortunately, soil conditions are known to vary with time and location, even within relatively small geographical areas. Therefore, except for special test conditions, it is necessary to standardize soil conditions for testing. In the absence of a specific soil, it is recommended that all features whose impact performance is sensitive to soil-structure interaction be tested with soil described in Section 2.2.1.1. Product developers and user agencies should assess the potential sensitivity of a feature to foundation conditions, and if the feature is likely to be installed in a soil that could be expected to degrade its performance, testing in one or more of the special soils described in Section 2.2.1.2 may be appropriate.

For tests involving soils recommended in Sections 2.2.1.1 and 2.2.1.2, the length, width, and depth of the embedment material should be sufficient to effectively eliminate influence of native soil on impact performance. If these dimensions cannot be determined, it is recommended the test soil depth be approximately 1.2 times the embedment length of the test article and that the lateral extent of the test soil, outside an envelope of the embedded portion of the test article, be approximately 1.3 times the embedment length.

2.2.1.1 Standard Soil

It is recommended that the standard soil meet AASHTO standard specifications for "Materials for Aggregate and Soil Aggregate Subbase, Base and Surface Courses," designation M 14765 (1990), grading A or B (see Appendix B, Part 1). It should be compacted in accordance with AASHTO *Guide Specifications for Highway Construction,* Sections 304.05 and 304.07 (see Appendix B, Part 3). The soil should be recompacted, as necessary, before each test to meet density requirements of the *Guide Specifications.* The soil should be well drained at the time of the crash test. The test should not be performed if the ground is frozen or if the soil is saturated unless the test is specifically designed to evaluate these conditions (see next section).

2.2.1.2 Special Soils

The following guidelines may be used to evaluate a feature in a weak, saturated, or frozen soil.

Weak Soil. It is recommended the weak soil meet AASHTO standard specification for "Fine Aggregate for Portland Cement Concrete," AASHTO designation M 6-87 (see Appendix B, Part 2). The soil should be compacted in accordance with AASHTO *Guide Specifications for Highway Construction,* Sections 304.05 and 304.07 (see Appendix B, Part 3). The soil should be recompacted, as necessary, before each test to meet density requirements of the *Guide Specifications.* The soil should be well drained at the time of the crash test.

Saturated Soil. The "standard soil" and "weak soil" previously described may be used to evaluate impact performance of a feature under saturated soil conditions. Moisture content of the soil should replicate expected in-service conditions.

Frozen Soil. The "standard soil" and "weak soil" described may be used to evaluate impact performance of a feature under frozen soil conditions. The degree to which the soil is frozen, measured in terms of depth and temperature, and its moisture content should replicate expected in-service conditions.

2.2.1.3. Embedment of Test Article

Within the range of expected in-service conditions, the depth and method of embedment should be those likely to reveal the poorest performance of the test article. This may not be predictable, in which case testing should be done under sufficient embedment conditions to reveal the worst performance. The method used in embedding the test article should replicate the method by which the feature will be embedded in service. A sign support is typically embedded by driving the post or stub directly into the soil, by inserting the support in a drilled hole and then backfilling the soil, or by placing the support or stub in a concrete footing. Similar methods are used for embedment of a longitudinal barrier post. Most luminaire support poles are supported by a concrete footing and most utility poles are placed in a drilled hole and then the soil is backfilled.

Some testers have developed universal foundations for testing breakaway devices consisting of multiple-use base plates supported on a very rigid concrete footing. While these foundations reduce testing costs, they effectively eliminate soil-foundation interaction. They also raise questions about feature-to-foundation interface friction and anchor bolt rigidity. If it cannot be shown that these effects are insignificant, the test report should, at a minimum, alert the user agency of potential problems and provide recommendations for foundation systems that will ensure proper breakaway performance.

2.2.2. Special Structures

Ends of roadside and median barriers must generally be anchored by terminal sections, bridge rails must be connected to a deck, and transitions are often attached to a rigid bridge end or wing wall on the downstream end of the transition. It is preferable that in-service designs of these auxiliary structures be used in the crash test when feasible. For example, when testing a roadside barrier, it is preferable that the ends be terminated as they would in service. When this is not feasible, special structures must be constructed. A key requirement of a special end anchorage device for a longitudinal barrier is that it have the capability to resist tensile loads developed in the rail.

Testing of a bridge rail will, in general, require a special support structure, i.e., a simulated bridge deck. For a nonrigid bridge rail where lateral deflection is of concern, the structure to which the railing is attached should simulate edge conditions so that the effect of vehicular penetration beyond the edge of the deck can be properly evaluated. Regardless of the rail's strength, it may be desirable to evaluate structural adequacy of the deck itself for the impact conditions, in which case the deck structure should have the same strength and properties as the in-service structure.

If a universal or generic deck is used in a bridge rail test, it is desirable that impact loads imposed on the deck be measured or computed from measured vehicular response and reported. By so doing, a user agency will have some guidance on how to design a deck that differs from the one used in the test. Procedures that may be used to estimate impact loads from measured vehicular accelerations are presented in references 17 and 72. For tests of a flexible-to-a-rigid longitudinal barrier transition, a prototype bridge end structure or wing wall structure must be constructed. Length, strength, and geometry of the prototype should be sufficient to approximate impact response expected of the in-service bridge end or wing wall. Section 2.3.2.1 contains further recommendation on the prototype bridge end.

2.3. TEST ARTICLE

2.3.1. General

All key elements or materials in the test article that contribute to its structural integrity or impact behavior should be sampled, tested, and the results documented in the test report. Physical and chemical material properties can generally be obtained from the supplier providing the test article(s). To ensure that all critical elements are considered, a careful after-test examination of the test article is essential. Materials should be tested independently when a failure occurs.

Material specifications such as ASTM, AASHTO, and so forth, should be reported for all key elements. Results of random sample tests should confirm that stated specifications have been met and that key elements in the test article were representative of normal production quality. The tester should offer a judgment on effects marginal materials or materials that significantly exceed minimum specifications might have on performance of the test article. In addition, specified but unverified properties of all other materials used in the test article should be reported.

The test article should be constructed and erected in a manner representative of in-service installations and should conform to specifications and drawings of the manufacturer or designer. To assure uniformity and integrity of structural connections, current American Welding Society specifications for highway bridges, Aluminum Association Specifications for Aluminum Bridges and Other Highway Structures, American Institute of Steel Construction bolting procedures, and other relevant documents should be used as appropriate. Deviations from fabrication, specification, or erection details should be delineated in the test report.

2.3.2. Installation Details

2.3.2.1. Longitudinal Barriers

For tests examining performance of the length-of-need section, rails or barrier elements should be installed straight and level and anchored. Horizontally curved installations, sloped shoulders, embankments, dikes, and curbs should be avoided for general performance tests; when used, nonstandard features should be reported. As a general rule, length of the test section excluding terminals or end anchorage devices should be at least three times the length in which deformation is predicted but not less than 23 m for a rigid barrier (one for which little, if any, lateral displacement is anticipated) and 30 m for a flexible barrier such as a metal beam and post roadside barrier. Length of the test section should be such that (1) terminals or end anchorage devices do not influence in an abnormal manner the dynamic behavior of the barrier and (2) the ability of the barrier to contain and redirect the test vehicle in the recommended manner can be clearly ascertained. Exceptions to recommended lengths can be made provided the installation satisfies these two requirements.

A free-standing, unanchored barrier, such as a precast, segmented concrete barrier whose impact performance depends in part on frictional resistance between it and the surface it is resting on, should be tested on a surface that replicates the type it will be placed on in service. If it will be placed on more than one surface in service, it should be tested on the one surface likely to have the most adverse effect on performance, usually the one having the least frictional resistance. Type of surface as well as end anchorages or terminals used should be reported.

The barrier system used for a transition test should be oriented as it would in service. As a general rule, transitions of most concern are those that serve to connect a less stiff barrier on the upstream side to a more stiff barrier on the downstream side, such as the transition from a flexible roadside or median barrier to a rigid bridge rail. In such cases, length, strength, and geometry of the prototype bridge rail or wing wall should be sufficient to approximate impact response expected of the in-service bridge rail or wing wall. It is recommended that the length of the prototype bridge rail or wing wall be 5 m at a minimum. A minimum of 15 m of the more flexible barrier is recommended, exclusive of a properly anchored end.

In some cases, the transition serves to connect longitudinal barriers with similar lateral stiffness but with different geometry. In such cases, a minimum of 15 m each of the adjoining barriers is recommended, exclusive of a properly anchored end.

2.3.2.2 Terminals and Crash Cushions

Reference should be made to Section 3.2.2 for recommendations relative to the manner in which a terminal or crash cushion should be oriented with respect to the vehicular approach direction. When testing terminals, the test article should be erected on level grade. As a rule, a 30-m length-of-need longitudinal barrier section should be attached to the terminal and anchored at the downstream end. If the terminal is designed for a specific longitudinal barrier, the length-of-need section used in the test should be composed of the specific barrier. Exceptions to the recommended length for the length-of-need section are permissible provided the ability of the terminal to stop, contain and redirect, or allow controlled penetration of the test vehicle in the recommended manner can be clearly ascertained.

A rigid, nonyielding backup structure should be used to simulate a highway feature (such as a bridge pier, elevated gore, or bridge end) when appropriate. For crash cushions having side hit redirection capability designed to be struck on one side by direct traffic and on the other side by opposing traffic, the test article should be installed with side hit deflector hardware oriented to accommodate both types of side hits. The crash cushion should be anchored as required by specifications or drawings.

2.3.2.3 Support Structures, Work Zone Traffic Control Devices, and Breakaway Utility Poles

Reference should be made to Section 3.2.3 for recommendations relative to the manner in which a support structure, a work zone traffic control device, or a breakaway utility pole should be oriented with respect to the vehicular approach direction. Testing should verify breakaway or yielding features designed to function identically when impacted from specific directions, such as a breakaway base designed for front or rear impacts, or those features designed to function identically when impacted from any direction, e.g., a breakaway base designed for omnidirectional impacts. Support systems should be fully equipped with full height structures, including sign, mailbox or mailboxes, call box, and mast arm for luminaire. For tests of a luminaire support, it is preferable that an actual luminaire be used rather than a substitute equivalent weight that has often been used in the past. Tests have shown the luminaire/ballast can break loose during impact and consequently may present a hazard to other motorists or to occupants of the impacting vehicle. A full-length utility pole should also be used together with associated cross-arm(s). down guys, and conductors (lines).

The test may involve multiple supports such as multiple mailbox supports or multiple, closely spaced drums in a work zone. Orientation and spacing of these supports should be representative of in-service conditions.

Occasionally, a work zone traffic control device such as a barricade or a plastic drum will overturn or will intentionally be placed in an overturned position along the shoulder in a work zone. A barricade may typically be placed so that its panels are parallel rather than perpendicular to traffic. As such, these devices may pose a greater risk to an errant motorist than in their normal or upright position. If there is a reasonable expectation that such a device will likely be in this orientation during service, it should be tested in the overturned position, provided this orientation poses a greater risk to the motorist than in its normal position. If it cannot be determined which position is more critical, tests in both the normal and the overturned or rotated position should be conducted.

For tests of a sign support system, the area of the sign panel should approximate that of the largest panel that would normally be used on the support system. Sign panel material should be that normally used or to be used on the support system. If panels of different materials, such as plywood, sheet metals, or fiber reinforced plastics, are used with the support system, the test should be conducted with the material expected to pose the greater risk to occupants of the impacting vehicle. If it cannot be determined which material is most critical, it is recommended the test be conducted with the panel of greatest mass. Aspect ratio of the sign (height-to-width ratio) should be typical of the largest panel that would normally be used on the support system. Mounting height of the sign panel (distance from ground to bottom of panel) should be the minimum height the panel would normally be mounted in service unless it can be shown that a higher mounting height would pose a greater risk to occupants of the impacting vehicle.

For tests of a mailbox support system with a single mailbox, the mailbox should be the largest that would normally be used on the support system. For tests of a mailbox support system with multiple mailboxes, the number and size of mailboxes should be the largest that would normally be used on the support system.

2.3.2.4 Truck-Mounted Attenuators (TMA)

For tests with the 700C or 820C vehicles, the supporting truck should be placed against a rigid barrier to prevent any forward movement.

TABLE 2.1. Recommended properties of 700C, 820C, and 2000P test vehicles

Property	700C (Small Car)	820C (Small Car)	2000P (Pickup Truck)
MASS (kg)			
Test Inertial	700 ± 25	820 ± 25	2000 ± 45
Dummy	75	75	---
Max. Ballast	70	80	200
Gross Static	775 ± 25	895 ± 25	2000 ± 45
DIMENSIONS (cm)			
Wheelbase	230 ± 10	230 ± 10	335 ± 25
Front Overhang	75 ± 10	75 ± 10	80 ± 10
Overall Length	370 ± 20	370 ± 20	535 ± 25
Track Width[b]	135 ± 10	135 ± 10	165 ± 15
CENTER OF MASS LOCATION[a] (cm)			
Aft of Front Axle	80 ± 15	80 ± 15	140 ± 15
Above Ground	55 ± 5	55 ± 5	70 ± 5
LOCATION OF ENGINE	Front	Front	Front
LOCATION OF DRIVE AXLE	Front	Front	Rear
TYPE OF TRANSMISSION	Manual or Automatic	Manual or Automatic	Manual or Automatic

[a] For "test inertial" mass
[b] Average of front and rear axles

For tests with the 2000P vehicle, the supporting truck should be placed on a clean, dry, paved surface. Asphaltic or portland cement concrete surfaces are recommended. Conditions such as a polished surface or a bleeding asphaltic surface that could lower available tire-pavement friction should be avoided.

For tests with the 2000P vehicle, the supporting truck should be in second gear with park brakes on. Front tires should have no steering angle, that is, they should not be turned to the left or to the right.

2.4. TEST VEHICLES

2.4.1. Description of Test Vehicles

Impact performance of a highway safety feature may be evaluated by use of a commercially available, production model vehicle or by a validated surrogate vehicle. To date, with the exception of breakaway support structures, safety features have been evaluated by use of production model vehicles.

2.4.1.1 Production Model Test Vehicles

Recommended properties of production model test vehicles are given in Tables 2.1 and 2.2. Vehicles 700C and 820C are small cars, vehicle 2000P is a pickup truck, vehicle 8000S is a single-unit truck, vehicle 36000V is a tractor/van-type trailer unit, and 36000T is a tractor/tank trailer unit. Note that the numeric portion of the test vehicle designation is the vehicle's mass in kilograms.

In general, any test vehicle should be in good condition and free of major body damage and missing structural parts (i.e., doors, windshield, hood, etc.). Special purpose vehicles are not generally acceptable because they do not possess suspension and handling characteristics found in typical vehicles. Any manufacturer-installed equipment (power brakes and steering, air conditioning, etc.) is permitted so long as the equipment is contained within the body shell. The vehicle fuel tank should be purged and the battery removed from remotely powered test vehicles to reduce exposure to needless hazards. The bumpers on vehicles 700C, 820C, and 2000P should be standard equipment and unmodified for the test; configuration and height above grade should be reported. Tire size should be in accordance with the manufacturer's suggested size for each respective test vehicle. Highway, all season tires should be used on test vehicles 700C, 820C, and 2000P; mud or snow tires should not be used.

It is recommended the 700C vehicle be selected from one of the top two models, in terms of sales for the given model year, for cars with a curb mass of approximately 750 kg or less. It is recommended that the 820C vehicle be selected from one of the top two models, in terms of sales for the given model year, for cars with a curb mass in the 750 kg to 845 kg range. Car sales data may be obtained from the annual "Market Data Book," *Automotive News,* or "Automotive Year Book," *Wards Reports, Inc.* Reference should be made to the commentary for further discussions relative to the 820C vehicle.

TABLE 2.2 Recommended properties of 8000S, 36000V, and 36000T test vehicles

Property	8000S (Single-Unit Van Truck)	36000V (Tractor/Van Trailer)			36000T (Tractor/Tank Trailer)		
		Tractor[g]	Trailer[e]	Combination	Tractor[g]	Trailer[f]	Combination
Mass (kg)							
Curb	5,450 ± 450	N/S	N/S	13,200 ± 1,400	N/S	N/S	13,200 ± 1,400
Ballast[d]	As Needed	N/A	As Needed	N/A	N/A	As Needed	N/A
Test Inertial	8,000 ± 200	N/S	N/S	36,000 ± 500	N/S	N/S	36,000 ± 500
Dimensions (cm)							
Wheelbase	535 (max)	480 (max)	N/S	N/A	480 (max)	N/S	N/A
Overall Length	870 (max)	N/S	1,525 (max)	1,985 (max)	N/S	N/S	1,985 (max)
Trailer Overhang[a]	N/A	N/A	220 (max)[c]	N/A	N/A	185 (max)	N/A
Cargo Bed Height[b] (Above Ground)	130 ± 5	N/A	132 ± 5	N/A	N/A	N/A	N/A
Center of Mass Location (cm)							
Ballast[d] (Above Ground)	170 ± 5	N/A	185 ± 5	N/A	N/A	205 ± 10	N/S
Test Inertial (Above Ground)	125 ± 5	N/S	N/S	N/S	N/S	N/S	N/S

a. Distance from rearmost part of trailer to center of trailer tandems.

b. Without ballast

c. If trailer equipped with slide axies, they should be set at rearmost position.

d. See section 2.4.2.2 for recommended ballasting procedures.

e. It is preferable that the trailer structure be of the "semi-monoscope" type construction. It is preferable that a sliding undercarriage (slide axles) be used to attach the trailer tandems to the taller frame.

f. It is preferrable that a gasoline tank with an elliptical cross section be used.

g. Tractor should be a cab-behind-engine model, not a cab-over-engine model.

N/A – Not Applicable. N/S-Not Specified

Although it may be possible to meet recommended properties of the 2000P vehicle with a $^{1}/_{2}$-ton pickup, a $^{3}/_{4}$-ton pickup is recommended. It should have a regular or conventional cab; it should not have an extended or "crew" cab. It should have a regular or conventional bed; it should not have a special bed such as a "Sportside," a "Stepside," or a "Flareside." It should have only a rear drive axle. It is acceptable for the pickup to have a heavy-duty suspension option as provided by the manufacturer. Special suspension systems that alter the as-manufactured center of-mass location are to be avoided.

Model year of the 700C, 820C, and 2000P vehicles should be within 6 years of the year of test unless otherwise specified by the user agency. Exceptions to the age limitations are acceptable if it can be demonstrated that key properties of the test vehicle are essentially the same as those of a vehicle meeting all above recommended requirements. Key properties include those given in Table 2.1 plus unspecified properties that may change with succeeding model years such as dynamic force-deformation properties of the bumper and frontal structure of the vehicle and vehicular profile as defined by bumper height, hood height, hood sweep, windshield sweep, and height of windshield.

Three heavy test vehicles are included in Table 2.2 along with recommended properties. Although tests have been conducted with each of these vehicles, experience accumulated to date is limited and insufficient to clearly establish appropriateness of these vehicles for testing safety features or to establish well defined specifications for these vehicles.

No age limit is specified for the heavy test vehicles. However, if feasible they should be of recent vintage. They should be representative of widely used designs. The bumper on the truck/tractor should be original equipment or one with minimal structural capacity. Large, stiff, custom-made bumpers should not be used.

The trailer for the 36000V test vehicle should have a semimonocoque structural design. It should have a sliding undercarriage (slide axles) to attach the trailer frame to the tandem wheels. These axles should not be altered in any way for the test.

2.4.1.2 Surrogate Test Vehicles

A surrogate test vehicle or device may be used in lieu of a production model test vehicle provided (1) it can be demonstrated that the surrogate possesses essential properties of the production model test vehicle it is intended to replicate and (2) the production model vehicle it is intended to replicate meets recommendations of Section 2.4.1.1. A method for identifying essential properties and for validating a surrogate device for evaluating breakaway supports is given in reference 2. Note, however, that at the time of this writing there was no widely accepted methodology whereby a surrogate device could be designed and validated to replicate a given vehicle impacting a given safety feature. In absence of such a methodology, determination of validity and relevance of a surrogate device must be made by the appropriate user agency, that is, the agency responsible for selection and installation of the safety feature being evaluated by the surrogate.

In recent years, two types of surrogate devices have been used in the evaluation of breakaway supports for luminaires and signs, namely, a four-wheeled bogie (2) and a pendulum (3). The bogie has been used as a surrogate for a small car and has been used for both low- and high-speed tests. The pendulum has been used as a surrogate for a small car for low-speed impacts.

Current bogies do not have the capability to assess effects of test article contact with the roof, windshield, or undercarriage of a vehicle. As such, they should not be used when such contacts are anticipated. Previous testing has shown that features such as a mailbox, emergency call-box, and flashing lights placed on drums or barricades tend to separate from their support during a high-speed impact and thus have the potential for impacting and penetrating the windshield. Bogie tests in which these types of contact occur, or would occur if a production model vehicle had been used, should be repeated with a production model vehicle.

It is recommended that the surrogate be constructed to replicate properties of a specific production model vehicle as opposed to a generic vehicle. Acceptable production model vehicles that may be replicated are those meeting requirements of Section 2.4.1.1.

2.4.1.3 Truck Used in TMA Test

The truck or supporting vehicle to which an attenuator or crash cushion is attached for a TMA test should be representative of the type and mass (including typical cargo) of the vehicle commonly used in service. If different types and masses of vehicles are used, consideration should be given to conducting the recommended test series (see Section 3.2.4) using a vehicle at both the lower and upper extremes in terms of mass. In absence of a common support vehicle, it is recommended that TMA tests be conducted with a support vehicle having a test inertial mass of 9000 + 450 kg.

It may be possible to extrapolate results of a TMA test for supporting vehicles of differing mass. However, at the time of this writing no known validated procedures exist to make such extrapolations.

2.4.2. Mass Properties

2.4.2.1. Test Vehicle

Vehicle mass properties are important factors in the performance of a highway safety feature. Properties of sprung and unsprung mass, curb mass, test inertial mass, dummy mass, ballast mass, and test equipment mass are normally considered in some aspect of vehicle testing. For this document, mass properties of most importance are:

1. *Curb mass-mass* of test vehicle in its standard manufacturer condition, in which all fluid reservoirs are filled and the vehicle contains no occupants and cargo. In general, curb mass of vehicles 700C, 820C, and 2000P should not vary significantly from test inertial mass.
2. *Test inertial mass-mass* of test vehicle (including sprung and unsprung mass) and all items including ballast and test equipment rigidly attached to the vehicle structure. Mass of dummies, irrespective of the degree of restraint, is not included in test inertial mass.
3. *Loose ballast mass-mass* of simulated, unrestrained cargo used in special tests to evaluate effects of loose cargo.

4. *Gross static mass-total* of test inertial and dummy masses. For special tests it is the sum of test inertial mass, dummy mass, and loose ballast mass.
5. *Dummy mass-mass* of surrogate occupant.

Test vehicles 700C, 820C, and 2000P should be selected so that minimal adjustments to the curb mass will be necessary to achieve the test inertial mass. With the exception of seats, spare tires, fuel tank, battery, fluids and optional equipment, components should not be removed from the vehicle to meet mass requirements.

2.4.2.2 Ballast

For vehicles 700C, 820C, and 2000P, fixed ballast may be added as necessary to bring test inertial mass within limits of Table 2.1. Total mass added should not exceed 10 percent of the nominal test inertial mass given in Table 2.2. Ballast may be added in the following manner:

700C and 820C Vehicles. Concrete or metal blocks may be placed in the passenger compartment and rigidly attached to the vehicle structure by metal straps or other devices capable of sustaining expected impact loads (if expected loads are unknown attachments may be designed for loads equal to a minimum of 20 times the blocks' mass). The ballast should be placed in a manner that will minimize changes to the location of the center of mass of the vehicle in its "curb mass" configuration, consistent with needs of test equipment and its placement.

2000P Vehicle. Concrete or metal blocks may be placed in the passenger compartment and rigidly attached to the vehicle structure by metal straps or other devices capable of sustaining expected impact loads (if expected loads are unknown, attachments may be designed for loads equal to a minimum of 20 times the blocks' mass). Ballast should be placed in a manner that will minimize changes to the location of the center of mass of the vehicle in its "curb mass" configuration, consistent with needs of test equipment and its placement. As an alternative, metal plates may be placed and rigidly attached to the bed of the pickup truck immediately aft of the passenger compartment by metal straps or other devices capable of sustaining expected impact loads. The vertical dimension (height) of the plates should not exceed 7.5 cm. The plates should be oriented symmetrically with respect to the fore-aft centerline of the vehicle and in a manner that will minimize changes to the position of the center of mass of the vehicle for its "curb mass" configuration.

For the 8000S, 36000V and 36000T vehicles, it will be necessary to add ballast to simulate cargo and to achieve recommended nominal test inertial mass given in Table 2.2. It is recommended this be done as follows:

8000S Vehicle. Typically, it will be necessary to add approximately 2600 kg of ballast to the 8000S vehicle to obtain the nominal test inertial mass given in Table 2.2. This can be achieved by sand bags on pallets, bales of hay, or other means. To the extent possible, the ballast should be uniformly distributed along the length and width of the van. The ballast should be firmly secured to prevent movement during and after impact.

36000V Vehicle. Typically, it will be necessary to add approximately 23,000 kg of ballast to the trailer of the 36000V vehicle to obtain nominal test inertial mass given in Table 2.2. This can be achieved by sand bags on pallets or other means. To the extent possible, the ballast should be uniformly distributed along the length and width of the trailer. The ballast should be firmly secured to prevent movement during and after impact.

36000T Vehicle. Water should be added to the tank trailer to achieve nominal test inertial mass given in Table 2.2. Most tank trailers are compartmentalized and, if all compartments are filled with water, the recommended test inertial mass may be exceeded. In such a case, it is recommended selected compartments *befilled* so as to provide an approximately uniform distribution of ballast along the length of the trailer and to provide the highest possible center of mass of the ballast.

Support Truck Used in TMA Test. Depending on the type of support truck used in a TMA test, it may be necessary to add ballast to achieve the desired test inertial mass. Necessary ballast should be firmly secured to prevent movement during and after impact.

2.4.3. Propulsion and Braking

The test vehicle may be pushed, towed, or self-powered to the programmed test speed. If pushed or towed, the prime mover should be disengaged prior to impact permitting the vehicle to be "free-wheeling" during and after the collision. For self powered vehicles, the ignition should be turned off just prior to impact. Application of brakes should be delayed as long as safely feasible to establish the unbraked runout trajectory. As a minimum, brakes should not be applied until the vehicle has traveled at least two vehicle lengths plus 25 m from the point of last contact with the test article. Position of the vehicle at the time of brake application should be reported for each test.

To a very limited extent, live drivers have been used to conduct crash tests when risks are believed to be very small. In those cases, the driver accelerates the vehicle to the desired impact speed and steers the vehicle into the test article or feature at the desired impact angle. Special precautions are usually taken to protect the driver from unexpected behavior of the vehicle or test article. However, experience has demonstrated that the vehicle or test article in even a seemingly simple test can respond in a totally unexpected manner. Therefore, unless risks are *known* to be extremely small or unless the test necessarily involves the effects of driver response (such as tests to evaluate driver response to edge-drop problems), use of a live driver is strongly discouraged.

2.4.4. Guidance

Method of guidance of the test vehicle prior to impact is optional, providing the guidance system or its components do not effect significant changes in the vehicle dynamics during and immediately after the collision. The steering wheel should not be constrained unless essential for test safety purposes. If the steering wheel is to be constrained, the nature of this constraint should be clearly documented.

2.5 SURROGATE OCCUPANTS

As discussed below, a surrogate occupant is recommended for use in certain tests to evaluate the added mass effect and/or to evaluate kinematics of the occupant. However, the *primary* measure of risk is determined by the hypothetical "flail space" model.

As discussed in Section 5.3, a dummy may be used to supplement other occupant risk measures; if this is desired, it is recommended the Hybrid III dummy be used. However, it must be noted that the Hybrid III dummy is valid for *frontal or head on impacts* only, that is, those in which dummy movement is essentially parallel to the longitudinal vehicular axis (x-axis as shown in Figure 4.6). There is no dummy capable of accurately simulating the kinetics and kinematics of an occupant for oblique movements, that is, those in which occupant movement has both x and y components. Oblique occupant movement typically occurs when the vehicle is redirected away from the feature being impacted, such as a longitudinal barrier. Instrumentation of the surrogate occupant is not recommended unless the Hybrid III dummy is used.

An anthropometric dummy, such as the Hybrid I or II dummy, or other means such as sand bags, a concrete block, and so on, may be used to simulate the added mass effect of an occupant. If sand bags are used, they should be packaged in cloth bags of sufficient strength to avoid rupture during the test. When used, a dummy should be restrained with the existing occupant restraint system in the test vehicle. When used, sand bags or concrete blocks should be restrained to prevent movement during impact. As stated in Table 2.1, the surrogate occupant should be representative of the 50th percentile male with a mass of approximately 75 kg.

With the exception of tests with the 700C and the 820C vehicles, use of a surrogate occupant is optional. For tests with the 700C and 820C vehicles, one surrogate occupant is specified primarily to evaluate typical unsymmetrical vehicle mass distribution and its effect on vehicle stability since the dummy's mass is 8 to 10 percent of the vehicle's mass. If used for the added mass effect only, the surrogate occupant should be placed in either the driver's seat or the passenger's seat, whichever position contributes most to the vehicle's post-impact instability, or yawing moment on the vehicle. For example, in an off-center, head on impact into a terminal or a breakaway/yielding support, with impact on the left front of the vehicle, the surrogate occupant should be in the passenger's seat. If the instrumented Hybrid III dummy is used to assess occupant risk in a head-on impact, it should be placed in the front seat on the side nearest the impact point for off-center impacts, and placed in the driver's seat for centered impacts.

When a surrogate occupant or the Hybrid III dummy is used in the 2000P vehicle, placement and restraint condition should be as those for the 700C and 820C vehicles.

A dummy should also be used if there is a reasonable expectation that occupant/test article interaction will occur during or after impact, or both. For example, during an impact with a "tall" longitudinal barrier, the occupant's head may protrude through the side window and strike the barrier. The dummy should be placed either in the driver's seat or the passenger side of the front seat, whichever is nearest the impact point. An onboard camera should be used to record dummy movement when occupant/test article interaction is expected.

For vehicles 8000S, 36000V, and 36000T, when a dummy is used, it should be placed in the driver's seat and it should be restrained with the existing restraint system.

Experience gained from limited side impact tests clearly shows the vulnerability of an occupant on the impact side of the vehicle, due typically to large intrusions of the test article into the occupant compartment. Experience has also shown risks to an occupant in such a collision may be assessed by a valid side-impact dummy (74). Reference may be made to Appendix G for a methodology for side impact testing.

CHAPTER 3

TEST CONDITIONS

3.1 GENERAL

Guidelines are presented for the impact performance evaluation of various safety features. Individual tests are designed to evaluate one or more of the principal performance factors: structural adequacy, occupant risk, and post-impact behavior of vehicle. These evaluation criteria are presented in Chapter 5.

Depending on the feature being evaluated, there are up to six test levels that can be selected. In general, the lower test levels are applicable for evaluating features to be used on lower service level roadways and certain types of work zones while the higher test levels are applicable for evaluating features to be used on higher service level roadways or at locations that demand a special, high-performance safety feature. It will be noted that test levels 4 through 6 are applicable to longitudinal barriers only.

Note that the requirements of test level 3 are similar to those defined in the "Crash Test Conditions for Minimum Matrix" given in *NCHRP Report 230 (1).* It is to this level that most crash-tested safety features in use on U.S. highways have been qualified. Since the issuance of *Report 230,* there has been a greater recognition of the merits of tailoring performance and cost of safety features to site requirements. This is the reason for the multiple test levels presented here. It is beyond the scope of this document to define warrants for the various test levels.

It is the responsibility of the user agency(s) to determine which of the test levels is most appropriate for a feature's intended application. Agencies should develop objective guidelines for use of roadway safety features, considering factors such as traffic conditions, site conditions, traffic volume and mix, and the cost effectiveness of candidate safety alternatives. However, it is anticipated that safety features qualified for test level 3 will remain acceptable for a wide range of high-speed arterial highways. Test level 2 qualified features are expected to be deemed acceptable for most local and collector roads and many work zones. Test level 1 qualified features are expected to be deemed acceptable for some work zones and very low-volume, low-speed local streets and highways. Applicability of test levels 4 through 6 will probably be determined by volume of truck and heavy vehicle traffic and/or the consequences of penetration beyond the longitudinal barrier.

Although tests with the 700C vehicle are desirable, they are optional because (1) this vehicular type represents only a very small portion of the vehicular mix and (2) there is no assurance that an existing feature will meet the recommended performance criteria or that new features can be found that will fully meet the recommended performance criteria for these tests. In the interim until sufficient testing experience is acquired with the 700C type vehicle, the test article should perform acceptably with all appropriate tests using the 820C and 2000P type vehicles and preferably should perform acceptably during tests with the 700C type vehicle. It may be assumed that test articles performing acceptably with 700C and 2000P type vehicles will also perform acceptably with the 820C vehicle; thus, the 820C vehicle tests need not be performed.

It is important to note that tests recommended herein are based in large part on past experience. It is not possible to anticipate the form that new designs will take nor the critical impact conditions of these new designs. As such, the test matrices presented in this section must not be viewed as all-inclusive. When appropriate, the responsible agency should devise other critical test conditions consistent with the range of expected impact conditions. Also, if warranted, additional tests can be conducted to evaluate a feature for nonidealized conditions, such as longitudinal barrier with curvilinear alignment, the placement of a feature on nonlevel terrain, or the placement of a feature behind a curb.

It is not uncommon for a designer/tester to make design changes to a feature during the course of conducting the recommended test series or after successful completion of the test series. Changes are often made to improve performance or to reduce cost of the design or both. Questions then invariably arise as to the need to repeat any or all of the recommended tests. Good engineering judgment must be used in such instances. As a general rule, a test should be repeated if there is a reasonable uncertainty regarding the effect the change will have on the test.

Note that each test in a given matrix has a specific "test designation" of the form "i-jk." The "i" refers to the test level and "jk" refers to the test number. Test designations preceded by an "S" refer to tests with the optional 700C vehicle.

3.2 TEST MATRICES

3.2.1 Longitudinal Barriers

3.2.1.1 General

Recommended tests to evaluate longitudinal barriers for six test levels are presented in Table 3.1. Reference should be made to the Glossary for definitions of length of need (LON) and transitions. These guidelines are applicable to both permanent barriers and temporary barriers used in work or construction zones. However, except under very unusual conditions, a temporary barrier would not normally be designed for impact conditions greater than test level 3.

TABLE 3.1. Test matrix for longitudinal barriers

Test Level	Barrier Section	Test Designation	Impact Conditions[c]			Impact Point	Evaluation Criteria[e] (See Table 5.1)
			Vehicle	Nominal Speed (km/h)	Nominal Angle, θ (deg)		
1	Length of Need	1-10	820C	50	20	(b)	A,D,F,H,I,(J),K,M
		S1-10[a]	700C	50	20	(b)	A,D,F,H,I,(J),K,M
		1-11	2000P	50	25	(b)	A,D,F,K,L,M
	Transition	1-20[d]	820C	50	20	(b)	A,D,F,H,I,(J),K,M
		S1-20[a]	700C	50	20	(b)	A,D,F,H,I,(J),K,M
		1-21	2000P	50	25	(b)	A,D,F,K,L,M
2	Length of Need	2-10	820C	70	20	(b)	A,D,F,H,I,(J),K,M
		S2-10[a]	700C	70	20	(b)	A,D,F,H,I,(J),K,M
		2-11	2000P	70	25	(b)	A,D,F,K,L,M
	Transition	2-20[d]	820C	70	20	(b)	A,D,F,H,I,(J),K,M
		S2-20[a]	700C	70	20	(b)	A,D,F,H,I,(J),K,M
		2-21	2000P	70	25	(b)	A,D,F,K,L,M
3 Basic Level	Length of Need	3-10	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S3-10[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		3-11	2000P	100	25	(b)	A,D,F,K,L,M
	Transition	3-20[d]	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S3-20[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		3-21	2000P	100	25	(b)	A,D,F,K,L,M
4	Length of Need	4-10	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S4-10[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		4-11[d]	2000P	100	25	(b)	A,D,F,K,L,M
		4-12	8000S	80	15	(b)	A,D,G,K,M
	Transition	4-20[d]	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S4-20[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		4-21[d]	2000P	100	25	(b)	A,D,F,K,L,M
		4-22	8000S	80	15	(b)	A,D,G,K,M
5	Length of Need	5-10	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S5-10[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		5-11[d]	2000P	100	25	(b)	A,D,F,K,L,M
		5-12	36000V	80	15	(b)	A,D,G,K,M
	Transition	5-20[d]	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S5-20[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		5-21[d]	2000P	100	25	(b)	A,D,F,K,L,M
		5-22	36000V	80	15	(b)	A,D,G,K,M
6	Length of Need	6-10	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S6-10[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		6-11[d]	2000P	100	25	(b)	A,D,F,K,L,M
		6-12	36000T	80	15	(b)	A,D,G,K,M
	Transition	6-20[d]	820C	100	20	(b)	A,D,F,H,I,(J),K,M
		S6-20[a]	700C	100	20	(b)	A,D,F,H,I,(J),K,M
		6-21[d]	2000P	100	25	(b)	A,D,F,K,L,M
		6-22	36000T	80	15	(b)	A,D,G,K,M

[a] Test is optional. See Section 3.1.

[b] See Figure 3.1 for impact point.

[c] See Section 3.3.2 for tolerances on impact condition

[d] Test may be optional. See Section 3.2.1.2.

[e] Criteria in parenthesis are optional.

3.2.1.2 Description of Tests

Test 10

Test 10 is conducted for the LON section for all test levels. The purpose of this small car test is to evaluate the overall performance of the LON section in general, and occupant risks in particular.

Tests 11 and 21

Test 11 for the LON section and Test 21 for the transition section are conducted for test levels 1 through 3. They are intended to evaluate strength of the section in containing and redirecting the 2000P test vehicle. Tests 11 and 21 are optional for test levels 4, 5, and 6. They should be conducted if a reasonable uncertainty exists regarding impact performance of the system for these tests. It is recommended that results of Tests 12 and 22 be carefully examined prior to conducting Tests 11 and 21. Tests 12 and 22 will establish basic structural adequacy of the barrier. However, satisfactory performance for Tests 12 and 22 does not assure satisfactory performance for Tests 11 and 21. For example, there may be geometric incompatibilities between the barrier and the 2000P vehicle that could result in excessive snagging or pocketing.

Test 20

Test 20 for a transition section is an optional test to evaluate occupant risk and post-impact trajectory criteria for all test levels. It should be conducted if there is a reasonable uncertainty regarding the impact performance of the system for this test. Results of Test 21 should be carefully examined prior to conducting Test 20. Test 21 will establish the structural adequacy of the transition. However, satisfactory performance for Test 21 does not assure satisfactory performance for Test 20. For example, there may be geometric incompatibilities between the transition and the 820C vehicle which could cause a failure from excessive snagging or pocketing.

Tests 12 and 22

Test 12 for the LON section and Test 22 for the transition section are conducted for test levels 4, 5, and 6. They are intended to evaluate strength of the section in containing and redirecting the heavy test vehicles.

As noted in Figure 3.1, Section 3.4.2 contains guidance on determination of the critical impact point (CIP). As discussed therein, depending on barrier design, there may be two CIPs. For example, a bridge rail with a splice located between support posts may have two CIPs: one that would produce maximum loading on the splice and another that would have the greatest potential for causing wheel snagging or vehicular pocketing. As another example, a transition may have a CIP in the vicinity of the upstream end and another in the vicinity of the downstream end. Therefore, if one test cannot evaluate both points of concern, it may be necessary to conduct the relevant test(s) at both points of concern. See further discussion on this matter in Section 3.4.2.

3.2.2 Terminals and Crash Cushions

3.2.2.1 General

Recommended tests to evaluate terminals and crash cushions are presented in Table 3.2. Reference should be made to the Glossary for definitions of these features. These guidelines are applicable to both permanent features and temporary features used in work or construction zones. Note that impact performance requirements of a terminal and a redirective crash cushion are the same.

Impact performance requirements, and hence capabilities, of a nonredirective crash cushion are considerably less than those for a redirective crash cushion. A redirective crash cushion is subjected to more tests, and the requirements of those tests are more rigorous. For example, it is recommended that Test 38 be conducted at the critical impact point of the redirective crash cushion. A similar test would be difficult to pass for a nonredirective crash cushion. As a consequence, conditions or sites at which a nonredirective crash cushion can be used may be limited. It is the responsibility of the user agency to determine where features addressed in this document have application, including redirective and nonredirective crash cushions.

Reference is made herein to "gating" and "nongating" features or devices. A gating device is one designed to allow controlled penetration of the vehicle when impacted between the end and the beginning of the length of need (LON) of the device. The widely used breakaway cable terminal (BCT) is a gating device. A nongating device is designed to contain and redirect a vehicle when impacted downstream from the end of the device. A terminal or crash cushion with redirection capabilities along its entire length is a nongating device.

3.2.2.2 Description of Tests

Following is a description of each test. Reference should be made to Figures 3.2 and 3.3 for vehicle/test article orientation at impact.

Tests 30 and 40

These tests are conducted with the vehicle approaching parallel to the roadway, with impact to the left or right of the vehicle's centerline. They are intended primarily to evaluate occupant risk and vehicle trajectory criteria. The vehicle should be offset to the most critical side, that is, the side which will result in the greatest occupant risk during and subsequent to impact, recognizing the direction the vehicle will tend to roll, pitch, and yaw subsequent to impact. If the impact is to the right of the vehicle's centerline, the vehicle will tend to rotate clockwise (as viewed from above) or counterclockwise if the impact is to the left. It may also roll and pitch depending on the geometry and impact behavior of the test article.

Tests 31 and 41

These tests are conducted with the vehicle approaching parallel to the roadway with impact at the vehicle's centerline. For a device designed to decelerate a vehicle to a stop, these tests are intended to evaluate the capacity of the device to

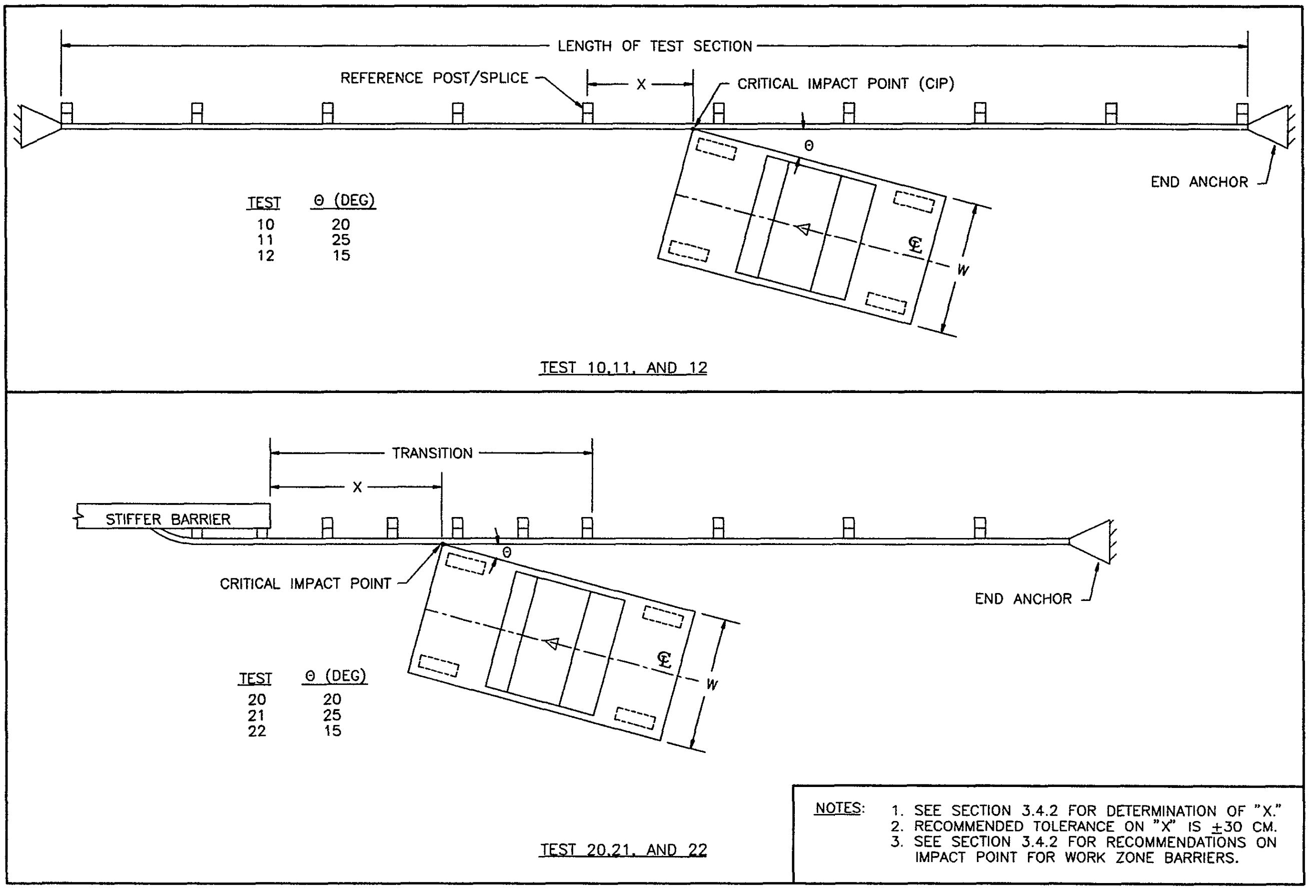

LENGTH OF TEST SECTION
REFERENCE POST/SPLICE
X
CRITICAL IMPACT POINT (CIP)
θ
END ANCHOR
TEST θ (DEG)
10 20
11 25
12 15
℄
W
TEST 10,11, AND 12
TRANSITION
X
STIFFER BARRIER
CRITICAL IMPACT POINT
θ
END ANCHOR
TEST θ (DEG)
20 20
21 25
22 15
℄
W
TEST 20,21, AND 22
NOTES:
1. SEE SECTION 3.4.2 FOR DETERMINATION OF "X."
2. RECOMMENDED TOLERANCE ON "X" IS ±30 CM.
3. SEE SECTION 3.4.2 FOR RECOMMENDATIONS ON IMPACT POINT FOR WORK ZONE BARRIERS.

Figure 3.1. Impact conditions for longitudinal barrier tests.

absorb the kinetic energy of the 2000P vehicle (structural adequacy criteria) in a safe manner (occupant risk criteria). For other types of devices these tests are intended primarily to evaluate occupant risk and vehicle trajectory criteria.

Tests 32 and 33

Tests 32 and 33 are intended primarily to evaluate occupant risk and vehicle trajectory criteria. For some devices, it may be possible to demonstrate through engineering analysis, with a high degree of confidence, that Tests 32 and 33 are less severe than Tests 30 and 31. For example, in all probability Tests 32 and 33 would be less severe than Tests 30 and 31, respectively, for the breakaway cable terminal (BCT) for W-beam guardrail; the BCT is a gating device. In such cases, Tests 32 and 33 may be optional. However, Tests 32 and 33 should be conducted for a gating device if there is a reasonable uncertainty about the impact performance of the system for these tests.

Tests 34 and 35

These tests are applicable to gating devices only. In Test 34, impact should be at a CIP (see definition in Glossary) between the end of the device and the beginning of the LON. Whereas definitive criteria are presented in Section 3.4.3 for estimating the CIP for selected devices, no such criteria are available for this particular application. Therefore, selection of the CIP for Test 34 should be based on test experience with similar devices, computer simulation if possible, and judgment. In selecting the CIP, consideration should not only be given to the point with the greatest potential for causing snagging or pocketing but also the point with the greatest potential for producing vehicular overturn. For example, in testing a sloped-end terminal, vehicular stability is the primary concern, not snagging or pocketing, and the CIP may not be midway between the end of the terminal and the beginning of the LON. In the absence of a determinable CIP, Test 34 may be conducted with the initial impact point midway between the end of the device and the beginning of the LON. Test 34 is intended primarily to evaluate occupant risk and vehicle trajectory criteria. Test 35 is intended primarily to evaluate the ability of the device to contain and redirect (structural adequacy criteria) the 2000P vehicle within vehicle trajectory criteria at the beginning of the LON.

Tests 36, 37, and 38

These tests are applicable to nongating devices only. In Tests 36 and 37, the impact point should be at the end of the

TABLE 3.2. Test matrix for terminals and crash cushions

Test Level	Feature	Feature Type[d]	Test Designation	Impact Conditions[c]			Impact Point	Evaluation Criteria[g] (See Table 5.1)
				Vehicle	Nominal Speed (km/h)	Nominal Angle, θ (deg)		
1	Terminals and Redirective Crash Cushions	G/NG	1-30	820C	50	0	(b,e)	C,D,G,H,I,(J),K,N
		G/NG	S1-30[a]	700C	50	0	(b,e)	C,D,G,H,I,(J),K,N
		G/NG	1-31	2000P	50	0	(b)	C,D,G,H,I,(J),K,N
		G/NG	1-32	820C	50	15	(b)	C,D,G,H,I,(J),K,N
		G/NG	S1-32[a]	700C	50	15	(b)	C,D,G,H,I,(J),K,N
		G/NG	1-33	2000P	50	15	(b)	C,D,G,H,I,(J),K,N
		G	1-34	820C	50	15	(b,e)	C,D,G,H,I,(J),K,N
		G	S1-34[a]	700C	50	15	(b,e)	C,D,G,H,I,(J),K,N
		G	1-35	2000P	50	20	(b)	A,D,G,K,L,M
		NG	1-36	820C	50	15	(b)	A,D,G,H,I,(J),K,M
		NG	S1-36[a]	700C	50	15	(b)	A,D,G,H,I,(J),K,M
		NG	1-37	2000P	50	20	(b)	A,D,G,K,L,M
		NG	1-38	2000P	50	20	(b)	A,D,G,K,L,M
		G/NG	1-39	2000P	50	20	(b)	C,D,G,K,L,M,N
	Nonredirective Crash Cushions[f]	G	1-40	820C	50	0	(e,h)	C,D,G,H,I,(J),K
		G	S1-40[a]	700C	50	0	(e,h)	C,D,G,H,I,(J),K
		G	1-41	2000P	50	0	(h)	C,D,G,H,I,(J),K
		G	1-42	820C	50	15	(h)	C,D,G,H,I,(J),K,N
		G	S1-42[a]	700C	50	15	(h)	C,D,G,H,I,(J),K,N
		G	1-43	2000P	50	15	(h)	C,D,G,H,I,(J),K,N
		G	1-44	2000P	50	20	(h)	C,D,G,K,N

[a] Test is optional. See Section 3.1.
[b] See Figure 3.2 for impact point.
[c] See Section 3.3.3 for tolerances on impact conditions.
[d] G/NG - Test applicable to gating and nongating devices.
G - Test applicable to gating device only.
NG - Test applicable to nongating device only.
[e] See discussion in Section 3.2.2.2.
[f] See discussion in Section 3.2.2.1 relative to nonredirective crash cushions.
[g] Criteria in parenthesis are optional.
[h] See Figure 3.3 for impact point.

terminal/crash cushion as illustrated in Figure 3.2. Tests 36 and 37 are intended to evaluate the redirectional capability of the device for impacts at or near the nose of the device. Test 36 is intended primarily to evaluate occupant risk and vehicle trajectory criteria. Test 37 is intended primarily to evaluate structural adequacy and vehicle trajectory criteria. Test 37 will create maximum loading on the anchorage system at the nose of the device.

Test 38 differs in purpose from Test 37 in that it is intended to evaluate the potential for pocketing or snagging at the juncture with the object the device is shielding or terminating. The impact point should be selected to maximize the potential for pocketing or snagging. Guidelines for the CIP are given in Section 3.4.3. Test 38 may not be necessary if the lateral stiffness of the terminal/crash cushion is greater than that of the object the device is shielding or terminating. For example, this test would be necessary for a terminal/ crash cushion used at the end of a rigid concrete barrier. It may not be necessary for a terminal/crash cushion used at the end of a semirigid guardrail barrier. Determination of need for this test should be based on test experience with similar devices, computer simulation if possible, and judgment.

Test 39

Test 39 is intended to evaluate the performance of a terminal/ crash cushion for a "reverse" hit. For roadside or median applications, this test is recommended if the terminal/crash cushion will commonly be placed within the clear zone of opposing traffic.

Tests 42, 43, and 44

These tests are applicable to a nonredirective crash cushion only. Tests 42 and 43 are intended to evaluate occupant risk and vehicle trajectory criteria for both a small and a large passenger vehicle for an angled impact into the nose of the cushion. Test 44 is intended to evaluate the ability of the cushion to safely stop a large passenger vehicle prior to a life-threatening impact with the corner of the hazardous object being shielded.

3.2.2.3 Orientation of Terminal V Crash Cushion

Implied in the terminal/crash cushion orientations shown in Figures 3.2 and 3.3 is that the centerline of the feature in service

TABLE 3.2. ***(Continued)***

Test Level	Feature	Feature Type[d]	Test Designation	Impact Conditions[c]			Impact Point	Evaluation Criteria[g] (See Table 5.1)
				Vehicle	Nominal Speed (km/h)	Nominal Angle, θ (deg)		
2	Terminals and Redirective Crash Cushions	G/NG	2-30	820C	70	0	(b,e)	C,D,F,H,I,(J),K,N
		G/NG	S2-30[a]	700C	70	0	(b,e)	C,D,F,H,I,(J),K,N
		G/NG	2-31	2000P	70	0	(b)	C,D,F,H,I,(J),K,N
		G/NG	2-32	820C	70	15	(b)	C,D,F,H,I,(J),K,N
		G/NG	S2-32[a]	700C	70	15	(b)	C,D,F,H,I,(J),K,N
		G/NG	2-33	2000P	70	15	(b)	C,D,F,H,I,(J),K,N
		G	2-34	820C	70	15	(b,e)	C,D,F,H,I,(J),K,N
		G	S2-34[a]	700C	70	15	(b,e)	C,D,F,H,I,(J),K,N
		G	2-35	2000P	70	20	(b)	A,D,F,K,L,M
		NG	2-36	820C	70	15	(b)	A,D,F,H,I,(J),K,M
		NG	S2-36[a]	700C	70	15	(b)	A,D,F,H,I,(J),K,M
		NG	2-37	2000P	70	20	(b)	A,D,F,K,L,M
		NG	2-38	2000P	70	20	(b)	A,D,F,K,L,M
		G/NG	2-39	2000P	70	20	(b)	C,D,F,K,L,M,N
	Nonredirective Crash Cushions[f]	G	2-40	820C	70	0	(e,h)	C,D,F,H,I,(J),K
		G	S2-40[a]	700C	70	0	(e,h)	C,D,F,H,I,(J),K
		G	2-41	2000P	70	0	(h)	C,D,F,H,I,(J),K
		G	2-42	820C	70	15	(h)	C,D,F,H,I,(J),K,N
		G	S2-42[a]	700C	70	15	(h)	C,D,F,H,I,(J),K,N
		G	2-43	2000P	70	15	(h)	C,D,F,H,I,(J),K,N
		G	2-44	2000P	70	20	(h)	C,D,F,K,N

[a] Test is optional. See Section 3.1.
[b] See Figure 3.2 for impact point.
[c] See Section 3.3.3 for tolerances on impact conditions.
[d] G/NG - Test applicable to gating and nongating devices.
G - Test applicable to gating device only.
NG - Test applicable to nongating device only.
[e] See discussion in Section 3.2.2.2.
[f] See discussion in Section 3.2.2.1 relative to nonredirective crash cushions.
[g] Criteria in parenthesis are optional.
[h] See Figure 3.3 for impact point.

will be parallel to the roadway and that the roadway is straight. It is likely that this will be the dominant orientation for most such features. However, if the feature will commonly be oriented otherwise, it should be so tested. In such a case the feature should be tested with its centerline oriented as it typically would be in service. Note that the impact angles recommended herein remain the same regardless of the orientation of the feature since they are implicitly measured with respect to the roadway tangent and not with respect to the feature. For example, Figure 3.4 illustrates the relative angle between a feature and the vehicle's approach for Test 31 (0-deg impact angle) for two different orientations of a feature. If the feature will typically be used in various orientations, worst-case test conditions should be selected within the recommended test matrices. For example, if a nongating, redirective crash cushion will typically be oriented in both configurations shown in Figure 3.4, redirective Tests 34, 35, and 36 should be conducted with the feature oriented as shown in configuration B.

3.2.2.4 Other Terminal/Crash Cushion Designs

Some current in-service terminals or crash cushions (such as short-radius, curved guardrail designs used in medians or at bridge ends near an intersecting access road, or the "dragnet arrestor" system, or the gravel-bed attenuators for run-away trucks) may not be specifically addressed by the test matrices in Table 3.2. In such cases, the designer and test agency should devise tests to explore critical elements of the feature within the framework and intent of the recommended tests and evaluation criteria. Recommended test vehicles, impact speeds, and impact angles should be used with appropriate critical impact points.

3.2.3 Support Structures, Work Zone Traffic Control Devices, and Breakaway Utility Poles

3.2.3.1 General

Shown in Table 3.3 are recommended tests to evaluate support structures, work zone traffic control devices and breakaway utility poles. Reference should be made to the Glossary for definitions of these features.

Support structures include sign supports, mailbox supports, light poles or luminaire supports, and emergency call-box supports. The guidelines are applicable to permanent support structures as well as those used in construction or work zones. Fire hydrants are another "man-made" device commonly found within

TABLE 3.2. (*Continued*)

Test Level	Feature	Feature Type[d]	Test Designation	Impact Conditions[c]			Impact Point	Evaluation Criteria[g] (See Table 5.1)
				Vehicle	Nominal Speed (km/h)	Nominal Angle, θ (deg)		
3 Basic Level	Terminals and Redirective Crash Cushions	G/NG	3-30	820C	100	0	(b,e)	C,D,F,H,I,(J),K,N
		G/NG	S3-30[a]	700C	100	0	(b,e)	C,D,F,H,I,(J),K,N
		G/NG	3-31	2000P	100	0	(b)	C,D,F,H,I,(J),K,N
		G/NG	3-32	820C	100	15	(b)	C,D,F,H,I,(J),K,N
		G/NG	S3-32[a]	700C	100	15	(b)	C,D,F,H,I,(J),K,N
		G/NG	3-33	2000P	100	15	(b)	C,D,F,H,I,(J),K,N
		G	3-34	820C	100	15	(b,e)	C,D,F,H,I,(J),K,N
		G	S3-34[a]	700C	100	15	(b,e)	C,D,F,H,I,(J),K,N
		G	3-35	2000P	100	20	(b)	A,D,F,K,L,M
		NG	3-36	820C	100	15	(b)	A,D,F,H,I,(J),K,M
		NG	S3-36[a]	700C	100	15	(b)	A,D,F,H,I,(J),K,M
		NG	3-37	2000P	100	20	(b)	A,D,F,K,L,M
		NG	3-38	2000P	100	20	(b)	A,D,F,K,L,M
		G/NG	3-39	2000P	100	20	(b)	C,D,F,K,L,M,N
	Nonredirective Crash Cushions[f]	G	3-40	820C	100	0	(e,h)	C,D,F,H,I,(J),K
		G	S3-40[a]	700C	100	0	(e,h)	C,D,F,H,I,(J),K
		G	3-41	2000P	100	0	(h)	C,D,F,H,I,(J),K
		G	3-42	820C	100	15	(h)	C,D,F,H,I,(J),K,N
		G	S3-42[a]	700C	100	15	(h)	C,D,F,H,I,(J),K,N
		G	3-43	2000P	100	15	(h)	C,D,F,H,I,(J),K,N
		G	3-44	2000P	100	20	(h)	C,D,F,K,N

[a] Test is optional. See Section 3.1.
[b] See Figure 3.2 for impact point.
[c] See Section 3.3.3 for tolerances on impact conditions.
[d] G/NG - Test applicable to gating and nongating devices.
G - Test applicable to gating device only.
NG - Test applicable to nongating device only.
[e] See discussion in Section 3.2.2.2.
[f] See discussion in Section 3.2.2.1 relative to nonredirective crash cushions.
[g] Criteria in parenthesis are optional.
[h] See Figure 3.3 for impact point.

the clear zone of urban roadways. Although specific tests are not presented, when feasible, these devices should be designed to the same safety standards as support structures. Work zone traffic control devices include plastic drums, barricades, cones, chevron panels and their supports, and delineator posts and lights that may be attached to drums or barricades.

Note there is no test level 1 for these features. This is because, with few if any exceptions, features of this type that are cost effective for test levels 2 and 3 will also be cost effective for low-speed applications. For most sign supports and moderately sized luminaire poles, the low-speed test (35 km/h) is more critical than the high-speed test (100 km/h).

The following notes are applicable to tests in Table 3.3.

- A critical impact angle (CIA) should be determined and used for each test, consistent with the manner in which the feature will commonly be oriented in service with respect to traffic. For a given test and the attendant range of vehicular impact angles, the CIA is the angle within this range judged to have the greatest potential for causing a failure when the test is assessed by the recommended evaluation criteria. To determine the CIA, the in-service orientation of the support(s) and attendant breakaway feature(s) if used, the work zone traffic control device(s), or the utility pole must first be established with respect to the normal direction of traffic. Then the CIA should be selected from a 0- to 20-deg impact angle envelope, as measured from the normal direction of traffic (see Figure 3.5). If the feature is commonly used at locations where impacts from a larger envelope are possible (such as at or near an intersection, or in a narrow median), it should be tested at the CIA within that envelope. If it is not apparent what the CIA is, it may be appropriate to test at each potential CIA. For example, if the feature can be struck from the front and rear, tests at CIAs consistent with these approach directions may be appropriate. (See discussion in Section 2.3.2.3 relative to special orientations for work zone traffic control devices.)
- For a single-support system, a single work zone traffic control device, or a utility pole, the center of the bumper should align with the vertical centerline of the feature. As an option, the left or right quarter point of the bumper may align with the vertical centerline of the device. Previous testing has shown that the offset option may be preferable in some cases because the same vehicle can be used to conduct both required tests (60 and 61) provided damage to the vehicle from the first test (usually the low-speed test) has no appreciable effect on impact performance of the vehicle/test article in the second test. How-

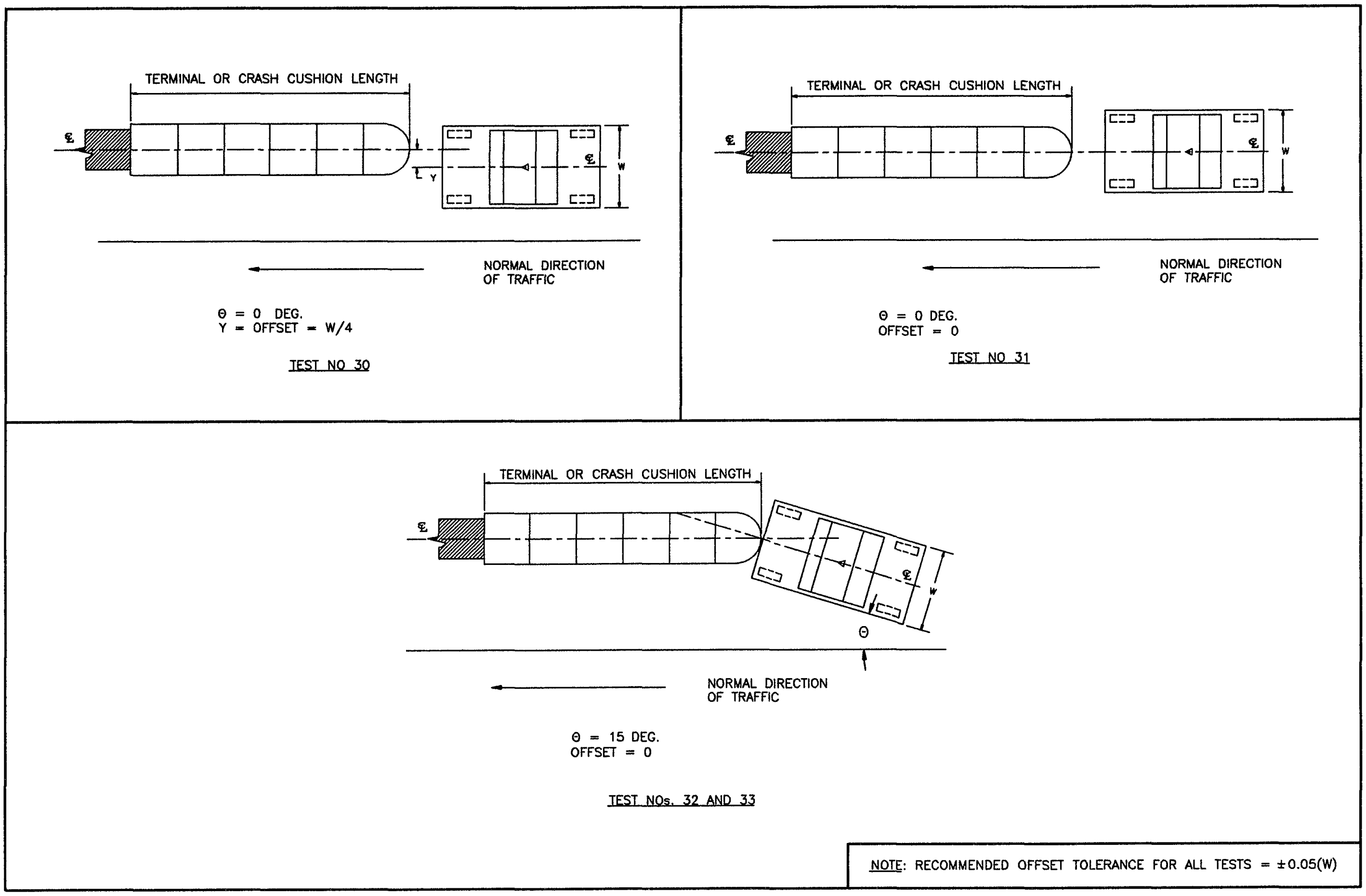

TERMINAL OR CRASH CUSHION LENGTH
℄
Y
W
NORMAL DIRECTION OF TRAFFIC
Θ = 0 DEG.
Y = OFFSET = W/4
TEST NO 30
TERMINAL OR CRASH CUSHION LENGTH
℄
W
NORMAL DIRECTION OF TRAFFIC
Θ = 0 DEG.
OFFSET = 0
TEST NO 31
TERMINAL OR CRASH CUSHION LENGTH
℄
W
Θ
NORMAL DIRECTION OF TRAFFIC
Θ = 15 DEG.
OFFSET = 0
TEST NOs. 32 AND 33
NOTE: RECOMMENDED OFFSET TOLERANCE FOR ALL TESTS = ±0.05(W)

Figure 3.2. Impact conditions for terminal and redirective crash cushion tests.

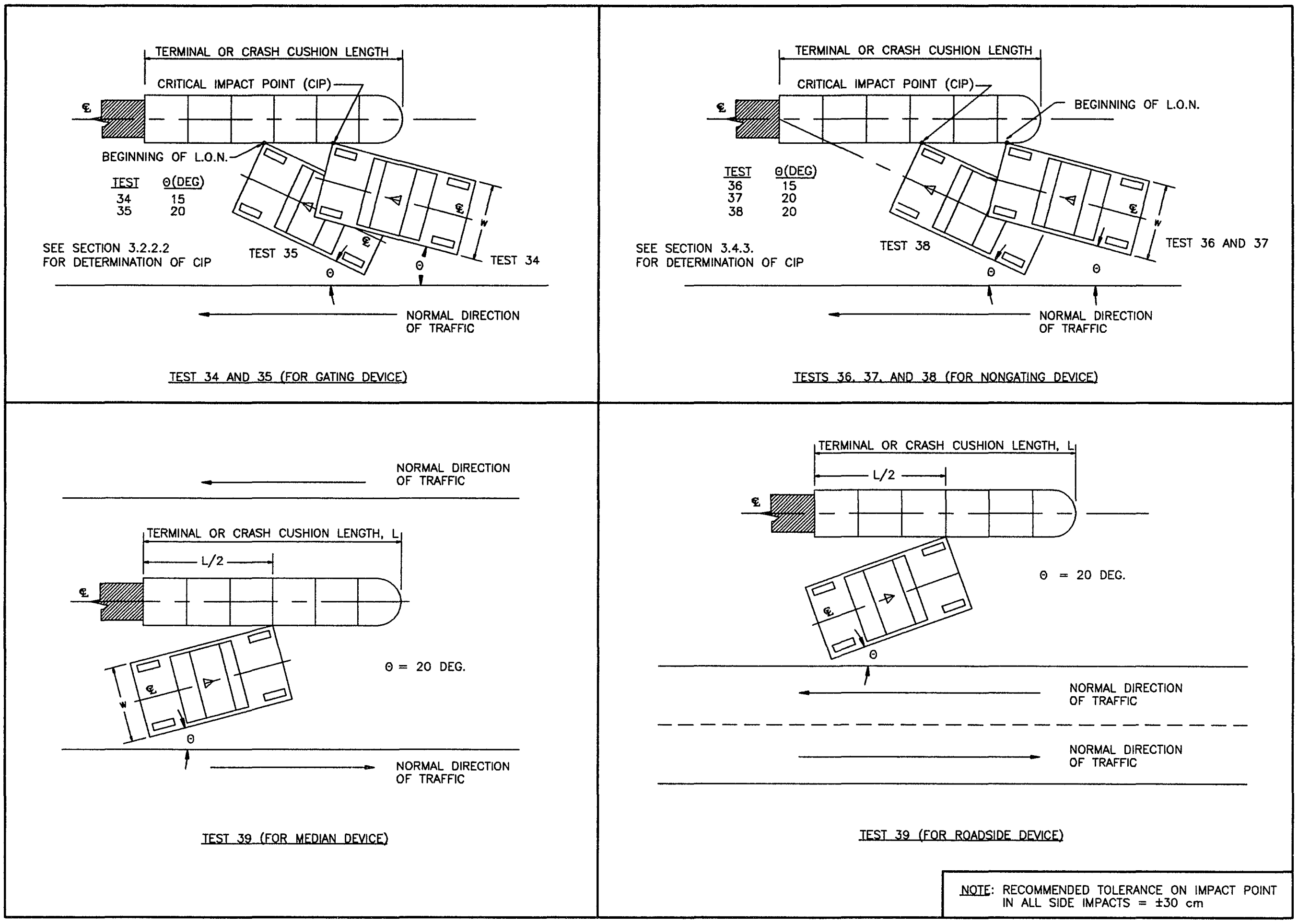

TERMINAL OR CRASH CUSHION LENGTH
CRITICAL IMPACT POINT (CIP)
℄
BEGINNING OF L.O.N.
TEST θ(DEG)
34 15
35 20
SEE SECTION 3.2.2.2
FOR DETERMINATION OF CIP
TEST 35
TEST 34
θ
W
NORMAL DIRECTION
OF TRAFFIC
TEST 34 AND 35 (FOR GATING DEVICE)
TERMINAL OR CRASH CUSHION LENGTH
CRITICAL IMPACT POINT (CIP)
BEGINNING OF L.O.N.
TEST θ(DEG)
36 15
37 20
38 20
SEE SECTION 3.4.3.
FOR DETERMINATION OF CIP
TEST 38
TEST 36 AND 37
NORMAL DIRECTION
OF TRAFFIC
TESTS 36, 37, AND 38 (FOR NONGATING DEVICE)
NORMAL DIRECTION
OF TRAFFIC
TERMINAL OR CRASH CUSHION LENGTH, L
L/2
θ = 20 DEG.
NORMAL DIRECTION
OF TRAFFIC
TEST 39 (FOR MEDIAN DEVICE)
TERMINAL OR CRASH CUSHION LENGTH, L
L/2
θ = 20 DEG.
NORMAL DIRECTION
OF TRAFFIC
NORMAL DIRECTION
OF TRAFFIC
TEST 39 (FOR ROADSIDE DEVICE)
NOTE: RECOMMENDED TOLERANCE ON IMPACT POINT
IN ALL SIDE IMPACTS = ±30 cm

Figure 3.2. (continued).

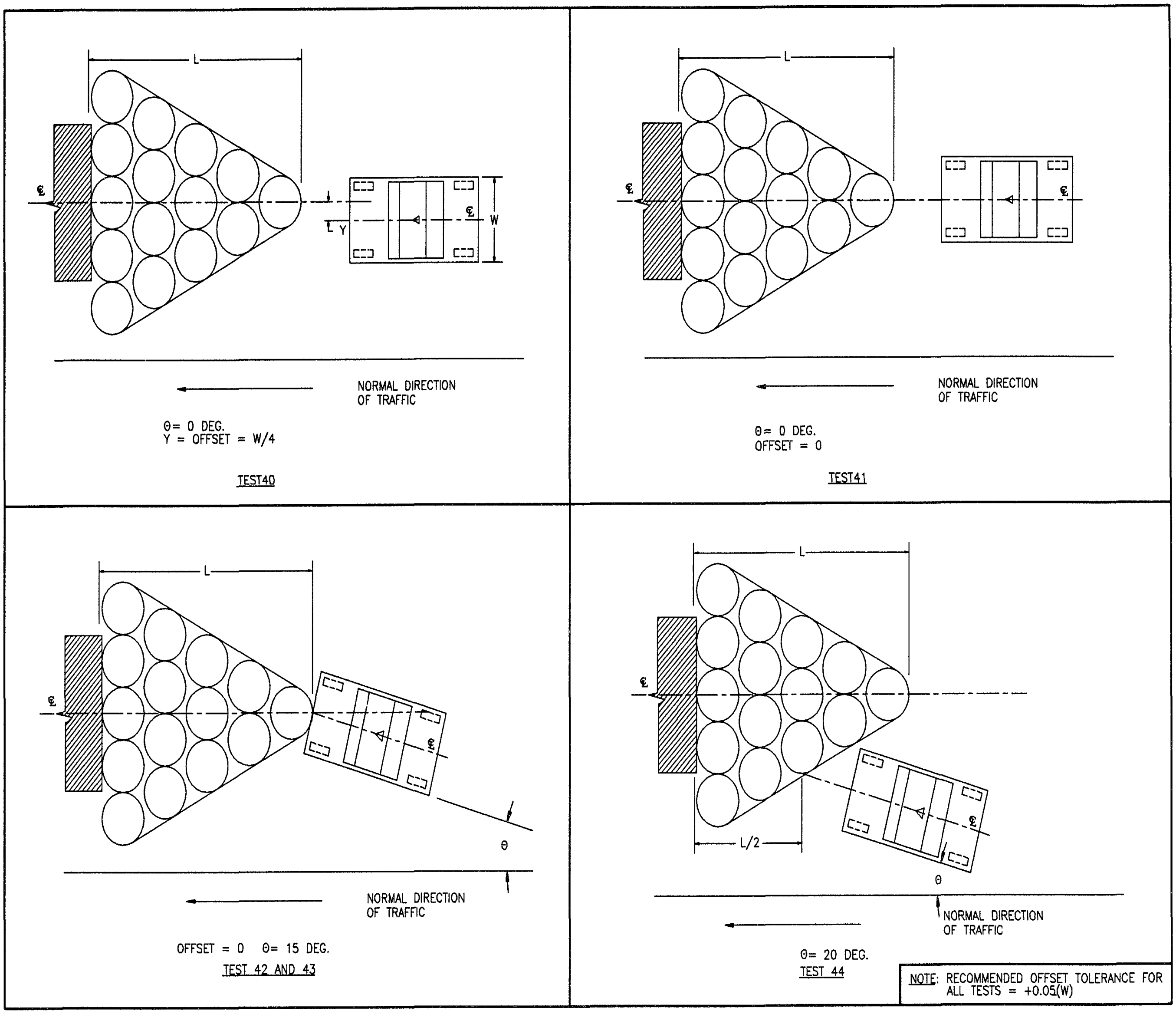

L
℄
Y
W
℄
NORMAL DIRECTION OF TRAFFIC
θ= 0 DEG.
Y = OFFSET = W/4
TEST40
L
℄
℄
NORMAL DIRECTION OF TRAFFIC
θ= 0 DEG.
OFFSET = 0
TEST41
L
℄
℄
θ
NORMAL DIRECTION OF TRAFFIC
OFFSET = 0 θ= 15 DEG.
TEST 42 AND 43
L
℄
L/2
℄
θ
NORMAL DIRECTION OF TRAFFIC
θ= 20 DEG.
TEST 44
NOTE: RECOMMENDED OFFSET TOLERANCE FOR ALL TESTS = +0.05(W)

Figure 3.3. Impact conditions for nonredirective crash cushion tests.

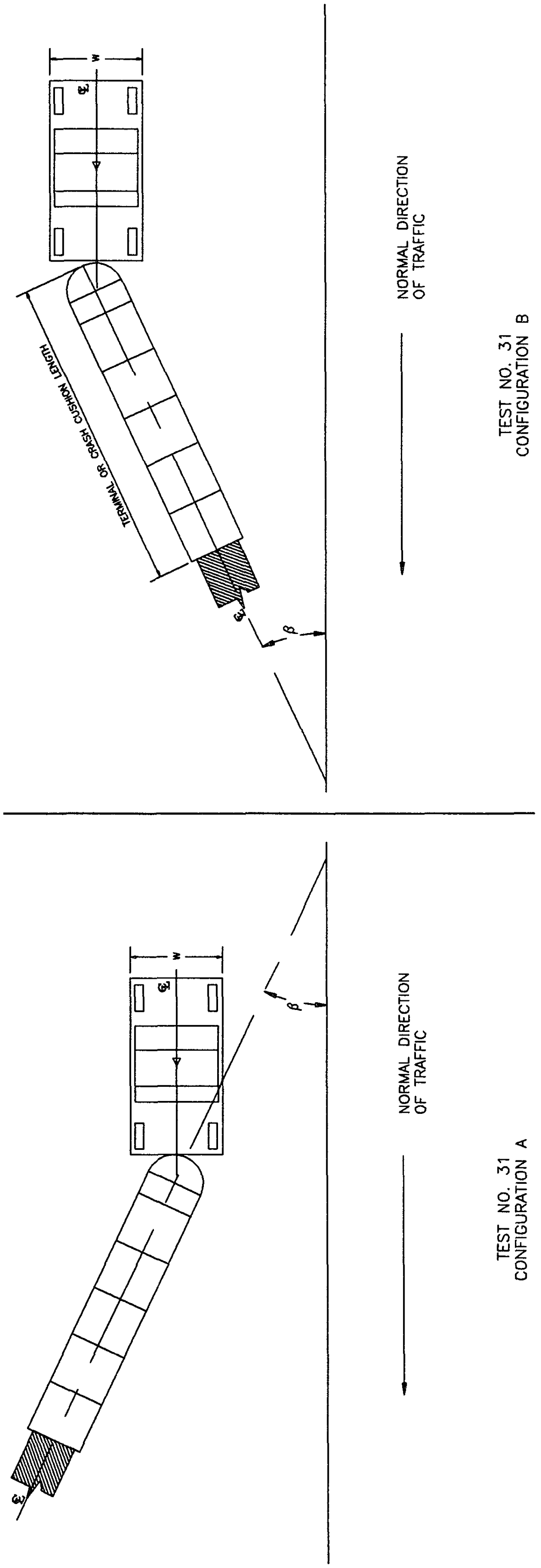

W
℄
TERMINAL OR CRASH CUSHION LENGTH
℄
β
NORMAL DIRECTION OF TRAFFIC
TEST NO. 31
CONFIGURATION B
W
℄
β
℄
NORMAL DIRECTION OF TRAFFIC
TEST NO. 31
CONFIGURATION A

Figure 3.4. Atypical orientation of terminal or crash cushion.

ever, the potential for vehicular overturn will be greater in the high-speed test due to the yawing moment on the vehicle created by the offset impact.

- For a multiple - support device, the vehicle's approach path should be aligned so that the maximum number of supports are impacted unless the test is intended to evaluate other conditions.
- Recommended tolerance on the impact point is ± 0.05(W), where *W* is the width of the vehicle.

3.2.3.2 Description of Tests

Tests 60 and 61-Support Structures

Two tests are recommended for support structures for each test level using either the recommended 820C vehicle or the optional 700C vehicle: a low-speed test and a high-speed test. The low-speed test is generally intended to evaluate the breakaway, fracture, or yielding mechanism of the support whereas the high-speed test is intended to evaluate vehicle and test article trajectory. Occupant risk is of concern in both tests. If the primary concern regarding the impact behavior of a support system is penetration of the test article or parts thereof into the occupant compartment as opposed to occupant impact velocity and ridedown acceleration and/or vehicular stability, it may be preferable to use the 2000P vehicle in lieu of or in addition to the 820C vehicle. The choice will depend on the front profile of the two vehicles in relation to the geometry of the test article and elements thereon that could potentially penetrate the occupant compartment. In evaluating support structures for test level 3, tests should also be conducted at speeds between 35 and 100 km/h if there is a reasonable potential for such tests to be more critical than those recommended.

Tests 70 and 71-Work Zone Traffic Control Devices

Two tests are recommended for work zone traffic control devices for each test level using either the recommended 820C vehicle or the optional 700C vehicle: a low-speed test and a high-speed test. The low-speed test is generally intended to evaluate the breakaway, fracture, or yielding mechanism of the device whereas the high-speed test is intended to evaluate vehicular stability and test article trajectory. Occupant risk is of concern in both tests. Test 70 may be omitted when it can be clearly determined that Test 71 is more critical. For example, Test 71 will be more critical than Test 70 for work zone traffic control devices having a relatively small mass, such as plastic drums used as channelization devices, lightweight barricades, and so on. If the mass of a free-standing (resting on but not attached to ground or paved surface) work zone traffic control device is 45 kg or less, evaluation criteria H and I of Table 5.1 are optional. For a device with sand bags or other ballasts at its base (for stability), the mass of the ballast need not be added to the mass of the device provided the ballast effectively does not contribute to the change in the vehicle's velocity upon impact with the device. If the primary concern regarding the impact behavior of a traffic control device is penetration of the test article or parts thereof into the occupant compartment as

TABLE 3.3. Test matrix for support structures, work zone traffic control devices, and breakaway utility poles

Test Level	Feature	Test Designation	Impact Conditions[c]			Impact Point	Evaluation Criteria[d] (See Table 5.1)
			Vehicle	Nominal Speed (km/h)	Nominal Angle[g], θ (deg)		
2	Support Structures	2-60	820C	35	0-20	(b)	B,D,F,H,I,(J),K,N
		S2-60[a]	700C	35	0-20	(b)	B,D,F,H,I,(J),K,N
		2-61	820C	70	0-20	(b)	B,D,F,H,I,(J),K,N
		S2-61[a]	700C	70	0-20	(b)	B,D,F,H,I,(J),K,N
	Work Zone Traffic Control Devices[h]	2-70[e]	820C	35	0-20	(b)	B,D,E,F,H,I,(J),K,N
		S2-70[a]	700C	35	0-20	(b)	B,D,E,F,H,I,(J),K,N
		2-71	820C	70	0-20	(b)	B,D,E,F,H,I,(J),K,N
		S2-71[a]	700C	70	0-20	(b)	B,D,E,F,H,I,(J),K,N
	Breakaway Utility Poles[i]	2-80	820C	50	0-20	(b)	B,D,F,H,I,(J),K,N
		S2-80[a]	700C	50	0-20	(b)	B,D,F,H,I,(J),K,N
		2-81	820C	70	0-20	(b)	B,D,F,H,I,(J),K,N
		S2-81[a]	700C	70	0-20	(b)	B,D,F,H,I,(J),K,N
3[f] Basic Level	Support Structures	3-60	820C	35	0-20	(b)	B,D,F,H,I,(J),K,N
		S3-60[a]	700C	35	0-20	(b)	B,D,F,H,I,(J),K,N
		3-61	820C	100	0-20	(b)	B,D,F,H,I,(J),K,N
		S3-61[a]	700C	100	0-20	(b)	B,D,F,H,I,(J),K,N
	Work Zone Traffic Control Devices[h]	3-70[e]	820C	35	0-20	(b)	B,D,E,F,H,I,(J),K,N
		S3-70[a,e]	700C	35	0-20	(b)	B,D,E,F,H,I,(J),K,N
		3-71	820C	100	0-20	(b)	B,D,E,F,H,I,(J),K,N
		S3-71[a]	700C	100	0-20	(b)	B,D,E,F,H,I,(J),K,N
	Breakaway Utility Poles[i]	3-80	820C	50	0-20	(b)	B,D,F,H,I,(J),K,N
		S3-80[a]	700C	50	0-20	(b)	B,D,F,H,I,(J),K,N
		3-81	820C	100	0-20	(b)	B,D,F,H,I,(J),K,N
		S3-81[a]	700C	100	0-20	(b)	B,D,F,H,I,(J),K,N

[a] Test is optional. See Section 3.1.
[b] See Figure 3.5 for impact point.
[c] See Section 3.3.4 for tolerances on impact conditions.
[d] Criteria in parenthesis are optional.
[e] See discussion in Section 3.2.3.2 relative to test 70.
[f] See discussion in Section 3.2.3.2 relative to impact speeds for test level 3.
[g] See discussion in Section 3.2.3.1 relative to the critical impact angle (CIA).
[h] See discussion in Section 3.2.3.2 relative to evaluation criteria for work zone traffic control devices.
[i] See discussion in Section 3.2.3.2 relative to evaluation criteria for breakaway utility poles.

opposed to occupant impact velocity and ridedown acceleration and/or vehicular stability, it may be preferable to use the 2000P vehicle in lieu of or in addition to the 820C vehicle. The choice will depend on the front profile of the two vehicles in relation to the geometry of the test article and elements thereon that could potentially penetrate the occupant compartment. In evaluating traffic control devices for test level 3, tests should also be conducted at speeds

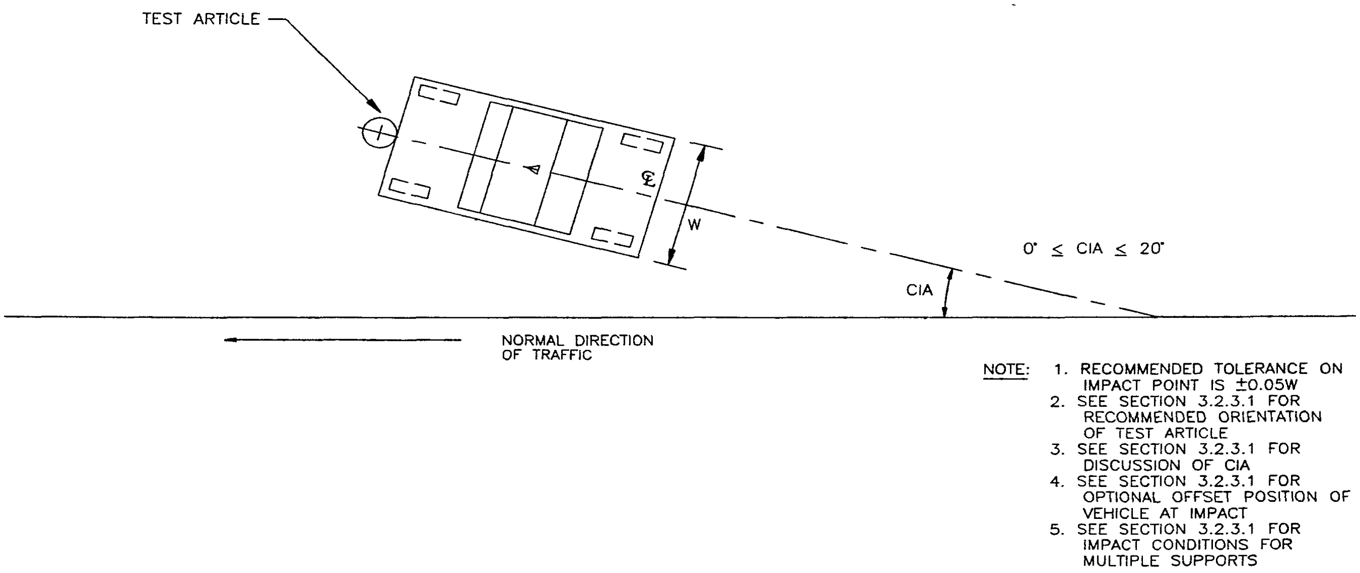

TEST ARTICLE
℄
W
0° ≤ CIA ≤ 20°
CIA
NORMAL DIRECTION OF TRAFFIC
NOTE:
1. RECOMMENDED TOLERANCE ON IMPACT POINT IS ±0.05W
2. SEE SECTION 3.2.3.1 FOR RECOMMENDED ORIENTATION OF TEST ARTICLE
3. SEE SECTION 3.2.3.1 FOR DISCUSSION OF CIA
4. SEE SECTION 3.2.3.1 FOR OPTIONAL OFFSET POSITION OF VEHICLE AT IMPACT
5. SEE SECTION 3.2.3.1 FOR IMPACT CONDITIONS FOR MULTIPLE SUPPORTS

Figure 3.5. Impact conditions for single-support structures, work zone traffic control devices, and breakaway utility poles.

between 35 and 100 km/h if there is a reasonable potential for such tests to be more critical than those recommended.

Tests 80 and 81-Breakaway Utility Poles

Two tests are recommended for breakaway utility poles for each test level using either the recommended 820C vehicle or the optional 700C vehicle: a low-speed test and a highspeed test. The low-speed test is generally intended to evaluate the breakaway, fracture, or yielding mechanism of the support whereas the high-speed test is intended to evaluate vehicular stability and test article trajectory. Occupant risk is of concern in both tests. In evaluating breakaway utility poles for test level 3, tests should also be conducted at speeds between 50 and 100 km/h if there is a reasonable potential for such tests to be more critical than those recommended. Note that it is recommended that occupant impact velocities in a breakaway utility pole test not exceed values given in the upper part of criterion "H" of Table 5.1. However, it is preferable that they not exceed the values given in the lower part of criterion "H." (See discussion in Appendix A, Section A3.2.3.)

3.2.4 Truck-Mounted Attenuators (TMAs)

3.2.4.1 General

Recommended tests to evaluate TMAs are given in Table 3.4. Impact conditions for a TMA are shown in Figure 3.6. To date most, if not all, TMAs have been designed to meet performance requirements of test level 2 or requirements similar thereto. Test level 3 requirements are considerably more demanding and there is no assurance that a TMA can be designed to these requirements without adversely affecting other factors that must be considered in the design and use of a TMA. Consequently, level 2 is considered the basic test level. Reference should be made to Section 2.3.2.4 concerning the manner in which the support truck is restrained or braked and to Section 2.4.1.3 for parameters of the support truck.

3.2.4.2 Test Descriptions

Tests 50 and 51

Test 50 is intended to evaluate risks to occupants of a small car impacting the TMA. Test 51 is intended to evaluate structural adequacy of the TMA, risks to occupants, and the roll-ahead distance of the supporting truck for impact with a heavy passenger vehicle. Roll-ahead distance is the distance the supporting truck moves after impact. Factors affecting this distance are discussed in Appendix A, Section A3.2.4. Roll-ahead distances are needed to select safe separation distances from the TMA vehicle and the workers it is intended to protect.

Tests 52 and 53 (optional)

Tests 52 and 53 are new tests and there is no assurance that current TMA designs can meet requirements of these tests. Moreover, there is no assurance that new TMA designs can be made to meet these test requirements without significant increases in cost or without detrimental effects on truck handling, TMA durability, or other operational considerations. For these reasons Tests 52 and 53 are optional. However, it should be a goal of developers and user agencies to develop a TMA capable of meeting requirements of these tests. Impact conditions for Tests 52 and 53 are believed to be representative of many collisions with TMAs.

TABLE 3.4. Test matrix for truck-mounted attenuators

Test Level	Test Designation	Impact Conditions[c]			Impact Point	Evaluation Criteria[e,g] (See Table 5.1)	EVALUATION CRITERIA[f,g] (See Table 5.1)
		Vehicle	Nominal Speed (km/h)	Nominal Angle, θ (deg)			
2 Basic Level	2-50	820C	70	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	S2-50[a]	700C	70	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	2-51	2000P	70	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	2-52[h]	2000P	70	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	2-53[h]	2000P	70	10	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
3[d]	3-50	820C	100	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	S3-50[a]	700C	100	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	3-51	2000P	100	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	3-52[h]	2000P	100	0	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K
	3-53[h]	2000P	100	10	(b)	D,F,H,I,(J),K	C,D,F,I,(J),K

[a] Test is optional. See Section 3.1.
[b] See Figure 3.6 for impact point.
[c] See Section 3.3.5 for tolerances on impact conditions. See Section 2.4.1.3 for recommendations relative to TMA support truck.
[d] See discussion in Section 3.2.4.1 relative to test level 3.
[e] For impacting vehicle and its occupants.
[f] For supporting truck and its driver. See discussion in Section 5.3.
[g] Criteria in parenthesis are optional.
[h] Test is optional. See Section 3.2.4.1.

Note that recommended tests only address safety performance of a TMA due to vehicular collisions. Predictable and continued safe performance of a TMA will require consideration of other factors such as durability, mobility of supporting truck, road-induced vibrations, maintainability, influence of temperature variations, influence of moisture, and other factors. Reference 53 provides a synthesis of practices relevant to the selection and use of TMAs.

3.2.5 Roadside Geometric Features

Occasionally it may be desirable to experimentally evaluate the impact performance of roadside geometric features, such as safety treatments of drainage structures (e.g., sloped-end culverts with or without safety grates), driveway slopes, median crossover slopes, curbs, ditches, embankment side slopes, rock cuts, and discontinuities between the edge of the paved road surface and the shoulder. Because of the special nature of these features, it is not feasible to develop specific test matrices.

Crash test studies of geometric features have generally involved a combination of computer simulations coupled with limited full-scale crash tests *(4-8)*. In most cases the HighwayVehicle-Object-Simulation Model (HVOSM) computer program valuate important parameters and to predict limits of impact performance of the feature. The accompanying crash test program is typically designed to verify and calibrate the results of the simulation study. This type of approach is usually necessitated by the number of variables that must normally be considered together with the limited number of tests that can be conducted. In the absence of a more rational procedure, a similar approach may be followed in future studies of these types of geometric features.

If feasible, these features should be designed and evaluated within the framework of one of the first three test levels for longitudinal barriers including test speeds and test vehicles. The recommended impact angles for longitudinal barrier tests may or may not be critical. It is possible that smaller angles will be more critical, something that can usually be determined by computer simulation. It is recommended that the impact angle not exceed 20 deg.

Previous studies (4) have shown that it is difficult from a practical and cost-effective standpoint to design certain geometric features, such as sloping culvert ends on driveways and the accompanying driveway slope, to meet evaluation criteria for high-speed (100 km/h) encroachments. Further studies are

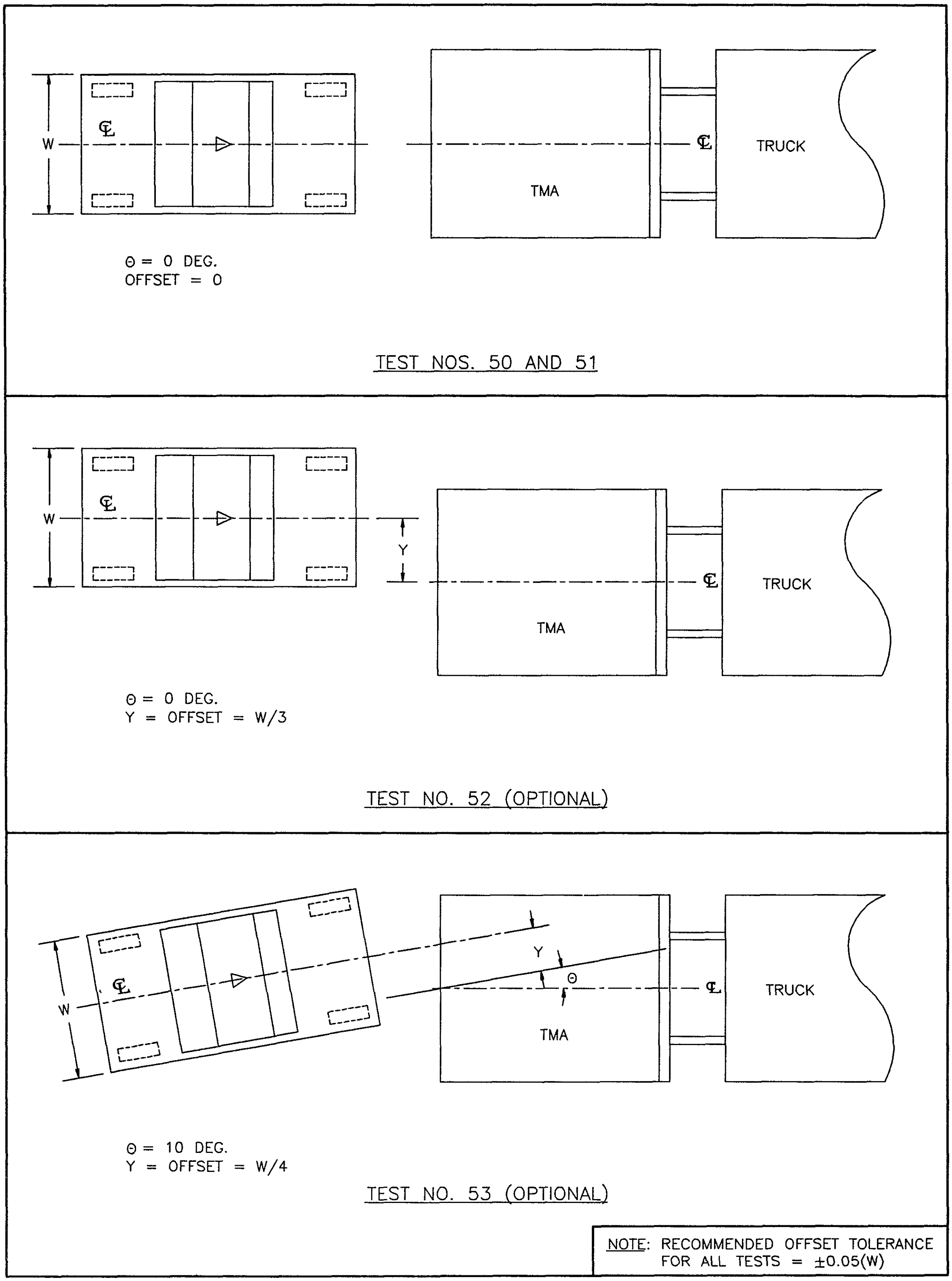

W
℄
TMA
℄
TRUCK
Θ = 0 DEG.
OFFSET = 0
TEST NOS. 50 AND 51
W
℄
Y
TMA
℄
TRUCK
Θ = 0 DEG.
Y = OFFSET = W/3
TEST NO. 52 (OPTIONAL)
W
℄
Y
Θ
TMA
℄
TRUCK
Θ = 10 DEG.
Y = OFFSET = W/4
TEST NO. 53 (OPTIONAL)
NOTE: RECOMMENDED OFFSET TOLERANCE FOR ALL TESTS = ±0.05(W)

Figure 3.6. Impact conditions for a TMA.

TABLE 3.5. Impact speed and angle tolerances[a]

Vehicle	Speed Tolerance (km/h)	Angle Tolerance (deg)
700C	± 4.0	± 1.5
820C	± 4.0	± 1.5
2000P	± 4.0	± 1.5
8000S	± 5.0	± 1.5
36000V	± 5.0	± 2.0
36000T	± 5.0	± 2.0

[a] See Section 3.3.1 for discussion of composite tolerance limits for impact speed impact angle, and vehicle mass.

needed to examine and develop cost-effective safety treatments for geometric features.

A limited number of tests have been conducted on special roadway and roadside conditions, such as potholes and pavement/ shoulder discontinuities *(11-14)*. Typically, the approach taken in these types of studies involves a determination of limiting vehicular response parameters for a given roadway or roadside condition. For example, in evaluation of pavement edge discontinuities, live drivers are typically used. The typical test procedure is, in sequential order, to (a) drop the right side tires, or all four tires, of the vehicle just off the edge of the pavement, (b) begin a steer-back-to-the-road maneuver until the tires mount the pavement, and (c) input a corrective steer to avoid intrusion into an opposite lane. The path of the vehicle is observed and recorded. The procedure is repeated at increasing increments of vehicular speed until a speed is reached at which vehicular intrusions into an opposite lane occur or vehicular loss of control occurs.

To the extent appropriate and practical, evaluation criteria for a longitudinal barrier should be used to evaluate the impact performance of geometric features. See Section 5.6 for further discussions of this matter.

3.3 TOLERANCES ON IMPACT CONDITIONS

3.3.1 General

Impact conditions as defined by the mass, speed, and angle of the impacting vehicle are sometimes difficult to control and may vary slightly from recommended nominal values. The ability to control impact conditions decreases somewhat for the heavy vehicle tests, that is, those with the 8000S, 36000V, and 36000T vehicles. It is therefore necessary to establish reasonable limits or tolerances on impact conditions.

Recommended tolerances on the test inertial mass of test vehicles are given in Tables 2.1 and 2.2. A survey of testing agencies was conducted to examine "target" versus "actual" impact speed and angle for a number and variety of tests. Based on this survey, recommended tolerances for speed and angle are given in Table 3.5.

In addition to recommended tolerances on vehicular mass, speed, and angle, a composite tolerance limit is presented for the combined effects of the test parameters as determined by an impact severity expression:

$$IS = \frac{1}{2} M(V \text{ Sin } \Theta)^2$$

Where *IS* is the impact severity in joules (J), *M* is the test inertial mass of the vehicle in kilograms (kg), V is the impact speed in meters/second (m/s), and 0 is the impact angle in degrees (deg). For tests of redirection features, e is the impact angle; for all other tests, Sin O should be set to 1.

In theory, either of the three parameters *M,* V, and O can be adjusted, within reason, to achieve a desired *IS.* In practice, it is recommended that the test inertial mass of the vehicle be adjusted, as necessary, within the recommended tolerances to be reasonably close to the recommended nominal value. Assuming the test inertial mass is near the nominal value, either speed or angle or both can be adjusted to achieve the recommended *IS.* However, as a general rule, the target impact angle should not be adjusted because impact severity is extremely sensitive to this parameter. Thus, test speed should be the primary parameter adjusted, if necessary, to achieve the recommended *IS.*

The following sections give recommended limits on *IS* for given classes of features. Values given in the tables were computed as follows:

Nominal IS. Computed from Equation 3.1 with *M, V,* and 0 set equal to nominal values as given in previous sections.

Suggested IS Tolerance. Two steps were used in computing negative and positive tolerances. For *negative tolerance: (1) IS* was computed from Equation 3.1 with *M* and 0 set equal to their nominal values and V set equal to its nominal value *minus* the tolerance given in Table 3.5, then (2) the suggested negative *IS* tolerance was determined by subtracting the nominal *IS* from the value of part 1. For *positive tolerance:* steps 1 and 2 were repeated except V was set equal to its nominal value *plus* the tolerance given in Table 3.5.

3.3.2 Longitudinal Barriers

Table 3.6 lists nominal or target *IS* values and recommended tolerances for tests of longitudinal barriers.

For a longitudinal barrier test, it is preferable that the actual *IS* be equal to or greater than the target value. Furthermore, if the actual *IS* exceeds the positive tolerance and if test results meet recommended evaluation criteria, the test need not be repeated. In such a case, it is not necessary to meet the positive tolerance on speed or angle recommended in Table 3.5.

3.3.3 Terminals and Crash Cushions

Table 3.7 lists nominal or target *IS* values and recommended tolerances for tests of terminals and crash cushions.

For a terminal or crash cushion test, it is preferable that the actual *IS* be equal to or greater than the target value. Furthermore, if the actual *IS* exceeds the positive tolerance and if test results meet recommended evaluation criteria, the test need not be repeated. In such a case it is not necessary to meet the positive tolerance on speed or angle recommended in Table 3.5.

TABLE 3.6. Nominal impact severity values and tolerances for longitudinal barriers

Test Designation[a]	Nominal IS (kJ)	Suggested IS Tolerance (kJ)	
Test Level 1			
1-10, 1-20	9.3	-1.4	+1.5
S1-10, S1-20	7.9	-1.2	+1.3
1-11, 1-21	34.5	-5.3	+5.7
Test Level 2			
2-10, 2-20	18.1	-2.0	+2.1
S2-10, S2-20	15.5	-1.7	+1.8
2-11, 2-21	67.6	-7.5	+7.9
Test Level 3			
3-10, 3-20	37.0	-2.9	+3.0
S3-10, S3-20	31.6	-2.5	+2.6
3-11, 3-2	138.1	-10.8	+11.3
Test Level 4			
4-10, 4-20	37.0	-2.9	+3.0
S4-10, S4-20	31.6	-2.5	+2.6
4-11, 4-21	138.1	-10.8	+11.6
4-12, 4-22	132.5	-16.1	+17.1
Test Level 5			
5-10, 5-20	37.0	-2.9	+3.0
S5-10, S5-20	31.6	-2.5	+2.6
5-11, 5-21	138.1	-10.8	+11.3
5-12, 5-22	596.2	-72.3	+76.9
Test Level 6			
6-10, 6-20	37.0	-2.9	+3.0
S6-10, S6-20	31.6	-2.5	+2.6
6-11, 6-21	138.1	-10.8	+11.3
6-12, 6-22	596.2	-72.3	+76.9

[a] See Table 3.1 for test descriptions.

3.3.4 Support Structures, Work Zone Traffic Control Devices, and Breakaway Utility Poles

Table 3.8 lists nominal or target IS values and recommended tolerances for tests of support structures, work zone traffic control devices, and breakaway utility poles.

For the *low-speed* test of a support structure, work zone traffic control device, or breakaway utility pole, it is preferable that the actual *IS* be equal to or less than the target value. However, it is recommended that the impact speed be within the recommended speed tolerances of Table 3.5 for the low-speed tests. For the *high-speed* test of a support structure, work zone traffic control device, or breakaway utility pole, it is preferable that the actual impact severity *IS* be equal to or greater than the target value. Furthermore, if the actual *IS* is greater than the recommended positive tolerance and if test results meet recommended evaluation criteria, the test need not be repeated. In such a case, it is not necessary to meet the positive tolerance on speed recommended in Table 3.5.

TABLE 3.7. Nominal impact severity values and tolerances for terminals and crash cushions

Test Designation[a]	Nominal IS (kJ)	Suggested IS Tolerance (kJ)	
Test Level 1			
1-30, 1-32, 1-40, 1-42	79.1	-12.1	+13.1
S1-30, S1-32, S1-40, S1-42	67.5	-10.4	+11.2
1-31, 1-33, 1-41, 1-43, 1-44	192.9	-29.6	+32.1
1-34, 1-36	5.3	-0.8	+0.9
S1-34, S1-36	4.5	-0.7	+0.8
1-35, 1-37, 1-38, 1-39	22.6	-3.5	+3.8
Test Level 2			
2-30, 2-32, 2-40, 2-42	154.9	-17.2	+18.2
S2-30, S2-32, S2-40, S2-42	132.3	-14.7	+15.5
2-31, 2-33, 2-41, 2-43, 2-44	377.9	-41.9	+44.4
2-34, 2-36	10.4	-1.2	+1.2
S2-34, S2-36	8.9	-1.0	+1.0
2-35, 2-37, 2-38, 2-39	44.2	-4.9	+5.2
Test Level 3			
3-30, 3-32, 3-40, 3-42	316.4	-24.8	+25.8
S3-30, S3-32, S3-40, S3-42	270.1	-21.2	+22.0
3-31, 3-33, 3-41, 3-43, 3-44	771.7	-60.4	+62.9
3-34, 3-36	21.2	-1.7	+1.7
S3-34, S3-36	18.1	-1.4	+1.5
3-35, 3-37, 3-38, 3-39	90.3	-7.1	+7.4

[a] See Table 3.2 for test descriptions.

TABLE 3.8. Nominal impact severity values and tolerances for support structures, work zone traffic control devices, and breakaway utility poles

Test Designation[a]	Nominal IS (kJ)	Suggested IS Tolerance (kJ)	
Test Level 2			
2-60, 2-70	38.7	-8.3	+9.4
S2-60, S2-70	33.1	-7.1	+8.0
2-61, 2-71, 2-81	154.9	-17.2	+18.2
S2-61, S2-71, S2-81	132.3	-14.7	+15.5
2-80	79.1	-12.1	+13.1
S2-80	67.5	-10.4	+11.2
Test Level 3			
3-60, 3-70	38.7	-8.3	+9.4
S3-60, S3-70	33.1	-7.1	+8.0
3-61, 3-71, 3-81	316.4	-24.8	+25.8
S3-61, S3-71, S3-81	270.1	-21.2	+22.0
3-80	79.1	-12.1	+13.1
S3-80	67.5	-10.4	+11.2

[a] See Table 3.3 for test descriptions.

TABLE 3.9. Nominal impact severity values for truck-mounted attenuators

Test Designation[a]	Nominal IS (kJ)	Suggested IS Tolerance (kJ)
Test Level 2		
2-50	154.9	-17.2 +18.2
S2-50	132.3	-14.7 +15.5
2-51, 2-52, 2-53	377.9	-41.9 +44.4
Test Level 3		
3-50	316.4	-24.8 +25.8
S3-50	270.1	-21.2 +22.0
3-51, 3-52, 3-53	771.7	-60.4 +62.9

[a] See Table 3.4 for test descriptions.

3.3.5 Truck-Mounted Attenuators (TMAs)

Table 3.9 lists nominal or target *IS* values and recommended tolerances for tests of TMAs.

For a TMA, it is preferable that the actual *IS* be equal to or greater than the target value. Furthermore, if the actual *IS* exceeds the positive tolerance and if test results meet recommended evaluation criteria, the test need not be repeated. In such a case, it is not necessary to meet the positive tolerance on speed recommended in Section 3.3.1.

3.4 IMPACT POINT FOR REDIRECTIVE DEVICES

3.4.1 General

Impact point is the initial point of contact on a test article by the impacting test vehicle. It is generally known that the location of the initial impact point on a redirective device such as a longitudinal barrier, a terminal, or a redirective crash cushion can affect its safety performance. The potential for wheel snag, vehicular pocketing, and structural failure of the device depends on the initial impact point. To the extent possible, the initial impact point for a redirective device should be selected to establish a worst-case testing condition, that is, the critical impact point (CIP) or the point with the greatest potential for causing failure of the test, whether this be by excessive wheel snag, excessive pocketing, or structural failure of the device.

The Barrier VII simulation program has been used to select the CIP for longitudinal barrier tests *(15)* and it has been validated for this purpose. Although other barrier simulation programs such as GUARD and NARD (see Appendix D) may have the capability for identifying the CIP, they have not been validated for this application and are generally more difficult and costly to use.

When practical, it is recommended that Barrier VII be used to identify the CIP for a given set of test conditions (vehicle, barrier, and test speed and angle). The procedure described in Appendix A, Section A3.4.2, may be used for this purpose. If this is not practical, the following guidelines, which were derived from the use of Barrier VII, may be used.

3.4.2 Longitudinal Barriers

In general there are two CIPs for a longitudinal barrier: one that produces the greatest potential for vehicular pocketing or wheel snagging and one that produces the greatest loading on a critical part of the barrier, such as at a rail splice. If the rail splice is located at a post, as is common in roadside and median barriers, both CIPs are coincident and one test can be used to evaluate all failure modes of concern. It is not uncommon in a bridge rail system for the splices to be located between posts. Fortunately, because the post spacing on most bridge rails is relatively small, all failure modes of concern can usually be evaluated with one test by placing a splice just upstream from the reference post (post from which the "x" distance is measured as described in the next section).

3.4.2.1 Tests with the 700C, 820C, and 2000P Vehicles

Computer simulations have demonstrated that critical impact points are controlled primarily by the post dynamic yield force per unit length of barrier, F_p, and the effective plastic moment of the barrier rail element(s), M_p, *(16)*. Post yield forces can be controlled by either post strength or soil confinement. F_p is calculated by dividing post dynamic yield force by post spacing. M_P is the effective plastic moment of all barrier rail elements. For a single rail barrier system, M_p is merely the plastic moment of the rail element. The effective plastic moment of a multiple rail system is the sum of the plastic moment of the highest beam and the plastic moments of lower beams reduced by a ratio of the heights of the highest and lower rail elements as given in Equation 3.2.

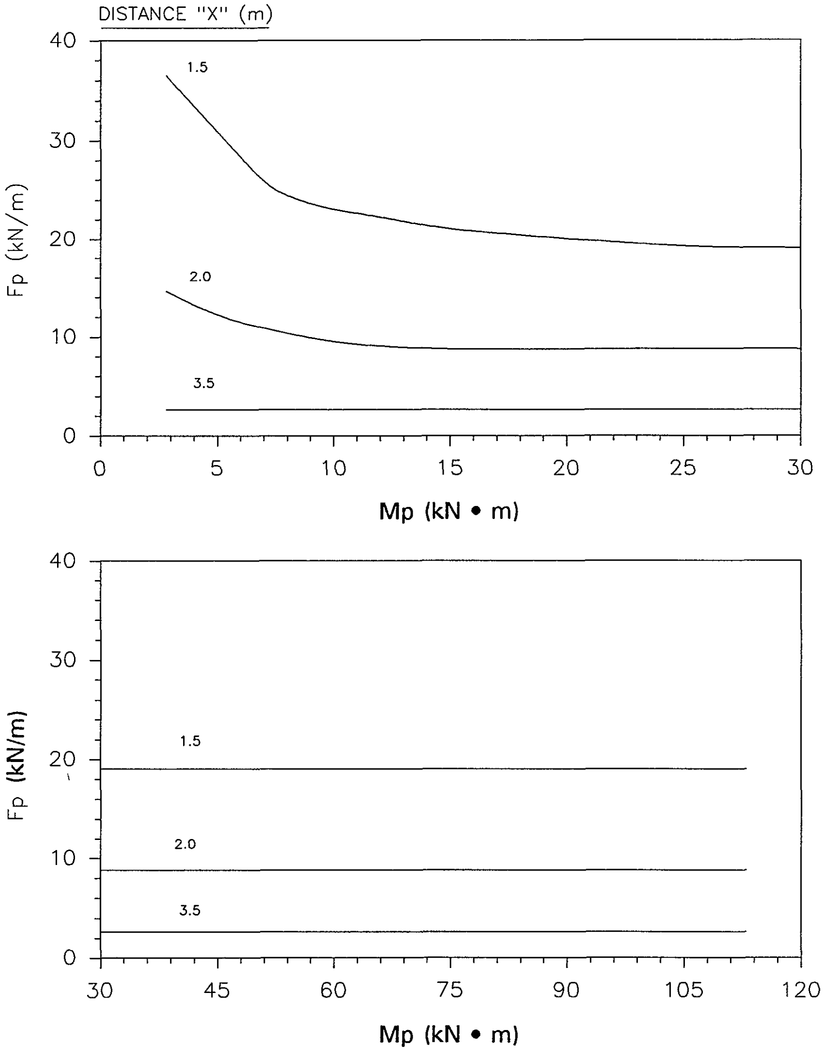
$$M_p = M_h + \sum M_i \times \left(\frac{H_i}{H_h}\right) \qquad (3.2)$$

where:

M_p = effective plastic moment of all barrier rail elements;
M_h = plastic moment of highest rail element above ground or deck;
M_i = plastic moment of a lower barrier rail element;
H_i = height of a lower rail element; and
H_h = height of highest rail element.

A more detailed discussion of F_p and M_p as well as tables of typical values can be found in Appendix A, Section A3.4.2.1.

Figures 3.7 through 3.10 are used to locate the critical impact point, as defined by the distance *"x,"* for the length-of-need portion of flexible, longitudinal barriers (Tests 10 and 11) for a given test level. The figures show plots of the critical impact distance x for values of F_p and M_p for a given barrier system. Distances shown are measured upstream from the reference post/splice as shown in Figure 3.1. A rail splice should be located at



SEE FIGURE 3.1 FOR "X"

Figure 3.7. Critical impact point for Test 10, test levels 1 and 2.

or just upstream of the reference post, provided this is consistent with in-service practice. Interpolation may be used to find values of x for points between the curves and extrapolation may be used to find values of x for points above the upper curve or below the lower curve.

Figures 3.11 through 3.14 are used to locate the CIP for transitions between longitudinal barriers having different lateral stiffness (Tests 20 and 21) for a given test level. The figures show plots of the critical impact distance x for values of F_p and M_p for a given barrier system. Distances shown are measured upstream from the end of the stiffer system as shown in Figure 3.1. Interpolation may be used to find values of x for points between the curves and extrapolation may be used to find values of x for points above the upper curve or below the lower curve.

Properties of the more flexible barrier should be used for determining F_p and M_p. Note that Figures 3.11 through 3.14 were developed with a transition to a rigid barrier. When the stiffer barrier system is not rigid, the distance x will increase slightly. However, crash testing and simulation have shown that this effect is relatively small and can usually be ignored. A more detailed discussion of the above procedures can be found in Appendix A, Section A3.4.2.1, and in reference 16.

When testing very stiff barrier systems (i.e., those that are essentially rigid such as concrete bridge rails), Table 3.10 may be used for x distances for the indicated tests. These numbers also represent minimum values for x distances. If extrapolations of curves shown in Figures 3.7 through 3.14 give CIP values lower than those shown in Table 3.10, "*x*" distances should be selected from Table 3.10.

Unlike beam and post systems, safety-shaped rigid barriers rely to a large extent on tire/barrier contact forces for redirecting impacting vehicles. As a result, critical impact distances for these barriers are somewhat different from the above values for rigid barriers. Table 3.11 shows CIPs for safety-shaped barrier impacts. Data shown in this table were derived from crash testing and simulation of impacts with both rigid and unanchored or

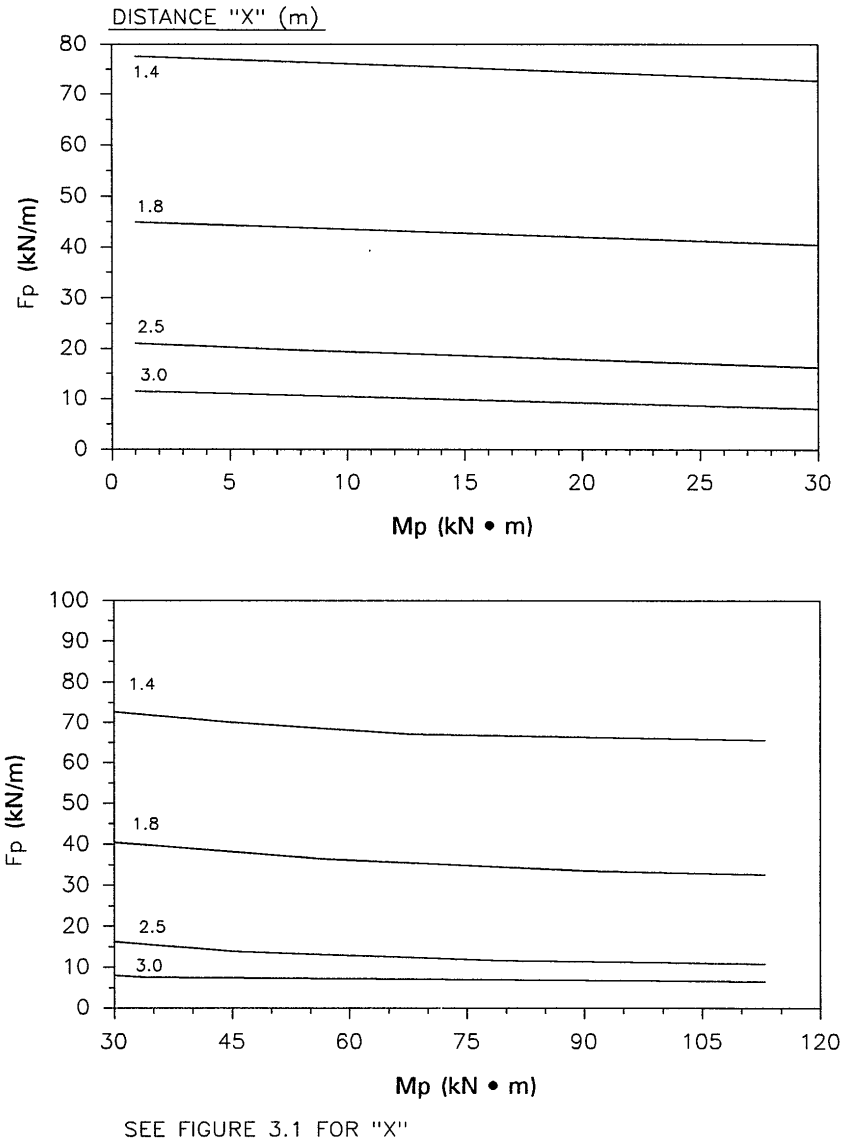


Figure 3.8. Critical impact point for Test 10, levels 3, 4, 5, and 6.

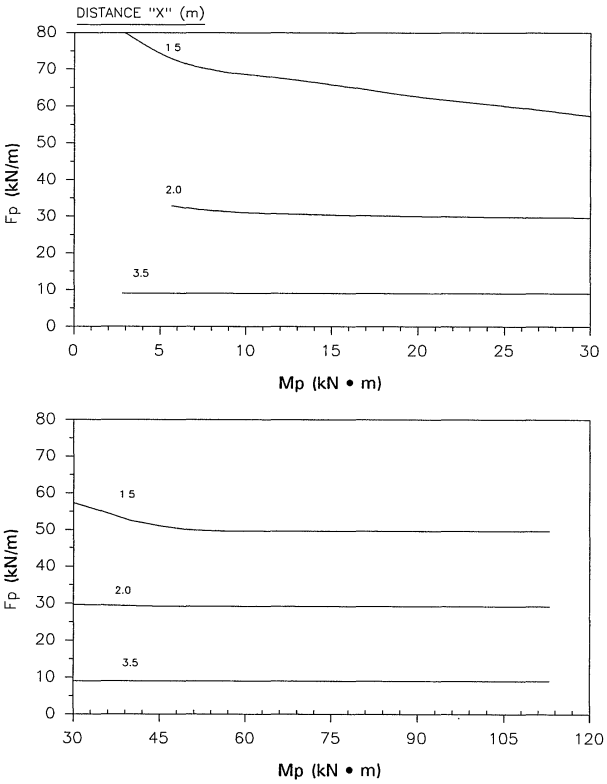
free-standing safety-shaped barrier systems. While testing and simulation results indicate these barriers exhibit slightly larger critical impact distances than their rigid counterparts, the differences are not significant. Therefore, Table 3.11 may be used for both rigid and free-standing safety-shaped barrier systems including precast concrete barriers used in work zones.

3.4.2.2 Tests with the 8000S, 36000V, and 36000T Vehicles

Large trucks do not exhibit the same sensitivity to wheel snag as automobiles and pickups. Therefore, the critical impact point for these vehicles should be chosen to maximize loading on critical barrier elements such as joints or splices. Because general procedures for determining critical impact points are not available for large trucks, these distances must be estimated from crash testing of rigid barriers. Table 3.12 shows the critical distance x from impact point to the location of maximum lateral loading for a number of heavy vehicle crash tests *(17)*. Note that a positive number indicates the maximum loading is *downstream* from the initial impact point and a negative number indicates the maximum loading is *upstream* from the initial impact point. Distances shown in this table were measured on a *rigid,* instrumented wall and, therefore, represent minimum values. Slightly larger distances can be expected for less rigid barriers as well as safety-shaped concrete barriers. Computer simulation is recommended to estimate the CIP for a truck test whenever practical.

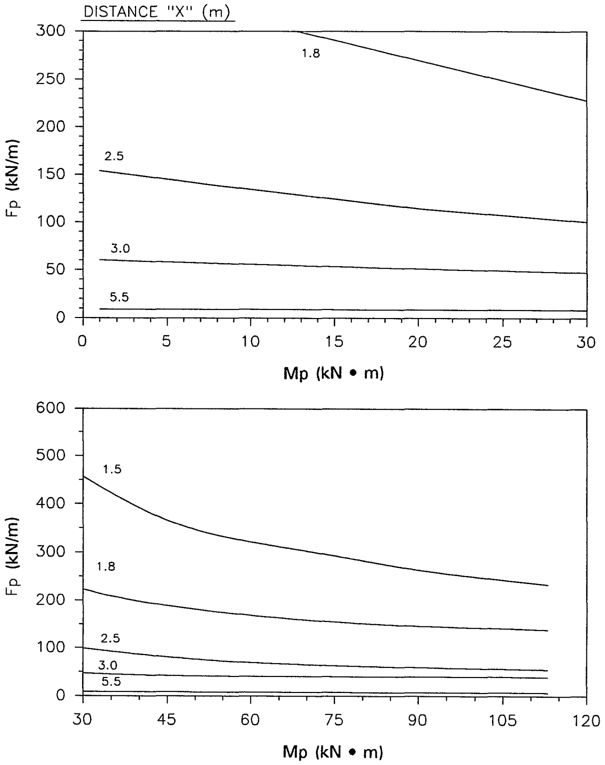
3.4.3 Terminals and Redirective Crash Cushions

Barrier terminals and redirective crash cushions are frequently used to prevent vehicles from impacting the end of a barrier or



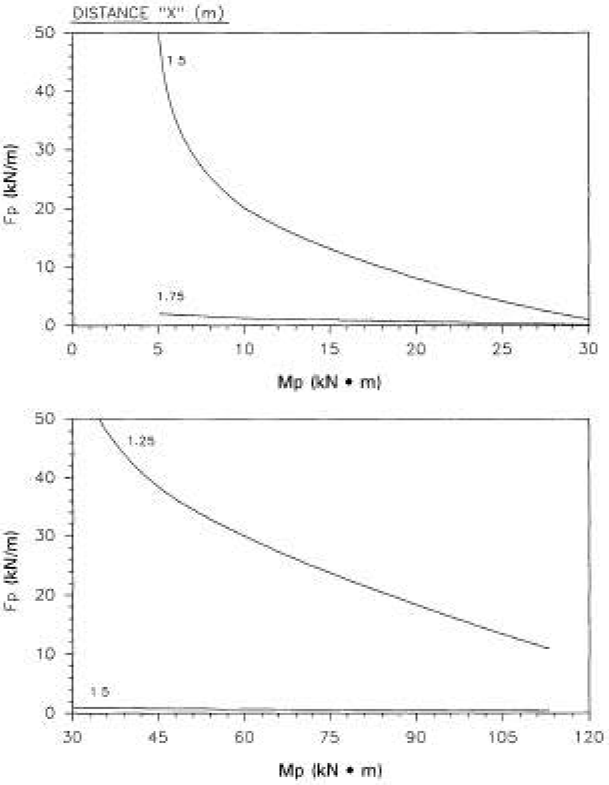
SEE FIGURE 3.1 FOR "X"

Figure 3.9. Critical impact point for Test 11, test levels 1 and 2.



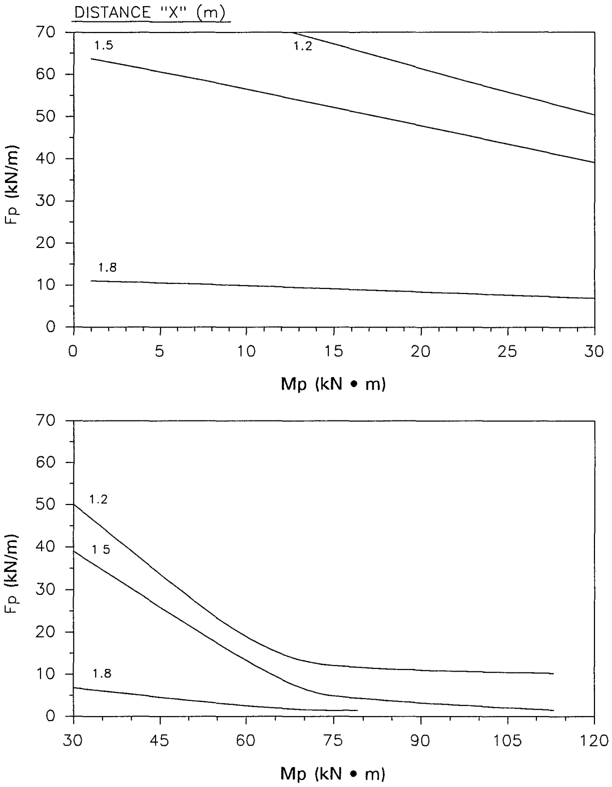
SEE FIGURE 3.1 FOR "X"

Figure 3.10. Critical impact point for Test 11, test levels 3, 4, 5, and 6.



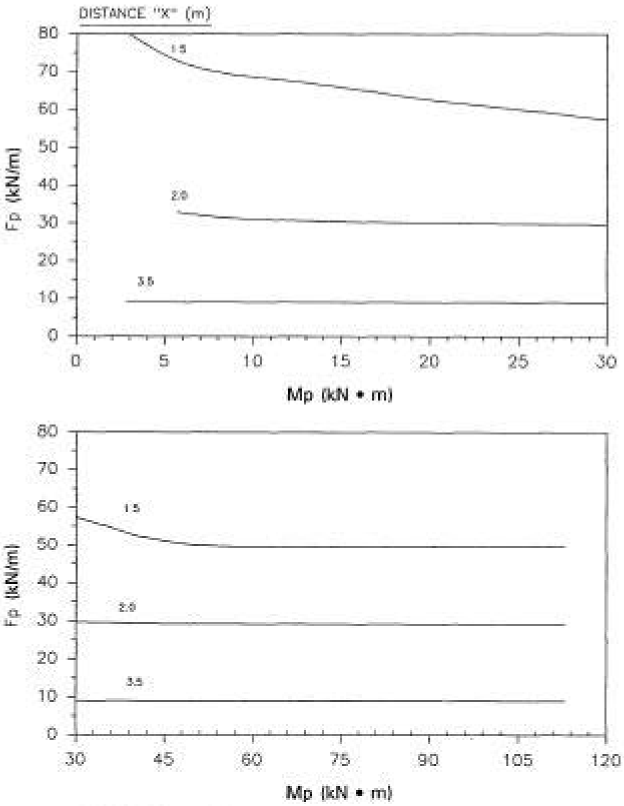
SEE FIGURE 3.1 FOR "X".

Figure 3.11. Critical impact point for Test 20, test levels 1 and 2.



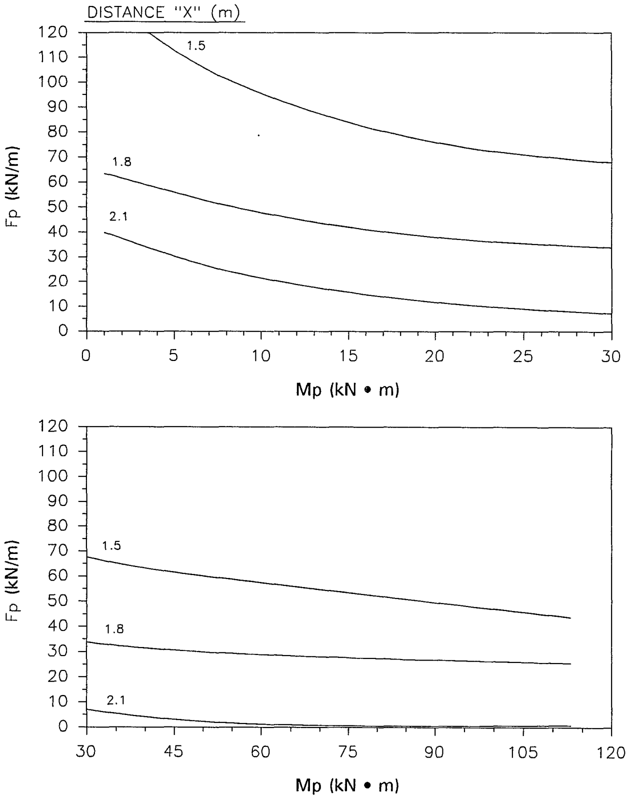
SEE FIGURE 3.1 FOR "X"

Figure 3.12. Critical impact point for Test 20, test levels 3, 4, 5, and 6.



SEE FIGURE 3.1 FOR "X"

Figure 3.9. Critical impact point for Test 11, test levels 1 and 2.



SEE FIGURE 3.1 FOR "X"

Figure 3.14. Critical impact point for Test 21, test levels 3, 4, 5, and 6.

TABLE 3.10. Critical impact point for rigid barrier tests with 700C, 820C, and 8000S vehicles

Test Designation[a]	"x" Distance[b] (m)
1-10, S1-10, 2-10, S2-10	1.0
3-10, S3-10, 4-10, S4-10 5-10, S5-10, 6-10, S6-10	1.1
1-11, 2-11	0.8
3-11, 4-11, 5-11, 6-11	1.3

[a] See Table 3.1 for test descriptions.

[b] See Figure 3.1 for "x."

TABLE 3.11. Critical impact point for concrete safety-shaped barrier tests

Test Designation[a]	"x" Distance[b] (m)
1-10, S1-10, 2-10, S2-10	0.7
3-10, S3-10, 4-10, S4-10 5-10, S5-10, 6-10, S6-10	0.8
1-11, 2-11	0.8
3-11, 4-11, 5-11, 6-11	1.2

[a] See Table 3.1 for test descriptions.

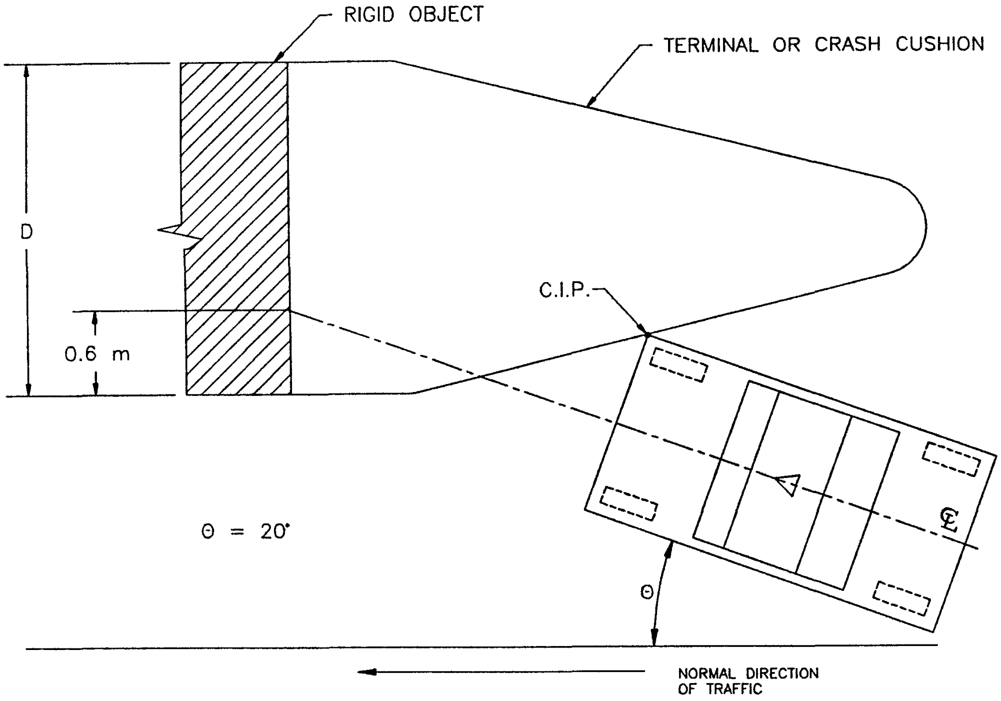
[b] See Figure 3.1 for "x."

TABLE 3.12. Critical impact point for heavy vehicle tests

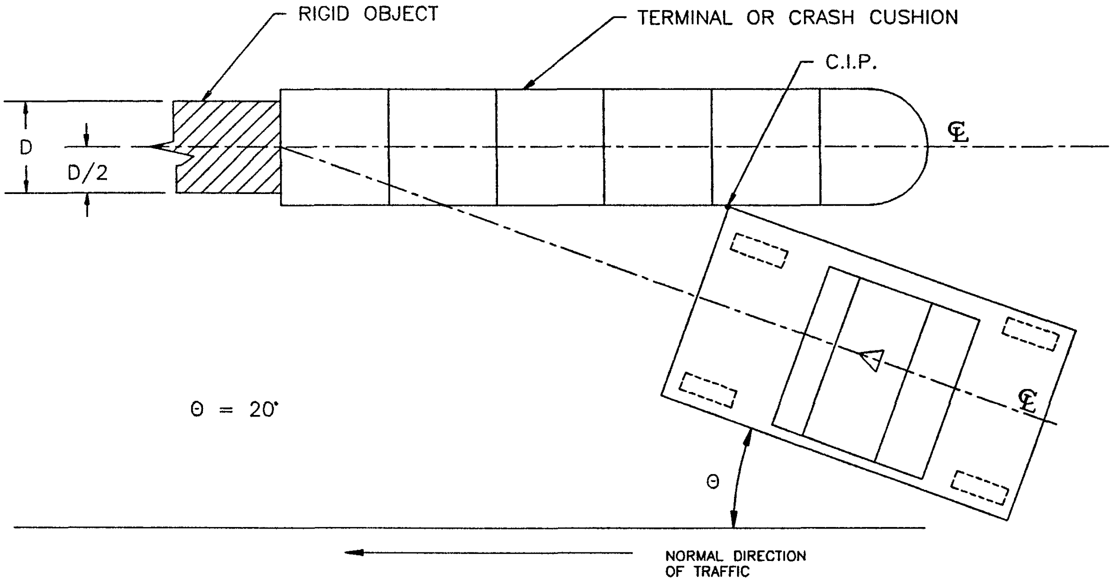
Test Designation[a]	"x" Distance[b] (m)
4-12	1.5
5-12	-0.3
6-12	0.6

[a] See Table 3.1 for test descriptions.

[b] See Figure 3.1 for "x."


RIGID OBJECT
TERMINAL OR CRASH CUSHION
D
0.6 m
C.I.P.
Θ = 20°
℄
Θ
NORMAL DIRECTION OF TRAFFIC

TEST 38 FOR NONGATING DEVICE WHEN D>1.2 m


RIGID OBJECT
TERMINAL OR CRASH CUSHION
C.I.P.
D
D/2
℄
Θ = 20°
℄
Θ
NORMAL DIRECTION OF TRAFFIC

TEST 38 FOR NONGATING DEVICE WHEN D≤1.2 m

Figure 3.15. Critical impact point for Test 38 of nongating device.

some other rigid obstacle. Proper performance of these systems requires redirection of vehicles impacting on the side of the feature. Although no studies have been conducted to identify critical impact points for such features, crash test experience has indicated that for a narrow rigid object the greatest potential for wheel snag or pocketing occurs when the center of the test vehicle is directed at the center of the rigid object. In the case of a very wide rigid object, the greatest potential for wheel snag or pocketing is believed to occur when the center of the vehicle is directed at a spot on the object approximately 0.6 m from the edge of the rigid object. These points are illustrated in Figure 3.15. In the absence of more definitive guidelines for determining the CIP for a terminal or a redirective crash cushion, the impact point for Test 38 may be determined as illustrated in Figure 3.15.

3.5 SIDE IMPACT TESTS

Previous testing guidelines for roadside safety features, as well as those presented herein, use a tracking test vehicle. As such, the vehicle approaches the test article with no appreciable yawing or sideslip motion and, hence, the rear tires essentially track the path of the front tires. However, a large percentage of errant vehicles leave the travelway and impact roadside features in a nontracking or yawed mode *(18)*. Furthermore, the impact performance of a safety or geometric feature is more likely to be unacceptable for a nontracking or side impact than for a tracking impact, all other conditions being the same. Evidence to this effect can be found in the literature *(7,19)*.

A limited number of tests have been conducted in recent years to evaluate side impacts into pole-type support structures *(20)*. Unfortunately, as results of these tests show, it is not feasible to design most roadside safety and geometric features to satisfy current evaluation criteria for side impacts. Consequently, no recommendations are made relative to the impact performance of a safety feature for side impacts.

Risks to occupants in a side impact are high due in large part to very limited vehicular crashworthiness for side impacts. The relatively "soft" side structure of passenger vehicles, coupled with the close proximity of the occupants to the side structure, presents formidable occupant protection problems. Breakaway devices do not readily activate due to the softer side structure and due to the resultant impact force being applied at a higher point than for a head-on impact. Also, there is an increased propensity for overturn when the vehicle is in a nontracking mode.

Concurrent with the effort to prepare this report, an FHWA sponsored study was being conducted to examine the side impact problem and to develop tentative side impact test procedures for sign and luminaire support structures *(21)*. Recommended test and evaluation procedures for side impact testing developed in the FHWA study are given in Appendix G. Until these or other guidelines are nationally accepted, developers of safety features for side impact capabilities may use the guidelines in Appendix G.

This discussion points to the need for further studies to (a) define the magnitude and nature of the nontracking impact problem and (b) develop strategies to mitigate the problem.

CHAPTER 4

DATA ACQUISITION

4.1 TYPICAL PARAMETERS

Proper documentation of a crash test involves three distinct phases. (1) Key properties of the test article and the test vehicle must be recorded in the pretest phase. (2) During the test itself, it is important that the dynamic behavior of the test article and the test vehicle be properly recorded. (3) Finally, it is essential that damage to the test article and test vehicle be documented in the post-test phase.

4.2 PRETEST PARAMETERS

In the pretest phase, the chief objective is to document properties of the test vehicle and the as-built test article. Use of photography is essential in this phase. Key parameters that should be documented in the pretest phase are given in the next section. These parameters are not necessarily all-inclusive and should be supplemented by other relevant parameters as necessary.

4.2.1 Test Vehicle

It is recommended that parameters identified in Figures 4.1 through 4.5 be measured and recorded for the respective test vehicles. It is also preferable that the parameters be recorded in the format shown to facilitate standard reporting among testing agencies. Pretest photographs of the exterior of the test vehicle as well as the occupant compartment of vehicles 700C, 820C, and 2000P should be made (see Section 4.4). Reference should be made to Section 4.3.3 for recommendations relative to accelerometer placement and documentation thereof. In absence of a more exact method, the procedure in reference 81 may be used to determine the center of mass location for two-axle vehicle.

In addition to the information of Figures 4.1 through 4.5, the use of ballasts in any of the test vehicles should be properly documented. This documentation includes ballast type, mass, location, center of mass, method by which it is secured, and photos of the ballast. Also, when feasible, the longitudinal and vertical position of the center of mass of the tractor and unballasted trailer for vehicles 36000V and 36000T should be measured and recorded.

In tests of longitudinal concrete barriers, some testing agencies have used chalk or paint on the sidewall of tires that contact the barrier. Different colored chalk or paint is used to determine length and height of each tire's contact with the barrier. This practice is acceptable provided that the chalk or paint does not appreciably alter the tire-barrier friction properties.

For a TMA test, the year, make, and model of the supporting truck should be documented. The ballast mass, test inertial mass of the supporting truck, and the distribution of the test inertial mass between the front and rear axles should be recorded. The location and manner in which the ballast was secured should be documented. The manner in which the supporting truck was braked should also be documented. See Section 2.3.2.4 for recommendations relative to truck braking.

4.2.2 Test Article

4.2.2.1 General

For any test, the type of paving or soil surface on which the article is mounted and on which the pre-impact approach and post-impact trajectory of the vehicle takes place should be reported. Material specifications should refer to a nationally recognized standard such as ASTM, AASHTO, ACI, AISC, or AISI. The research report should document any pertinent details of the construction of the test article, including special equipment used, problems encountered, and so on.

Reported information should be based on a careful review of the actual test article installation and not just the design drawings and specifications. The following are believed to be key parameters for the respective test articles.

4.2.2.2 Longitudinal Barriers

Geometrics. Mounting heights and lengths of rail elements; post spacing; length of test installation; alignment and orientation of barrier relative to vehicular approach; targeted point of impact relative to end of barrier.

Foundation. Manner in which barrier was supported, including description of structural details of deck and railing-to-deck attachment for bridge railings; details of prototype wing wall or bridge end and its foundation details used in a transition test; embedment procedures for posts embedded in soil (driven, drilled and backfilled, placed in driven inserts, concrete footing, etc.); embedment depth of test article; description of soil and its properties when relevant; description of surface on which free-standing temporary barrier was resting and its frictional properties.

End Conditions. Manner in which the ends of the barrier were anchored, if anchored, including conventional end treatments such as a breakaway cable terminal, or by special end anchorage designs. Some cable barrier systems use a temperature-compensating end anchorage to maintain a specified amount of cable tension. When used, details of these anchorages should be documented including cable tension at the time of the test.

DATE: ____________ TEST NO.: ____________ VIN NO.: ____________ MAKE: ____________

MODEL: ____________ YEAR: ____________ ODOMETER: ____________ TIRE SIZE: ____________

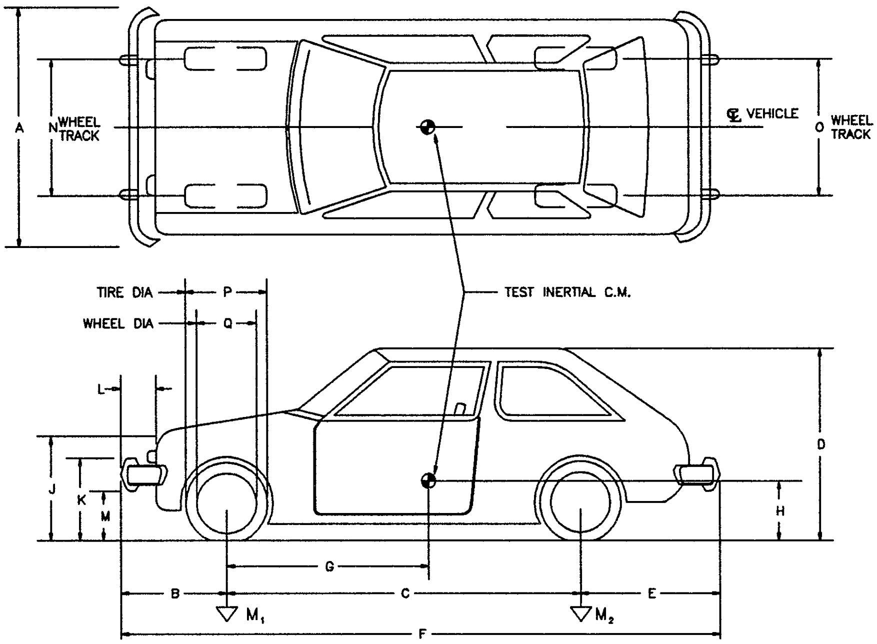
TIRE INFLATION PRESSURE: ____________

MASS DISTRIBUTION (kg) LF ____________ RF ____________ LR ____________ RR ____________

DESCRIBE ANY DAMAGE TO VEHICLE PRIOR TO TEST:

____

____


A
N
WHEEL TRACK
℄ VEHICLE
O
WHEEL TRACK
TIRE DIA
P
WHEEL DIA
Q
TEST INERTIAL C.M.
L
D
J
K
M
H
G
B
C
E
M_1
M_2
F

ENGINE TYPE: ____________

ENGINE CID: ____________

TRANSMISSION TYPE:

— AUTO

— MANUAL

OPTIONAL EQUIPMENT:

____

____

____

DUMMY DATA:

TYPE: ____________

MASS: ____________

SEAT POSITION: ____________

GEOMETRY – (cm)

A ________ D ________ G ________ K ________ N ________ Q ________

B ________ E ________ H ________ L ________ O ________

C ________ F ________ J ________ M ________ P ________

MASS – (kg)	CURB	TEST INERTIAL	GROSS STATIC
M_1			
M_2			
M_T			

Figure 4.1. 700C and 820C parameters.

DATE: ____________ TEST NO.: ____________ VIN NO.: ____________ MAKE: ____________

MODEL: ____________ YEAR: ____________ ODOMETER: ____________ GVW: ____________

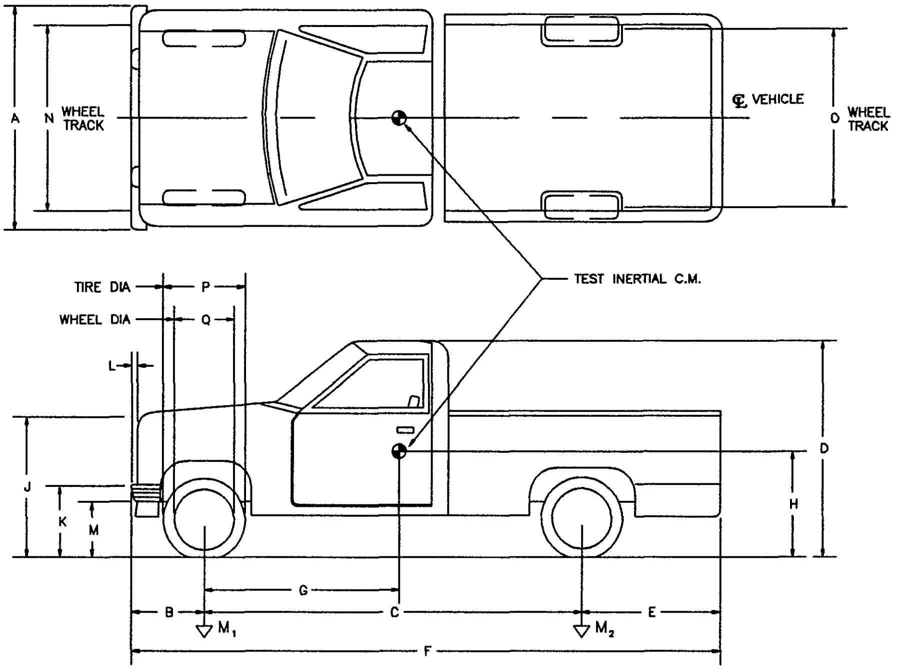
TIRE SIZE: ____________ TIRE INFLATION PRESSURE: ____________ TREAD TYPE: ____________

MASS DISTRIBUTION (kg) LF ____________ RF ____________ LR ____________ RR ____________

DESCRIBE ANY DAMAGE TO VEHICLE PRIOR TO TEST:

____

____


A
N
WHEEL TRACK
℄ VEHICLE
O
WHEEL TRACK
TIRE DIA
P
WHEEL DIA
Q
L
TEST INERTIAL C.M.
D
J
H
K
M
G
B
C
E
M_1
M_2
F

ENGINE TYPE: ____________

ENGINE CID: ____________

TRANSMISSION TYPE:

— AUTO

— MANUAL

OPTIONAL EQUIPMENT:

____

____

____

DUMMY DATA:

TYPE: ____________

MASS: ____________

SEAT POSITION: ____________

GEOMETRY – (cm)

A ________ D ________ G ________ K ________ N ________ Q ________

B ________ E ________ H ________ L ________ O ________

C ________ F ________ J ________ M ________ P ________

MASS – (kg)	CURB	TEST INERTIAL	GROSS STATIC
M_1	________	________	________
M_2	________	________	________
M_T	________	________	________

Figure 4.2. 2000P parameters.

DATE: ____________ TEST NO.: ____________ VIN NO.: ____________ MAKE: ____________

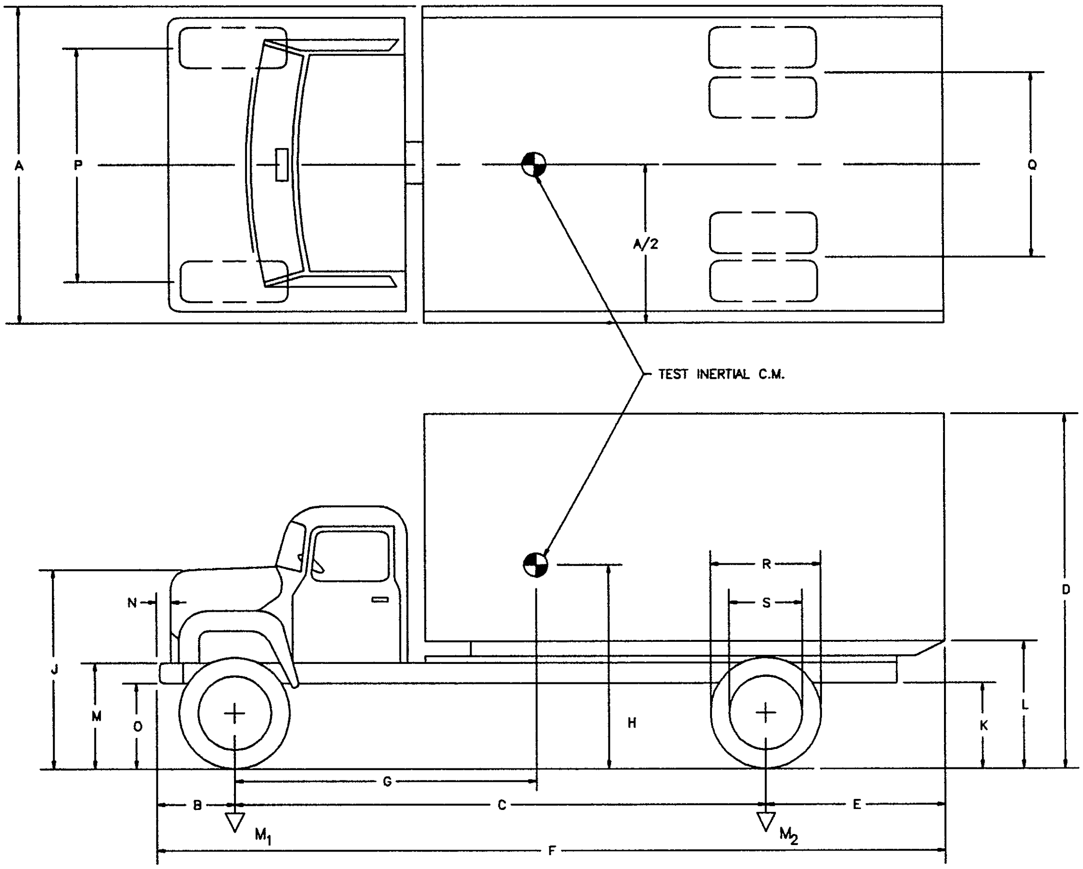
MODEL: ____________ YEAR: ____________ ODOMETER: ____________ TIRE SIZE: ____________

MASS DISTRIBUTION (kg) LF________ RF________ LR________ RR________

DESCRIBE ANY DAMAGE TO VEHICLE PRIOR TO TEST:

____

____


A
P
Q
A/2
TEST INERTIAL C.M.
N
J
M
O
H
R
S
D
L
K
G
B
C
E
M_1
M_2
F

GEOMETRY – (cm)

A________
B________
C________
D________
E________
F________
G________
H________
J________
K________
L________
M________
N________
O________
P________
Q________
R________
S________

MASS – (kg)	CURB	TEST INERTIAL	GROSS STATIC
M_1			
M_2			
M_T			

Figure 4.3. 8000S parameters.

DATE: ____________________ TEST NO.: ________________

TRACTOR:

VIN NO.: ____________________ MAKE: ____________________ MODEL: ____________________

YEAR: ____________________ ODOMETER: ________________

TRAILER:

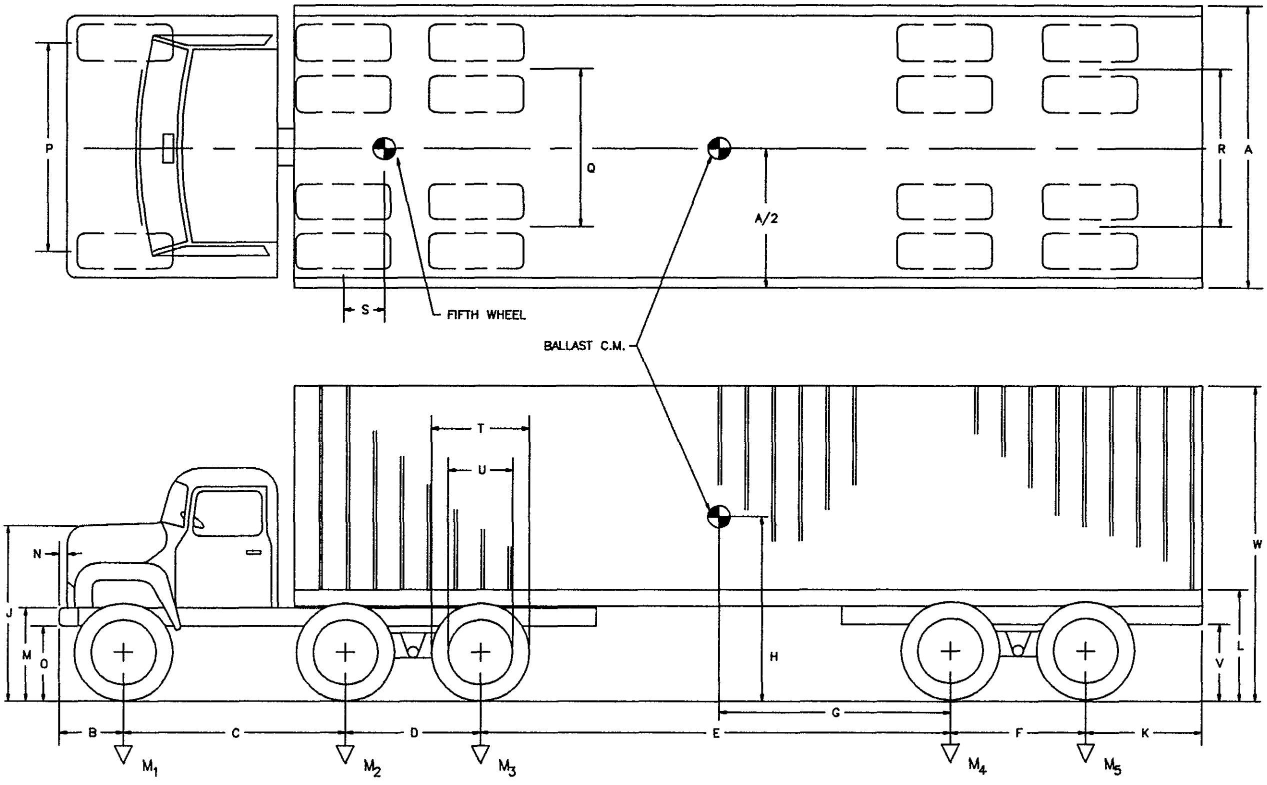
VIN NO.: ____________________ MAKE: ____________________ MODEL: ____________________

YEAR: ____________________

DESCRIBE ANY DAMAGE TO VEHICLE PRIOR TO TEST:

____

____


P
Q
R
A
A/2
S
FIFTH WHEEL
BALLAST C.M.
T
U
W
N
J
M
O
H
L
V
B
C
D
E
G
F
K
M_1
M_2
M_3
M_4
M_5

GEOMETRY – (cm)

A____________

B____________

C____________

W____________

D____________

E____________

F____________

G____________

H____________

J____________

K____________

L____________

M____________

N____________

O____________

P____________

Q____________

R____________

S____________

T____________

U____________

V____________

MASS – (kg)	CURB	TEST INERTIAL	GROSS STATIC
M_1			
M_2			
M_3			
M_4			
M_5			
M_T			

Figure 4.4. 36000V parameters.

DATE: ____________ TEST NO.: ____________

TRACTOR:

VIN NO.: ____________ MAKE: ____________ MODEL: ____________

YEAR: ____________ ODOMETER: ____________

TRAILER:

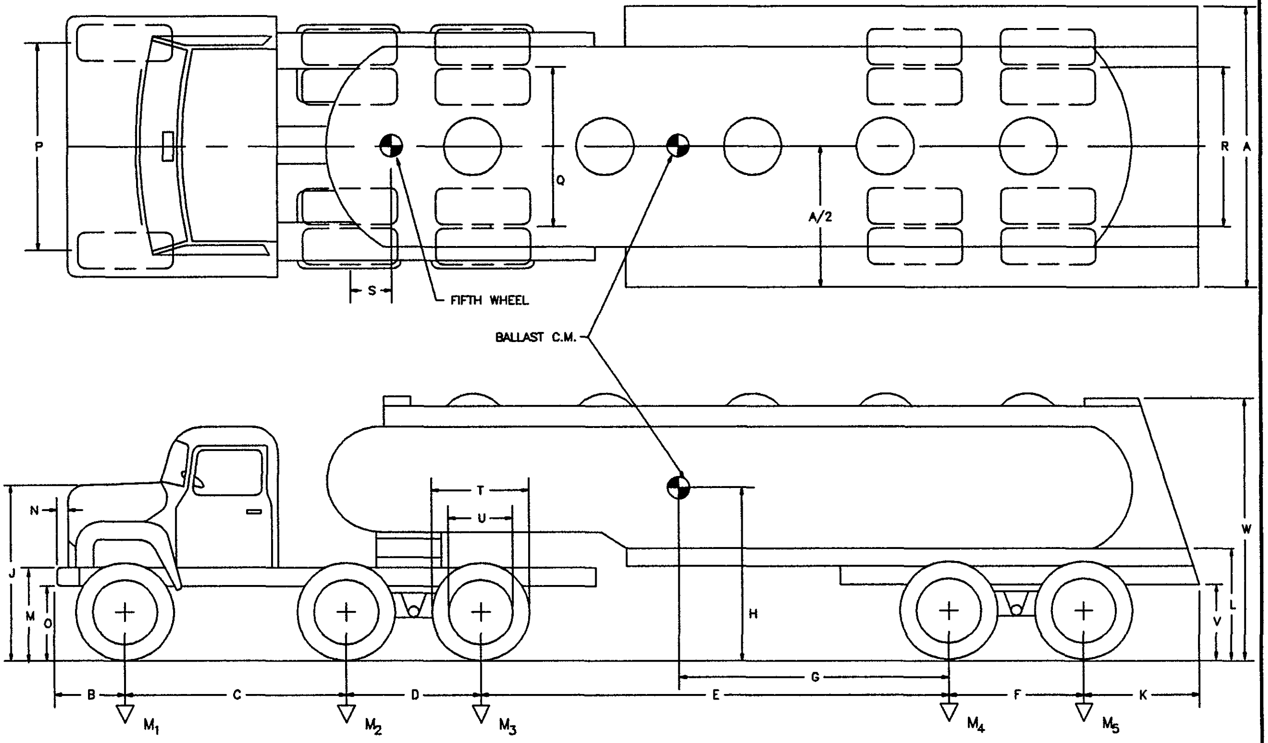
VIN NO.: ____________ MAKE: ____________ MODEL: ____________

YEAR: ____________

DESCRIBE ANY DAMAGE TO VEHICLE PRIOR TO TEST:

____

____


P
Q
R
A
A/2
S
FIFTH WHEEL
BALLAST C.M.
T
U
N
J
M
O
H
W
L
V
B
C
D
E
G
F
K
M_1
M_2
M_3
M_4
M_5

GEOMETRY – CM

A______ D______ G______ K______ N______ Q______ T______

B______ E______ H______ L______ O______ R______ U______

C______ F______ J______ M______ P______ S______ V______

W______

MASS – Kg	CURB	TEST INERTIAL	GROSS STATIC
M_1			
M_2			
M_3			
M_4			
M_5			
M_T			

Figure 4.5. 36000T parameters.

Material and Finish Specifications. Specifications for as-tested rail and post material and other major structural elements and for associated hardware (fasteners, nuts, washers, back-up plates, etc.) used in the test article. Actual sizes and physical and chemical properties of hardware items should also be reported.

Connection Details. Connection details of precast concrete barrier elements used in work zones; bolts, nuts, and washers used to anchor a bridge railing post to the deck or abutment; bolts, nuts, and washers and other hardware used to connect rail elements to each other, to supporting posts, or to a bridge abutment; bolt torques when relevant.

4.2.2.3 Terminals and Crash Cushions

Geometrics. Mounting heights of rail elements; post spacing; length of test installation, including back-up structure if used; position of energy absorbing elements; targeted point of impact relative to end of article.

Foundation. Manner in which device was supported, including embedment procedures for posts embedded in soil (driven, drilled and backfilled, placed in driven inserts, concrete footing, etc.); description of soil, and soil properties, when relevant; description of surface on which free-standing device was resting, including its frictional properties when relevant.

End Conditions. Manner in which the ends of the device were anchored, including cable anchors and special back-up structures when used.

Material and Finish Specifications. Specifications for as-tested rail and post material, other structural elements, energy absorbing elements, and associated hardware (fasteners, nuts, washers, backup plates, etc.) used in the test article. Sizes of hardware elements should also be reported.

4.2.2.4 Support Structures, Work Zone Traffic Control Devices, and Breakaway Utility Poles

Geometrics and Mass. Height and width of sign panel; mounting height of sign panel above grade; spacing of multiple-post sign supports; mass of all breakaway support components and utility pole systems; targeted impact point and direction of vehicular approach relative to test article(s); orientation of breakaway mechanism relative to vehicular approach; position of fasteners used in mounting sign panel to wind beams (if used) and in connecting wind beams (or panel) to support posts; dimensions of traffic control devices; mounting height and connection details of lights used on traffic control devices; location of slip planes and hinge points relative to the ground for breakaway supports; height of luminaire pole and luminaire; dimension and orientation of cantilevered luminaire support arm; height of pole and crossarm for utility pole; size and location of all elements mounted on call-box support or mailbox support.

Foundation. Embedment procedures for post or pole embedded in soil (driven, drilled and backfilled, placed in driven inserts, concrete footing, etc.); embedment depth of test article; description of soil and its properties if relevant; description of surface on which free-standing device was resting and its frictional properties if relevant.

Material and Finish Specifications. Specifications for as-tested support structure, sign panel, traffic control device, utility pole, and associated hardware (fasteners, nuts, washers, back-up plates, etc.) used in the test article; size of hardware elements; physical, chemical and dynamic impact properties if relevant of as-tested devices for which impact performance depends on material fracture or separation, such as certain base bending signs and frangible bases on luminaire poles; results of fatigue and strength tests of key elements.

Connection Details. Bolt torques in slip bases and hinges of breakaway devices, and other connections if relevant.

4.2.2.5 Truck-Mounted Attenuators (TMAs)

Geometrics and Mass. Length, width, and height of attenuator; height of attenuator with respect to the ground; length of supporting truck; mass of TMA and supporting hardware; detailed drawings of mounting hardware; targeted impact point on TMA.

Test Surface. Description of test surface, including type (asphalt or portland cement concrete), finish, and other features that could effect its frictional properties.

Material and Finish Specifications. Specifications for as-tested attenuator, other structural elements, and associated hardware (fasteners, nuts, washers, back-up plates, etc.) used in the test article; sizes of hardware elements.

Supporting Truck. Manner in which the wheels of the supporting truck are braked (i.e., conventional brakes, mechanical lock-up of wheels).

4.3 TEST PARAMETERS

4.3.1 General

Vehicular impact speed and angle, impact point on vehicle and test article, dynamic displacement of the test article, exit speed and angle of the vehicle, vehicular accelerations, and the three-dimensional response of the vehicle are important parameters. Exit speed and angle are measured at the time the vehicle loses contact with the test article. For some support structures such as a yielding sign support, the test article, or parts thereof, may stay in contact with the vehicle for a considerable distance beyond the initial impact point. In those instances, the exit speed and angle may be measured at the time the vehicle clears the foundation or footing supporting the test article. It is also desirable to measure and record the contact length of vehicle with the barrier, airborne distance of vehicle, and maximum rise of vehicle's bumper and wheels. Dynamic strains of the test article may also be of interest in some tests. Key parameters that should be documented during the test phase are given in Table 4.1.

4.3.2 Electronic and Photographic Instrumentation Specifications

Except as noted in this section, it is recommended that SAE J211 OCT88 be used for electronic and optical instrumentation specifications. A copy of this document is included in Appendix C. Note that Part 1 relates to electronic instrumentation and Part 2 relates to photographic instrumentation. Included in Part 1 are specifications on data channel performance requirements, data channel selection, mounting of transducers, sign convention, data

TABLE 4.1. Key test parameters

Parameter	Recommended Measurement Tolerance	Acceptable Measurement Technique	Remarks
Impact/Exit Speed	See Section 4.3.2	(a) Speed Trap	For impact speed only.
		(b) High-speed Cine	Minimum film speed of 400 frames/s.
		(c) Radar	
		(d) Fifth Wheel	
Impact/Exit Angle	± 0.1 deg	(a) High-speed Cine	Minimum film speed of 200 frames/s. with overhead camera.
		(b) Rate Gyro	Integrate rate gyro data to obtain exit angle.
Impact Point			
Test Article	± 30 cm	Conventional Scales	
Vehicle	± 0.05 W[a]	Conventional Scales	
Vehicular Acceleration	See Section 4.3.2 and 4.3.3	Accelerometers	See Sections 4.3.2 and 4.3.3.
Vehicular Trajectory	± 30 cm	High-speed Cine	Minimum film speed of 200 frames/s. Overhead and end views of installation preferred.
Vehicular Roll, Pitch, and Yaw Rates	± 1% of Full Scale	Rate Gyros	Sensitivity should be ≤ 0.02 deg/s/G.
Test Article Dynamic Deformation	± 5 cm	High-speed Cine Displacement Potentiometers	Minimum film speed of 200 frames/sec overhead camera
Occupant (optional)			
(a) Kinematics	(Not Applicable)	Anthropomorphic or Anthropometric Device and On-board Cine	Onboard movie camera should have minimum film speed of 64 frames/s; camera should be positioned to capture expected dummy motion. As a minimum, the dummy should have gross mass distribution and gross joint movements of a 50th percentile male surrogate.
(b) Kinetics: Accelerations, Forces, and Displacements	See Section 4.3.2	Anthropomorphic Device	50 percentile Hybrid III conforming to Part 572, Subpart E, Title 49 of Code of Federal Regulations, Chapter V (10-1-88 Edition) (Limited to frontal impacts).

[a] W = width of vehicle.

recording, digital data processing, timing marks, time of initial contact, and presentation of results. With regard to testing of highway safety features, Part 1 of the specifications applies primarily to the determination of impact speed of the test vehicle and to factors related to the measurement, recording, and reduction of vehicular accelerations. Part 1 also applies to factors related to the measurement, recording, and reduction of a surrogate occupant's accelerations, forces, and displacements. To the extent that photographic instrumentation is used for these same purposes, Part 2 of the specifications is applicable.

With regard to SAE J211 OCT88, the following exceptions and additions are made:

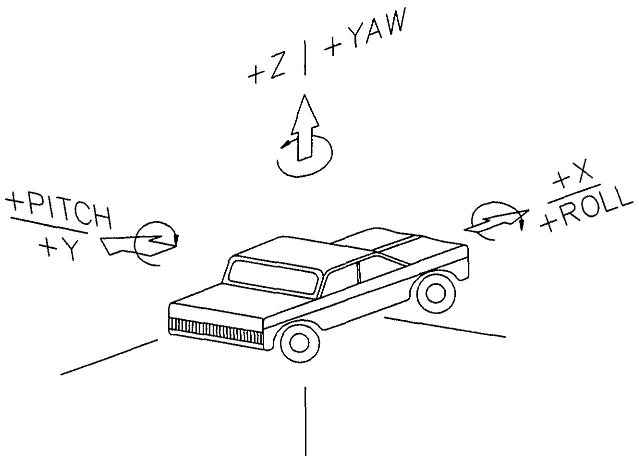
- Contrary to the apparent recommendation in Part 1, Section 5, it is not necessary that all data be gathered at class 1000 or higher. Rather it is recommended that the frequency response classes given in Table 1 of Part 1, Section 5, be used. Specifically, note that data for "vehicle structural accelerations for use in integration for velocity or displacement" should be gathered at channel class 180.
- With regard to Part 1, Section 12, it is recommended that vehicular accelerations be filtered at a "channel frequency class" of 60 Hz prior to plotting acceleration versus time data presented in the test report.
- Section A2.2 is optional and Section A.3 is not applicable.
- Part 2, Section 3.1.1, is not applicable if a wide-angle lens is used because most such lenses cannot meet "distortion index."

With regard to the measurement of accelerations, it will be noted that SAE J211 OCT88 does not specify a Channel Amplitude Class (CAC), that is, the maximum acceleration level for the accelerometer. Selection of the CAC must be made by the testing agency. However, the CAC should be selected so as to maximize the accuracy of the expected results without exposing the accelerometer to undue risk of damage. In general, good accuracy can be expected if measured accelerations are near the midrange of the accelerometer limits. To the extent possible the CAC should be selected based on a review of results from similar tests and the best judgment of the testing agency.

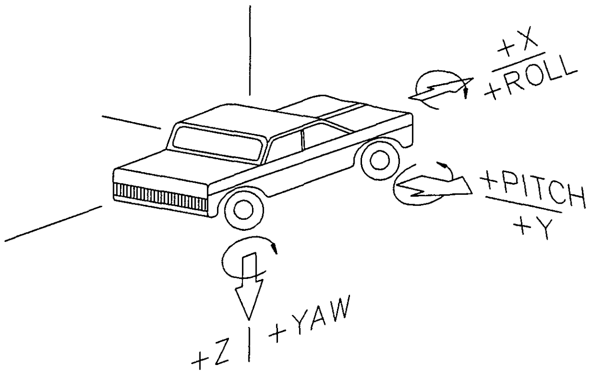
It should also be noted that the sign convention in SAE J211 OCT88 differs from that in *NCHRP Report 230.* The recommended convention, in accordance with SAE J211 OCT88, for a vehicle is as follows:

x-Positive in the normal forward motion direction,
y-Positive toward the right, and
z-Positive vertically downward.

The origin of these axes is typically placed at the center of mass of the vehicle. The positive axes are shown in Figure 4.6.



ACCORDING TO REPORT 230



CURRENT

Figure 4.6. Positive vehicular sign convention.

4.3.3 Accelerometer Placement and Data Reduction for Test Vehicles 700C, 820C, and 2000P

It is important that vehicular accelerations be measured or calculated with respect to a common point so that meaningful and consistent results can be expected from various testing agencies. Vehicular accelerations at the center of mass of the vehicle are used in the computation of one of the critical evaluation criteria, that is, occupant risk. It is therefore important that a tri-axial set of accelerometers be placed at or very near the vehicle's center of mass. If the accelerometers cannot be placed within a ±5 cm radial, measured in the *x-y* plane, of the center of mass, it is recommended the instrumentation and data reduction method given in Appendix A, Section A4.3.3, be used.

Accelerometers should be mounted on a major structural element of the vehicle so that "rigid" vehicular body motions are measured. A metal block is recommended for combining accelerometers on a common structure; the block can then be attached to the vehicle. The mass of the block should be minimal so as not to appreciably alter the frequency response of the structural element to which it is attached.

Although occupant risk evaluation using the flail space model is not required for the 8000S, 36000V, and 36000T vehicles, it is recommended that a procedure similar to that in Section A4.3.3 be used to measure accelerations in the subject vehicles. Note that for the 8000S vehicle two sets of longitudinal and lateral

TABLE 4.2. Key post-test parameters

Parameter	Suggested Measurement Tolerance	Acceptable Measurement Technique	Remarks
Test Article			
Permanent Deformation or Displacement (if applicable)	± 5 cm	Conventional Scales Transit	See discussion in Section 5.2.
Final Rest Position (if applicable)	± 30 cm	Conventional Scales Transit	Locate and report significant debris. See discussion in Section 5.3.
General Damage	(Not Applicable)	Visual Inspection	Identify and report significant damage.
Test Vehicle			
Final Rest Position	± 30 cm ± 5.0 deg	Conventional Scales Transit	
Exterior Damage	(Not Applicable)	Visual Inspection and Photos	VDS standard photos should be shown in report.
Interior Damage	(Not Applicable)	Visual Inspection and Photos See Appendix E	See discussion in Section 4.4.
Undercarriage Damage	(Not Applicable)	Visual Inspection and Photos	See discussion in Section 4.4.

accelerometers are needed forward of the cab/cargo van interface to determine accelerations at any point along the x-axis forward of the interface, and two sets are needed aft of the interface to determine accelerations at any point aft of the interface. Also, two sets are needed in the tractor and two sets are needed in the trailer to determine accelerations at any point along the x-axis in the tractor and trailer of vehicles 36000V and 36000T.

4.4 POST-TEST PARAMETERS

After the test, deformation, damage and final rest position of both the test article (or portions thereof) and the vehicle should be documented and reported. This includes the roll-ahead distance for the support truck in a TMA test. Both the "vehicle damage scale" (VDS)(75) and "collision damage classification" (CDC) *(76)* should be determined and reported. Key parameters that should be documented in the post-test phase are given in Table 4.2. In addition to the recommended measurements, it is important that these parameters be fully documented photographically.

Collapse and failure modes of the test article and components thereof should be described to the extent possible. This includes the foundation when applicable, fasteners, and other hardware items. Follow-up material testing to determine physical and chemical properties is recommended for an unexpected component failure. Results of the follow-up tests should be discussed and presented in the test report.

One of the factors considered in the evaluation of a crash test is the structural integrity of the occupant compartment (see Chapter 5). These criteria require that the test article not penetrate the occupant compartment and that there be no deformations of or intrusions into the occupant compartment that could cause a disabling injury. In the absence of an acceptable measure of such deformations or intrusions, it is essential that adequate documentation in the form of photographs of occupant compartment damage be made and reported. Photographs of the interior prior to the test should also be made to permit direct comparisons of before and after conditions. Until an acceptable methodology is developed, it is recommended that the procedure given in Appendix E be used to compute and document an Occupant Compartment Deformation Index (OCDI). Although it is recommended that the OCDI be used for information purposes only and that it not be used to determine acceptance of a test, its use will permit some degree of quantification of occupant compartment damage. As experience is gained with its use, definitive acceptance criteria may be established in the future. Note that the OCDI will also be used by the European Committee for Standardization to quantify occupant compartment deformation *(77)*.

Occasionally the undercarriage of the vehicle will make contact with the test article. Damage to the oil pan, gas tank, or other components may occur. Such instances should be documented and reported.

4.5 ADDITIONAL PARAMETERS

Recommendations given in the foregoing sections relative to data acquisition are based on tests to date of known test articles. As new safety features are developed with special design and performance characteristics, there may be a need to amend these recommendations. These recommendations should therefore not be considered all-inclusive. Other parameters peculiar to a test article or to its expected application or parameters peculiar to the test vehicle may require additional data acquisition techniques.

CHAPTER 5

EVALUATION CRITERIA

5.1 GENERAL

Recommended safety performance evaluation criteria for safety features are given in Table 5.1. Three dynamic performance evaluation factors are given together with recommended evaluation criteria and applicable tests. The factors are (1) structural adequacy, (2) occupant risk, and (3) post-impact vehicular response.

To the extent possible and practicable, limiting values recommended for the respective evaluation criteria are based on current technology and, when necessary, on the collective judgment of experts in roadside safety design. Establishment of performance criteria has been based on a "state-of-the-possible" philosophy since the late 1960s. That philosophy basically contends that as technological and economic conditions permit, higher levels of safety performance should be expected of certain safety features than others. Thus, impact performance requirements of a breakaway sign or luminaire support are more demanding than a crash cushion. Recommended values were also made in consideration of the limitations of the recommended test procedures and methodologies used to estimate occupant risk. As a consequence and in view of the very complex nature of vehicular collisions, the complex manner in which an occupant responds dynamically to the collision, and the complex nature of human tolerances to impact, the recommended criteria should be treated as general guidelines and not as absolute criteria. The adequacy of these or other criteria must ultimately be established by the agency responsible for the implementation of the safety device being evaluated.

Note that the evaluation criteria relate to the impact performance of the safety feature. Costs (i.e., installation, maintenance, damage repair, etc.), aesthetics, maintainability, durability, and other service requirements are not evaluated.

5.2 STRUCTURAL ADEQUACY

Structural adequacy is generally the first factor to be evaluated, and the safety feature should perform successfully according to the requirements presented in Table 5.1. Depending on its intended function, the feature may satisfy structural adequacy by redirecting the vehicle, by stopping the vehicle in a controlled manner, or by permitting the vehicle to break through the device.

It should be noted that structural adequacy criteria refer to the structural requirements associated with the impact itself. The criteria do not imply, for example, that a sign support system that meets the structural adequacy requirements of a test will meet the structural adequacy requirements of wind and ice loads or other environmental considerations when applicable.

Wide use is now made of temporary longitudinal barriers in work zones. Free-standing, unanchored precast concrete barriers are most commonly used. Increasing use is being made of a "moveable," precast concrete barrier in work zones to separate traffic in high-occupancy vehicle lanes. When used for the latter, the moveable barrier is typically moved laterally from one lane to another, one or more times per day. A primary concern for barriers of this type is the deflection they undergo during a vehicular impact. Because the amount a given installation can deflect without adverse consequences depends on site conditions, it is not feasible to establish limiting deflection values for crash tests of these barriers. Rather, it is important to accurately measure and report barrier displacement that occurs during the test so that a user agency can make an objective assessment of the appropriateness of the barrier for its intended application.

5.3 OCCUPANT RISK

Risk of occupant injury during impact with a highway safety feature depends to a large extent on the crashworthiness of the impacting vehicle. Crashworthiness depends in large part on the design of the occupant compartment including factors such as structural integrity, padding, restraint system, and so on. However, to the extent possible, the variability of vehicular crashworthiness has been removed from safety feature evaluation. Occupant risk is appraised according to gross vehicular accelerations because they are primarily functions of the safety feature design and the external structural design of the test vehicle. Whereas the highway engineer is ultimately concerned with safety of the vehicle's occupants, the occupant risk criteria of Table 5.1 should be considered as guidelines for generally acceptable dynamic performance.

Detached elements, fragments, or other debris from the test article should not penetrate or show potential for penetrating the occupant compartment or present an undue hazard to other traffic, pedestrians, or workers in a construction zone if applicable. The degree to which detached elements, fragments, or other debris and the degree to which the displacement of a temporary barrier presents a hazard to other traffic, pedestrians, and workers in a construction zone will depend on the location of the feature and the impact conditions. A sign in a median when struck by a vehicle may pose a hazard if detached elements encroach into opposing traffic lanes. The same sign when struck on the roadside may be of little concern except to occupants of the impacting vehicle. Fragments and debris from an impact with a traffic control device in a construction zone may or may not pose a hazard to workers in the zone, depending on their location relative to the device and the impact conditions. Consequently, it is not practical to establish absolute limits on test article trajectory, debris scatter, or barrier displacement. Rather, it is important to accurately record and report test article trajectory and debris scatter so that a user agency can make an objective assessment of the appropriateness of the safety feature for the intended application.

A factor listed in item D concerns deformations and intrusions into the occupant compartment. Of necessity, this factor must be assessed in large part by the judgment of the test agency and the user agency, or both. Risk of injury from a deformation depends on location, extent, and rate of deformation. In the absence of a widely accepted measure of risks associated with deformations or intrusions, it is essential that adequate documentation in the form of photographs and measurements of occupant compartment damage be made and reported. Photographs of the interior prior to the test should also be made to permit direct comparisons of before and after conditions. Until an acceptable methodology is developed, the procedure given in Appendix E may be used to compute and document an Occupant Compartment Deformation Index (OCDI). Although the OCDI should be used for information purposes only and should not be used to determine acceptance of a test, its use will permit some degree of quantification of occupant compartment damage. As experience is gained with its use, definitive acceptance criteria may be established in the future.

Although not a specific factor in assessing test results, integrity of the test vehicle's fuel tank is of concern. It is preferable that the fuel tank remain intact and unpunctured. Damage to or rupture of the fuel tank, oil pan, floor pan, or other features that might serve as a surrogate of a fuel tank should be reported.

For the majority of tests, a key requirement for occupant risk evaluation is for the impacting vehicle to remain upright during and after the collision, although moderate rolling, pitching, and yawing are acceptable. This requirement has the effect of minimizing the vertical component of vehicular acceleration; thus, this component is not normally evaluated in a typical crash test. Although it is preferable that all vehicles remain upright, this requirement is not applicable to tests involving the 8000S, 36000V, and 36000T vehicles, and all tests within test level 1 for terminals and crash cushions. See Appendix A, Section A3.2.2, for a discussion of these exceptions.

Occupant risk is also assessed by the response of a hypothetical, unrestrained front seat occupant whose motion relative to the occupant compartment is dependent on vehicular accelerations. The "point mass" occupant is assumed to move through space until striking a hypothetical instrument panel, windshield, or side structure and subsequently is assumed to experience the remainder of the vehicular acceleration pulse by remaining in contact with the interior surface. The two performance factors are (1) the lateral and longitudinal component of occupant velocity at impact with the interior surface and (2) the highest lateral and longitudinal component of resultant vehicular acceleration averaged over any 10-ms interval for the collision pulse subsequent to occupant impact. Performance factor two is referred to as the ridedown acceleration. Methods for calculating the impact velocity and ridedown acceleration components are given in Appendix A, Section A5.3. Generally, low values for these factors indicate less hazardous safety features. While a surrogate occupant is required in tests with the 820C and 700C vehicles and is optional in other tests, its dynamic and kinematic responses are not required or used in occupant risk assessment; hypothetical occupant compartment impact velocity and ridedown accelerations are calculated from vehicular accelerations.

It is also necessary to assess risk of injury to the driver of a supporting truck in a TMA system. Because the types of impacts in this case are primarily unidirectional and the supporting truck is accelerated forward, the driver will not move forward, at least initially, and is restrained from flailing rearward by the seat and headrest, which should be standard on these vehicles. As such, the primary risk of injury would stem from ridedown accelerations as the vehicle is accelerated forward. It is therefore recommended that ridedown acceleration criteria be used as the primary assessment of the risk of injury to the driver of a supporting truck in a TMA system.

TABLE 5.1. Safety evaluation guidelines

Evaluation Factors	Evaluation Criteria	Applicable Tests[a]
Structural Adequacy	A. Test article should contain and redirect the vehicle; the vehicle should not penetrate, underride, or override the installation although controlled lateral deflection of the test article is acceptable.	10, 11, 12, 20, 21, 22, 35, 36, 37, 38
	B. The test article should readily activate in a predictable manner by breaking away, fracturing, or yielding.	60, 61, 70, 71, 80, 81
	C. Acceptable test article performance may be by redirection, controlled penetration, or controlled stopping of the vehicle.	30, 31, 32, 33, 34, 39, 40, 41, 42, 43, 44, 50, 51, 52, 53

[a] Test numbers refer to last two digits in Test Designation for each Test Level unless otherwise noted.

TABLE 5.1. *(Continued)*

Evaluation Factors	Evaluation Criteria				Applicable Tests[a]
Occupant Risk	D. Detached elements, fragments or other debris from the test article should not penetrate or show potential for penetrating the occupant compartment, or present an undue hazard to other traffic, pedestrians, or personnel in a work zone. Deformations of, or intrusions into, the occupant compartment that could cause serious injuries should not be permitted. See discussion in Section 5.3 and Appendix E.				All
	E. Detached elements, fragments or other debris from the test article, or vehicular damage should not block the driver's vision or otherwise cause the driver to lose control of the vehicle.				70, 71
	F. The vehicle should remain upright during and after collision although moderate roll, pitching and yawing are acceptable.				All except those listed in Criterion G.
	G. It is preferable, although not essential, that the vehicle remain upright during and after collision.				12, 22, 30^b, 31^b, 32^b, 33^b, 34^b, 35^b, 36^b, 37^b, 38^b, 39^b, 40^b, 41^b, 42^b, 43^b, 44^b
	H. Occupant impact velocities (see Appendix A, Section A5.3 for calculation procedure) should satisfy the following:	Occupant Impact Velocity Limits (m/s)			
		Component	Preferred	Maximum	
		Longitudinal and Lateral	9	12	10, 20, 30, 31, 32, 33, 34, 36, 40, 41, 42, 43, 50, 51, 52, 53, 80, 81
		Longitudinal	3	5	60, 61, 70, 71
	I. Occupant ridedown accelerations (see Appendix A, Section A5.3 for calculation procedure) should satisfy the following:	Occupant Ridedown Acceleration Limits (G's)			
		Component	Preferred	Maximum	
		Longitudinal and Lateral	15	20	10, 20, 30, 31, 32, 33, 34, 36, 40, 41, 42, 43, 50, 51, 52, 53, 60, 61, 70, 71, 80, 81
	J. (Optional) Hybrid III dummy. Response should conform to evaluation criteria of Part 571.208, Title 49 of Code of Federal Regulation, Chapter V (10-1-88 Edition). See Section 5.3 for limitations of Hybrid III dummy.				10, 20, 30, 31, 32, 33, 34, 36, 40, 41, 42, 43, 50, 51, 52, 53, 60, 61, 70, 71, 80, 81

[a] Test numbers refer to last two digits in Test Designation for each Test Level unless otherwise noted.
[b] For Test Level 1 only.

TABLE 5.1. ***(Continued)***

Evaluation Factors	Evaluation Criteria	Applicable Tests[a]
Vehicle Trajectory	K. After collision it is preferable that the vehicle's trajectory not intrude into adjacent traffic lanes.	All
	L. The occupant impact velocity in the longitudinal direction should not exceed 12 m/sec and the occupant ridedown acceleration in the longitudinal direction (see Appendix A, Section A5.3 for calculation procedure) should not exceed 20 G's.	11, 21, 35, 37, 38, 39
	M. The exit angle from the test article preferably should be less than 60 percent of test impact angle, measured at time of vehicle loss of contact with test device.	10, 11, 12, 20, 21, 22, 35, 36, 37, 38, 39
	N. Vehicle trajectory behind the test article is acceptable.	30, 31, 32, 33, 34, 39, 42, 43, 44, 60, 61, 70, 71, 80, 81

[a] Test numbers refer to last two digits in Test Designation for each Test Level unless otherwise noted.

Recommended limits for occupant impact velocity and ridedown acceleration are given in Table 5.1. Note that two values are given for each parameter, a "preferred" limit and a "maximum" limit. As implied, it is desirable that the occupant risk indices not exceed the preferred values, and it is recommended that they not exceed the maximum values. Reference should be made to Appendix A, Section A5.3, for the rationale used in selecting these values. Establishment of absolute occupant risk limits is a policy decision and accordingly must be made by the user agency responsible for the implementation of the recommendations contained herein.

As indicated in Table 5.1, if a dummy is to be used to supplement the assessment of occupant risk, it is recommended the Hybrid III dummy be used. However, note that the Hybrid III dummy is valid for *frontal or head-on impacts* only, that is, those in which dummy movement is essentially parallel to the longitudinal vehicular axis (x-axis, as shown in Figure 4.6). Specifications, calibration, and instrumentation of the Hybrid III dummy should be in accordance with Part 572, Subpart E, Title 49 of the *Code of Federal Regulations,* Chapter V (10-1-88 Edition). Response data should conform to Part 571.208, Title 49 of the CFR, Chapter V (10-1-88 Edition). There is no dummy capable of accurately simulating the kinetics and kinematics of an occupant for oblique movements, that is, those in which occupant movement has both x and y components. Oblique occupant movement typically occurs when the vehicle is redirected away from the feature being impacted, such as a longitudinal barrier.

Although not required, testing agencies are encouraged to calculate and report the Theoretical Head Impact Velocity (THIV), the Post-Impact Head Deceleration (PHD), and the Acceleration Severity Index (ASI), as described in Appendix F. The THIV, PHD, and the ASI have been adopted by the European Committee for Standardization (CEN) (77) as measures of occupant risks. At some time in the future, it is expected that the U.S. and CEN will develop common impact performance standards for highway features. By calculating and reporting the THIV, PHD, and the ASI, a database will be developed from which comparisons can be made relative to the flail space model and from which decisions can be made as to appropriate measures of occupant risk.

5.4 POST-IMPACT VEHICULAR TRAJECTORY

Vehicular trajectory hazard is a measure of the potential of the post-impact trajectory of the vehicle to cause a subsequent multivehicle accident, thereby subjecting occupants of other vehicles to undue hazard or to subject the occupants of the impacting vehicle to secondary collisions with other fixed objects. As indicated in Table 5.1, it is preferable that the vehicle trajectory and final stopping position intrude a minimum distance, if at all, into adjacent or opposing traffic lanes. Criterion "L" is included to limit pocketing or snagging of the vehicle and the post-impact consequences of excessive pocketing or snagging, such as a high vehicular exit angle or spin-out of the vehicle. It is preferable that the vehicle be smoothly redirected (for redirective devices), and this is typically indicated when the exit angle is less than 60 percent of the impact angle. Acceptable post-impact behavior may also be achieved if the vehicle is decelerated to a stop while vehicular-barrier contact is maintained, provided all other relevant criteria of Table 5.1 are satisfied. Note that if the barrier is within a lane width of adjacent traffic, the slowed or stopped vehicle may pose risks to oncoming motorists. As indicated for certain classes of safety features, vehicular trajectory behind the test article is acceptable.

5.5 GEOMETRIC FEATURES

Tests of a geometric feature, such as a ditch, driveway, embankment, or curb, typically involve three-dimensional vehicular motions. Also, the duration of the test is usually long (up to 5 s or more) in comparison to a test of a barrier or other highway safety feature (typically 0.30 s or less). As a consequence, an unrestrained occupant can be expected to flail about the occupant compartment in three dimensions over an extended time duration, possibly contacting a given surface(s) more than once. For these reasons, the flail space model is generally not applicable.

Fortunately, in tests of most geometric features there are no design elements that would cause sudden and large vehicular velocity changes. The primary concern is overturn of the vehicle as it traverses the feature.

In the absence of more objective criteria, the following procedures and evaluation criteria may be used for a geometric feature:

(a) Part F of Table 5.1 must be satisfied.

(b) Compute average accelerations in the longitudinal and lateral directions for each consecutive 50-ms period for the duration of the event.

(c) If the average longitudinal or lateral acceleration computed instep b exceeds 2 Gs during any 50-ms period, apply the flail space model at the beginning of the period over which the average acceleration was computed. Evaluate the results of the flail space model according to Parts H and I of Table *5.1*.

CHAPTER 6
TEST DOCUMENTATION

6.1 GENERAL REPORTING RECOMMENDATIONS

Of primary importance in the guidelines presented herein is the preparation of a comprehensive test report. The test(s) should be documented in sufficient detail so that, if necessary, others could repeat the test(s) and obtain similar results. Liberal use of photographs is encouraged to document before, during, and after test conditions. Reference should be made to Chapter 4 for key pretest, test, and post-test parameters.

A recommended table of contents for the test report is given in Figure 6.1. In general, the report should include at a minimum the following information:

Identification. The test report should include the name of personnel responsible for the test(s), the name and address of the testing organization, location of test facility, and the date of the test(s).

Test Vehicle. A description of the test vehicle should be given in the report. Data described in Sections 4.2.1 and 4.3.3 should be included.

Test Article. The test article should be fully described with engineering drawings and material specification. Key parameters that should be recorded are given in Section 4.2.2. It is preferable that engineering drawings be made with a computerized drafting system to facilitate their use by others. As appropriate, revisions to the design made during the course of the test program should be fully documented. Of particular importance is the delineation of special fabrication and installation procedures (such as heat treatment, weldments, bolt tension, galvanizing in critical stressed areas, etc.) that may influence impact performance.

Test Soil. If relevant, the type of soil used, why it was selected, its adherence to recommended specifications, and its properties at the time of the test should be reported.

Test Procedures. Key test procedures are given in Section 4.3 and Table 4.1. A description of the test facility and associated equipment should be contained in the report. It is desirable that weather conditions that may affect test results be reported, including those at the time of the test as well as those preceding the test (e.g., prolonged duration of subfreezing weather or prolonged duration of rainy weather). The data acquisition systems should be fully described, together with the procedures used in calibrating and processing the data.

Findings. To facilitate comparison of findings from two or more testing agencies, a findings presentation format, as described in Table 6.1, is recommended. It is also recommended that the report contain a summary page with the information given in Figure 6.2. A brief description of the items in Figure 6.2 follows:

I. INTRODUCTION

- A. Problem
- B. Background/Literature Search
- C. Objectives/Scope of Research

II. TECHNICAL DISCUSSION

- A. Test Parameters
  1. Test Facility
  2. Test Article--Design and Construction
  3. Test Vehicles
  4. Soil Conditions

- B. Test Conditions and Results
  1. Impact Description/Vehicle Behavior
  2. Barrier Damage/Debris Patterns
  3. Vehicle Damage
  4. Dummy Behavior (optional)

- C. Assessment of Test Results
  1. Occupant Risk
  2. Structural Adequacy
  3. Vehicle Trajectory Hazard

III. CONCLUSIONS AND RECOMMENDATIONS

APPENDIX:

- A. Test Vehicle Equipment and Guidance Methods (Details)
- B. Photo Instrumentation (Diagrams, specifications, details and analysis)
- C. Electronic Instrumentation (Diagrams, specifications, details, and analysis)
- D. Detailed Drawings of Test Article
- E. Material Sample Test Results
- F. Test Article Construction Experience and/or Installation Procedures (if unusual)
- G. References

Figure 6.1. Recommended table of contents for crash test report.

1. Sequential photos may be taken from selected frames of high-speed film or video, or from frames of a sequence camera. They should begin at impact and should show the response of the vehicle/test article during the contact phase. For longitudinal barrier tests, an overhead view and/or a view parallel to the barrier is preferred. For terminal or crash cushion tests, an overhead view is preferred. For other features, a view perpendicular to the vehicle's approach path is preferred.

2. The plan view of the installation should show the overall layout of the installation, the impact point of the vehicle with the test article, and the post-impact trajectory of the vehicle and the test article.

3. An elevation view of the test article should show basic dimensions, heights, and if applicable the depth of embedment of the test article.

4. The test agency, test number, and date of the test should be given here.

TABLE 6.1. Recommended format for reporting of findings

Item	Description	Format
Photography		
Still	Before and after test of vehicle and test article	Photographs
High-Speed Cine	Selected sequential frames (8 minimum) during impact	Photographs
Accelerations		
Vehicle	x,y,z components; filtered (see Section 4.3.2)	Plots[b]
Dummy[a]	x,y,z components of head and chest (see Section 4.3.2)	Plots[b]
Angular Displacements	Roll, pitch, and yaw vehicular displacements	Plots[b]
Force[a]		
Seat Belt	Seat belt load cell	Plots[b]
Dummy Femur	Femur load cell	Plots[b]
Test Article Deformation		
Permanent	Profile of deformation	Plot[b]/Table
Dynamic	Maximum deformation	Text
Damage		
Test Article	Damage to key elements	Photos and Narrative
Vehicle	Damage to exterior and interior	Photos and Narrative VDS Scale CDC Scale OCDI Scale

[a] Optional.

[b] Plots should be scaled to maximize resolution of parameters.

5. To the extent possible and as space permits, the test article should be described here. *Type-Give* basic type of article tested (e.g., longitudinal barrier/bridge rail). *Installation length-For a* longitudinal barrier test, this is the length of the standard barrier section, exclusive of the end terminals. For a terminal test, it is the length of the terminal and the adjoining longitudinal barrier; lengths of each should be given separately. For a crash cushion test, it is the length of the cushion and backup structure, if necessary; lengths of each should be given separately. For a TMA, it is the length of the cushion and the support truck; lengths of each should be given separately. It is not applicable for tests of support structures, work zone traffic control devices, and breakaway utility poles. *Size, dimensions, and material of key elements-Sizes* and dimensions and material for key elements of the test article, such as rails, posts, support structures, traffic control devices, and utility poles, should be given.

6. If applicable, reference should be made to the type of soil used (e.g., "standard soil of Section 2.2.1.1"), and a note should be made if soil conditions differed from those recommended therein.

7. Test vehicle parameters of interest are: *Type-Indicate* whether a production model vehicle or a surrogate (bogie or pendulum) test vehicle was used. *Designation-Indicate* which of the six test vehicles was used in the test (i.e., 700C, 820C, 2000P, 8000S, 36000V, or 36000T). *Model-Indicate* actual make and model year if a production model vehicle was used or the make and model year simulated by the surrogate. *Mass-See* Section 2.4.2.1 for definition of these parameters.

8. Terms are self-explanatory.

9. The exit conditions should be measured at the time the vehicle loses contact with the test article. See Section 4.3.1 for further discussion of exit conditions.

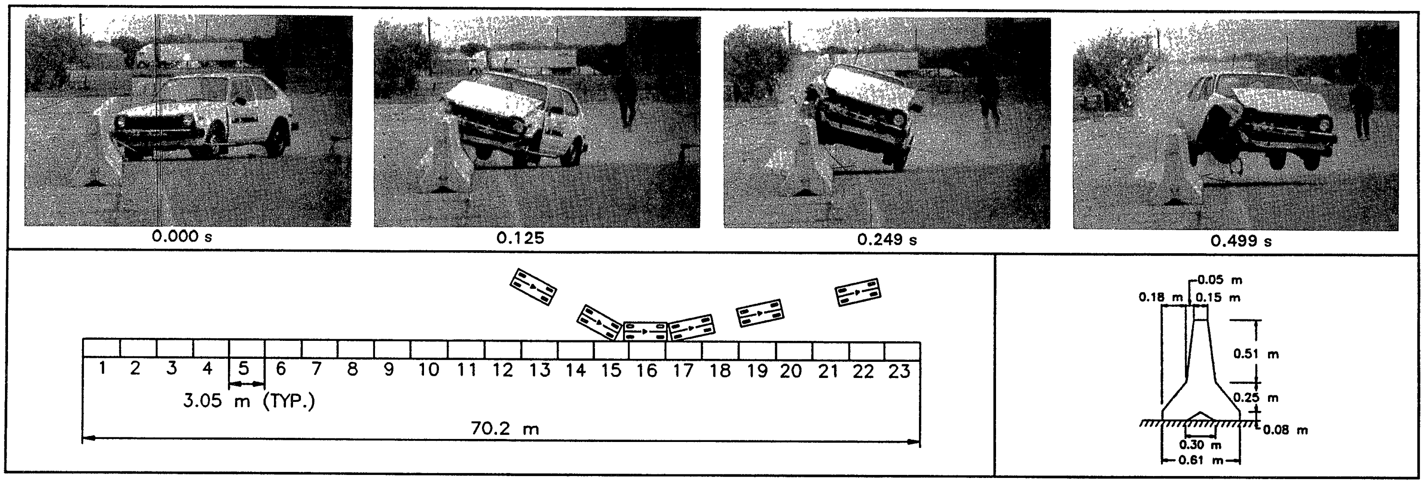
10. Occupant risk values are computed as described in Section A5.3 of Appendix A, and in Appendix F.

11. Dynamic deflection of a test article is the maximum deflection that occurs during the impact. Permanent deflection is the residual deflection remaining after the impact. These measurements normally apply to longitudinal barriers, terminals, crash cushions and TMAs.

12. See Section 4.4 for discussion of VDS and CDC. See Section 4.4 and Appendix E for discussion of OCDI.

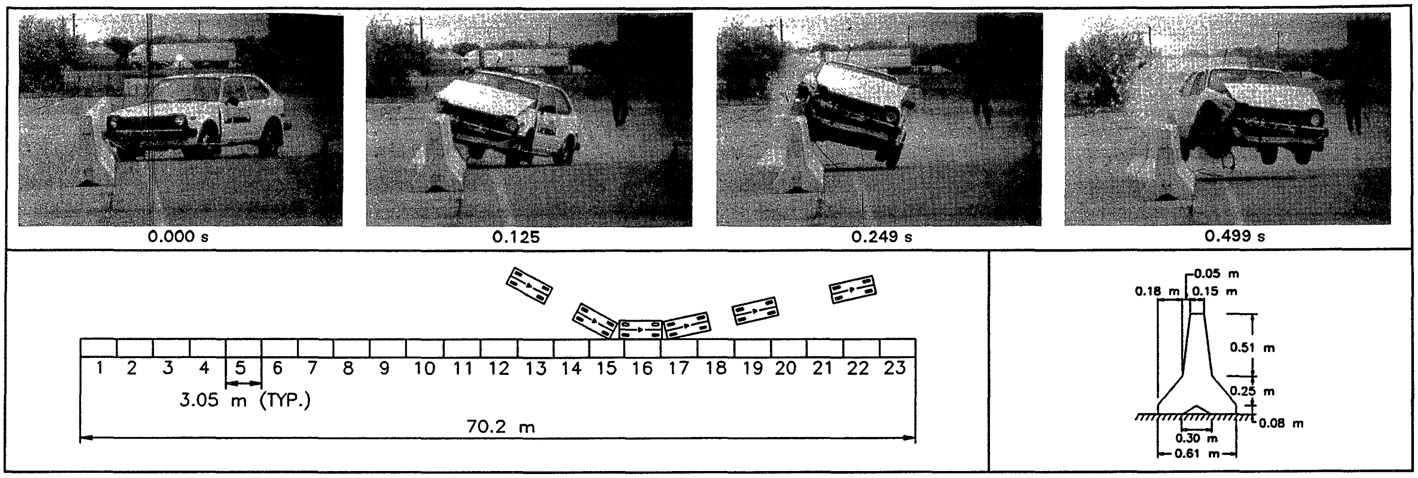
13. Indicate the maximum roll, pitch, and yaw angles of the vehicle during the test. See Figure 4.6 for definition and sign convention of these angles.

Shown in Figure 6.3 is an example of the use of Figure 6.2.


0.000 s
0.125
0.249 s
0.499 s
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
3.05 m (TYP.)
70.2 m
0.05 m
0.18 m
0.15 m
0.51 m
0.25 m
0.08 m
0.30 m
0.61 m

4. General Information
    Test Agency .... ________
    Test No. .... ________
    Date .... ________
5. Test Article
    Type .... ________
    .... ________
    Installation Length (m) .... ________
    Size and/or dimension and material
      of key elements .... ________
    .... ________
6. Soil Type and Condition .... ________
7. Test Vehicle
    Type .... ________
    Designation .... ________
    Model .... ________
    Mass (kg) Curb .... ________
      Test Inertial .... ________
      Dummy(s) .... ________
      Gross Static .... ________
8. Impact Conditions
    Speed (km/h) .... ________
    Angle (deg) .... ________
9. Exit Conditions
    Speed (km/h) .... ________
    Angle (deg) .... ________
10. Occupant Risk Values
    Impact Velocity (m/s)
      x-direction .... ________
      y-direction .... ________
    THIV (optional) .... ________
    Ridedown Acceleration (g's)
      x-direction .... ________
      y-direction .... ________
    PHD (optional) .... ________
    ASI (optional) .... ________
11. Test Article Deflections (m)
    Dynamic .... ________
    Permanent .... ________
12. Vehicle Damage
    Exterior
      VDS .... ________
      CDC .... ________
    Interior
      OCDI .... ________
13. Post-Impact Vehicular Behavior
    Maximum Roll Angle (deg) .... ________
    Maximum Pitch Angle (deg) .... ________
    Maximum Yaw Angle (deg) .... ________

Figure 6.2. Recommended summary sheet for crash test results.



General Information	
Test Agency	Texas Transportation Institute
Test No.	7110-1
Date	03/22/90
Test Article	
Type	Longitudinal Barrier/ Temporary Barrier
Installation Length (m)	70.2
Size and/or dimension and material of key elements	3.05 m Precast Concrete Sections
Soil Type and Condition	N/A
Test Vehicle	
Type	Production Model
Designation	820 C
Model	1981 Honda Civic
Mass (kg) Curb	830
Test Inertial	817
Dummy(s)	77
Gross Static	894
Impact Conditions	
Speed (km/h)	71.8
Angle (deg)	28.0
Exit Conditions	
Speed (km/h)	36.2
Angle (deg)	12.5

Occupant Risk Values	
Impact Velocity (m/s)	
x-direction	6.2
y-direction	4.9
THIV (optional)	7.3
Ridedown Acceleration (g's)	
x-direction	-3.7
y-direction	-8.2
PHD (optional)	8.4
ASI (optional)	0.8
Test Article Deflections (m)	
Dynamic	0.43
Permanent	0.43
Vehicle Damage	
Exterior	
VDS	02FR4
CDC	02FREW3
Interior	
OCDI	RS0000011
Post-Impact Vehicular Behavior	
Maximum Roll Angle (deg)	19
Maximum Pitch Angle (deg)	11
Maximum Yaw Angle (deg)	35

Figure 6.3. Example of recommended summary sheet for crash test results.

TABLE 6.2. Example of recommended assessment summary page

Test No. 7110-1
Date 03/20/90
Test Agency Texas Transportation Institute

<table>
<tr><th>Evaluation Criteria</th><th>Test Results</th><th>Assessment</th></tr>
<tr><td>Structural Adequacy
A. Test article shall contain and redirect the vehicle; the vehicle shall not penetrate, underride, or override the installation although controlled lateral deflection of the test article is acceptable.</td><td>Vehicle contained and smoothly redirected.</td><td>Pass</td></tr>
<tr><td>Occupant Risk
H. Occupant impact velocities (see Appendix A5.3 for calculation procedure) shall satisfy the following:<table>
<tr><th colspan="3">Occupant Impact Velocity (m/s)</th></tr>
<tr><th>Component</th><th>Preferred</th><th>Maximum</th></tr>
<tr><td>Longitudinal and Lateral</td><td>9</td><td>12</td></tr>
</table></td><td>Longitudinal Impact Vel. = 7 m/s
Lateral Impact Vel. = 10 m/s</td><td>Pass
Marginal</td></tr>
<tr><td>Vehicle Trajectory
M. The exit angle from the test article preferably should be less than 60 percent of test impact angle, measured at time of vehicle loss of contact with test device.</td><td>Exit angle less than 60% of impact angle.</td><td>Pass</td></tr>
</table>

Note: Table entries are not all inclusive and are for illustration purposes only. This table will typically require more than one page to complete.

As a part of the documentation, a 16-mm movie or a video composite of the test(s) may be prepared to include a title block identifying the test, test conditions, date, and sponsoring agency, before-and-after documentary coverage of the test article and vehicle, and high-speed views of the impact (both profile and overhead).

It is important that the test report contain an ample number of photographs of the pretest, test, and post-test conditions. Additionally, because it is not feasible to include actual photographs in the test report, reproduction techniques used to copy them should produce high quality copies.

Assessment. The impact performance of the test article should be discussed with regard to the three evaluation factors: structural adequacy, occupant risk, and post-impact vehicular trajectory. It is recommended that a summary page(s) be prepared to address each relevant evaluation criteria of Table 5.1. It is recommended that the summary page be as shown in Table 6.2. Note that example entries shown in Table 6.2 are for illustrative purposes only and are not all-inclusive.

A conclusion should be presented as to acceptability of the impact performance of the test article. Recommendations should be offered as to modifications that may improve the impact performance and cost effectiveness of the test article. Recommendations should be categorized as either desirable or essential. Known or predictable limitations of the test article, such as sensitivity to foundation conditions or hazards that would exist if the test article were improperly oriented, should be given. Recommended applications may also be identified.

6.2 ELECTRONIC DATA

Requirements for documenting and recording electronic test data on a magnetic cartridge have been developed for use on projects sponsored by the Federal Highway Administration *(22).* Basic test data, such as test number, contract number, and date, and digitized electronic data from onboard transducers are recorded in a specified format. A software package is available from FHWA for entry of the data on the cartridge. These requirements are intended to facilitate and standardize the documentation of data stored in FHWA's Roadside Safety Library (RSL), housed at the Turner-Fairbank Highway Research Center in McLean, Virginia. It is recommended that these requirements be followed, when feasible, and that the data, along with film or video documentation, be submitted to FHWA for entry into the RSL. It is planned that the RSL will serve as a repository for roadside safety data, which can be readily accessed by the highway safety community. The address is as follows:

Federal Highway Administration
Roadside Safety Library (HSR-20)
Turner-Fairbank Highway Research Center 6300 Georgetown Pike
McLean, VA 22101-2296

CHAPTER 7

IMPLEMENTATION AND IN-SERVICE EVALUATION

7.1 PURPOSE

As mentioned in Chapter 1, in-service evaluation is the final and a key step in the development of a new or extensively modified safety feature. The purpose of in-service evaluation is to determine and document the manner in which the safety feature performs during a broad range of collision, environmental, operational, and maintenance situations for typical site and traffic conditions. The in-service evaluation step is necessary as analytical and crash test experiments described in previous chapters only partially assess the efficacy of a feature and a more thorough and in-depth knowledge of the feature is needed.

Figure 1.1 of Chapter 1 presents a flow chart depicting steps that generally should be followed in the development of a safety feature. Chapter 7 basically covers guidelines for step 9. However steps 7 and 8 should be done prior to initiation of the in-service evaluation. Step 10 is a policy decision made by the appropriate agency (see Section 7.4).

The following sections describe goals and suggested procedures for in-service evaluation. However, it is well understood that the random and extremely complex nature of vehicular accidents coupled with resource limitations of transportation agencies greatly restrict the extent to which these goals can be met and the procedures can be carried out. With few exceptions these studies will be clinical in nature; it will not be feasible to collect sufficient data to develop statistically valid findings. For these reasons, sites must be picked to gain the greatest information within limited time periods and resources.

7.2 OBJECTIVES

In-service evaluation involves the installation of a number of the candidate safety features at carefully selected sites and then the monitoring of the features' performance under "real world" conditions for a period of time. The sites, type, and frequency of information to be acquired and the length of observation period must be selected judiciously and planned to satisfy six objectives:

1. Demonstrate that design goals are achieved in the field and identify modifications that might improve performance.
2. Acquire a broad range of collision performance information on features installed in typical and special situations. Desirably the information would include exposure data and data on occupant injuries and vehicular impact conditions from which severity index values could be defined. In addition to "reported accidents," a measure of the more numerous brush hits and drive-away collisions should be monitored to establish failure/ success ratio and collision damage repair costs.
3. Identify factors that may compromise or defeat a feature's performance. Examples of such factors include vulnerability of the feature to pilferage or vandalism, accelerated corrosion or degradation of materials due to de-icing salts and other contaminants, and so on.
4. Examine the influence of climate/environment on collision performance. To be determined are the effects, if any, of extremes in heat and cold, ice, snow, rain, wind, and dust on the collision performance and maintenance of the safety feature.
5. Examine the influences that the feature may exhibit on other highway conditions that, in turn, may adversely affect highway operations and traffic. Such factors to be monitored are traffic congestion, change in accident rates or patterns, disruption of surface drainage, or the cause of snow or debris buildup.
6. Acquire routine maintenance information. As a part of this effort, the feature's design and layout should be examined for possible modifications that would lower installation, maintenance, and damage repair costs. Problems encountered during routine maintenance and damage repair should be documented and reported. Note that frequency of repair and repair demand (after both nominal and severe impacts) are critical factors. Systems that can sustain numerous or severe impacts while remaining serviceable offer substantially better protection to motorists than those that are rendered out of service by virtually every impact. This becomes especially critical on high-volume roadways, on roadways where maintenance activities cause congestion and increased risks of accidents, and at problem or high-accident locations. Information of this type often becomes the primary consideration in selection of a barrier system for such locations.

These objectives are general and all may or may not be applicable to a candidate safety feature. Their delineation here is to illustrate the scope and possible type of information that should be acquired.

7.3 CHARACTERISTICS OF TRIAL INSTALLATIONS

To acquire sufficient field information on experimental safety features that will adequately demonstrate their in-service performance, the trial installations may have the following characteristics:

1. The trial period should extend for a minimum of 2 years. This period will expose the hardware to two complete annual climate/environmental cycles. During early stages of the trial,

the local traffic should become familiar with unique appearance of novel designs; thereafter, the affected traffic pattern can return to a more normal state.

2. A sufficient length of installation for a longitudinal barrier, or sufficient numbers of a device for a feature such as an end treatment or a crash cushion, with carefully selected sites, should be determined to provide a number of impacts during the trial period. Potential sites for the new device should be examined and those with a high probability for a collision(s) should be considered for the trial installations. Generally, collision probability increases with traffic volume, proximity of the device to the travel lane, and adverse highway geometrics such as horizontal curvature and grade. The sites should also be selected to expose the feature to the spectrum of impact conditions for which it was intended. Of course, the service requirements of the site must not exceed the service expectations of the device. All collisions, both reported and unreported, are of importance.

3. Each installation should be examined at frequent intervals for the duration of the trial period. The purpose of these site visits is to detect and record minor impacts that might otherwise go unreported. Also to be noted is the state of readiness of the device. Highway maintenance, traffic operations, and law enforcement agencies should be alerted to the test installations and requested to report changes in traffic accident patterns.

4. A new safety feature may be evaluated on a "before/after" basis (a) by gathering accident data for a time period prior to and after installation of the feature, or (b) by gathering accident data for a time period after installation of the feature, and by gathering accident data at a controlled site (without the new feature) during the same time period.

5. An accident/collision reporting technique should be established that will trigger on all impacts even drive-aways. This may entail such techniques as reporting and then painting over or erasing scuff marks.

6. Maintenance forces should perform a field evaluation immediately after construction to determine ease of meeting installation specifications. Maintenance forces should keep costs and labor records on test and control sections. In addition, maintenance personnel could be used to gather drive-away and scuff mark information.

7. At the conclusion of the trial period, an in-service evaluation report should be prepared that presents findings and recommendations. The evaluation report should include a description of site conditions such as roadway geometrics, device location, vehicle operating speeds, vehicle mix, and some measure of exposure. It should also include comments on observed field performance.

7.4 DISCUSSION

Guidelines for in-service performance evaluation of roadside safety features were presented in *NCHRP Report 230* in 1981. It was recognized that the guidelines were general in nature and would require modification and amplification to suit the type of safety feature and local conditions. Since 1981, Solomon and Boyd *(23)* have developed a model procedure for evaluating roadside safety hardware that provides more in-depth instructions for in-service evaluation. In addition, Leonin and Powers *(24)* reported on the in-service evaluation of eight safety features in 14 states.

In the past, FHWA was a key arbiter in establishing acceptability and operational status of new safety features, especially those used on federal-aid highways. Current FHWA policy establishes acceptability of a new safety feature for use on federal-aid projects based on design details, specifications, and crash test results. The decision whether FHWA-accepted features should be deployed as "experimental" or "operational" must now be made by the states.

It is recommended that the developer and user agency of the safety feature develop a detailed plan for the in-service evaluation for review and approval by appropriate authorities prior to initiating the evaluation. Depending on the importance of the device, extent of potential application to a regional or nationwide basis, and funding priorities, the evaluation may be conducted under an extensive federal contract. A cooperative effort of two or more state highway agencies is another feasible evaluation plan. For proprietary devices that are developed using nongovernment funding, it may be appropriate for the owner to sponsor or contribute to the in-service evaluation.

It is recognized that certain design details may be identified during the in-service evaluation that, if properly modified, might improve some aspect of the feature's performance. Such modifications must not be made before their effect on feature safety performance is carefully verified through vehicular crash testing or other appropriate means (see item 10 in Figure 1.1). Past research has shown that seemingly minor variations in design details can adversely affect the safety performance of a feature.

Even after a new or extensively modified feature has successfully passed the in-service evaluation and has been accepted for general use, the operational performance of the feature should continue to be monitored to a lesser degree to enable any flaws or weakness to be corrected or controlled as soon as possible (see item 12 of Figure 1.1). Such weaknesses may be due to conditions that were not anticipated, such as vehicle design changes or different installation site conditions.

Finally, it is important to note that Section 1034 of the Intermodal Surface Transportation Efficiency Act of 1991 (ISTEA) mandates that all states must implement, among other things, a safety management system by fiscal year 1996. It is expected that an in-service evaluation procedure for new and existing roadside safety features will be an integral part of this management system.

APPENDIX A

COMMENTARY

Further discussion and elaboration are provided on certain sections in the text. Those sections for which commentary is given correspond to section numbers in the text preceded by the letter "A." For example, "A3.2.1" refers to Section 3.2.1 in the text.

CHAPTER ONE

A .1.1

Vehicle crash tests are complex experiments that are not easily replicated because of difficulties in controlling critical test conditions such as speed, angle, and condition of test vehicle and the sometimes random and unstable behavior of dynamic crush and fracture mechanisms. Testing guidelines are intended to enhance precision of these experiments while maintaining their costs within acceptable bounds. User agencies should recognize the limitations of these tests and exercise care in interpreting the results.

It is impractical or impossible to duplicate the innumerable highway site and safety feature layout conditions that exist in a limited number of standardized tests. Accordingly, the aim of the guidelines is to normalize or idealize test conditions. Hence, straight longitudinal barriers are tested, although curved installations exist; a flat grade is recommended, even though installations are sometimes situated on sloped shoulders and behind curbs. These normalized factors have significant effect on a barrier's performance and may obscure serious safety deficiencies that exist under more typical but less ideal conditions. However, these normalized factors are thought to be secondary in importance when the object of a test program is to compare the results of two or more systems. Moreover, the normalized conditions are more easily duplicated by testing agencies than say, a unique feature. Consequently, they should promote correlation of results from different groups. Nevertheless, when the highway engineer requires the performance of a system for specified site conditions (such as a unique soil or curb layout) or the performance of a safety feature is suspected of being unacceptable under some likely conditions, it is important that these conditions be used instead of, or in addition to, the idealized conditions.

These guidelines are intended for use with highway safety features that will be permanently or temporarily installed along the highway. Temporary features are generally used in work or construction zones or other temporary locations and their duration of use is normally relatively small. An important additional characteristic of a work zone is the exposure of work zone personnel to errant traffic. Thus, a barrier in a work zone may be required to (1) redirect errant traffic away from a roadside hazard of other traffic and (2) to shield workers from errant vehicles. Depending on specific site conditions, potential collision severity may equal or even exceed conditions found at typical nonconstruction zone sites.

A1..2

The intent of this section is to make the designer/developer of a safety feature aware of the process that must typically be followed and aware of available tools so that the most cost-effective approach or combination of tools may be selected. This report does not endorse or approve any one design method, procedure, or analytical technique or suggest that one procedure is equivalent to another. Rather, the basic purpose of the report is to provide guidelines for the full-scale crash testing and in-service evaluation of a feature once a decision has been made to proceed with step 4 of Figure 1.1. Users of the document should follow as closely as practical the recommended guidelines. Acceptability of a complementary technique in lieu of, or in addition to, full-scale crash testing is a policy decision beyond the purview of this report.

As indicated in steps 2 and 3 of Figure 1.1, there are various design and analysis tools and experimental tools that can be used in the initial development of a safety feature. Reference should be made to Appendix D for information on these tools. Also, as indicated, the developer should be cognizant of the various factors, other than those related to impact performance, that must be considered in the development of a feature. Conventional structural analysis and design techniques are most useful in early development stages of a safety feature. Computer simulations of the vehicle/feature dynamic interactions are useful in gaining insight for a wide range of impact conditions. The potential for obtaining valuable insight from simple static tests of components and assemblies should not be overlooked.

CHAPTER TWO

A2.2.1

Impact performance of many longitudinal barriers and breakaway or yielding support structures depends on strength and fixity of the soil foundation. Soil foundation is an integral part of such systems. For example, displacement and/or rotation of a breakaway device footing during collision can adversely affect the fracture mechanism. Insufficient soil support can lead to excessive guardrail post movements and guardrail lateral deflection during vehicle collision and result in a lower system capacity to contain and redirect errant vehicles. Insufficient soil strength can also be a critical and limiting factor for the anchoring function of a longitudinal barrier terminal. On the other hand, an unusually firm soil can increase the lateral stiffness of a longitudinal barrier and subject occupants of a colliding vehicle to undue hazard.

Soil conditions along the highway are variable. Strength is a function of soil type and ranges from soft sand materials to hard rock materials; moreover, the soil type may vary considerably within a locale as well as from region to region. Soil strength may also be a function of the season as it can be significantly affected by moisture content and whether it is frozen. The testing agency should be aware of the importance of soil strength and select the most appropriate soil type consistent with potential application of the feature.

Recommended soils are well-graded materials that should be readily available to most testing agencies. The standard soil of Section 2.2.1.1 is a selected AASHTO material that compacts to form a relatively strong foundation. The weak soil of Section 2.2.1.2 is a typical

AASHTO fine aggregate. These soils are essentially the same as the "strong" and "weak" soils of NCHRP Report 230.

The following general guidance is offered the user agency and the testing agency in soil selection:

Standard Soil

Unless the test article is limited to areas of weak soils, the standard soil should be used with any feature whose impact performance is sensitive to soil-foundation or soil-structure interaction. A large percentage of previous testing has been performed in similar soil and a historical tie is needed. Although it is probably stronger than the average condition found along the roadside, it is still representative of a considerable amount of existing installations.

Weak Soil

The weak soil should be used, in addition to the standard soil, for any feature whose impact performance is sensitive to soil-foundation or soil-structure interaction if identifiable areas of the state or local jurisdiction in which the feature will be installed contain soil with similar properties, and if there is a reasonable uncertainty regarding performance of the feature in the weak soil. Tests have shown that some base-bending or yielding small sign supports readily pull out of the weak soil upon impact. For features of this type, the strong soil is generally more critical and tests in the weak soil may not be necessary.

In addition to soil selection, the footing or foundation used in a test of a breakaway support structure should be designed for the minimum wind conditions permitted, thus yielding a minimum footing mass and size; a larger footing will yield a greater breakaway device fixity and, hence, is less critical.

It has been shown that the standard soil of Section 2.2.1.1 is especially sensitive to moisture content. The testing agency should sample and test the soil to insure moisture content is within recommended limits given in the specification at the time of the test.

A2.3.1

Failure or adverse performance of a highway safety feature during crash testing can often be attributed to seemingly insignificant design or construction details, something as innocuous as a substandard washer. For this reason it is most important to assure that the test article has been properly assembled and erected and that critical materials have the specified design properties. Details of most concern are those that are highly stressed (such as welded and bolted connections, anchor cables, cable connections, and concrete footings) or those that must fracture or tear away during impact (such as transformer bases or weakened barrier posts). Compressive tests of concrete cylinders, proof tests of cable assemblies, and physical and chemical properties of materials, in general, should be performed on a random sample of the test article elements or obtained from the supplier of the material. Even though well-defined material specifications and appropriate fracture modes may not be fully developed, the properties of all material used in the test article should be documented in detail in the test report.

A2.3.2.1

Proper judgment must be exercised in establishing test installation length. In specifying minimum length of a longitudinal barrier installation, the intent is to minimize influence of terminals and thereby simulate a long barrier. Also to be considered is the possible need to extend the barrier installation to observe a second collision between vehicle and barrier.

A2.3.2.4

See commentary in Section A3.2.4.

A2.4.1

The vehicle's design and condition at the time of testing can have major influence on the dynamic performance of a feature. Among the more important parameters are vehicle bumper height, configuration, and stiffness; vehicle mass distribution and suspension system; and vehicle structure. For these reasons the test vehicles should correspond closely to the recommended vehicle properties.

A2.4.1.1

Changes have occurred in the vehicle fleet since publication of NCHRP Report 230. Automobiles with curb masses of 725 kg and less are now operating on U.S. highways. The typical family sedan now has a mass somewhere between 900 kg and 1800 kg with only the expensive luxury cars and a few station wagons weighing more than 1800 kg. The mix also contains a significant number of "light-duty trucks," such as pickups, vans, Suburbans, Blazers, Broncos, etc., and "recreational vehicles," such as van conversions (customized vans), motor homes, etc. A significant portion of the light-duty trucks has a mass over 1800 kg. Many, if not most, of the light-duty trucks, in fact, serve as a passenger vehicle as opposed to a commercial or utility vehicle. Cars and light-duty trucks are combined herein into a single "passenger vehicle" category.

Sales of trucks above the light-duty category totaled 368,703 in 1988. These are trucks with a gross vehicle mass in excess of 4500 kg. Of this total, 40% were in the 15,000 kg and over category.

Unfortunately it is difficult, at best, to project even short term trends in the vehicle mix due to the volatile and unpredictable nature of factors that influence vehicle design. Due to the intense competitive and proprietary nature of the automobile industry, future vehicular design data are simply not available. Further, the development cycle time from inception to production of a new model has been decreasing over the years and is projected to continue to decrease. Perhaps the best available source for projected trends in automotive design is a recent report prepared by the University of Michigan (54). The study was conducted in 1988 and it was the fifth in a series of Delphi surveys of high-level automotive industry leaders. According to the report, "More than 300 CEO's, presidents, vice presidents, directors, and managers - organized into three panels - responded to 160 questions concerning technological, marketing, and materials developments within the automotive industry through the year 2000." One of the many projections of this study was that the "expected total vehicle weight will remain fairly constant through the year 2000 based on present fuel economy regulations and vehicle sizes." However, it was predicted that there will be a slight downward trend in weights due primarily to the increased use of structural composites. The total curb mass for the average North American produced passenger car was predicted to be between 1375 kg and 1430 kg for model year 1985 and between 1335 kg and 1380 kg for model year 2000. The average weight for the 1987 model year was estimated to be 1440 kg.

Upon review of available data and after careful review of various options, the recommended test vehicles listed in Tables 2.1 and 2.2 were selected. Consideration was given to including a vehicle more representative of the automobile population, i.e., a car with a mass of approximately 1550 kg. In some special cases it may be possible to design a feature to meet the performance recommendations for the 820C and 2000P test vehicles, leaving a potential problem for the typical car. A crash cushion is one such feature. It may also be possible for a longitudinal barrier to perform satisfactorily for the small and large vehicles but exhibit a snagging problem for the typical car. However, tests with a typical car were ultimately dismissed for three basic reasons: (1) they would not, in most cases, reveal problems that would not be identified with the 820C and 2000P vehicles, (2) technology has advanced to the point that potential problems with a typical car can usually be foreseen prior to the testing phase of the feature's development or can be inferred upon careful review of the recommended tests, and (3) they would significantly increase the cost of testing. If after exhausting all design and evaluation procedures a reasonable doubt remains, additional tests with the typical car may be warranted.

Test Vehicle 700C

This is a new test vehicle and it represents the very low end of the passenger car spectrum in terms of mass. Test experience with this vehicle is limited. Reference should be made to a study of the impact performance of widely-used highway safety features for a vehicle similar to 700C (7). In general this study found that acceptable impact performance can be expected for impacts with rigid and semirigid longitudinal barriers. As expected, marginal or unacceptable performance can be expected for most in-service features sensitive to the mass of the impacting vehicle, such as terminals, crash cushions, and some support structures. It was predicted that most support structures utilizing a slip-base breakaway device or other devices with similar behavior would have acceptable impact performance. Even so, it was pointed out that impact performance of slip-base devices for a 700C type vehicle is much more sensitive to factors such as bolt torques (and, hence, breakaway force), frontal crush stiffness of vehicle, and proper activation of the upper hinge in a sign support than a larger vehicle. It was also reported that more overturns can be expected with a vehicle of this size due to its inherently lower stability. Roadway or roadside surface discontinuities or irregularities such as curbs, ruts, and vegetation that would not upset a sliding larger vehicle would cause this vehicle to overturn.

Although agencies should consider the 700C vehicle in safety feature development, its use in lieu of the 820C vehicle should be viewed as desirable but not required. A suggested policy is to design for the 700C vehicle when it can be done cost effectively.

Test Vehicle 820C

The recommended test inertial mass of the 820C test vehicle (820 kg or approximately 1800 lb) is essentially the same as the 1800S vehicle of NCHRP Report 230. It was chosen as a basic test vehicle for the following reasons: (1) it is representative of a reasonable portion of passenger cars at the lower end of the spectrum in terms of mass, (2) although tenuous, predictions suggest minor changes in passenger car sizes over the next 10 years, and (3) its use will provide linkage with a considerable crash test database accumulated since publication of Report 230.

Although several makes of cars were used for the 1800S vehicle of Report 230, the vast majority of tests were conducted with Honda Civics of various model years. In recent years the mass of the Civic has increased to the point that it can no longer be used as an 1800S test vehicle. While a specific vehicle make was not recommended in Report 230, the unofficial adoption of the Honda Civic as the standard for the 1800S vehicle had its obvious advantages.

A specific make for the 820C vehicle is not required herein either. However, in view of the diversity of key vehicular properties of small cars (see, for example, the variations in frontal crush stiffness of small cars reported in reference 7) and in view of the population differences of available makes, consideration should be given to the selection of an "unofficial" make for the 820C vehicle. After reviewing candidate vehicles that would meet recommended properties for the 820C vehicle for at least through 1996, it was concluded that the Ford Festiva should be given strong consideration as an unofficial 820C test vehicle. Sales of the Festiva exceeded other cars in the sub-820 kg category for the 1987-1990 model years. It should therefore be readily available.

Test Vehicle 2000P

A pickup truck was selected to replace the full-size automobile widely used in the past (4500S vehicle in Report 230) for the following reasons:

(1) Sales of light-duty trucks, in general, and pickup trucks, in particular, have increased to the point that they now constitute a significant portion of all passenger vehicles operating on U.S. highways.

(2) Full-size automobiles with the mass of the 4500S test vehicle (2040 kg) are no longer sold in the U.S. with the exception of a few expensive luxury cars. The nominal mass of a full-size family sedan now being sold in the U.S. is about 1350 kg.

(3) Although there are structural and profile differences, the recommended 2000 kg pickup will produce impact loading reasonably similar to the 4500S vehicle of Report 230. Limited full-scale crash tests with an instrumented wall (17) indicate that a pickup will produce a maximum impact force slightly less than that of an automobile

of equal mass, whereas the effective height of the impact force will be slightly higher for the pickup, all other conditions being equal. Consequently, the 2000P test vehicle is expected to provide linkage with the numerous tests conducted with the 4500S vehicle.

A 3/4-ton pickup is recommended for the following reasons:

(1) Section 1073 of the Intermodal Surface Transportation Efficiency Act of 1991 mandated the development of standards for roadside barriers and other safety appurtenances "...which provide an enhanced level of crashworthy performance to accommodate vans, minivans, pickup trucks, and 4-wheel drive vehicles..." The 3/4-ton pickup is believed to be representative of a large segment of the light-duty truck population. The light-duty truck population includes large numbers of conversion vans on 3/4-ton chassis, Blazers, Broncos, and pickups with and without 4-wheel drive, pickups with campers, minivans, etc., whose mass and center of mass above ground approximate those of the 3/4-ton pickup. However, the exact degree to which features designed to meet test and evaluation requirements recommended herein will satisfy the intent of Section 1073 is not known at this time. Impact performance of any given feature is known to be sensitive to small changes in test parameters, especially those associated with the test vehicle. It must also be noted that some 4-wheel drive vehicles, as well as some conventional-drive vehicles, are either manufactured or customized by their owners to have oversized tires, extended suspension systems, small track widths, etc.. These design features can greatly diminish a vehicle's stability, i.e., its resistance to overturn. It is not economically feasible to design safety features to accommodate vehicles of this type.

(2) Very little, if any, ballast will be needed to meet the recommended test inertial mass.

(3) Use of a specific pickup type will enhance test standardization.

Test Vehicles 8000S, 36000V, and 36000T

These three heavy vehicles were selected for use in crash test evaluation of longitudinal barriers designed for the higher test or service levels. Several tests have been conducted with each of the three vehicles. Studies have indicated heights of approximately 81 cm, 107 cm, and 205 cm will be required for rigid barriers for the 8000S, 36000V, and 36000T vehicles, respectively, when ballasted as recommended.

Note in Table 2.2 that some of the dimension parameters have a suggested maximum value but no minimum value. This was done for two basic reasons: (1) it allows the testing agency more flexibility in purchasing the test vehicle and (2) impact loads will tend to increase as the value of these parameters decrease. Thus, although it is preferable to select vehicles with parameters near the maximum permissible values, lower values will provide an added factor of safety in the test.

Testing and user agencies should be aware of potential problems that may occur with a test using the 36000V test vehicle. In particular, the undercarriage attachment of the trailer tandems to the trailer frame may not be of sufficient strength to provide necessary restraint during the specified test. This problem is believed to be peculiar to sliding undercarriage or sliding axle designs. In at least one test, the attachment (which was the sliding undercarriage type) failed due to an inability to transfer lateral impact loads and the trailer went over the barrier. In a similar test with a fixed undercarriage attachment, no such failure occurred and the trailer did not go over the barrier. A sliding attachment is recommended for the test trailer since it is widely used in the industry. While it is desirable to test with widely-used vehicles and equipment, the primary purpose of the test is to demonstrate structural adequacy of the barrier, not the trailer. A barrier capable of containing a trailer with a sliding axle may have to be considerably taller than one capable of containing a trailer with a fixed axle. Nevertheless, public safety requires effective containment of vehicles on the road. If testing reveals this defect in trailer design will cause significant increases in the cost of effective barrier designs, support should be sought from appropriate officials and agencies to develop improved trailer designs and possibly the retrofitting of existing designs.

Each of the above test vehicles should be in sound structural shape without major sheet metal damage. Use of a vehicle for more than one crash test without repairs should be avoided because vehicle damage may affect performance in a subsequent test. This is particularly important in evaluating safety features such as a breakaway support where vehicle crush significantly affects the fracture mechanism.

A2.4.1.2

An emerging trend in evaluating impact performance of selected features is the use of surrogate test devices such as a bogie vehicle or a pendulum. A bogie vehicle is now being used by FHWA at the Federal Outdoor Impact Laboratory (FOIL) facility for compliance testing of breakaway sign and luminaire supports. It has exhibited a good degree of repeatability in replicating the response of a small car. Another key attribute is its relatively low cost of operation.

A wheeled bogie vehicle and a swinging pendulum with a crushable nose are the two primary types of surrogates used to date. While the pendulum can be used to evaluate certain aspects of impact performance, it is limited in terms of impact speed and replication of the postimpact behavior of an actual vehicle. It is also limited basically to single-support type structures. For certain features the bogie can replicate the full, three-dimensional dynamic behavior of an actual vehicle for the full range of design impact speeds. Although the following discussion will concentrate on the wheeled bogie, issues relevant to the pendulum are also relevant to the bogie.

A bogie is defined as a surrogate vehicle mounted on four wheels whose mass and other relevant characteristics match a particular vehicle or are representative of a typical or generic vehicle. It can be directed into the test article by a guide rail or cable, by remote control, or other means. It can be accelerated to impact speeds up to about 100 km/h by a push or tow vehicle, by self-power, or by a stationary windlass. The cost of operation is low since it can be reused without major repairs.

In addition to mass, properties such as frontal crush stiffness, weight distribution and center-of-mass location, dimensional properties including wheelbase and track width, and tire properties can be adjusted to represent a selected vehicle. Frontal crush stiffness can be simulated by adjusting the size and type of crushable modules.

Currently, the bogie at the FOIL is the only operational one in the U.S. designed for roadside safety studies. It is presently configured and has been validated to replicate a 1979 Volkswagen Rabbit. However, automobiles weighing from 635 kg up to 1,020 kg can be modeled. At the time of this writing, the California Department of Transportation (Caltrans) was developing two bogies for its California Automotive Research Test Site (CARTS) located at the University of California at Davis. One will cover weight ranges of 680-1,360 kg and the other will cover the range of 1,350-2,725 kg. The CARTS bogies can be configured with or without a suspension system (i.e., springs, shock absorbers, and suspension stops). The FOIL bogie does not have a suspension system. The FOIL bogie vehicle has exhibited a good degree of repeatability in replicating the response of a small car impacting breakaway support structures.

It is recommended that the surrogate be configured to model a specific vehicle, as opposed to a generic vehicle, with the stipulation that the vehicle being modeled meet specifications for production model test vehicles, i.e., specifications that define tolerances on age, weight, etc. This is by far the least expensive of the two options since properties of only one vehicle have to be measured and the validation process involves crash testing with only one vehicle model.

It would be desirable for FHWA or NCHRP to establish a project in which all bogie properties would be updated and validated periodically to keep current bogies within specifications. This would not only be the most efficient approach since each testing agency would not have to do it independently, it would insure uniformity throughout the testing community.

A2.4.1.3

See commentary in Section A3.2.4.

A2.4.2.2

Ballast for test vehicles that is free to shift or that can break loose during impact may be totally ineffective or only partially effective in initial loading of the feature because it tends to move independently of the vehicle. Unless specifically designed to evaluate effects of cargo shifting, tests with the 8000S and 36000V vehicles are to be conducted with a firmly secured ballast. The tie-down system should preferably be capable of resisting a lateral load equal to approximately ten times the weight of the ballast.

It must be noted, however, that test experience has shown that it is quite difficult to design a ballast tie-down system for a van truck or trailer with sufficient strength to resist typical impact loads for two reasons: (1) the absence of lateral stiffness in the walls of the van and (2) the height the ballast must be placed above the floor of the van to achieve the recommended center of mass of the ballast. For reasons of economy and convenience, sand bags on pallets are commonly used as a ballast in tests of van trucks or van trailers. While this achieves the required mass and center-of-mass height, it is difficult to secure this type of ballast and it creates a concentrated lateral load at some height above the floor of the van during impact. It would be preferable to use a ballast material with a density as low as possible so that the ballast would be uniformly distributed along the length, width, and height of the van, thus minimizing the need and structural requirements of the tie-down system. Bales of hay have been used as a relatively low density ballast.

A2.4.3

Because front wheel brakes of the test vehicle are sometimes damaged during impact, remotely actuated brakes are generally applied to the rear wheels only. This braking mode may cause the vehicle to yaw or spin during after-collision trajectory. For this reason braking should be delayed as long as safely feasible so that the unbraked after-collision trajectory can be observed. One suggestion is to use diagonal wheels, the front wheel away from impact and the impact-side rear wheel for braking in order to reduce vehicle spin. This practice would also be representative of brake designs on many automobiles. In any case, vehicle position at the time of brake application should be noted in the report.

A2.5

The automobile manufacturers and the National Highway Traffic Safety Administration (NHTSA) have devoted considerable effort in upgrading responsiveness and measurement techniques for dummies, primarily in the chest and head/neck regions. New and highly advanced dummies such as Hybrid III and Eurosid (developed in Europe) have been developed with 40 or more channels of data. However, it was concluded that the greatly increased cost of acquiring, maintaining, and applying dummies of this type and the added complexity of and demands on data acquisition and data reduction systems would more than offset the added benefits that may be realized in roadside safety design. Use of these dummies is therefore optional. Effectiveness of dummies that preceded Hybrid III and Eurosid in accurately quantifying the severity of a crash test has been found to be very limited. They are, therefore, not recommended except for use in studying the gross motion of an occupant and/or in studying the added mass effects of an occupant.

Also, expanded use is being made of sophisticated collision victim simulation computer simulation models. The CVS model developed under sponsorship of **NHTSA** is a three dimensional model with many features and complexities. To use it to evaluate a crash test one would input the response of the test vehicle into CVS, and the program would compute the dynamic response of an occupant positioned anywhere in the passenger compartment for either a restrained or an unrestrained condition. However, the amount and complexity of input data required for its use, the cost of running the program and, more importantly, the absence of any past record of performance and demonstrated efficacy of the program in the assessment of crash tests by the roadside safety community essentially precludes its application at this time.

CHAPTER THREE

A3.1

The "multiple service level" (MSL) concept for highway safety features was first introduced in NCHRP Report 239 (55). This study only addressed bridge railings. Report 230 adopted the MSL concept to a degree. Table 3 in NCHRP Report 230, "Crash Test Conditions for Minimum Matrix," provided testing for a MSL of 2. Table 4 of Report 230, "Typical Supplementary Crash Test Conditions," provided testing for an MSL of 1 and 3. The supplementary matrix applied primarily to longitudinal barriers. In recent years there has been an increased interest in the concept, not only for longitudinal barriers but for other features. AASHTO recently endorsed the MPL concept for the design of bridge railings (29). Three performance levels were selected for bridge railings.

For the present document it was decided that "test level" would be a more appropriate term than "service level." Selection of the recommended set of test levels and associated test conditions was based on the collective judgment of the researchers and the advisory panel after carefully reviewing past and current practices and anticipated future needs. It was also made in close collaboration with those responsible for developing AASHTO policies relevant to the safety performance of bridge rails and sign and luminaire support structures. The advantage of this approach is that it utilized current information and it reflects the combined expertise of a cross-section of disciplines and agencies directly involved in the full array of roadside safety issues. The disadvantage is that some of the decisions were, of necessity, based on limited quantitative data. As noted in the text, there are no warrants or criteria that identify roadway classifications, traffic conditions, traffic volumes, etc., for which a safety feature meeting a given test or performance level should be used. Given the choice, it would be preferable to first establish conditions or warrants for which features having given capabilities would be cost effective and thereby define appropriate test levels than to first establish a set of test levels with the uncertainty as to where features developed to meet these levels would have application. If and when warrants for multiple test level features are developed, it is possible that some of the levels will prove to be unnecessary or redundant and/or that other levels are needed.

A3.2

Errant vehicles of all sizes and classes leave the travelway and strike highway safety features with a wide range of speeds, angles, and attitudes. It should be a goal of transportation officials to design safety features that will satisfactorily perform for this range of impact conditions. Combinations of vehicle speed, mass, and approach angle that occur are unlimited. However, impact conditions must be reduced to a very limited number to keep an evaluation test series within economic and practical bounds. The approach used in formulating the recommended test conditions is to evaluate the devices for cases that are very severe, yet practical. Accordingly, there is no assurance that a safety feature will perform acceptably with other vehicle types presently in service or those vehicle types that may come into use during the normal service life of the device.

For test levels 3 through 6 and for the passenger test vehicles (700C, 820C, and 2000P), features are tested at a 100 km/h speed instead of the 89 km/h limit applicable to all highways other than rural freeways (most of which have a 105 km/h limit). Since a large percentage of high-speed travel occurs on other than rural freeways, the 100 km/h test speed should provide some degree of additional conservatism to the design of a feature. In addition to examining safety features for a range of impacts, the low-speed tests are important for certain features since they explore the activation of fracture or breakaway devices at relatively low kinetic energy levels.

A3.2.1

For test levels 4 through 6 and for the truck test vehicles (8000S, 36000V, and 36000T), longitudinal barriers are tested at a speed of 80 km/h. This is in recognition that speed limits for trucks are generally lower than for passenger vehicles. Also, most truck tests have been conducted at 80 km/h and linkage to past practices is desirable.

While vehicles leave the travelway and impact barriers within a wide spectrum of angles, most reported collisions with longitudinal barriers occur at impact angles less than 25 deg with the majority less than 15 deg. Historically, the 25 deg impact angle has been accepted as a practical worst case and the 15 deg approach angle as a more typical collision condition. The 25 deg angle test with the 2000P vehicle has been retained in the present document and is intended as a strength test for test levels 1 through 3. Tests with vehicles 8000S, 36000V, and 36000T are at 15 deg and are intended as strength tests for test levels 4, 5, and 6, respectively.

Since publication of Report 230 there has been a recognition and acceptance that while the 15 deg impact angle is more typical, a 20 deg angle is more discerning for tests with the 820C vehicle. The 20 deg angle has therefore been adopted for evaluating the impact severity of longitudinal barriers.

Critical impact point (CIP) is a new concept in testing of longitudinal barriers and other features. Rather than requiring the initial impact point to be at a specified point, e.g., midway between posts in Report 230, it is recommended that an effort be made to determine the CIP or the point with the greatest potential for causing a failure of the test. Failure can be caused by excessive snagging or pocketing of the vehicle, fracture of the barrier, vehicular override or underride of the barrier, vehicular overturn, etc. Suggested procedures for determining the CIP are given in Sections 3.4 and A3.4.

A3.2.2

Terminals and crash cushions function in the same or similar manner, i.e., they either bring the vehicle to a controlled stop, redirect the vehicle, allow controlled penetration of the vehicle, or a combination thereof. However, since any given design will generally not function in all three of these modes, it was necessary to categorize the recommended test matrices.

The two major categories are "terminals and redirective crash cushions" and "nonredirective crash cushions." This was done in recognition that there are two distinct types of crash cushions widely used in the U.S., i.e., those that redirect a vehicle if impacted along their length and those that do not, and in recognition that both types have proven very effective in reducing roadside hazards when properly applied. As noted in the text, terminals and redirective crash cushions are subjected to a more rigorous and demanding test series than nonredirective crash cushions. Consequently, impact performance capabilities of a redirective crash cushion will generally be greater than for a nonredirective cushion. Determining site conditions for which each type would have application is the responsibility of the user agency and was beyond the scope of this document.

Terminals and redirective crash cushions are further categorized into "gating" and "nongating" devices. As a general rule, a gating device is designed to allow controlled penetration of the vehicle for impacts upstream of the beginning of the length of need (LON). The breakaway cable terminal and its successor, the eccentric loader terminal, are examples of a gating device. As a general rule, a nongating device will redirect the vehicle if impacted along its side and brings the vehicle to a controlled stop if impacted on its end. The "GREAT" terminal (56) is considered to be a nongating device. Unfortunately, these subcategories do not uniquely describe the manner in which all terminals and redirective crash cushions function. For example, the "ET-2000" terminal (57) permits controlled penetration along portions of its length and it brings the vehicle to a controlled stop when impacted on its end if the impact angle is within a certain envelope.

The above discussion underscores an important point made in Section 3.1, i.e., the recommended test matrices cannot and should not be expected to be an all inclusive set of standardized procedures. When appropriate, the testing agency and/or the user agency should devise other critical test conditions consistent with the range of expected impact conditions.

In comparison to Report 230, additional tests are recommended for terminals and/or redirective crash cushions. Tests 32, 33, 37 and 39 are new tests. As indicated in Table 3.2 and in the text, some of these tests may not be required, depending on the design of the terminal and on its intended application. Note in the redirective "strength" tests (35, 37, and 38), the recommended impact angle is now 20 deg compared to 25 deg in Report 230. Selection of 20 deg was based on a recognition that design of a terminal or crash cushion is very sensitive to the redirective requirements of the 2000P vehicle and the associated impact angle, and crash cushions have historically been designed for a 20 deg side impact angle.

While it is preferable that the test vehicle remain upright after each test described herein, exceptions are made for all heavy vehicle tests and for tests of crash cushions and terminals within test level 1 (see Criterion G of Table 5.1). Overturn is permitted in the heavy vehicle tests since the primary goal in these tests is to demonstrate that the longitudinal barrier being evaluated can contain and redirect the vehicle. Crash test experience with heavy vehicles has shown that if overturn occurs, the vehicle usually undergoes only a 90 degree roll, remaining on its side while coming to rest. Exceptions are made for tests of crash cushions and terminals for test level 1 since most overturns at 50 km/h are not believed to be life-threatening. Exceptions are also made to permit and encourage the development and use of cost-effective crash cushion and terminals for low-speed applications. For example, a concrete, sloped-end terminal section can probably be designed to satisfy test level 1 criterion. Note that even though overturn is permitted for all heavy vehicle tests and level 1 tests of crash cushions and terminals, evaluation criterion D of Table 5.1 must be satisfied, i.e.. the overturn must not result in deformations of the occupant compartment that could cause serious injuries.

A3.2.3

Test matrices for work zone traffic control devices and breakaway utility poles are new. Limited testing has been conducted on both types of features and the recommended tests and evaluation criteria were based in part on these studies. As more experience is gained in the test and evaluation of these features, it may be desirable to amend the recommendations.

The energy or force required to fracture a breakaway device or support structure, in general, may be sensitive to its orientation with respect to direction of impact or the impact angle. For example, tests have indicated a breakaway transformer base breaks more readily when struck on a corner than on a flat side. Because errant vehicles may approach a support structure, work zone traffic control device, or a breakaway utility pole at various angles, it is recommended that the device be tested assuming the most severe direction of vehicle approach consistent with expected traffic conditions or at the critical impact angle (CIA) discussed in Section 3.2.3 For instance, the transformer base should be oriented so the vehicle strikes a flat side. Moreover, because the energy required to fracture a device can be increased due to buckling of the support at the point of contact with the vehicle, the handhold in the luminaire shaft should be positioned during a test so that probability of local collapse of the shaft is maximized.

Development of an energy-absorbing, yielding luminaire support pole was under way at the time of this writing. It is designed to decelerate the vehicle to a safe stop, similar to a crash cushion, rather than permit the vehicle to break through and continue with minimal reduction in speed. Rigid, nonbreakaway supports are often used in urban areas where encroachment of the vehicle beyond the pole could endanger pedestrians or other innocent bystanders. While this practice may offer protection for the innocent bystander, it may also increase risks to errant motorists. The yielding pole may have application in these areas, and/or areas where trees or other hazards exist just beyond the pole line that could endanger occupants of the encroaching vehicle. However, since such a design would not pass occupant risk criteria recommended for breakaway support structures, it would be necessary to use criteria recommended for a crash cushion. Recommendations on use of such features is beyond the purview of this document. Their use must be based on policy decisions by the user agency.

Breakaway utility poles are tested and evaluated somewhat differently from other support structures. A higher occupant impact velocity is permitted in a utility pole test. This is-in recognition of the relatively high change in vehicular velocity and, hence, occupant impact velocity that occurs during impact with commonly used wooden utility poles with the 820C test vehicle, irrespective of the breakaway mechanism. The change in vehicular velocity occurs in large part as a result of momentum transfer caused by the mass of the pole. Since a higher occupant impact velocity is permitted, the impact speed for the "low speed" test was set at 50 km/h, or 13.9 m/s. Note that for an impact speed of 35 km/h or 9.7 m/s (as used for other support structures), the vehicle could come to an abrupt stop and still pass the 12 m/s maximum occupant impact velocity criterion. Recommended tests and assessment criteria notwithstanding, it should be a goal of the designer to develop breakaway utility pole systems that minimize vehicular velocity change and, when possible, limiting occupant velocities should equal those for other support structures. Replacement of solid timber poles with lighter structures, if feasible, could reduce or eliminate problems associated with the relatively large mass of timber poles. Utility poles could then be expected to meet the same safety standards as other support structures.

A special feature somewhat related to support structures and not specifically included in the test guidelines is a fire hydrant. When it is required to evaluate the impact performance of a fire hydrant, it is recommended that it be tested to an appropriate test level and evaluated in accordance with the recommendations for a support structure.

A3.2.4

There are three basic areas of concern in an impact with a TMA: (1) risks to occupants of the impacting vehicle, (2) risks to occupants of the support truck to which the attenuator is attached, and (3) risks to workers if the support truck is pushed or rolls forward into the area occupied by the workers. All other factors being equal, risks to the occupants of the impacting vehicle generally increase as the mass of the support truck increases and/or as the degree of braking of the support truck increases. On the other hand all other factors being equal, risks to the occupants of the support truck and to workers ahead of the truck generally increase as the mass of the support truck decreases and/or the degree of braking decreases. Roll-ahead distance, the distance the support truck will advance upon impact, increases as the mass of the support truck decreases and as the degree of braking of the support truck decreases. Preferably, all these areas of concern would be evaluated in a given test series. It was concluded that at a minimum the recommended test series should focus on the first area of concern identified above, and additional optional tests could be conducted to evaluate other areas if necessary. Furthermore, it was concluded that the recommended tests should be standardized to the extent practicable, recognizing the rather wide variance in TMA specifications, support truck sizes and weights, and operating conditions (53). Thus, for test with the 820C (or the 700C) vehicle the support truck is braced against a rigid wall to prevent movement, thereby eliminating support truck mass effects. It can be shown that the small car test when conducted in this manner in most cases will not produce results significantly different from those with a braked support truck, considering the mass of most support trucks now in use. It is believed that this test will have major safety implications since it will require that all TMA's meet a minimum performance standard, regardless of support truck mass, and since risks to occupants of a small car impacting a TMA are generally greater than occupants of a larger vehicle, all other factors being equal. For test with the 2000P vehicle, the standardized truck mass (see Section 2.4.1.3) is representative of the heavier trucks used by state transportation agencies. The recommended braking is believed to be representative of typical in-service conditions. Test with the 2000P vehicle are designed to assess occupant risks and the roll-ahead distance of the support truck. It is noted that roll-ahead distances can be accurately estimated from the "conservation of momentum" principle of mechanics, knowing the frictional resistance of the support truck to forward movement. Reference 53 contains a description of the use of this principle in calculating roll-ahead distances.

A3.4.2

Longitudinal barriers generally fail due to structural inadequacies that allow snagging or pocketing on stiff points in the barrier systems or rupture of one of the "weak points" in the barrier system, such as a connection point. Thus, most barrier systems have one or more critical locations where failure is likely to take place, whether it be through wheel snag or rupture of a barrier element. The potential for each type of failure is affected to some extent by the selected impact point.

Report 230 recommended that the impact point be selected to provide a worst-case loading on the redirective device. However, in the absence of guidelines, most testing agencies used the default recommendation, i.e., impacting midway between posts for a length-of-need test, 4.6 m upstream from stiffer system for transitions, midway between nose and beginning of length of need for terminals, and at midlength for crash cushions. Recent studies have developed procedures for quantifying the critical impact point for certain devices.

Whenever possible, Barrier VII or another simulation program should be used to identify CIP's for longitudinal barrier tests. The following procedure may be followed to identify the CIP for snagging:

(a) Input the appropriate barrier and vehicular properties.

(b) Select an impact point with respect to the reference post. It is preferable, although not necessary, that this point be in reasonable proximity to the expected CIP so as to minimize the number of computer runs necessary to converge on the CIP.

(c) Determine vehicular and barrier response for the impact conditions of concern. The primary measure of snagging potential is the degree of wheel overlap with the reference post. Reference 15 discusses the manner in which the overlap is measured.

(d) Make incremental changes in the location of the impact point, repeating step c for each increment. Sufficient runs must be made to clearly bracket and then determine the CIP, i.e., the point that produces the greatest wheel overlap with the reference post. Experience has indicated the distance from the reference post to the CIP, denoted as "x," ranges from approximately 1 m for stiff systems to approximately 6 m for flexible systems.

A3.4.2.1

The small mass and low crush stiffness of passenger vehicles increases the likelihood and severity of wheel snag or pocketing on stiff elements of longitudinal barriers. Therefore, testing of longitudinal barriers with the 700C, 820C, and 2000P vehicles must be planned to examine the potential for wheel snag and pocketing as well as structural failure of the barrier elements. Wheel snagging and vehicular pocketing are the two barrier failure modes that exhibit the greatest sensitivity to impact point selection. When an impact point is too close to a post or other stiff point in a barrier system, the vehicle will not penetrate into the barrier prior to reaching the snag point. Conversely, when the selected impact point is too far from a snag point, the vehicle will redirect and begin to exit the barrier prior to snagging.

Connection loading is another important test parameter that is affected by impact location. Fortunately, impact locations that maximize wheel snagging or pocketing at one point in the barrier will also maximize connection loads near that same point in the barrier. Therefore, whenever rail splices or other critical connections fall at or near (within 1 meter) a snag point such as a barrier post, the impact location can be chosen to maximize both the potential for snagging and connection loadings. Since barrier loadings are generally higher upstream of the snag point, critical connections should be placed at or just upstream of the snag point, provided the connection locations are consistent with in-service locations. Rail tensile loads are maximized all along the length of the first span upstream from the snag point. Thus, the potential for rail splice tensile failure can generally be maximized by choosing the CIP for snagging if the connection is placed at the snag point or anywhere within the first span upstream from the snag point.

However, when a barrier connection is not located within approximately one meter of a snag point, bending moment and shear in the connection will not be maximized by an impact location chosen to maximize snagging. When barrier connections are not within one meter of a snag point and when wheel snag or pocketing as well as connection loading in bending and/or shear are significant concerns, the designer may consider conducting two tests with different impact locations. Barrier VII or a similar simulation program is recommended to investigate the need for two tests and to select CIP's.

It has been found that the CIP with regard to snagging is sensitive primarily to dynamic yield force of barrier posts, plastic moment of rail elements, and post spacing (16). Post yield forces and spacing were then combined into a single parameter, Fp, by dividing the dynamic post yield forces by the post spacing. CIP selection curves were then developed as a function of plastic moment of rail elements, M_p, and post yield force per unit length of barrier, F_p. Reference 16 contains a more detailed description of the development of CIP selection curves shown in Figures 3.7 through 3.14.

The plastic moment of a barrier rail element is merely the product of the beam's plastic section modulus and the material yield stress. Procedures for calculating plastic section modulus are presented in many textbooks on plastic design of steel structures (64). The plastic section modulus can be estimated with a reasonable degree of accuracy by multiplying the elastic section modulus by a form factor. Form factors for common beam shapes vary from a low of about 1.1 to a maximum of 2.0. As the fraction of a beam's cross section located near the neutral surface increases, the form factor of the cross section increases. Wide flange beams have very little material near the neutral surface and, as a result, generally have form factors less than 1.18 with an average near 1.14. Form factors for square box beams range from a low of 1.13 for a very thin-walled tube to a high of 1.5 for a solid rectangular rod. Form factors and plastic moments for some common barrier rail elements are shown in Table A3.1.

Barriers with multiple rail elements complicate the selection of an appropriate plastic moment for the barrier. When this type of barrier deflects during an impact, the upper rail deflection is much higher than that of lower rail elements. A simple energy analysis indicates that the total energy absorbed by each rail element is roughly proportional to the mounting height of the element. Equation 3.2 was then developed to estimate an equivalent plastic moment for multiple rail systems. A limited sensitivity study using Barrier VII revealed that

TABLE A3.1. PROPERTIES OF COMMON BARRIER RAIL ELEMENTS

Rail[a]	Elastic Section Modulus (cm^3)	Form Factor	Plastic Section Modulus (cm^3)	Plastic Moment (kN-m)
12 ga. W-Beam	22.5	1.41	31.6	10.9
10 ga. W-Beam	28.8	1.41	40.6	14.0
12 ga. Thrie-Beam	35.7	1.40	50.0	17.2
10 ga. Thrie-Beam	45.9	1.40	64.2	22.1
W6x15	.159	1.11	177	43.9
TS 6x6x3/16 Box Beam	134	1.16	156	49.5
TS 6x6x3/8 Box Beam	244	1.20	293	93.0
TS 8x6x1/4	257	1.19	306	97.1

[a] Post sizes are in English units.

the **CIP** determined by use of Equation 3.2 accurately estimates the CIP for most multiple rail barrier systems. This study indicated that the procedure was somewhat less accurate for barriers that have relatively stiff rail elements well above the impacting vehicle. For this situation, barrier posts will yield above the impacting vehicle and the upper rails will not deflect as much as the lower rails. Although the CIP selection procedures do give reasonable estimates of critical impact locations for most of these barriers, a simulation program should be used when possible to verify the findings.

Prior to determining F_p it is necessary to determine the dynamic yield force of the post. The post yield force will be governed by the smaller of two values: that necessary to yield the post itself assuming it is rigidly anchored at its base, or that necessary to yield the soil in which the post is embedded.

When barrier posts are rigidly anchored, yield forces are controlled by the material properties of the post. A dynamic magnification factor is normally applied to the plastic section modulus of metal posts to estimate the dynamic yield force for a post as given in Equation A3. 1.

$$F_y = D \left(\frac{\sigma_y Z_p}{H_r}\right) \qquad \text{(A3.1)}$$

where:

F_y = dynamic post yield force for a rigid anchor;
D = dynamic magnification factor;
ρ_y = post yield stress;
Z_p = post plastic section modulus; and
H_r= height of highest rail above base of post.

The accuracy of Equation A3.1 can be demonstrated by comparing a measured value of F_y for a rigidly anchored W6X9 steel post with the calculated value. A dynamic magnification factor of 1.5 is typically used for steel posts and a W6X9 beam has a plastic section modulus of 103 cm^3 and a yield stress of 248 MPa. For a 0.53 meter mounting height, Equation A3.1 gives an F_y of 71.9 kN compared to a measured value of 74.7 kN from reference 58.

Wood materials exhibit a brittle failure mechanism and therefore the plastic section modulus in Equation A3.1 is replaced by the elastic section modulus. Reference 58 reported that pendulum tests of a 6 inch X 8 inch (15.2 cm X 20.3 cm) Douglas Fir post have an average failure force of 72.1 kN when mounted in a rigid support. Southern Douglas Fir has an average modulus of rupture of 46.8 MPa (59). Using a dynamic magnification factor of 1.0, Equation A3.1 predicts failure forces of 91.9 kN and 74.0 kN for rough cut and finished posts with a nominal 6 inch X 8 inch (15.2 cm X 20.3 cm) size. Although it is unclear whether posts used in the pendulum tests were rough cut or finished size, the test results do indicate that the dynamic magnification factor from Equation A3.1 should be no more than 1.0 for wood materials. Table A3.2 shows the modulus of rupture for some common wood post materials.

Dynamic yield forces for posts embedded in soil are generally more difficult to estimate. Soil yield forces are usually measured through pendulum or instrumented cart testing at speeds near 32 km/h. A number of guardrail posts have been tested for various soil embedment conditions (34,58,60,61,62,63). Dynamic yield forces for common guardrail posts embedded in strong soils are shown in Table A3.3. The testing programs referenced above have shown that post yield forces can be approximated as a linear function of the square of the embedment depth. Thus, yield forces from Table A3.3 can be extrapolated for other embedment depths by multiplying the forces shown by the square of the ratio of the two embedment depths as given in Equation A3.2.

$$F_s' = F_s \times (\frac{D_e'}{D_e})^2 \qquad \text{(A3.2)}$$

where

$F_{s'}$ = soil dynamic yield force at alternate embedment depth, D_e';
F_s = soil dynamic yield force shown in Table A3.3;
D_e' = alternate embedment depth; and
D_e = post embedment depth shown in Table A3.3.

Some pendulum tests have been conducted in soft soils and are reported in reference 58. Analytical procedures for estimating the yield forces of other post sizes and soil conditions are discussed in reference 60.

The application of CIP selection curves is demonstrated in the following example:

Barrier	Rail	10 ga. thrie-beam mounted 0.58 m above ground
	Post:	W6X9 Steel with 1.5 meter embedment
	Spacing	2.5 meters
Test 3-11	Vehicle	2000P
	Impact Cond:	100 km/h, 25 deg

From Table A3.1 the plastic moment of a 10 ga. thrie-beam is found to be 22.1 kN-m. From Table A3.3 the dynamic yield force for a W6X9 steel post embedded 1.12 m in soil is approximately 55.2 kN. The approximate soil yield force for a W6X9 steel post embedded 1.5 m in soil can be estimated using Equation A3.2.

$$F_s' = 55.2\ KN \times (\frac{1.5\ m}{1.1\ m})^2 = 103\ KN$$

The yield force for a rigidly anchored, W6X9 steel post can be calculated from Equation A3. 1. A W6X9 beam has a section modulus of 102 cm3 and a yield stress of 248 MPa (64). The post yield force then becomes:

TABLE A3.2. WOOD POST PROPERTIES

Wood	Modulus of Rupture (MPa)	Shear Strength (MPa)
Douglas Fir	46.8	6.6
Southern Yellow Pine	50.4	5.9
Redwood	40.8	6.2

TABLE A3.3. DYNAMIC YIELD FORCES OF POSTS EMBEDDED IN STRONG SOIL

Post Type[a]	Embedment Depth (m)	Maximum Soil Limit (kN)	Maximum Post Limit (kN)
6 inch x 8 inch Wood Post	0.91	50.2	72.1
8 inch x 8 inch Wood Post	0.91	55.2	101.0[b]
10 inch x 10 inch Wood Post	0.91	72.5	205.0[b]
W6x9 Steel Post	1.12	55.2	65.0
W6x15 Steel Post	1.12	81.4	105.2

[a] Post sizes are in English units.

[b] Estimated for Douglas Fir using Equation A3.1.

$$F_y = 1.5\,(\frac{248\ MPa\ 102\ cm^3}{0.58}) \times (\frac{1m}{100cm})^3 = 65\ KN$$

The numerator of F_p in this case is the lower of the above two values, or 65 kN. Th post yield force per unit length for this barrier then becomes:

$$F_p = \frac{65\ KN}{2.5\ m} = 26\ KN/m$$

The CIP distance, "x," for this test is found from Figure A3.1. For an M_p of 22.1 kN-m, "x" distances of 3.0 m and 5.5 m correspond to F_p values of 50 kN-m and 8 kNm, respectively. Linear extrapolation can be used to estimate "x" for this example as follows:

$$\frac{5.5\ -\ 3.0}{50\ -\ 8} = \frac{x\ -\ 3.0}{50\ -\ 26}$$

$$x = 4.4\ m$$

Thus, the impact point for this test should be 4.4 m upstream from the reference post. Note that the reference post should be located at or just downstream from a rail splice.

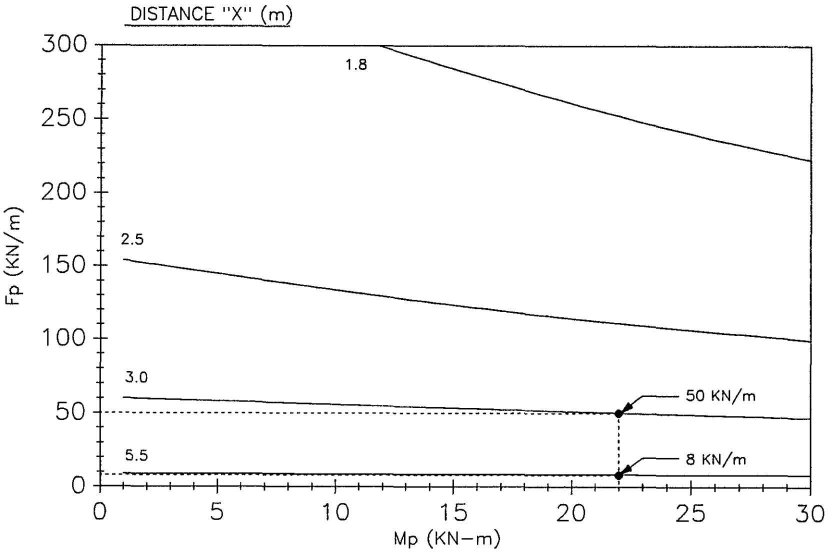
A3.4.2.2

Connection loading is the test parameter of primary importance for selecting impact points for heavy vehicle crash tests. Impact point selection guidelines presented in Section 3.4.2.1 are based on the distance from initial contact to the location of maximum lateral force. When possible, the impact point should be selected to generate maximum lateral loading at all important connection points including rail splices, rail-to-post connections, and post-to-base or post-to-deck connections. If the primary concern is for the truck to roll over the top of the barrier, the impact point should be selected to maximize lateral loading at midspan where the top barrier rail would be expected to deflect downward and increase rollover potential. Note that since heavy trucks spread impact loads over a larger area, a single test can usually be devised to apply near maximum loadings on all critical connections and adequately investigate the potential for post failure as well as rollover.

CHAPTER FOUR

A4.1

Proper documentation of key test details is often missing in a test report. For those not directly involved in the test program, assessment of a test and its results and development and implementation of standards for the test article cannot easily be done without good documentation. Sections 4.2, 4.3, and 4.4 describe important pretest, test, and post-test parameters.



SEE FIGURE 3.1 FOR "X".

FIGURE A3.1. EXAMPLE OF CRITICAL IMPACT POINT SELECTION

A4.3.2

Although not required at this time, the testing agency is encouraged to develop the capability to determine the six components of accelerations for the sprung mass (assumed to be a rigid body): translational accelerations in the x, y, and z vehicular axes and angular accelerations about these axes. These data, as well as corresponding velocities and displacements, should be shown in the report in plots or tables as a function of time.

High-speed cine is essential for study of crash dynamics to determine behavior of the test vehicle and the test article. In addition, high-speed cine has been used by some agencies as a backup system for determining vehicular accelerations and kinematics. Guidance for this secondary system consists of (1) minimum film speed (see Table 4.1), (2) internal or external timing device, and (3) stationary references located in the field of view of at least two cameras positioned 90 degrees apart. Layout and coordinates of references, camera positions, and impact point should be reported. Reference targets should be located on the side and the top of the test vehicle and should be of sufficient size and distance apart to allow accurate interpretation of the film. The instant of impact should be denoted by a flash unit placed in view of data cameras. The instant of impact should also be recorded on the electronic recording device(s).

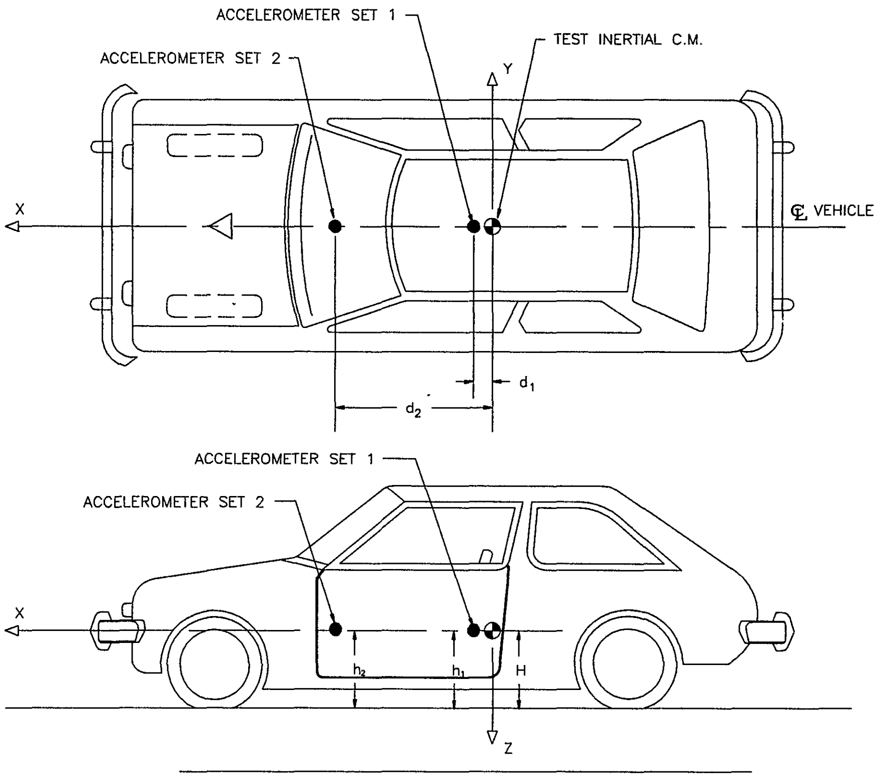
A4.3.3

Vehicular accelerations are used in the assessment of test results through the occupant flail space model. Accelerations may also be used to estimate impact forces between the vehicle and the test article.

Implicit in the flail space model is the assumption that accelerations are measured at the center of mass of the vehicle. NCHRP Report 230 recommended that a set of accelerometers be placed at or near the center of mass. However, experience has shown that this cannot always be done due to physical constraints within the vehicle. As a result, actual placement of the set of accelerometers may be offset a significant distance from the center of mass. Depending on the offset, major differences can occur between measured accelerations and those at the center of mass for redirection impacts (such as impacts with a longitudinal barrier) or impacts which cause angular vehicular motions. The following procedure is recommended if accelerometers cannot be placed within +5 cm of the center of mass as measured in the x-y plane. Although roll motions (rotations about the vehicle's x-axis) of the vehicle are not accounted for, the method has been shown to give acceptable levels of accuracy even for moderate roll motions.

Procedure:

(1) A triaxial set of accelerometers, set 1 in Figure A4.1, is mounted on a common block and placed as close to the vehicle's center of mass as practical with their positive directions corresponding to the positive sign convention given in Figure 4.6. Measurement of the vertical (z direction) acceleration is optional, but preferred. The set must be mounted along the fore-aft centerline (along x axis) of the vehicle. Theoretically, it is not necessary that set 1 be placed near the center of mass; however, this is recommended in the event accelerometer set 2 malfunctions. It is preferable that distance h_1 be within ±3 cm of distance H.


ACCELEROMETER SET 1
TEST INERTIAL C.M.
ACCELEROMETER SET 2
Y
X
℄ VEHICLE
d1
d2
ACCELEROMETER SET 1
ACCELEROMETER SET 2
X
h2
h1
H
Z

DIMENSIONS, cm

NOTE: h_1 AND h_2 SHOULD PREFERABLY EQUAL H

d_1 ______ h_1 ______ H ______

d_2 ______ h_2 ______

FIGURE A4.1 ACCELEROMETER PLACEMENT

(2) A triaxial set of accelerometers, set 2 in Figure A4. 1, is mounted as far as practical from set 1, preferably 60 cm or greater, either in front of or behind set 1. Note that both sets must be mounted forward of the cab/bed interface for the 2000P vehicle. The separation distance of the two sets should be as large as practical to reduce computational errors provided the accelerometers are not placed in an area that would be expected to undergo significant local dynamic deformations. Set 2 must also be mounted along the fore-aft centerline of the vehicle. It is preferable that distance h_2 be within +2 cm of distance h1.

(3) Using output from the above two accelerometer sets and distances d_1 and d_2, lateral, longitudinal, and vertical accelerations at the center of mass are computed by Equations A4.3, developed below. Note that d_1 and d_2 and their signs are measured with respect to the origin of the x,y,z axes located at the center of mass. For positions shown in Figure A4.1, both d_1 and d_2 are positive. However, it is not necessary that either be positive.

(4) Values of d_1, d_2, h_1, h_2, and H should be recorded and reported as shown in Figure A4.1.

Derivations of Equations:

Accelerations in the longitudinal direction a_x, lateral direction a_y, and vertical direction a_z measured by accelerometers located on the x axis a distance d forward from the center of mass are given by

$$a_x = a_{xg} - d(\omega_y^2 + \omega_z^2)$$

$$a_y = a_{yg} + d\dot{\omega}_z$$

$$a_z = a_{zg} - d\dot{\omega}_y$$

where, a_{xg}, a_{yg}, a_{zg} = longitudinal, lateral, and vertical accelerations at the center of mass; and

ω_y, ω_z, $\dot{\omega}_y$, $\dot{\omega}_z$ = pitch and yaw rates, and pitch and yaw accelerations.

Thus, the accelerations at points 1 and 2 of Figure A4. 1 are given by

$$
\begin{aligned}
a_{x_1} &= a_{xg} - d_1(\omega_y^2 + \omega_z^2) \\
a_{y_1} &= a_{yg} + d_1\dot{\omega}_z \\
a_{z_1} &= a_{zg} - d_1\dot{\omega}_y \\
a_{x_2} &= a_{xg} - d_2(\omega_y^2 + \omega_z^2) \\
a_{y_2} &= a_{yg} + d_2\dot{\omega}_z \\
a_{z_2} &= a_{zg} - d_2\dot{\omega}_y
\end{aligned}
\qquad \text{(A4.2)}
$$

Equatioons A4.2 can be solved to obtain the desired accelerations at the center of mass,$a_{xg,}$.$a_{yg,}$ and a_{zg} as follows:

$$
\begin{aligned}
a_{xg} &= \frac{d_2 a_{x1} - d_1 a_{x2}}{d_2 - d_1} \\
a_{yg} &= \frac{d_2 a_{y1} - d_1 a_{y2}}{d_2 - d_1} \\
a_{zg} &= \frac{d_2 a_{z2} - d_1 a_{z1}}{d_2 - d_1}
\end{aligned}
\qquad \text{(A4.3)}
$$

Note that the second and fifth equations and the third and sixth equations of set A4.2 can be solved to yield an explicit solution for pitch and yaw acceleration as follows:

$$
\begin{aligned}
\dot{\omega}_y &= -\frac{a_{z1} - a_{z2}}{d_1 - d_2} \\
\dot{\omega}_z &= \frac{a_{y1} - a_{y2}}{d_1 - d_2}
\end{aligned}
\qquad \text{(A4.4)}
$$

Pitch rate ,$\omega_{y,}$ at any time T after impact can be obtained by adding the pitch rate at impact to the integral of the first equation of set A4.4 with respect to time from impact to time T. Yaw rate, $\omega_{z,}$ can be similarly computed using the second equation of set A4.4

A4.4

Measuring and recording both the vehicle damage scale (VDS), formerly the traffic accident data scale (TAD), and the collision damage classification (CDC) are recommended for the following reasons. First, VDS has been in use for a number of years by various accident investigation agencies, and a considerable bank of data exists relating VDS to occupant injuries. Hence, by not reporting VDS, the tie of future tests with these historical data would be lost. And second, the National Highway Traffic Safety Administration (NHTSA) has standardized on the CDC for its multidisciplinary accident investigations. Therefore, CDC is needed to tie test vehicle damage (in which vehicle accelerations are measured) to real accidents in which occupant injury is documented.

CHAPTER FIVE

A5.1

Recommended evaluation criteria are limited to appraising safety performance of highway features for idealized vehicle crash test conditions. The basic purpose of crash tests is to screen out those candidate systems with functional deficiencies and to compare the relative merits of two or more promising candidate safety features. The test results are insufficient to project the overall performance of a safety feature for in-service use or in an actual collision situation. Final evaluation of a safety feature should be based on carefully documented in-service use.

Criteria for evaluating a vehicular crash test of a safety feature are patterned after those in Report 230 and consist of three interrelated factors: structural adequacy, occupant risk, and vehicle trajectory. In comparison to Report 230, the present criteria presented in Table 5.1 incorporate the following changes and/or modifications (further discussion of these items are given in following sections):

(a) Item D was moved from the "Structural Adequacy" category to the "Occupant Risk" category.

(b) Item E was added for evaluation of work zone traffic control devices.

(c) Item G was added for evaluation of heavy vehicle tests and test level 1 terminals and crash cushions.

(d) Under item H in the upper part of the table, the lateral occupant impact velocity limit was set equal to the longitudinal limit.

(e) The Hybrid III dummy is recommended as an optional measure of occupant risk for frontal impacts.

(f) Item L replaced item I of Report 230.

A5.2

The "structural adequacy" factor essentially assesses the feature from a structural and mechanical aspect. Depending on the feature, conditions to be examined include:

1. *Strength*. For longitudinal barriers, this requires containment and redirection of the design vehicles. Terminals and redirective crash cushions should develop necessary anchoring forces for anticipated site conditions.

2. *Geometry*. Longitudinal barrier rail members should engage the colliding vehicle at proper height to prevent the vehicle from underriding or overriding the installation. As a general rule, the vehicle-barrier contact surface should facilitate a smooth redirection. However, controlled stopping of the vehicle in a safe manner while the vehicle remains in contact with the rail is also satisfactory performance.Rail discontinuities such as splices and transitions and other elements such as support posts should not cause snagging to the extent that occupant risk criteria would not be met, or another failure mode would occur. Shaped rigid barriers, such as the New Jersey concrete barrier, should be designed to consider the stability of design vehicles.

3. *Mechanisms*. Stiffness, deformation, yielding, fracture, energy absorption and/or dissipation, etc., are characteristics of a feature that should be verified over the range of design vehicles.

In general, a safety feature should perform its function of redirecting, containing, or permitting controlled penetration of the test vehicles in a predictable and safe manner. Violent roll or rollover, pitching, or spinout of the vehicle reveal unstable and unpredictable dynamic interaction, behavior that is unacceptable.

A5.3

Relationships between occupant risk and vehicle dynamics during interaction with a high-way safety feature are extremely difficult to quantify because they involve such important by widely varying factors as occupant physiology, size, seating position, attitude and restraint, and vehicle interior geometry and padding. Advances have been made in recent years in better defining these relationships through development and application of sophisticated analytical and experimental tools, such as the collision victim simulation (CVS) computer program (66) and the Hybrid III dummy. Use of these tools would undoubtedly enhance assessment of occupant risk in tests of safety features. However, for the present document this was ruled unfeasible because of (a) costs associated with their purchase and/or use, (b) level of instrumentation and expertise needed, and (c) the absence of experience by testing agencies involved in evaluating highway safety features. Studies are needed to better define feasibility and effectiveness of tools of this type in improving occupant risk assessment in crash tests.

Flail-Space Model

Report 230 adopted the simplified point mass, flail-space model for assessing risks to occupants within the impacting vehicle due to vehicular accelerations. Two measures of risk are used; (1) the velocity at which a hypothetical occupant impacts a hypothetical interior surface and (2) "ridedown" acceleration experienced by the occupant subsequent to contact with the interior surface. Reference should be made to Report 230, the section on "Occupant Risk" in Chapter Four, in particular, for the underlying reasons for its adoption and its description, limitations, and assumed limiting risk factors. Based in part on reasons given in Section A5.3, it was concluded that the flail-space model should be retained for the present document.

Furthermore, it has served its intended purpose well and there are no indications that features designed and assessed thereby have performed adversely in service. Consideration was given to upgrading the flail-space model to better track the "occupant" as it flailed about the "occupant compartment." Assumptions made in the current model were:

(a) Occupant positioned at the vehicle's center of mass;

(b) Yaw motions of vehicle are ignored and, consequently, motion of the occupant in the lateral direction is completely independent of motion in the longitudinal direction,

(c) Vehicular and occupant motion is planar (in x-y plane); and

(d) Occupant contained in a compartment such that ± 0.3 m lateral movement can occur before impact with the sides of the compartment (idealized vehicular side structure), and 0.6 m longitudinal (forward) movement can occur before impact with the front of the compartment (idealized instrument panel/dash/windshield).

Options considered in updating the model all concerned changes that would affect results of redirection impacts. These included (a) positioning the occupant at the driver's and/or right-front passenger's seated position, (b) properly accounting for yaw motion of vehicle, and (c) changing the dimensions of the compartment to better represent the actual occupant compartment, e.g., this would allow the driver to flail 0.3 m to the left and in excess of 1.0 m to the right. After further study and careful review, it was concluded that the current model would be retained without changes for the following reasons:

(a) For typical redirection impacts, incorporation of options a and b, while effecting noncontrolling factors, would not have significant effects on controlling factors.

(b) If option c were incorporated, practically all redirection features would not meet limiting risk factors. This could be interpreted to mean one of several things including: most in-service redirection features developed according to Report 230 guidelines are unsafe; limiting occupant impact velocities and ridedown accelerations are too low; impact conditions of most accidents are not as severe as test conditions; occupants do not flail about the seats as would be assumed by option c; or a combination of these and/or other things. Since most redirection features designed according to Report 230 appear to be performing satisfactorily and since the flailspace model is actually an index or measure of occupant risk as opposed to an absolute measure, incorporation of option c does not appear warranted.

(c) Incorporation of these options would require use of a rather complex, standardized computer program and standardized input.

(d) A problem to date with determination of occupant risks through the flail-space model has been inconsistencies in positioning accelerometers used in measuring accelerations, i.e., they are not being placed at the vehicle's center of mass.

Recommendations contained in Section 4.3.3 should greatly reduce or eliminate this problem.

In the flail-space approach, lateral and longitudinal but not vertical vehicular accelerations measured at the vehicle's center-of-mass are used. By requiring that the vehicle in the occupant risk test remain upright throughout the collision, it is believed that the vertical component of vehicle acceleration becomes of secondary importance with regard to occupant kinematics for the level terrain tests described in this document and for most roadside features. Consequently, the vertical acceleration is considered an optional factor at present and has been neglected in the flail-space calculations.

The performance design strategy for a feature should be to (1) keep the occupant-vehicle interior impact velocity low by minimizing average vehicle accelerations or vehicle velocity change during the time the occupant is traveling through the occupant space and (2) limit peak vehicle accelerations during occupant ridedown.

Limiting Values for Impact Velocity and Ridedown Acceleration

The following items are to be noted:

(a) Report 230 presented "threshold" values and suggested feature-dependent factors of safety to apply to the threshold values. Table 8 of Report 230 contains values thusly obtained. In the present document, two sets of limiting values are given in Table 5.1: "preferred," which with some exceptions correspond to values in Table 8, and "maximum," which with some exceptions correspond to the threshold values.

(b) Based on consultations with biomechanics experts in the automotive industry and based on a review of the literature (67,68,69), it was concluded that Report 230 threshold values for occupant longitudinal impact velocity, and lateral and longitudinal ridedown accelerations should be retained. Based on information from these same sources, it was concluded that the threshold value for lateral occupant impact velocity should be increased to equal the value used for the longitudinal component. Note that reference 68 reported on a study that addressed the efficacy of the flail-space model and limiting values used therein. Among other things, this study found that recommended limiting occupant risk values of Report 230 were conservative, i.e., they overstated the risk level.

(c) Report 230 does not have a criterion that corresponds to the "preferred" limit for occupant impact velocity for support structures and work zone traffic control devices. The preferred limiting value of 3 m/sec and the maximum limiting value of 5 m/sec are approximately the same as those adopted by AASHTO (40). The maximum limiting value is slightly higher than the recommended value of Report 230.

(d) Due to conversions to the SI system, limiting occupant impact velocities and ridedown accelerations were rounded and consequently are not precisely the same as those in Report 230 or AASHTO.

"Maximum" limiting values of Table 5.1 should be treated as threshold limits. Test results should fall below these limits and desirably should not exceed the "preferred" values to promote safer performing features. In developing appropriate acceptance values, consideration should be given to the art-of-the-possible (i.e., can a device be made, regardless of cost, to perform to the requirements?) and cost effectiveness (i.e., can the increase in safety performance level justify the added cost?). Establishment of acceptance values is a policy decision and, therefore, beyond the purview of this report.

Procedures for acquiring and reducing vehicular accelerations used in determining occupant risk should follow recommended specifications given in Sections 5.3.2 and 5.3.3.

Calculation Procedures

The expression for occupant impact velocity is

$$V_{I_{x,y}} = \int_0^{t*} a_{x,y}\, dt \qquad \text{(A5.1)}$$

Where $V_{1_{xy}}$ is occupant-car interior impact velocity in the x or y directions, $a_{x,y}$ is vehicular acceleration in x or y direction, and t* is time when the occupant has traveled either 0.6 m forward or 0.3 m lateral, whichever is smaller. Time t* is determined by incremental integration as follows:

$$X, Y = \int_0^{t*} \int_0^{t*} a_{x,y}\, dt^2 \qquad \text{(A5.2)}$$

where, X = 0.6 m and Y = 0.3 m. Acceleration in the x direction is integrated twice with respect to time to find the value of time, t_x*, at which the double integration equals 0.6 m. Acceleration in the y direction is integrated twice with respect to time to find the value of time, t_y*, at which the double integration equals 0.3 m. Time t* is the smaller of t_x* and t_y*.

In tests of breakaway features the impulse on the vehicle may be relatively small and of short duration. It is not unusual in such tests for X and Y to be less than 0.6 m and 0.3 m, respectively, during the period in which accelerations are recorded or up to the time brakes are applied to the test vehicle. In such cases it is recommended that the occupant impact velocity be set equal to the vehicle's change in velocity that occurs during contact with the test article, or parts thereof. If parts of the test article remain with the vehicle after impact, the vehicle's change in velocity should be computed at the time the vehicle clears the footing or foundation of the test article.

For the ridedown acceleration to produce occupant injury, it should have at least a minimum duration ranging from 0.007 to 0.04 sec, depending on body component (70). Thus, vehicular acceleration "spikes" of duration less than 0.007 s are not critical and should be averaged from the pulse. An arbitrary duration of 0.010 s has been selected as a convenient and somewhat conservative time base for averaging accelerations for occupant risk assessment. This is accomplished by taking a moving 10-ms average of vehicular "instantaneous" accelerations in the x and y directions, subsequent to t*.

The occupant impact velocity and the highest 10-ms average acceleration values are then compared to recommended limits; it is desirable that both values be below the "preferable" limits; values in excess of the "maximum" limits are considered to be unacceptable.

Recommendations relative to the measurement of accelerations are given in Section 4.3.3 and in Appendix C. Further, for purposes of standardization of occupant risk calculation procedures, the following are recommended:

(1) Prior to integration using above formulas, accelerometer analog data should be digitized at 1,500 samples per second. This is consistent with recommendations of Appendix C, Section 9.2. It is recommended therein that the sample rate be, at a minimum, eight times F_h, where $F_h = 180$ for measurement of vehicular response. Note that F_h X 8 = 1,440, which is rounded to 1,500 for convenience and ease of integration.

(2) It is recommended the "linear acceleration" assumption or the equivalent "trapezoidal rule" be used to integrate the digitized accelerometer data. As such, accelerations are assumed to vary linearly over each time step t_i to t_{i+1} . Description of the trapezoidal rule can be found in most numerical methods textbooks.

A5.4

In general the ideal after-collision vehicular trajectory performance goal for all features should be that the vehicle trajectory and final stopping position should not intrude into the adjacent or opposing traffic stream. For breakaway or yielding supports the trajectory of a vehicle after it has collided with a test article that satisfies structural adequacy and occupant risk requirements is generally away from the traffic stream and, hence, is normally noncritical. For end-on impacts into crash cushions and barrier terminals that function as crash cushions, preferably the final position of the vehicle should be next to the test device.

For redirectional performance tests of length of need, transitions, terminals and redirective crash cushions, the after-collision trajectory is more difficult to assess. The after-collision trajectory may be one of the least repeatable performance factors because of variation in method and timing of brake application. Further, variables that are in part related to the specific model of vehicle selected for tests such as damage to vehicle suspension, tires, etc., may alter the vehicle's stability and path. Moreover, because driver response in avoiding secondary collisions is not simulated in the crash tests, it seems inappropriate to predict in-service performance based on the complete test trajectory. For these reasons trajectory evaluation for the redirectional type of tests is focused on the vehicle during contact (criterion L of Table 5.1). At the time it loses contact with the test article (criterion M of Table 5.1) and the subsequent part of the trajectory is only subjectively evaluated (criterion K of Table 5.1).

A study (71) conducted since publication of Report 230 found that risks to occupants of the impacting vehicle or to other motorists, as a consequence of the vehicle being redirected, are not as great as previously believed, provided the impacting vehicle does not collide subsequently with other roadside objects. The study pointed out that secondary collisions with roadside objects such as trees, other barriers, etc., is of major concern after a redirected impact. Hence, the 24.2 km/h (15 mi/h) velocity change criterion of Report 230 (item I of Table 6) was not adopted in the present document. Criterion L of Table 5.1 is intended to allow controlled deceleration of the vehicle without excessive pocketing or snagging of the vehicle. Note that for most redirectional devices, a 12 m/sec occupant impact velocity equates to a vehicular velocity change of approximately the same magnitude or 43.2 km/h (26.8 mi/h).

While the above cited study suggests that trajectory of the vehicle into adjacent or opposing lanes may not be as critical as previously thought, it remains preferable that the vehicle exit the device at a low angle. Although ideal performance would be for the vehicle to exit with a path parallel to the installation, an upper limit of 60 percent of the impact angle is recommended.

A5.5

Specific test and evaluation guidelines for geometric features are not provided due to the largely nonstandard and variable nature of such features. However, it should be a goal of transportation agencies to design and implement geometric features that meet the spirit, if not the specifics, of safety recommendations for the more well-defined roadside safety features.

Evaluation guidelines given in this section were derived from a review of past practices and the collective expertise of those involved in preparing the document. They are, of necessity, general and may be amended as necessary to accommodate special designs or test conditions.

CHAPTER SIX

(No commentary is provided for this chapter)

CHAPTER SEVEN

A7.1

In-service evaluation guidelines are intended to encourage a cautious, systematic introduction of a new safety feature. With careful monitoring, unanticipated problems and design deficiencies can be identified before the feature has been installed in an excessive number of sites. Moreover, all the affected departments will have an opportunity to observe the performance of the device with respect to their operations. For instance there may be minor design changes recommended by the maintenance groups that may reduce normal maintenance or damage repair costs. Also, substitution of material or fasteners could ease the problem of a large inventory of spare parts. Care should be taken not to make changes in design details that could adversely affect safety performance without verification of adequate performance through full-scale crash testing or other acceptable means.

The in-service evaluation guidelines are intended to encourage a more consistent and thorough implementation of new devices and to promote a more direct and systematic process in demonstrating the operational status of ,safety features.

APPENDIX B

SOIL SPECIFICATIONS

APPENDIX B

PART 1

STANDARD SOIL

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APPENDIX B

PART 2

WEAK SOIL

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APPENDIX B

PART 3

COMPACTION GUIDELINES

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PART 1

Standard Specification
for

Materials for Aggregate and Soil-Aggregate Subbase, Base and Surface Courses

AASHTO DESIGNATION: M 147-65 (1990)

1. SCOPE

1.1 This specification covers the quality and grading of sand-clay mixtures; gravel, stone or slag screenings; or sand, crusher run coarse aggregate consisting of gravel, crushed stone or slag with or without soil mortar or any combination of these materials for use in the construction of subbase, base and surface courses. The requirements are intended to cover only materials having normal or average specific gravity, absorption and gradation characteristics. Where other materials are to be used, appropriate limits suitable to their use must be specified.

2. GENERAL REQUIREMENTS

2.1 *Coarse Aggregate*:

2.1.1 Coarse aggregate retained on the 2.00 mm (No. 10) sieve shall consist of hard, durable particles or fragments of stone, gravel or slag. Materials that break up when alternately frozen and thawed or wetted and dried shall not be used.

2.1.2 Coarse aggregate shall have a percentage of wear, by the Los Angeles test, AASHTO T 96, of not more than 50.

NOTE 1-A higher or lower percentage of wear may be specified by the Engineer, depending upon the materials available for the work.

2.2 *Fine Aggregate*:

2.2.1 Fine aggregate passing the 2.00 mm (No. 10) sieve shall consist of natural or crushed sand, and fine mineral particles passing the 0.075 mm (No. 200) sieve.

2.2.2 The fraction passing the 0.075 mm sieve shall not be greater than two-thirds of the fraction passing the 0.425 mm (No. 40) sieve. The fraction passing the 0.425 mm sieve shall have a liquid limit not greater than 25 and a plasticity index not greater than 6

2.3 All material shall be free from vegetable matter and lumps or balls of clay. The soil-aggregate material shall conform to the grading requirements of Table 1. The grading requirements for composite aggregate material will be specified by the Engineer.

3. SUBBASE MATERIALS

3.1 Materials for subbase shall conform to the requirements of sections 2 and 3 for gradings A, B, C, D, E, or E The type and grading desired shall be specified.

NOTE 2-Where local experience has shown that lower percentages passing the 0.075 (No. 200) sieve than are required in Table 1 are necessary for subbase materials in order to prevent damage by frost action, the Engineer should specify such lower percentages.

4. BASE COURSE MATERIALS

4.1 Materials for base course shall conform to the requirements of section 2 for gradings A, B, C, D, E, or F The grading desired shall be specified.

NOTE 3-Where local experience has shown that lower percentages passing the 0.075 mm (No. 200) sieve than are required in Table 1 are necessary for base course materials in order to prevent damage by frost action, the Engineer should specify such lower percentages.

5. SURFACE COURSE MATERIALS

5.1 Materials for surface course shall conform to the requirements of sections 2 and 3 for gradings C, D, E, or F The gradings desired shall be specified.

NOTE 4-Where it is planned that the soil aggregate surface course is to be maintained for several years without bituminous surface treatment or other superimposed impervious surfacing, the Engineer should specify a minimum of 8 percent passing 0.075 mm (No. 200) sieve in lieu of the minimum percentages shown in Table I for grading C, D, or E, and should specify a maximum liquid limit of 35 and plasticity index range of 4 to 9 in lieu of the limits given in 2.2.2.

TABLE 1 Grading Requirements for Soil-Aggregate Materials

Sieve Designation		Mass Percent Passing					
Standard mm	Alternate	Grading A	Grading B	Grading C	Grading D	Grading E	Grading F
50	2 in.	100	100	—	—	—	—
25.0	1 in.	—	75-95	100	100	100	100
9.5	⅜ in.	30-65	40-75	50-85	60-100	—	—
4.75	No. 4	25-55	30-60	35-65	50-85	55-100	70-100
2.00	No. 10	15-40	20-45	25-50	40-70	40-100	55-100
0.425	No. 40	8-20	15-30	15-30	25-45	20-50	30-70
0.075	No. 200	2-8	5-20	5-15	5-20	6-20	8-25

6. MOISTURE CONTENT

6.1 All materials shall contain moisture equal to or slightly below the optimum necessary to insure that the design density requirements are obtained when materials are compacted.

7. ADMIXTURE

7.1 Calcium chloride used for the control of moisture shall meet the requirements of Standard Specifications for Calcium Chloride (AASHTO M 144).

8. METHODS OF SAMPLING AND TESTING

8.1 Sampling and testing shall be in accordance with the following standard methods of the American Association of State Highway and Transportation Officials:

Sampling	T 2
Sieve analysis	T 27 or T 88
Surveying and sampling soils for highway subgrades	T 86
Preparing samples	T 87
Liquid limit	T 89
Plastic limit and plasticity index	T 90
Percentage of wear	T 96
Passing 0.075 mm ..	T 11

PART 2

Standard Specification
for

Fine Aggregate for Portland Cement Concrete

AASHTO DESIGNATION: M 6-87

1. SCOPE

1.1 This specification covers the quality and grading of fine aggregate for port land cement concrete used in pavements or basics, highway bridges, and incidental Structures.

1.2 *Units of Measurement:*

1.2.1 For sieve sizes and the size of aggregate as determined by the use of testing sieves, the values in inch-pound units are shown for the convenience of the user; however, the standard sieve designation shown in parentheses is the standard value as stated in AASHTO M 92.

1.2.2 For other units of measure, the values stated in inch-pound units are to be regarded as standard.

2. REFERENCED DOCUMENTS

2.1 AASHTO Standards:

M 80 Coarse Aggregate for Portland Cement Concrete
M 92 Wire Cloth Sieves for Testing Purposes
T 2 Sampling Aggregates
T 11 Amount of Material Finer Than 75-mu m Sieve in Aggregate
T 21 Organic Impurities in Fine Aggregate for Concrete
T 27 Sieve Analysis of Fine and Coarse Aggregates
T 71 Effect of Organic Impurities in Fine Aggregate on Strength of Mortar
T 103 Soundness of Aggregates by Freezing and Thawing
T 104 Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
T 112 Clay Lumps and Friable
T 113 Lightweight Pieces in Aggregate
T 161 Resistance of Concrete to Rapid Freezing and Thawing

Particles in Aggregate

3. ORDERING INFORMATION

3.1 The purchaser shall include the following information in the purchase order or contract when applicable:

3.1.1 Reference to this Specification, AASHTO M 6, and year of issue,

3.1.2 Grading requirements for No. 8 (2.36 mu m) and No. 30 (600 mu m) sieves, when required (5.1 and Note 1),

3.1.3 Whether the deleterious substances limits for Class A or Class B apply, and limits on other deleterious substances (7.1 and Note 3),

3.1.4 If the sulfate soundness requirement is waived (8.5),

3.1.5 In the case of the sulfate soundness test (8.1), which salt is to be used. If none is stated, either salt may be used.

3.1.6 If the supplementary requirement for reactive aggregates applies, (see S 1), and

3.1.7 Any exceptions or additions to this specification.

4. GENERAL REQUIREMENTS

4.1 Fine aggregate shall consist of natural sand or manufactured sand or combinations thereof, having hard, strong, durable particles.

4.2 Fine aggregate from different sources of supply shall not be mixed or stored in the same pile.

5. GRADING

5.1 Fine aggregate, when tested by means of laboratory sieves, shall conform to the following requirements, except as provided in 5.2 and 5.3:

NOTE 1-The purchaser or specifier may insert specific values for percent passing the No. 8 and No. 30 sieves to further control the grading of the material.

Sieve	Mass Percent Passing
3/8 in. (9.5 mm)	100
No. 4 (4.75 mm)	95-100
No. 8 (2.36 mm)	--
No. 16 (1.18 mm)	45-80
No. 30 (600 µm)	--
No. 50 (3(x) µm)	10-30
No. 100 (150 µm)	2-10

5.2 The minimum percent shown above for material passing the No. 50 (300-µm) and No. 100 (150-µm) sieves may be reduced to 5 and 0, respectively, if the aggregate is to be used in air-entrained concrete containing more than 400 lb of cement per cubic yard (237 kg/m^3) or in nonair-entrained concrete containing more than 500 lb of cement per cubic yard (297 kg/m^3) or if an approved mineral admixture is used to supply the deficiency in percent passing these sieves. Air-entrained concrete is here considered to be concrete containing air-entraining cement or an airentraining agent and having an air content of more than 3%.

5.3 The fine aggregate shall have not more than 45% passing any sieve and retained on the next consecutive sieve of those shown in 5. 1.

6. UNIFORMITY OF GRADING

6.1 The grading requirements given in 5.1 represent the extreme limits which shall determine suitability for use from all sources of supply. The grading from any one source shall be reasonably uniform and not subject to the extreme percentages of grading specified above. For continuing shipments of fine aggregate from a given source, the fineness modulus shall not vary more than 0.20 from the base fineness modulus. The base fineness modulus shall be that value that is typical of the source. If necessary, the base fineness modulus may be changed when approved by the purchaser.

NOTE 2-The base fineness modulus should be determined from previous tests, or if no previous tests exist, from the average of the fineness modulus values for the first ten samples (or all preceding samples if less than ten) on the order. The proportioning of a concrete mixture may be dependent on the base fineness modulus of the fine aggregate to be used. Therefore, when it appears that the base fineness modulus is considerably different from the value used in selecting proportions for the concrete mixture, a suitable adjustment in the mixture may be necessary.

7. DELETERIOUS SUBSTANCES

7.1 The amount of deleterious substances shall not exceed the following limits (see table entitled "Deleterious Substances Limits"):

NOTE 3-The purchaser or specifier, due to knowledge of the requirements of the work and the constituents of locally available aggregate, should insert appropriate requirements when needed.

7.2 Organic Impurities:

7.2.1 Fine aggregate shall be free of injurious amounts of organic impurities. Except as herein provided, aggregates subjected to the test for organic impurities and producing a color darker than the standard shall be rejected.

7.2.2 A fine aggregate failing in the test may be used, provided that the discoloration is due principally to the presence of small quantities of coal, lignite, or similar discrete particles.

7.2.3 A fine aggregate failing in the test may be used, provided that, when tested for the effect of organic impurities on strength of mortar, the relative strength at 7 days calculated in accordance with AASHTO T 71 is not less than 95%.

8. SOUNDNESS

8.1 Except as provided in 8.2 through 8.5, fine aggregate subjected to five cycles of the soundness test shall have a weighted average loss not greater than 10% when sodium sulfate is used or 15% when magnesium sulfate is used.

8.2 Fine aggregate failing to meet the requirements of 8.1 may be accepted, provided that concrete of comparable properties, made from similar aggregate from the same source, has given satisfactory service when exposed to weathering similar to that to be encountered.

8.3 Fine aggregate not having a demonstrable service record and failing to meet the requirements of 8.1 may be accepted, provided it gives satisfactory results in concrete subjected to freezing and thawing tests. (See AASHTO T 161.)

8.4 Fine aggregate failing to meet the requirements given in 8.1 may, at the option of the purchaser or specifier, be subjected to an alternate freezing and thawing test of unconfined aggregate and may be accepted provided it gives satisfactory results.

NOTE 4-The purchaser or specifier should determine the details of the evaluation and criteria for determining satisfactory performance in 8.2, 8.3, and 8.4.

8.5 The requirements for soundness given in 8.1 may be waived in the case of aggregate for use in structures or portions of structures not exposed to weathering.

9. METHODS OF SAMPLING AND TESTING

9.1 Sampling and testing of fine aggregate shall be in accordance with the following methods of the American Association of State Highway and Transportation Officials:

9.1.1 Sampling-T 2

9.1.2 Sieve analysis and fineness modulus-T 27

9.1.3 Clay lumps and friable particles-T 112

9.1.4 Coal and Lignite-T 113, using a liquid of 2.0 specific gravity to remove the particles of coal and lignite. Only material that is brownish-black, or black, shall be considered coal or lignite. Coke shall not be classed as coal or lignite.

9.1.5 Material finer than No. 200 (75-µm)-T 11

9.1.6 Organic impurities-T 21

9.1.7 Effect of organic impurities on strength-T 71

9.1.8 Sulfate soundness-T 104

9.1.9 Soundness (unconfined freezing and thawing)-T 103

9.1.10 Freezing and thawing of concrete-T 161

SUPPLEMENTARY REQUIREMENT

The following supplementary requirement applies only when specifically stated in the order or contract.

S1. REACTIVE AGGREGATE

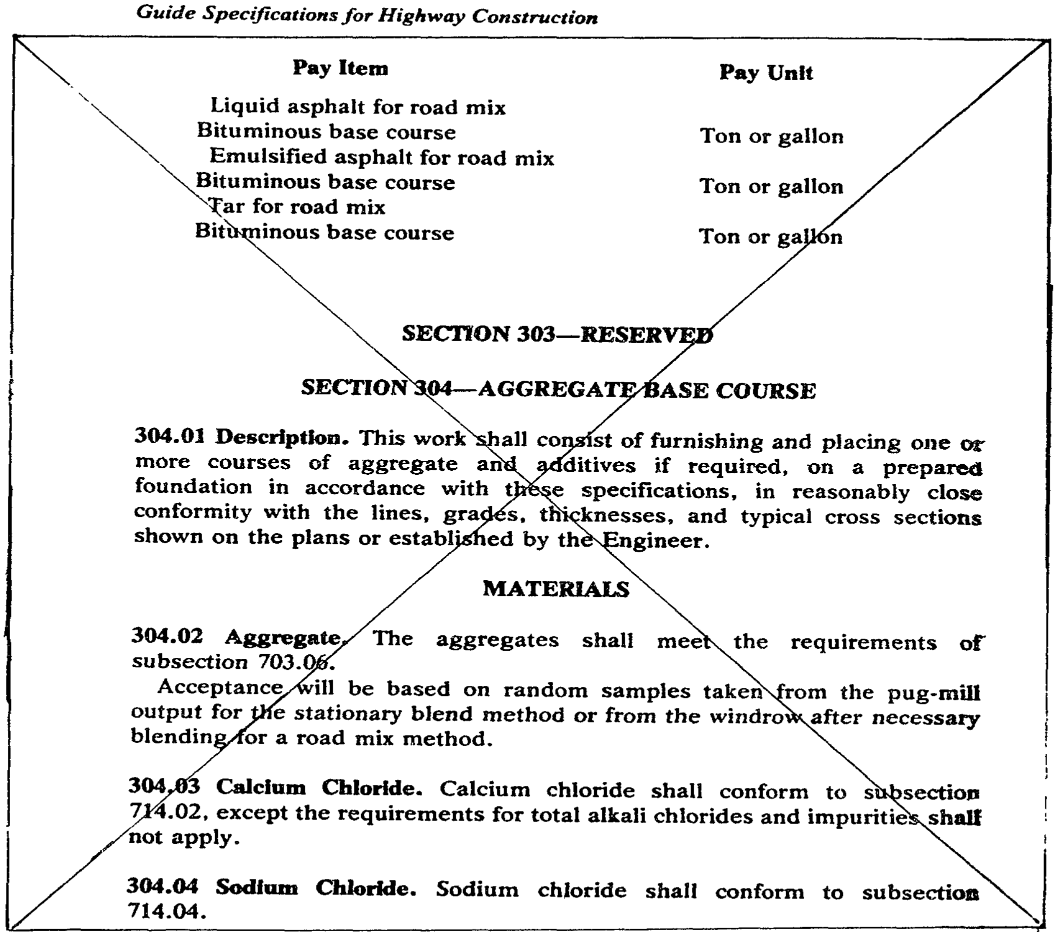
S1.1 Fine aggregate for use in concrete that will be subject to wetting, extended exposure to humid atmosphere, or contact with moist ground shall not contain any materials that are deleteriously reactive with the alkalies in the cement in an amount sufficient to cause

Deleterious Substances Limits

	Class A, max, Mass Percent	Class B, max, Mass Percent
Clay lumps and friable particles	3.0	3.0
Coal and Lignite	0.25	1.0
Material finer than No. 200 (75-µm) sieve:		
(a) In concrete subject to surface abrasion, not more than	2.0	4.0
(b) All other classes of concrete, not more than	3.0	5.0
Other deleterious substances (such as shale, alkali, mica, coated grains, soft and flaky particles)	Note 3	Note 3

excessive expansion of mortar or concrete, except that if such materials are present in injurious amounts, the fine aggregate may be used with a cement containing less than 0.60% alkalies calculated as sodium oxide equivalent ($Na_2O + 0.658K_20$) or with the addition of a material that has been shown to prevent harmful expansion due to the alkali-aggregate reaction. (See Appendix X1 of AASHTO M 80.)

NOTE SI-This supplementary requirement would normally be specified only in areas having a history of reactive aggregate, except for unusually critical structures, to save costs and delays due to unnecessary testing.



~~Guide Specifications for Highway Construction~~

~~**Pay Item**~~	~~**Pay Unit**~~
~~Liquid asphalt for road mix Bituminous base course~~	~~Ton or gallon~~
~~Emulsified asphalt for road mix Bituminous base course~~	~~Ton or gallon~~
~~Tar for road mix Bituminous base course~~	~~Ton or gallon~~

~~SECTION 303—RESERVED~~

~~SECTION 304—AGGREGATE BASE COURSE~~

~~**304.01 Description.** This work shall consist of furnishing and placing one or more courses of aggregate and additives if required, on a prepared foundation in accordance with these specifications, in reasonably close conformity with the lines, grades, thicknesses, and typical cross sections shown on the plans or established by the Engineer.~~

~~MATERIALS~~

~~**304.02 Aggregate.** The aggregates shall meet the requirements of subsection 703.06.~~

~~Acceptance will be based on random samples taken from the pug-mill output for *the stationary blend method or from the windrow after necessary* blending for a road mix method.~~

~~**304.03 Calcium Chloride.** Calcium chloride shall conform to subsection 714.02, except the requirements for total alkali chlorides and impurities shall not apply.~~

~~**304.04 Sodium Chloride.** Sodium chloride shall conform to subsection 714.04.~~

CONSTRUCTION REQUIREMENTS

304.05 Placing. If the required compacted depth of the base course exceeds 6 inches, the base shall be constructed in two or more layers of approximate equal thickness. The base material shall be laid by means of an approved mechanical spreader capable of placing the material to a uniform depth without segregation. The maximum compacted thickness of any one layer shall not exceed 6 inches, except a single layer upon approval may be increased to a depth of 8 inches when vibratory equipment or special compaction equipment is used.

~~**304.06 Mixing.** Unless otherwise specified, the Contractor shall mix the base course by one of the following methods and incorporate any additives that may be required as shown on the plans.~~

~~(a) *Stationary Plant Method* — The material shall be mixed in an approved mixer and water added during the mixing operation to provide the optimum moisture content for compacting. Immediately after mixing, the base material shall be transported to the job site and placed on the roadbed by means of an approved aggregate spreader.~~

~~(b) *Travel Plant Method* — After the material has been placed through an aggregate spreader or windrow sizing device, the material shall be uniformly mixed by a traveling mixing plant and water added during the mixing operation to provide optimum moisture content for compacting.~~

~~(c) *Road Mix Method* — After the material has been placed, it shall be uniformly mixed by means of motor graders or other approved equipment and water added during the mixing operation to provide optimum moisture content.~~

304.07 Shaping and Compaction. The material shall be shaped to the required section and water applied or aerated as necessary to provide the optimum moisture content for compaction. Compaction shall continue until a density of not less than ..... percent (95 percent suggested) of the maximum density determined in accordance with AASHTO T 180 Method D has been achieved. The surface shall be maintained during the compaction operations in such a manner that a uniform texture is produced and the aggregates firmly keyed. Water shall be uniformly applied over the base materials during compaction in the amount necessary for proper consolidation.

In-place density will be determined in accordance with AASHTO T 191, T 205, or other approved method. The use of AASHTO T 224 (alternative) to correct for oversize particles may be required. AASHTO T 238 may be used if the limitations of the method as cited in the appendix to the standard are fully recognized.

~~**304.08 Method of Measurement.** Aggregate base course will be measured by the ton, cubic yard, or square yard in accordance with Section 109—Measurement and Payment. The weight of moisture (surface and hygroscopic) will be deducted when the aggregate is measured by weight.~~

APPENDIX C

ELECTRONIC AND PHOTOGRAPHIC INSTRUMENTATION SPECIFICATIONS

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HIGHWAY
VEHICLE
PRACTICE

Submitted for recognition as an American National Standard

SAE J211

Issued October 1970
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Superseding J211 JUN80

Ø INSTRUMENTATION FOR IMPACT TEST

SCOPE:

This recommended practice outlines a series of performance recommendations which concern the whole data channel. These recommendations are not subject to any variation and all of them should be adhered to by any agency conducting tests to this practice. However, the method of demonstrating compliance with the recommendations is flexible and can be adapted to suit the needs of the particular equipment the agency is using.

It is not intended that each recommendation be taken in a literal sense, as necessitating a single test to demonstrate that the recommendation is met. Rather, it is intended that any agency proposing to conduct tests to this practice should be able to demonstrate that if such a single test could be and were carried out, then their equipment would meet the recommendations. This demonstration should be undertaken on the basis of reasonable deductions from evidence in their possession, such as the results of partial tests.

In some systems it may be necessary to divide the whole channel into subsystems, for calibration and checking purposes. The recommendations have been written only for the whole channel, as this is the sole route by which subsystem performances affect the quality of the output. If it is difficult to measure the whole channel performance, which is usually the case, the test agency may treat the channel as two or more convenient subsystems. The whole channel performance could then be demonstrated on the basis of subsystem results, together with a rationale for combining the subsystem results together.

Part 1 of this recommended practice covers electronic instrumentation and Part 2 covers photographic instrumentation.

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PART 1 - ELECTRONIC INSTRUMENTATION

1. PURPOSE:

The purpose of this SAE Recommended Practice is to provide guidelines and recommendations for the techniques of measurement used in impact tests. The aim is to achieve uniformity in instrumentation practice and in reporting test results. Use of this recommended practice will provide a basis for meaningful comparisons of test results from different sources.

2. FIELD OF APPLICATION:

The instrumentation as defined in this recommended practice applies in particular to impact tests for road vehicles, including tests of their sub-assemblies, and occupant surrogates.

3. DEFINITIONS:

The definitions in 3.5 to 3.14 apply to the whole data channel, as defined in3.1.

3.1 Data Channel: All of the instrumentation from and including a single transducer (or multiple transducers whose outputs are combined in some specified way) up to and including any analysis procedures that may alter the frequency content or the amplitude content or the timing of data. It also includes all cabling and interconnections.

3.2 Transducer: The first device in a data channel, used to convert a physical quantity to be measured into a second quantity (such as an electrical voltage) which can be processed by the remainder of the channel.

3.3 Full Scale: The maximum usable linear range of a data channel.

3.4 Data Channel Full Scale: That value of a data channel determined by the component of the channel with the lowest full scale level. This is expressed in terms of the measured variable (input). For example, F.S.=50 G, 1000 N, 100 cm/s, etc.

3.5 Channel Amplitude Class, CAC: The designation for a data channel that meets certain amplitude characteristics as specified by this recommended practice.

The CAC number is numerically equal to the upper limit of the measurement range, (that is, equivalent to the data channel full scale).

3.6 Characteristic Frequencies, Fh, F1, Fn: These frequencies are defined in Fig. 1.

3.7 Channel Frequency Class, CFC: The channel frequency class is designated by a number indicating that the channel frequency response lies within limits specified by Fig. 1.

This number and the value of the frequency Fh in Hz are numerically equal.

3.8 Calibration Value: The value measured and read during the calibration of a data channel (see 4.6).

3.9 Sensitivity Coefficient: The slope of the straight line representing the best fit to the calibration values determined by the method of least squares within the channel amplitude class.

3.10 Calibration Factor of a Data Channel: The arithmetic mean of the sensitivity coefficients evaluated over frequencies which are evenly spaced on a logarithmic scale between Fl and Fh/2.5.

3.11 Linearity Error: The ratio, in percent, of the maximum difference between the calibration value and the corresponding value read on the straight line defined in 3.9 at the upper limit of the channel amplitude class (data channel full scale).

3.12 Sensitivity: The ratio of the output signal (in equivalent physical units) to the input signal (physical excitation), when an excitation is applied to the transducer. (Example: 10.24 mV/G/V for a strain gage accelerometer.)

3.13 Phase Delay Time: The phase delay time of a data channel is equal to the phase delay (in radians) of a sinusoidal signal, divided by the angular frequency of that signal (in radians per second).

3.14 Environment: The aggregate, at a given moment, of all external conditions and influences to which the data channel is subjected.

3.15 Transverse Sensitivity (of a rectilinear transducer): The sensitivity to excitation in a nominal direction perpendicular to its sensitive axis.

Note: The transverse sensitivity is usually a function of the nominal direction of the axis chosen.

3.16 Transverse Sensitivity Ratio (of a rectilinear transducer): The ratio of the transverse sensitivity to its sensitivity along its sensitive axis.

4. DATA CHANNEL PERFORMANCE REQUIREMENTS:

4.1 Linearity Error: The absolute value of the linearity error of a data channel at any frequency in the CFC, shall be less than or equal to 2.5% of the value of the CAC, through the whole measurement range. In general a sufficient number of measurements should be carried out in order to ensure the linearity in the range of interest, that is, between Fl and Fh. For a transducer, linearity at DC is sufficient.

4.2 Amplitude Against Frequency: The frequency response of a data channel shall lie within the limiting curves given in Fig. 1. The zero dB line is defined by the calibration factor.

4.3 Phase Delay Time: The phase delay time between the input and the output of a data channel shall be determined, and shall not vary more than 1/(lO*Fh) seconds between 0.03*Fh and Fh. This includes the transducer, that is the input is the excitation to the transducer.

4.4 Time:

4.4.1 Time Base: A time base shall give at least 1/100 second resolution with an error of less than 1/10000 seconds.

4.4.2 Relative Time Delay: The relative time delay between the signals of two or more data channels regardless of their frequency class, must not exceed 1 millisecond excluding phase delay caused by phase shift. Two or more data channels of which the signals are combined shall have the same frequency class and shall not have a relative time delay greater than l/(10*Fh) seconds.

This requirement applies to analog-signals as well as synchronization pulses and digital signals.

4.5 Transducer Transverse Sensitivity Ratio: The transverse sensitivity ratio of all transducers shall be less than 5% in any direction.

4.6 Calibration: Values in this section apply to reference equipment or "standards" against which a data channel is "calibrated", that is, its performance is determined.

4.6.1 General: A data channel shall be calibrated at least once a year against reference equipment traceable to known standards. The methods used to carry out a comparison with reference equipment shall not introduce an error greater than 1% of the CAC. The use of the reference equipment is limited to the range of frequencies for which they have been calibrated.

Subsystems of a data channel may be evaluated individually and the results factored into the accuracy of the total data channel. This can be accomplished for example by an electrical signal of known amplitude simulating the output signal of the transducer which allows a check to be made on the gain of the data channel, except the transducer.

4.6.2 Accuracy of Reference Equipment for Calibration: The accuracy of the reference equipment shall be certified or endorsed by an approved metrology service (for example, traceable to the National Bureau of Standards).

4.6.2.1 Static Calibration:

4.6.2.1.1 Accelerations: The error shall be less than 1.5% of the channel amplitude class.

4.6.2.1.2 Forces: The error shall be less than 1% of the channel amplitude class.

4.6.2.1.3 Displacements: The error shall be less than 1% of the channel amplitude class.

4.6.2.2 Dynamic Calibration:

4.6.2.2.1 Accelerations: The error in the reference accelerations expressed as a percentage of the channel amplitude class shall be less than 1.5% below 400 Hz, less than 2% between 400 and 900 Hz, and less than 2.5% between 900 Hz and the maximum frequency at which the reference acceleration is utilized (see 4.6.4).

4.6.2.2.2 Forces and Displacements: (No method for the evaluation of the dynamic response during calibration of data channels for forces and displacements is included since no satisfactory method is known at present.)

4.6.2.3 Time: The error in the reference time shall be less than 0.00001 seconds.

4.6.3 Sensitivity Coefficient and Linearity Error: The sensitivity coefficient and the linearity error shall be determined by measuring the output signal of the data channel against a known input signal, for various values of this signal. (The input signal is referenced to well known physical data, that is, a load or acceleration, but not voltage.)

The calibration of the data channel shall cover the whole range of the amplitude class. (This is between Fl and Fh/2.5.)

For bi-directional channels, both the positive and negative values shall be evaluated.

If the calibration equipment cannot produce the required input, due to excessively high values of the quantity to be measured, calibrations shall be carried out within the limits of these calibration standards and these limits shall be recorded in the report.

A total data channel shall be calibrated at a frequency or at a spectrum of frequencies with its significant values comprised between Fl and Fh/2.5.

4.6.4 Calibration of the Frequency Response: The response curves of phase and amplitude against frequency for the data channel shall be determined by measuring the output signals of the data channel in terms of phase and amplitude against a known input signal, for various values of this signal varying between Fl and 10 times the CFC or 3000 Hz whichever is the lower value.

4.7 Environmental Effects: The presence of any environmental effects shall be checked (that is, electric or magnetic flux, cable velocity, etc.). This can be done for instance by recording the output of spare channels equipped with dummy transducers.

If an output signal is greater than 2% of the expected data peak value, corrective action shall be taken, for instance re-allocation or replacement of cables.

4.8 Choice and Designation of the Data Channel: The CAC and CFC define a data channel.

A data channel consistent with the specifications of this recommended practice shall be designated according to the following codes:

SAE J211 XXXXX - (number of this recommended practice)
CAC ... - (channel amplitude class)
CFC ... - (channel frequency class)

If the calibration of the amplitude or frequency response does not cover the complete CAC or CFC owing to limited properties of the calibration equipment, then the CAC or CFC shall be marked with an asterisk.

Example, SAE J211 XXXX
CAC* 200 m/s^2
CFC 1000 Hz

means that:

-this measurement has been carried out according to this recommended practice:

-the channel amplitude class was 200 m/s^2;

-the channel frequency class was 1000 Hz;

-the calibration of the amplitude response did not cover the complete CAC.

The test report shall indicate the calibration limits.

5. DATA CHANNEL SELECTION:

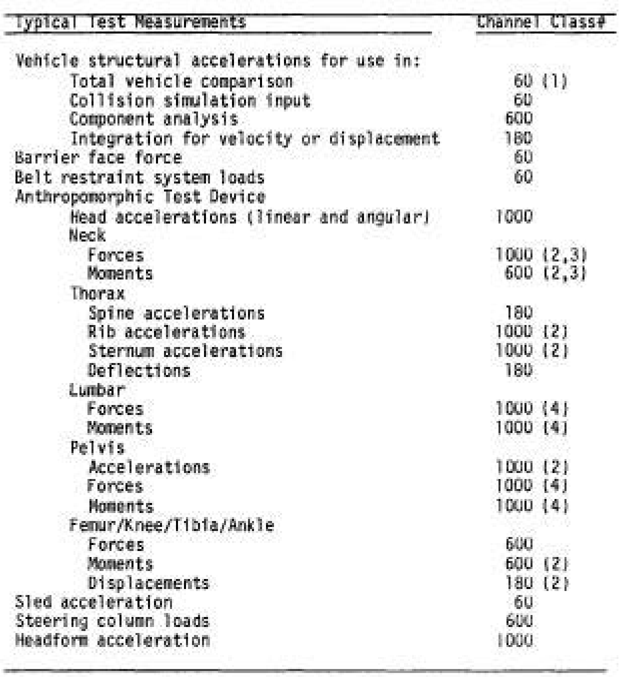
The selection of a frequency response class is dependent upon many considerations, some of which may be unique to a particular test. The ultimate usage of the data and good engineering judgment will determine what portions of the frequency spectrum are significant or useful. The various classes of frequency response in Fig. 1 are intended to permit appropriate choices for different engineering requirements.

It is important to note that valid comparisons using different frequency response classes may be difficult to make. It is useful to establish specific frequency response classes when comparing test results from different sources. The frequency response classes in Table 1 are recommended for that purpose. These recommendations reflect current practices and equipment. However, it is recognized that other considerations (for example, biomechanics) may impose special instrumentation requirements.

5. DATA CHANNEL SELECTION: (continued)

The channel class recommendations for a particular application should not be considered to imply that all the frequencies passed by that channel are significant for the application. In several cases, such as occupant head accelerations, headform accelerations, and femur force, the recommendation may be higher than necessary, but current biomechanical knowledge will not permit a closer specification. All data is to be gathered at class IOO0 or higher, for any purpose.

TABLE 1 - Frequency Response Classes



Typical Test Measurements	Channel Class#
Vehicle structural accelerations for use in:	
Total vehicle comparison	60 (1)
Collision simulation input	60
Component analysis	600
Integration for velocity or displacement	180
Barrier face force	60
Belt restraint system loads	60
Anthropomorphic Test Device	
Head accelerations (linear and angular)	1000
Neck	
Forces	1000 (2,3)
Moments	600 (2,3)
Thorax	
Spine accelerations	180
Rib accelerations	1000 (2)
Sternum accelerations	1000 (2)
Deflections	180
Lumbar	
Forces	1000 (4)
Moments	1000 (4)
Pelvis	
Accelerations	1000 (2)
Forces	1000 (4)
Moments	1000 (4)
Femur/Knee/Tibia/Ankle	
Forces	600
Moments	600 (2)
Displacements	180 (2)
Sled acceleration	60
Steering column loads	600
Headform acceleration	1000

5. DATA CHANNEL SELECTION: (continued)

#Filtering can cause appreciable time lag (for example, approximately 2.5 ms with class 60 channel). These effects should be considered when comparing film and electronic data, or when performing integration.

1. When overall acceleration of the frame or body in a given direction is desired and a higher frequency response class is used, readability of the data may be improved by averaging outputs of two or more transducers at different locations prior to recording of the output.

2. References are listed in Section A.4.

3. These classifications are needed to calculate head impact forces based n neck forces and head accelerations when using the Hybrid III crash test dummy.

4. No rationale for this classification. By default, class 1000 was chosen.

6. MOUNTING OF TRANSDUCERS:

Mechanical resonances associated with transducer mounting should not distort readout data.

Transducers should be mounted on dummies using a support specially provided for this purpose. In cases where properties of non-mechanical test subjects preclude rigid transducer mounting, an analytical or experimental evaluation of mounting effects on the data should be provided.

Acceleration transducers, in particular, should be mounted in such a way that the initial angle of the actual measurement axis to the corresponding axis of the reference axis system is not greater than 5 deg unless analytical or experimental assessment of the effect of the mounting on the collected data is made. When multi-axial accelerations at a point are to be measured, each acceleration transducer axis should pass within 10 mm of that point, and the center of seismic mass of each accelerometer should be within 30 mm of that point.

7. SIGN CONVENTION:

A standardized coordinate system and axes designation for major body segments of dummies or other test surrogates as well as for test vehicles and sleds is needed throughout the crash testing community. The results of computer modeling and biomechanical research have been hard to interrelate with test results due to different coordinate and sign conventions.

The sign convention for dummies and other test surrogates should be consistent with Fig. 2 for applicable instrumentation. Table 2 provides a method for determining positive output from dummy transducers when a force is applied to its various body segments.

TABLE 2

The directions are defined in relation to a seated dummy

Body Segment - Measured Force	Positive Output Direction
Neck - FX shear	Chest forward or Head rearward
FY shear	Chest right or Head left
FZ axial	Chest down or Head up
MX moment (roll)	Left shoulder to left ear
MY moment (pitch)	Chest to chin
MZ moment (yaw)	Left shoulder to chin
Femur - FX shear	Knee up
FY shear	Knee right
FZ axial	Knee forward (tension)
MX moment (roll)	Knee left
MY moment (pitch)	Knee up
MZ moment (yaw)	Knee rotated CCW when facing front of dummy
Knee clevis - FZ axial	Tibia down (tension)
Upper tibia - MX moment	Ankle left
MY moment	Ankle forward
Lower tibia - FX shear	Ankle forward
FY shear	Ankle right
FZ axial	Ankle down (tension)
MX moment	Ankle left
MY moment	Ankle forward
Chest displacement	Chest expansion (1)
Knee shear displacement	Pull tibia from femur (2)

(1) Chest is normally compressed for negative output.

(1) Chest is normally compressed for negative output.

(2) Push on tibia for negative output.

For vehicles and sleds the positive X-axis should be in the normal forward motion direction of the vehicle or sled. The positive Z-axis should be vertically downward (+G) and, following the right hand rule, the positive Y-axis should be towards the right.

Inertial coordinate systems should be chosen with positive Z-axis vertically downward (+G) and the positive Y-axis toward the right from the chosen X-axis. The directional reference of the X-axis should be reported.

8. RECORDING:

8.1 Analogue Magnetic Recorder: Tape speed should be stable to within less than 0.5% of the tape speed used. The signal-to-noise ratio of the recorder should not be less than 42 dB at the maximum tape speed.

The total harmonic distortion should be less than 3% and the linearity error should be less than 1% of the measurement range.

It is suggested that standard tape speeds be used: 15/16, 1 7/8, 3 3/4, 1.5, 15, 30, 60, 120 ips. Conformance to Inter-Range Instrumentation Group (IRIG) specifications is desirable.

8.2 Digital Magnetic Recorder: Tape speed should be stable to within less than 10% of the tape speed used. It is suggested that standard tape speeds be used: 75 or 100 ips. Tapes should be 0.5 in wide and recorded at standard 800, 1600 or 6250 bpi on 7 or 9 track formats.

8.3 Paper Tape Recorder: In the case of direct data recording, the paper speed in millimeters per second should be at least 1.5 times the number expressing Fh in Hz.

In other cases, the paper speed should be such that an equivalent resolution should be obtained.

9.. DIGITAL DATA PROCESSING:

This section establishes guidelines for digital data processing equipment used by crash testing agencies.

9.1 Presample Filtering: Filtering corresponding to the frequencies of the data channel class may be carried out during processing of data.

However, before recording, analog filtering at a higher level than CFC-1O00 should take place in order to use at least 50% of the dynamic range of the recorder and to reduce the risk of high frequencies saturating the recorder. Since crash test data may have high-frequency components above the channel class Fh, presample filtering should be used to keep these components from causing aliasing errors in the sampling process. The user is cautioned to examine the unfiltered data for signal overloads, since the filtering process can mask certain overload conditions. Since Class 1000 data is generally the highest frequency data required in crash testing, many laboratories set the presample filter to Class 1000 and use digital filtering for lower classes. Digital filtering should only be done once per data channel; that is, do not digital filter a digitally filtered signal.

The maximum error induced by aliasing at the Fh frequency shall not exceed 0.1% of the CAC.

9.2 Sampling Rate: The minimum acceptable sampling rate is a function of many variables, particularly sophistication of the reconstruction method used in the processing software. For those installations utilizing only simple reconstruction software, the sample rate should be a minimum of eight times the Fh. In installations with Class lO00 presample filters, this corresponds to a minimum sampling rate of approximately 8000 samples per second per channel. In the case of analog recording, when the recording and playback speeds are different, the sampling frequency can be divided by the speed ratio.

9.3 Resolution: Digital word lengths of at least 10 bits (including sign) should be used to be assured of reasonable accuracy in processing. In those systems in which the dynamic range of the data is less than 5U% of the A/D converter full-scale, a higher resolution may be required. The least significant bit should correspond to approximately 0.2% of the CAC.

9.4 Data Processing: Processing software is typically used to scale and filter data, determine zero levels, perform mathematical operations and prepare data plot formats.

9.4.1 Digital Filtering: Filtering may be either phase-shifting or phaseless. Phase-shifting filters will cause time offsets and phaseless filters will cause time uncertainty; either of which will cause problems in comparing data to film, and comparing data to data if the class filters are different. Filtering should precede all non-linear operations, such as calculation of resultant vectors and injury indices. Any filtering algorithm can be used as long as the results conform to the data channel performance requirements as given in Section 4. The type of digital filter used should be reported.

9.4.2 Scaling and Zeroing: Software should be used to determine zero levels and calibration factors rather than relying on set gains and expecting no zero drift. Zero offset errors in orthogonal components cause comparable errors in resultant computation that are often difficult to detect.

9.4.3 Injury Index Calculations: Injury index calculations should use all sampled data points. Head Injury Criterion calculations should use all data points for the integration. However, the maximizing time intervals need be no more precise than 1 ms.

10. TIMING MARKS:

Timing marks are essential in data analysis and correlation of high-speed film to other data channels. Timing frequency error should be less than 1% of the chosen or designated frequency. Timing synchronization should be within +/-1 ms.

11. TIME OF CONTACT:

Time of initial contact (real or simulated) should be known within +/- 1 ms and can be accomplished by recording a switch actuated by the impact or by observing the instant the test acceleration exceeds a predetermined value (for example, 0.5 G). It should also be recorded in film data through strobe lights or timing mark channels.

12. PRESENTATION OF RESULTS:

In reporting results of tests, the following information should be provided with data tabulations, time history traces, etc.:

(a) The data channel designations.
(b) Description of designated reference points and locations of vehicle accelerations.
(c) Transducer mounting analysis, if required by Section 6.
(d) Type of digital filter used.
(e) Method of combining sub-systems for calibration.
(f) Inertial coordinate system definition.

12. PRESENTATION OF RESULTS: (continued)

The results should be presented on A4 (210 x 297 mm) size paper (ISO 216) or 8.5 x 11 in paper. Results presented as diagrams should have axes scaled with one measurement unit corresponding to a suitable multiple of the chosen unit (for example 1, 2, *5,* 10, 20 millimeters). SI units shall be used, except for vehicle velocity where km/h may be used and for accelerations due to an impact where G may be used (with G = 9.81 m/s/s).

ANNEX TO PART 1 - SPECIFIC MEASUREMENTS

A.1 Impact Velocity: This can be calculated by measuring the time required to traverse a known distance prior to impact. Determination of impact velocity should be with an error of less than 1% of the actual velocity.

A.2 Test Specimen Crush:

A.2.1 Residual Crush: Residual crush is specified by one or more single-valued data points, with respect to a designated reference points. Determination of residual crush should be with an error of less than 5% of the actual crush.

A.2.2 Dynamic Crush: Maximum dynamic crush is a measurement of the maximum deformation of the test specimen during the impact. This is also measured with respect to one or more designated reference points. Contingent on the size of the specimen and the magnitude of the expected dynamic crush, the following are possible measurement methods:

(a) High-speed motion picture photography.
(b) Double integration of acceleration data.
(c) Use of a specific displacement transducer.

The error should be less than 5% of the actual crush.

A.3 Steering Column Displacement: Displacements relative to designated reference points on the vehicle can be measured by various techniques. The coordinate system in which displacement is measured should be indicated. Determination of steering column displacement should be accurate to +/-0.5 in (+/-1.27 cm).

A.4 References: Information pertaining to the justification of some of the specified filter frequency classes is contained in the minutes of the SAE Safety Test Instrumentation Subcommittee meetings held on April 16, September 24, and November 21, 1986.

PART 2 - PHOTOGRAPHIC INSTRUMENTATION

1. SCOPE:

The purpose of this recommended practice is to define criteria of performance for an optical data channel when numerical time and space data are taken from the images to analyze impact test results.

The requirements are to facilitate comparison between results obtained by different laboratories.

2. DEFINITIONS:

2.1 Optical Data Channel: A system composed of an image taking device (for example, camera and lenses), a recording medium for these images (film, disc, magnetic tape . . .), an optical path (for example, fiber optic cable) and a system for analyzing the images including any analysis procedure that may modify the content of the data.

2.2 Distortion Index: The distortion index is a quality parameter of the optical data channel.

2.3 Analysis System: An analysis system is composed of a system for measuring and collecting the coordinates of image points as a function of time.

2.4 Time Base System: A device to enable the determination of the time interval elapsed between any two recorded events.

2.5 Time Origin Identification Device: A device for identifying the instant chosen as the time origin, usually the beginning of the impact.

2.6 Imaging Rate: This is the frequency of renewal of information for a given point expressed in renewals per second or in images per second when all the points of the image are renewed simultaneously.

3. PERFORMANCES:

The performances of the optical data channel should be evaluated initially to establish performance levels and repeated whenever the system is modified to an extent which could cause a change in accuracy.

The following measures of performance may be evaluated as detailed in the annex or in an equivalent manner.

3.1 Optical Performance: The optical performance is determined by analyzing the variation of apparent length of 40 diameters according to the annex.

3.1.1 Distortion Index: The distortion index is assessed by using the photographic target and procedure described in the annex. It shall be evaluated as detailed in the annex and shall not exceed 1%.

3.2 Time Base: A time base is required. It shall permit the determination of the time between recorded events with an error of less than the reciprocal of the imaging rate or 1% of the actual time, whichever is greater

3.3 Time Origin Identificaton Device: The accuracy of the device shall be equal to the value expressed in s of the inverse of the imaging rate.

3.4 Imaging Rate: The imaging rate shall be left to the user's initiative, taking account of the following main three factors:

- the goal to be attained
- the limitations due to the equipment, (blur, etc...)
- the need to combine data from several image taking devices and rom electronic recordings of the impact test.

The user's choice shall be guided by the distance the recorded point moves between two analyzed images. This distance should not exceed the accuracy required for its positional determination.

3.5 Length Reference: A length calibration shall be performed which permits the determination of lengths within any requirements there may be on their accuracies.

In lieu of such requirements it is recommended to be able to perform length determinations with an error of less than 1% of the diagonal of the picture.

ANNEX TO PART 2

Recommended procedure for determining the optical performance of an optical data channel.

A.1 Test Equipment: A rectangular target having dimensions in conformity with Fig. 3 shall be used.

This target shall be divided into four parts by joining the center points of two opposite sides. In each quadrant thus obtained and at the target center, circles 400 +/- 1 mm in diameter shall be drawn.

The five circles shall be marked on their circumference with sixteen equally spaced
reference marks, thus defining eight diameters for each, and hence forty diameters for the whole target.

A.2 Test Procedure: The target shall be exposed "full frame" by the camera forming one of the elements of the data channel to be tested.

The film or tape to be used for calibration shall be of the same type and quality as that used for impact testing. It shall be analyzed using the analysis system.

The coordinates of the marks of a diameter, d_{ij}, must be measured on the same frame while the coordinates of the marks of different diameters must be measured on consecutive analyzed images.

For each circle, j, j=l to 5, label the 16 marks, i, i=l to 16, consecutively around a circle. Then the forty diameters (d_{ij}) shall be measured as

$$d_{ij} = [(x_{ij} - x_{i+8,j})^{2} + (y_{ij} - y_{i+8,j})^{2}]^{1/2},$$

where i=1,2...8 and j=1,2...5.

The mean diameter (D) and the standard deviation (S) shall be calculated taking the 40 diameter values (d_{ij}) into account:

$$D = \text{sum}^{8}_{i=1} (\text{sum}^{5=}_{j\ =1dij}/N) \qquad \text{with } N=40,$$

$$S = [\text{sum}^{8}_{i=1} (\text{sum}^{5}_{j=1} ((d_{ij} - D)^{2})/(N-1))]^{1/2}$$

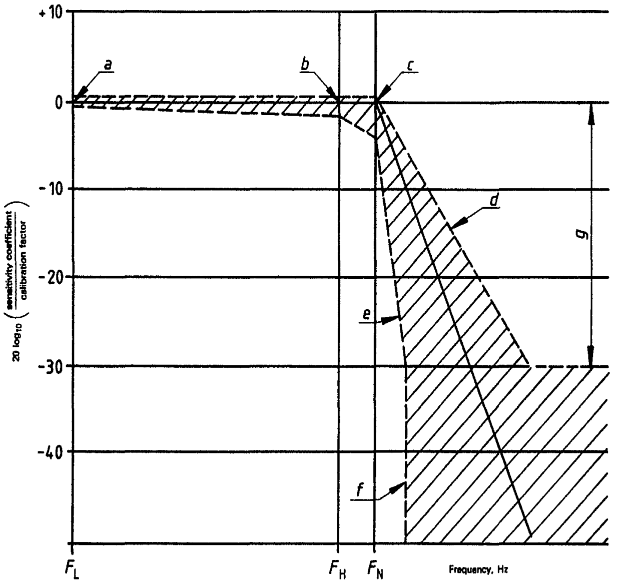
The distortion index is equal to the ratio (S/D) of the standard deviation (S) to the mean diameter (D).

A.3 Alternate Procedure: Other targets with different sizes or patterns of reference marks may be used, in which case the user has to determine the distortion index in an indirect way.

There is then also a need to show that the indirect method gives a result equivalent to the use of the pattern specified in this annex.

This may be of advantage when for instance a target with a rectangular pattern of reference marks is employed for the determination of lens corrections, which are needed in case of some wide angle lenses.

The phi (∅) symbol is for the convenience of the user in locating areas where technical revisions have been made to the previous issue of the report. If the symbol is next to the report title, it indicates a complete revision of the report.

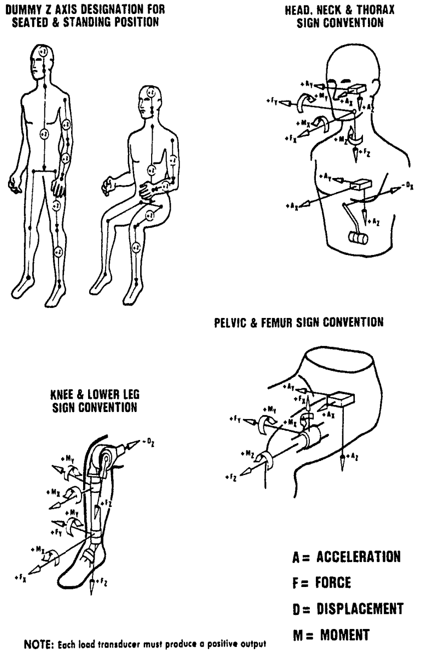


CFC	F_L Hz	F_H Hz	F_N Hz
1 000	< 0,1	1 000	1 650
600	< 0,1	600	1 000
180	< 0,1	180	300
60	< 0,1	60	100

Logarithmic scale

a	± 0,5 dB
b	+ 0,5; − 1 dB
c	+ 0,5; − 4 dB
d	− 9 dB/octave
e	− 24 dB/octave
f	∞
g	− 30 dB

FIGURE 1 - Data Channel Dynamic Accuracy


DUMMY Z AXIS DESIGNATION FOR SEATED & STANDING POSITION
HEAD, NECK & THORAX SIGN CONVENTION
+A_Y
+M_Y
+A_X
+A_Z
+F_Y
+M_X
+F_X
+M_Z
+F_Z
-D_X
PELVIC & FEMUR SIGN CONVENTION
KNEE & LOWER LEG SIGN CONVENTION
-D_X
A= ACCELERATION
F= FORCE
D= DISPLACEMENT
M= MOMENT

NOTE: Each load transducer must produce a positive output when force is applied to structure below it, in the direction of arrow point.

FIGURE 2 - Hybrid III Positive Sign Convention

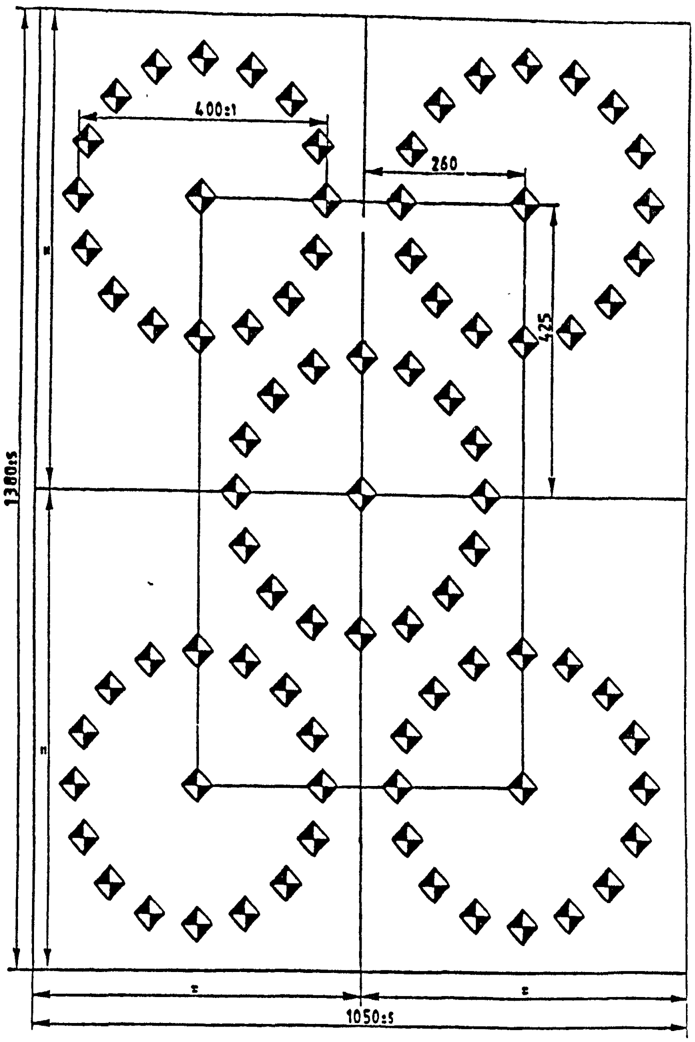


FIGURE 3

J211 OCT 88

RATIONALE:

Not applicable.

RELATIONSHIP OF SAE STANDARD TO ISO STANDARD:

Not applicable.

REFERENCE SECTION:

Not applicable.

APPLICATION:

The purpose of this SAE Recommended Practice is to provide guidelines and recommendations for the techniques of measurement used in impact tests. The aim is to achieve uniformity in instrumentation practice and n reporting test results. Use of this recommended practice will provide a basis for meaningful comparisons of test results from different sources.

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APPENDIX D

ANALYTICAL AND EXPERIMENTAL TOOLS

Design, synthesis, and development of a new safety feature is not a straightforward procedure but is an iterative process requiring tradeoffs among sometimes conflicting safety performance requirements, environmental considerations, and costs. In this appendix, analytical and experimental tools (excluding full-scale crash tests) that are typically used in developing and evaluating new safety features are presented and discussed. Also, application and limitations of these techniques are presented.

D1 Useful Techniques

D1.1 Structural Design

The primary design objective of longitudinal barriers and crash cushions is the safety performance with errant vehicles, but there are other design considerations pertaining to economy, environment, maintenance, and operational needs.

On the other hand, safety performance during vehicle collision is not the reason for installing supports for signs and luminaires. If lighting or information for the motorist is not needed, the installation should not be located on the roadside. Hence, the primary design objective is to support the luminaire or sign blank for environmental loading, and safety performance is a secondary objective. Fortunately, engineers have been able to develop breakaway and yielding supports that will satisfy both the primary and secondary objectives.

Structural loading and design procedures are contained in civil engineering textbooks, AASHTO design manuals, and research publications. Primary references are given in Table D-l for each type of safety feature. The designer/developer should consult these references as a first step along with the appropriate safety feature manufacturer. These references will aid the developer in estimating design loads and proportioning a new design for subsequent evaluation steps.

1.2 Static Tests

During an early stage of development, certain critical details and connections of a safety feature may require strength/deflection evaluation. Safety features are often designed to function at or near ultimate capacity of the materials. Since the materials will be loaded well beyond typical design ranges such as elastic limits, these tests are often very specialized and do not conform to standard tests suggested by ASTM.

Most static tests will have one of the following objectives:

* Demonstrate safety feature performance under simulated environmental loading.
* Evaluate ultimate strength of critical connections.
* Develop load/deflection properties for subsequent computer model simulations.

TABLE D-1. SOURCES FOR SAFETY FEATURE INFORMATION

Feature	Principal Reference
I. Longitudinal Barriers	
A. Bridge Rails	17, 27, 28, 29, 30, 31, 32
B. Guardrails	27, 30, 31, 33, 34, 35, 36
C. Median Barriers	7, 27, 30, 31, 33, 34, 37
II. Crash Cushions and Terminals	27, 31, 37
III. Breakaway or Yielding Supports	
A. Luminaire	39, 40, 41, 42
B. Sign Supports	40, 43
C. Utility Poles	44, 45
IV. Truck Mounted Attenuators	46, 47, 48
V. Roadside Geometric Features	4, 5, 6, 7, 8, 18, 19, 27, 35

Static testing is often used to compare the performance of competing design details. When such tests are used to evaluate safety feature components that must perform under dynamic loading, developers should be aware of the many problems that can arise from material load rate sensitivity. A primary concern in the design of many roadside safety features is the energy absorbed as a component fails. Static testing generally allows a component to fail at the lowest possible load. However, the lowest failure load often will not correspond to the lowest energy failure mode. For example, wood posts embedded in soil seldom fracture under static loading and energy dissipation is usually high as the post rotates in the soil. Under dynamic loading, soils can generate much higher loadings and a wood post can fracture prematurely with little energy dissipation.

In general it is anticipated that quality control of materials used in the operational system may vary extensively. Where possible, safety performance behavior should not rely heavily on material properties that cannot be carefully controlled. For instance, energy to fragment a frangible transformer base can vary more than 100 percent with even minor changes in heat treatment of the aluminum alloy. Soil conditions can exhibit even wider seasonal variations as a soil goes through saturated, dry, and frozen situations. In contrast, tensile strength and elastic modulus of wire rope vary within a narrow performance band.

Even at this stage the developer should be aware of value engineering by avoiding overspecifying materials, especially components that are noncritical. Moreover, he should ensure that materials used in the prototypes are typical and routinely acquired materials and not "Sunday samples." Where possible, the developer should use standard hardware elements for initial economy and to minimize costs associated with inventory maintenance (31).

D1.3 Computer Simulations

A number of computer programs have been developed that simulate vehicle dynamics and kinematics during interactions with highway safety features. Also, several models have been developed to simulate occupant dynamics during impact. These models vary in complexity of mathematical analog, type of safety feature investigated, and class of vehicle. Simulation results are sensitive to vehicle parameters that are sometimes difficult to obtain or approximate. Vehicle moments of inertia, crush properties, and suspension stiffness properties are generally not published and must be determined experimentally. Moreover, barriers often exhibit large deflections that are difficult to predict because of uncontrolled features such as joint slack, soil variation, and unstable structural behavior.

Most of the available simulation programs have been correlated to some degree with crash tests. For the validated cases, simulation results can be most helpful to the safety feature designer by providing sometimes unique insight into the collision event. Where the program has been validated for multiple impact conditions, it can be used with some confidence in investigating conditions that are bracketed by the validated conditions. Although investigation of impact conditions outside the validated range can provide some insight, the engineer should use care in these extrapolations and view the findings with caution. The most important simulation programs are given in Table D-2 and are described below.

TABLE D-2. SUMMARY OF HIGHWAY SAFETY COMPUTER PROGRAMS

Name	Developer/Date of Last Mod.	Principal Application	Model	Validation	Documentation Availability	Comment
HVOSM	CALSPAN/1989	Simple vehicle/rigid directive barrier	3D Lump Mass Vehicle	Extensive	FHWA (R & D)	Excellent wheel/suspension system analog; simplified vehicle body crush analog
BARRIER VII	Univ. of Cal./1973	Simple vehicle/flexible redirective barrier	2D Finite Element Model (FEM)	Extensive	FHWA (R & D	For cases where roll and pitch of vehicle are negligible.
GUARD	HTRI/1989	Simple vehicle with detailed bumper modeling capability; flexible or rigid barrier systems	3D FEM Vehicle and Barrier Model	Limited	FHWA (R & D)	Developed in part from CRUNCH program--simplistic wheel/suspension model
NARD	HTRI/1989	Articulated vehicle/flexible or rigid barrier capabilities	3D FEM	Limited	FHWA (R & D)	Similar to GUARD program but with articulated vehicle
SMAC	CALSPAN/1975	Rigid body motion--2D vehicle (no barrier)	Vehicle	Limited	NHTSA	Model used for accident reconstruction

HVOSM - The Highway Vehicle Object Simulation Program (HVOSM) (6,7) is a very sophisticated and widely used vehicle handling model. HVOSM incorporates an 11 degree of freedom (DOF) vehicle model with relatively sophisticated suspension and tire models. This program has been extensively validated against a wide variety of crash tests involving many different terrain configurations. The program has demonstrated validity for modeling vehicle traversals of ditches, driveways, and a variety of roadside slopes. HVOSM is especially suited for evaluation of roadway and roadside geometrics where vehicle stability is a primary concern.

This program has also been successfully used for simulating rigid barrier impacts (7). This version of the program incorporates a relatively sophisticated sheet metal crush model and barrier/vehicle interaction routines. However, the program continues to be limited by its thin disk tire model and an inability to adequately predict or model suspension and wheel damage. Nonetheless, HVOSM is the best available model for simulating rigid barrier impacts.

Barrier VII - Barrier VII program *(49)* is a widely used model for simulating impacts with flexible barriers. This program incorporates a beam and column finite element model (FEM) of the barrier and a two-dimensional vehicle model. The FEM code incorporates both geometric and material nonlinearities as well as a number of specialized barrier elements including nonlinear springs and dashpots. Although the vehicle model incorporates relatively simple bilinear spring elements and is limited to three degrees of freedom (DOF), this program has been successfully validated for a wide variety of flexible barriers and a number of different vehicles. The primary limitation of this program is that it cannot be used to predict vehicle stability. However, the program is especially suited for use as a tool for barrier design in predicting maximum loads on and strains in barrier components. Further, the program has proven useful for identification of critical impact locations as well as predicting vehicle snagging and pocketing.

GUARD - The GUARD program (50) utilizes a three dimensional finite element barrier model and a 6 DOF vehicle model. The finite element barrier model incorporates both translational and rotational degrees of freedom for all elements. Although this feature makes the program versatile and should improve accuracy for many structures, the low rotational stiffnesses of many barrier elements, such as W-beam guardrail, tend to destabilize the solution routine. As a result, this program is very difficult to use and has not been adequately validated for very many barriers or impact conditions. The program is also severely handicapped by the limited number of barrier nodes and elements that can be accommodated. GUARD's sheet metal crush model incorporates deformable panels that interact with predescribed lines on the barrier surface. Special care must be taken when describing panel surfaces to make sure that panel contact lines are properly positioned relative to barrier contact lines. Further, GUARD's suspension and tire models are relatively crude and severely limit the program's usefulness for modeling safety-shaped barrier impacts.

GUARD has yet to be adequately validated for any general class of simulations. Although the program is ostensibly capable of simulating impacts with virtually any barrier, it is best suited for relatively strong beam barriers where torsional stiffnesses are high. This program is not recommended for the inexperienced user and even experienced users should view simulation findings with extreme caution.

NARD - The Numerical Analysis for Roadside Design (NARD) program (51) has evolved from **GUARD**. The program is designed to be both a handling and impact model and can simulate a variety of vehicles including passenger cars, single unit trucks, and combination trucks with as many as three trailers. The program is also capable of simulating impacts with a variety of roadside objects including breakaway devices and crash cushions as well as longitudinal barriers. Unfortunately, **NARD** incorporates the same basic program structure as the GUARD program from which it evolved. As a result, NARD suffers from virtually all of the same problems as **GUARD**. Nard does incorporate a relatively sophisticated suspension and tire model similar to **HVOSM** that should allow it to simulate safety-shaped barrier impacts.

Like **GUARD**, the **NARD** program is both unvalidated and very difficult to use. After proper validation, the program should offer a reasonably accurate handling model for automobiles and trucks. However, an extensive effort is required before the program can be considered to be validated for barrier impacts. Users are cautioned against using NARD without proper validation.

SMAC - The Simulation Model for Automobile Collisions (**SMAC**) (52) is a two dimensional program for modeling two car collisions. The vehicles are modeled as uniformly crushable blocks with linear stiffness coefficients. Although this program was designed as a tool for reconstructing traffic accidents, the program has proven to be quite useful for analyzing impacts with crash cushions and barrier end treatments. For this purpose a roadside object can be modeled as one of the two vehicles. The program has been shown to be capable of predicting the extent of vehicle crush and planar rotations during impacts with guardrail terminals and roadside poles. The gross vehicle model incorporated by the program and problems associated with modeling roadside objects as a vehicle are significant limitations of this program. Nonetheless, SMAC can be a useful design tool that merits further consideration.

D1.4 Laboratory Dynamic Tests

In addition to full-scale crash tests procedures presented elsewhere in this report, there are four types of dynamic test methods to evaluate and study safety features: gravitational pendulum, drop mass/dynamic test device, scale model, bogie vehicle.

D1.5 Gravitational Pendulum

The facility is characterized by a striking mass that swings in a circular arc suspended by cables or by rigid arms from a main frame. The specimen is generally mounted in an upright manner. Mass velocity at impact is governed by the formula

$$V_i = \sqrt{2gh}$$

subject to minor corrections due to friction and aerodynamic losses, where h is the drop height of the mass. As an example, for an impact speed of 35 km/h (9.7 m/s), a drop height of 4.8 m is required. The swing radius is usually considerably larger than the drop height. Gravitational pendulums are commonly used to evaluate performance at impact speeds of approximately 40 km/h or less. A gravitational pendulum capable of high speed impacts would be very large and unwieldy.

A primary problem associated with this type of testing is the type of impact surface or crushable nose used on the pendulum. A rigid nose greatly increases the impact forces applied to the pendulum while reducing the energy dissipation during the test. An excessively soft nose will minimize impact forces and maximize energy dissipation associated with the tested feature. Although simulated soft noses have been developed for subcompact and minisize vehicles (2,30), these devices were developed to simulate vehicle models that are well over 10 years old. Nose assembly systems must be revalidated periodically if pendulum testing is to yield accurate predictions of safety feature performance during full-scale testing.

Pendulum testing is frequently used to evaluate the performance of breakaway structures such as luminaire and sign supports. Such systems often absorb more impact energy during low speed crashes than during high speed impacts. As a result, pendulums are an inexpensive method for evaluating the low-speed performance of prototype design alternatives. Some breakaway systems have been placed into service based solely on pendulum testing. As discussed in Section 2.4.1.2, the acceptance of safety features based on such testing is left to the discretion of the user agency.

Pendulums can also be used for dynamic testing of various safety feature components. For example, pendulums are often used for dynamic testing of barrier posts embedded in soil and crash cushion attenuator elements. This type of testing is not sensitive to the design of the pendulum's crushable nose and can yield valuable information with a rigid impact surface.

D1.6 Drop Mass/Dynamic Test Device

These facilities generally involve a rigid striking mass or plate that strikes a test specimen at prescribed velocities. Drop mass devices can be used to test large components and assemblies under low-speed dynamic conditions. Dynamic test devices are not limited to low test speeds, but specimen sizes are generally very limited. As a result, these devices are limited to tests of scale models or relatively small components of a safety feature. Although these test methods have proven to be quite valuable, developers should be aware of the problems associated with both test methods. Low test speeds associated with drop test facilities can lead to the same strain rate sensitivities associated with static testing. Further, since dynamic testing devices accelerate and decelerate the impact plate over relatively short distances, the velocity of the strike plate can vary significantly during the test event. In this case, strain rate sensitivities can make test results virtually useless since the test velocity is no longer constant.

D1.7 Scale Model

Scale model testing involves constructing models of safety features and test vehicles to a reduced scale. The complexity of modeling automobile sheet metal crush, tire-pavement interaction, and suspension behavior has limited the application of these procedures for development of most roadside safety features. However, scale modeling can be useful during the development phases of safety features where most vehicle properties are of secondary importance, such as impact attenuation devices (73). This technique may yield useful information about the gross motion of a vehicle during impact with selected safety features. Uncertainties associated with modeling of connection designs and material properties have continued to limit the usefulness of these procedures.

D1.8 Bogie Test

The bogie vehicle is defined as a structure mounted on four wheels and with mass equivalent to that of a selected passenger vehicle. The bogie vehicle is steered by rails, guide cable, remote control, or other means to strike the specimen. The bogie vehicle may be accelerated to impact speed by a push or tow vehicle, by self power, or by stationary windlass. A crushable or otherwise deformable nose is mounted on the front of the bogie.

As discussed in Section 2.4.1.2, bogie vehicles must be revalidated periodically if the devices are to be representative of modern vehicles. Existing bogies have been designed to replicate vehicular crush characteristics and inertial properties. Such a bogie has been shown to be capable of simulating impacts with breakaway structures. However, significant improvements to the vehicle crush and suspension models must be made before existing bogies can be expected to replicate impacts with other safety features such as longitudinal barriers.

D2.0 Comparison of Techniques

Applications and limitations of safety feature development techniques are given in Table D-3.

TABLE D-3. APPLICATIONS AND LIMITATIONS OF SAFETY FEATURE DEVELOPMENT TECHNIQUES

Development Technique	Principal Areas of Application	Possible Limitations
1. Structural Design Methods	• Preliminary and final design of feature for environment and non-collision performance • Preliminary design of feature for vehicle collision performance • Analysis of connections, material properties requirement and foundation design	• Dynamics and kinematics of feature and collision vehicle are not addressed • Collision severity in terms of occupant injuries and fatalities is not addressed
2. Static Tests (quasi-static)	• Mechanical properties of unique shapes, connections, new materials • Validation of structural design features • Quality control of critical material properties • Develop input values for computer programs	• Dynamic properties not examined • Generally applicable to samples, connections, and small subassemblies; entire system is not accommodated
3. Computer Simulations	• Study interrelations of feature and vehicle dynamics and kinematics • Study interrelations of vehicle dynamics and occupant dynamics • Study sensitivity of feature, vehicle and site conditions on vehicle/feature dynamic interactions	• Program should be validated by full-scale crash tests for specific conditions that bracket the conditions under study • Input parameters are sometimes not available and must be estimated • For practical and economic reasons, programs model only major feature/vehicle properties • Sometimes minor features decide the performance
4. Laboratory Dynamic Tests		
A. Gravitational Pendulum	• Compliance test for luminaire and single-leg sign breakaway supports • Evaluation of breakaway mechanisms • Force/deformation properties of guardrail post/soil interaction • Dynamic strength of anchor systems • Dynamic properties of barrier subsystems	• Impact speed 40 km/h or less • For dual-leg supports, upper-hinge mechanism are not examined • Does not simulate off-center impacts • Trajectory of article not reproduced • Base-bending support not applicable • Crushable nose must be tuned for type and width of specimen and recalibrated periodically • Cannot properly evaluate criterion D, Table 5.1
B. Drop Mass	• Quality control test of breakaway component • Test can be performed in a confined, indoor space	• Same limitations as for pendulum • For breakaway base, attached pole introduces artifact moment into base due to gravity
C. Scale Model	• Development testing of feature	• Difficulties and uncertainties in modeling vehicle and safety feature components
C. Bogie Vehicle Test	• Compliance test for single or multi-leg breakaway support • Repeatable test vehicle suspension, nose crash, and other dynamic properties • Low-cost, high-speed (0-60 mph) experiments	• Must be carefully designed and calibrated to represent vehicle characteristic of interest, which is often a long and expensive process • Designs have been appropriate for testing only limited variations in feature • Must be updated and recalibrated periodically.
D. Vehicle Crash Test	• Compliance test for all features • Investigation of unusual conditions • Most direct tie to actual highway collisions • Final proof test	• Relatively expensive to perform • Requires extensive capital facilities • Deliberate and slow to perform • Test results pertain to the specific vehicle model tested and may not be applicable to other vehicles

APPENDIX E

OCCUPANT COMPARTMENT DEFORMATION INDEX

El. Definition

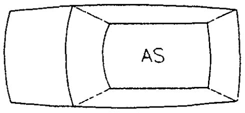
The Occupant Compartment Deformation Index (OCDI) described herein was taken from reference 77, and modified slightly. As recommended herein, it will be used by the European Committee for Normalization (CEN) for information purposes only and for the purpose of creating a data base from which future indices can be developed. It is referred to as the Vehicle Cockpit Deformation Index in reference 77.

This index designates both location and extent of deformation of the occupant compartment. It consists of two alphabetic characters plus seven numeric characters, in the form:

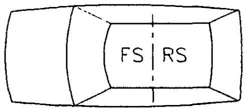
XXabcdefg

E2. Location of the Deformation

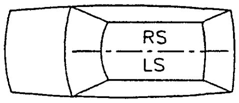
The location of occupant compartment deformation is indicated by two alphabetic characters XX, as follows:



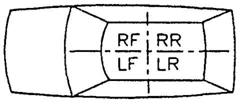
All Seats: XX = AS



Front Seats: XX = FS Rear Seats: XX = RS



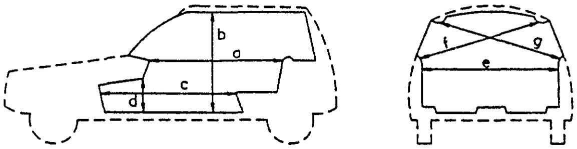
Right Seats: XX = RS **Left Seats:** XX = LS



Right Front: XX = RF **Right Rear:** XX = RR
Left Front: XX = LF **Left Rear:** XX = LR

3. Extent of the Deformation

The seven subindices a, b, c, d, e, f and g indicate the percentage of reduction of seven interior dimensions shown on the following figure:



where,

a = distance between the dashboard and a reference point at the rear of the occupant compartment, such as top of rear seat, or the rear part of the cab on a pickup;

b = distance between the roof and the floor panel;

c = distance between a reference point at the rear of the occupant compartment and the motor panel;

d = distance between the lower dashboard and the floor panel;

e = interior width;

f = distance between the lower edge of right window and the upper edge of left window; and g = distance between the lower edge of left window and the upper edge of right window.

All pre-impact reference points used in the above measurements should be marked and documented so that post-impact measurements can be made between the same points. To the extent possible, pre-impact measurements should be made in the area where maximum occupant compartment deformations are expected to occur. For example, the right-front part of the occupant compartment has the highest potential for damage from an impact with a longitudinal barrier, when the right-front part of the vehicle makes initial contact. In such a case, a set of a, b, c, and d measurements should be made on the right side of the occupant compartment, near the right door. Another set should probably be made midway between the sides of the occupant compartment. The value of each subindex should be based on the greatest reduction in each respective set of values. For example, the greatest reduction in c may occur at the center of the compartment and the greatest reduction in d may occur at the right side of the compartment. For the above longitudinal barrier impact, measurements e, f, and g should be made at the front windows of the vehicle.

Pre-impact reference points may not always coincide with the area of maximum compartment damage. In such a case it is recommended that pre-impact measurements be made on a vehicle of the same make and model year as the one tested. If this is not practical, estimates can be made from pre-impact measurements.

If resources permit, accurate measurements can be obtained by making sets of pre-impact subindices measurements along several planes that dissect the occupant compartment as one moves from the front to the rear of the compartment, and as one moves from one side of the compartment to the other.

The value of each of the seven numeric subindices is determined by the following scale:

0 - If the reduction is less than 3%
1 - If the reduction is more than 3% and less or equal to 10%
2 - If the reduction is more than 10% and less or equal to 20%
3 - If the reduction is more than 20% and less or equal to 30%
4 - If the reduction is more than 30% and less or equal to 40%, etc.

E4. Examples

(1) If an impact causes deformations to the right side of the occupant compartment, wherein e and f are reduced by 14% and g by 7% for the right seats (deformation centered approximately midway between the front and rear seats), the reduction of all the remaining dimension being below 3%, the OCDI will be: RS0000221.

(2) If in an impact distance a is reduced by 8% and c by 12% at the right front seat, all other reductions remaining below 3%, the OCDI will be: RF1200000.

APPENDIX F

DETERMINATION OF THE THIV, THE PHD, AND THE ASI

Fl. Introduction

The European Committee for Normalization (CEN) has adopted the Theoretical Head Impact Velocity (THIV) and associated Post-Impact Head Deceleration (PHD), and the Acceleration Severity Index (ASI) as measures of occupant risks for purposes of evaluating results of a crash test (77). They are presented herein with the hope and expectation that U.S. testers will determine and report these indices. The goal of this effort is to (a) develop a data base from which comparisons can be made between the THIV, ASI, the flail space indices recommended herein, and other measures of occupant risk, and (b) to provide a basis from which future test and evaluation procedures can be formulated by and harmonized between the U.S., CEN, and other countries.

F2. A Guide to the Measurement of the Theoretical Head Impact Velocity (THIV) and the Post-Impact Head Deceleration (PHD)

F2.1 General

The Theoretical Head Impact Velocity (THIV) concept has been developed for assessing occupant impact severity for vehicles involved in collisions with road vehicle restraint systems (78). The occupant is considered to be a freely moving object (head) that, as the vehicle changes its speed during contact with the safety feature, continues moving until it strikes a surface within the interior of the vehicle. The magnitude of the velocity of the theoretical head impact is considered to be a measure of the impact severity.

The head is presumed to remain in contact with the surface during the remainder of the impact period. In so doing it experiences the same levels of acceleration as the vehicle during the remaining contact period (Post-impact Head Deceleration - PHD) (79).

F2.2 Theoretical Head Impact Velocity (THIV)

It can be assumed that at the beginning of vehicular contact with the test article, both the vehicle and the theoretical head have the same horizontal velocity V_O, vehicular motion being purely translational.

During impact, the vehicle is assumed to move only in a horizontal plane, because high levels of pitch, roll, or vertical motion are not of prime importance unless the vehicle overturns. This extreme event does not need to be considered, as in this case the decision to reject the candidate system will be taken on the basis of visual observation or photographic recording.

Two reference frames are used, as indicated in Figure F2.1. The first of these is a vehicular reference Cxy, x being longitudinal and y transversal; the origin C is a point at or near the vehicle's center of mass, where two accelerometers and a rate gyroscope are typically

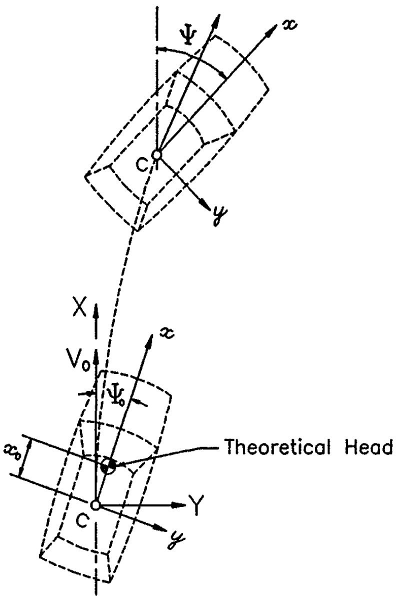


FIGURE F2.1: VEHICLE AND GROUND REFERENCE FRAMES

installed (see Section 4.3.3 for recommended procedures to determine accelerations and yaw rate at C if the instrumentation cannot be placed at or near the center of mass). Let $\ddot{x}_c$ and $\ddot{y}_c$ be the accelerations of point *C* (in m/s^2), respectively, along the x and y vehicle axes, recorded from the two accelerometers, and $\dot{\Psi}$ the yaw rate (in radians per second), recorded from the gyroscope ($\ddot{x}$ positive forward, $\ddot{y}$ positive to right hand side and $\dot{\Psi}$ positive clockwise looking from above).

The second reference frame is a ground reference *OXY,* with the X axis aligned with the initial vehicular velocity V_O, and the origin O coinciding with the initial position of the vehicular datum point C.

$X_c(t)$, $Y_c(t)$ are the ground coordinates of the vehicle reference *C,* while $X_b(t)$, $Y_b(t)$ are the ground coordinates of the theoretical head (see Figure F2.2).

With these definitions and simplifying hypotheses, vehicle and theoretical head motion can be computed as follows.

VEHICULAR MOTION

Initial conditions:

$$\text{at time } t = 0, \quad \begin{cases} X_c = 0 \quad Y_c = 0 \quad \Psi = \Psi_0 \\ \dot{X}_c = V_0 \quad \dot{Y}_c = 0 \quad \dot{\Psi} = 0 \end{cases} \tag{F2.1}$$

The yaw angle Ψ is computed by integration of the yaw rate Ψ:

$$\Psi(t) = \int_0^t \dot{\Psi}\, dt + \Psi_0 \tag{F2.2}$$

Then, from the components of vehicular acceleration in ground reference,

$$\begin{cases} \ddot{X}_c = \ddot{x}_c \cos\Psi - \ddot{y}_c \sin\Psi \\ \ddot{Y}_c = \ddot{x}_c \sin\Psi + \ddot{y}_c \cos\Psi \end{cases} \tag{F2.3}$$

Vehicular velocity and position are computed by integration:

$$\begin{cases} \dot{X}_c = \Delta\dot{X}_c + V_0 \quad \Delta\dot{X}_c = \int_0^t \ddot{X}_c dt \\ \dot{Y}_c = \Delta\dot{Y}_c \qquad\quad \Delta\dot{Y}_c = \int_0^t \ddot{Y}_c dt \end{cases} \tag{F2.4}$$

$$\begin{cases} X_c = \int_0^t \Delta\dot{X}_c dt + V_0 t \\ Y_c = \int_0^t \Delta\dot{Y}_c dt \end{cases} \tag{F2.5}$$

THEORETICAL HEAD MOTION RELATIVE TO GROUND

Initial condition: time=0

$$\begin{cases} X_b = x_0 \cos\Psi_0 = X_0 \quad Y_b = x_0 \sin\Psi_0 = Y_0 \\ \dot{X}_b = V_0 \qquad\qquad\qquad \dot{Y}_b = 0 \end{cases} \tag{F2.6}$$

Then, if the theoretical head continues its uniform motions:

$$X_b = V_0 t + X_0 \qquad Y_b = Y_o \tag{F2.7}$$

THEORETICAL HEAD MOTION RELATIVE TO VEHICLE

Vehicular components of the relative velocity of the theoretical head are:

$$\begin{cases} v_x(t) = -\Delta\dot{X}_c\cos\Psi - \Delta\dot{Y}_c\sin\Psi + y_b\dot{\Psi} \\ v_y(t) = \Delta\dot{X}_c\sin\Psi - \Delta\dot{Y}_c\cos\Psi - x_b\dot{\Psi} \end{cases} \qquad \text{(F2.8)}$$

Coordinates of the theoretical head with respect to the vehicle's frame can be computed by the formula:

$$\begin{cases} x_b(t) = \Delta X_b\cos\Psi + \Delta Y_b\sin\Psi \\ y_b(t) = -\Delta X_b\sin\Psi + \Delta Y_b\cos\Psi \end{cases} \quad \textit{where} \quad \begin{matrix} \Delta X_b = X_0 - \int_0^t \Delta\dot{X}_c\,dt \\ \Delta Y_b = Y_0 - \int_0^t \Delta\dot{Y}_c\,dt \end{matrix} \qquad \text{(F2.9)}$$

TIME OF FLIGHT

Notional impact surfaces inside the vehicle are assumed to be flat and perpendicular to the x and y vehicular axes (see Figure F2.2). The distances of such surfaces from the original head position (flail distances) are D_x forward and D_y laterally on both sides.

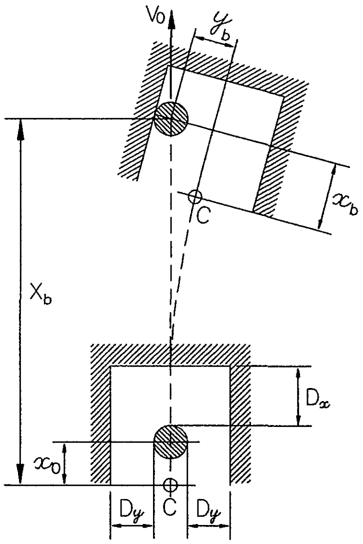


FIGURE F2.2: IMPACT OF THE THEORETICAL HEAD ON THE LEFT SIDE

The time of flight of the theoretical head is the time of impact on one of the three notional surfaces in Figure F2.2, i.e., the shortest time *T* when one of the three following equalities is satisfied:

$$x_b(T) = D_x + x_0; \quad or \quad y_b(T) = D_y; \quad or \quad y_b(T) = -D_y \qquad \text{(F2.10)}$$

The standard values of the flail distances are:

$$D_x = 0.6 \text{ m} \qquad D_y = 0.3 \text{ m}$$

THIV

Finally, the Theoretical Head Impact Velocity is the relative velocity at time T, i.e.:

$$THIV = \left[v_x^2(T) + v_y^2(T)\right]^{1/2}$$

THIV shall be reported in m/s.

F2.3 Post-Impact Head Deceleration (PHD)

Post-impact Head Deceleration (PHD) is the maximum value of the acceleration filtered by a 10 Hz low-pass filter, occurring after the time *T* of the collision of the theoretical head. If $\mathscr{F}_{10}$ represents the filtering, then:

$$PHD = MAX\left(\mathscr{F}_{10}(\ddot{x}_c^2 + \ddot{y}_c^2)^{1/2}\right) \qquad (for\ t > T)$$

PHD shall be reported in g units

F2.4 Summary of Procedure to Compute THIV and PHD

1. Record vehicular accelerations and yaw rate, and store in digital form at the sample rate S; let the data in the three record files be ${}^k\ddot{x}_c$, ${}^k\ddot{y}_c$, ${}^k\dot{\Psi}$ $(k = 1,2,...,N)$. The time interval between two subsequent data in the record file is $h = {}^kt - {}^{k-1}t = 1/S$. For example, if S = 500 samples per second, h = 2 ms.

2. Integrate the yaw rate by the recurrent formula (from Equation F2.2):

$${}^1\Psi = \Psi_0; \quad {}^2\Psi = {}^1\Psi + h\frac{{}^1\dot{\Psi} + {}^2\dot{\Psi}}{2} \quad ;...; \quad {}^{k+1}\Psi = {}^k\Psi + h\frac{{}^k\dot{\Psi} + {}^{k+1}\dot{\Psi}}{2}$$

3. Compute vehicular acceleration in ground reference (from Equation F2.3):

$${}^k\ddot{X}_c = {}^k\ddot{x}_c\cos{}^k\Psi - {}^k\ddot{y}_c\sin{}^k{}_{\Psi} \qquad {}^k\ddot{Y}_c = {}^k\ddot{x}_c\sin{}^k\Psi + {}^k\ddot{y}_c\cos{}^k\Psi$$

4. Integrate vehicular acceleration in ground reference (from Equations F2.4 and F2.9)

$$\begin{cases} {}^{1}\Delta\dot{X}_c = 0; & {}^{k+1}\Delta\dot{X}_c = {}^{k}\Delta\dot{X}_c + h\,\dfrac{{}^{k}\ddot{X}_c + {}^{k+1}\ddot{X}_c}{2}; \\ {}^{1}\Delta\dot{Y}_c = 0; & {}^{k+1}\Delta\dot{Y}_c = {}^{k}\Delta\dot{Y}_c + h\,\dfrac{{}^{k}\ddot{Y}_c + {}^{k+1}\ddot{Y}_c}{2}; \end{cases}$$

$$\begin{cases} {}^{1}\Delta X_b = X_0; & {}^{k+1}\Delta X_b = {}^{k}\Delta X_b - h\,\dfrac{{}^{k}\Delta\dot{X}_c + {}^{k+1}\Delta\dot{X}_c}{2} \\ {}^{1}\Delta Y_b = 0; & {}^{k+1}\Delta Y_b = {}^{k}\Delta Y_b - h\,\dfrac{{}^{k}\Delta\dot{Y}_c + {}^{k+1}\Delta\dot{Y}_c}{2}; \end{cases}$$

5. Compute relative position and relative velocity of the theoretical head as functions of time (from Equations 8 and 9):

$$\begin{cases} {}^{k}x_b(t) = {}^{k}\Delta X_b \cos {}^{k}\Psi + {}^{k}\Delta Y_b \sin {}^{k}\Psi \\ {}^{k}y_b(t) = -{}^{k}\Delta X_b \sin {}^{k}\Psi + {}^{k}\Delta Y_b \cos {}^{k}\Psi \end{cases}$$

$$\begin{cases} {}^{k}v_x = -{}^{k}\Delta\dot{X}_c \cos {}^{k}\Psi - {}^{k}\Delta\dot{Y}_c \sin {}^{k}\Psi + {}^{k}y_b\,{}^{k}\dot{\Psi} \\ {}^{k}v_y = {}^{k}\Delta\dot{X}_c \sin {}^{k}\Psi - {}^{k}\Delta\dot{Y}_c \cos {}^{k}\Psi - {}^{k}x_b\,{}^{k}\dot{\Psi} \end{cases}$$

6. Find the minimum value of *j* for which one of the three equalities is satisfied.

$${}^{j}x_b = D_x + X_0; \qquad or\ {}^{j}y_b = D_y; \qquad or\ {}^{j}y_b = -D_y$$

7. Compute

$$THIV = \left[{}^{j}v_x^2 + {}^{j}v_y^2\right]^{1/2}$$

8. Compute the resultant vehicular acceleration in g as a function of time

$${}^{k}A = \frac{1}{g}\left({}^{k}\ddot{x}_c^2 + {}^{k}\ddot{y}_c^2\right)^{1/2}$$

9. Filter the sequence ${}^{k}A$ with a digital Butterworth low-pass filter, having a cut-off frequency of 10 Hz and a roll-off of 48 dB/octave; PHD is the maximum of such filtered sequence.

F3. A Guide to the Measurement of the Acceleration Severity Index (ASI)

F3.1 Procedure

(F3. 1)

The Acceleration Severity Index (ASI), developed by TTI (80), is a function of time, computed with the following formula:

$$k_{ASI} = [(k_{\bar{a}_x}/12)^2 + (k_{\bar{a}_y}/9)^2 + (k_{\bar{a}_z}/10)^2]^{1/2}$$

(F3.2)

where $\hat{a}_x$, $\hat{a}_y$, *and* $\hat{a}_z$, are limit values for the components of the acceleration along the body axes x, y, and z; $\bar{a}_x$, $\bar{a}_y$, *and* $\bar{a}_z$ are the components of the acceleration of a selected point P of the vehicle, averaged over a moving time interval delta $\delta = 50$ ms, so that:

$$\bar{a}_x = \frac{1}{\delta}\int_t^{t+\delta} a_x dt; \quad \bar{a}_y = \frac{1}{\delta}\int_t^{t+\delta} a_y dt; \quad \bar{a}_z = \frac{1}{\delta}\int_t^{t+\delta} a_z dt \qquad \text{(F3.3)}$$

The index ASI is intended to give a measure of the severity of the vehicular motion during an impact for a person seated in the proximity of point P.

Averages computed in Equations F3.2 are equivalent to what would be obtained by a low pass filter, and take into account the fact that vehicular accelerations can be transmitted to the occupant body through relatively soft contacts which cannot pass the highest frequencies. Direct use of vehicular accelerations, even if averaged, implies that the parts of occupant body than can be injured are continuously in contact with some part of the vehicle.

Note that Equation F3.1 is a basic interaction formula of three variables. If any two components of vehicular acceleration are null, ASI reaches its limit value of 1 when the third component reaches its limit acceleration. When two or three components are non null, ASI may be 1 with the single components well below the relevant limits. Limit accelerations are interpreted as the values below which occupant risk is very small (light injuries, if any).

In Europe (France, Germany, and Netherlands), for occupants wearing safety belts, the generally used limit accelerations are:

$$\hat{a}_x = 12g\ ,\ \hat{a}_y = 9g\ ,\ \hat{a}_z = 10g$$

where $g = 9.81\ ms^{-2}$ is the acceleration of Earth gravity at sea level.

With the above definition ASI is a nondimensional quantity, which is a scalar function of time and, in general, of the selected vehicular point, having only positive values. Occupant risk is assumed to be proportional to ASI. Therefore, the maximum value attained by ASI in a collision is assumed as a single measure of the severity, or:

$$\text{ASI} = \max[\text{ASI}(t)] \qquad \text{(F3.4)}$$

Vehicular accelerations in the x, y, and z directions are measured at or near the center of mass of the vehicle (see Section 4.3.3 for recommended procedures to determine accelerations in the x and y directions at the center of mass if the accelerometers cannot be placed at or near the center of mass).

F3.2 Summary

In summary, the following steps are used to compute the ASI.

1. Record vehicular accelerations in the x, y, and z directions at or near the vehicle's center of mass (see Section 4.3.3 if accelerometers can not be placed at or near the center of mass). In general, accelerations are stored on a magnetic tape as three series of N numbers, sampled at a certain sampling rate S (samples/sec).

For such three series of measures,

$$\begin{aligned}&{}^{1}a_x, {}^{2}a_x, \dots, {}^{k-1}a_x, {}^{k}a_x, {}^{k+1}a_x, \dots, {}^{N}a_x\\&{}^{1}a_y, {}^{2}a_y, \dots, {}^{k-1}a_y, {}^{k}a_y, {}^{k+1}a_y, \dots, {}^{N}a_y\\&{}^{1}a_z, {}^{2}a_z, \dots, {}^{k-1}a_z, {}^{k}a_z, {}^{k+1}a_z, \dots, {}^{N}a_z\end{aligned}$$

where acceleration of gravity g is the unit of measurement.

2. Find the number m of samples in the averaging window delta o = 0.05s: m = INT(o *S) =INT(.05*S), where INT(R) is the integer nearest to R. For example, if S = 500 samples per sec., m = 25.

3. Compute the average accelerations from Equation F3.2:

$$\begin{aligned}k_{\bar{a}_x} &= \frac{1}{m}(k_{a_x} + k{+}1_{a_x} + k{+}2_{a_x} + \dots + k{+}m_{a_x}) = \frac{1}{m}\sum_{j=k}^{k+m} j_{a_x}\\k_{\bar{a}_y} &= \frac{1}{m}(k_{a_y} + k{+}1_{a_y} + k{+}2_{a_y} + \dots + k{+}m_{a_y}) = \frac{1}{m}\sum_{j=k}^{k+m} j_{a_y}\\k_{\bar{a}_z} &= \frac{1}{m}(k_{a_z} + k{+}1_{a_z} + k{+}2_{a_z} + \dots + k{+}m_{a_z}) = \frac{1}{m}\sum_{j=k}^{k+m} j_{a_z}\end{aligned}$$

Functions of time ${}^{k}t = h\,(k+m/2)$.

4. Compute ASI as a function of time from Equation F3. 1:

$$^{k}ASI = [(k_{\bar{a}_x}/12)^2 + (k_{\bar{a}_y}/9)^2 + (k_{\bar{a}_z}/10)^2]^{1/2}$$

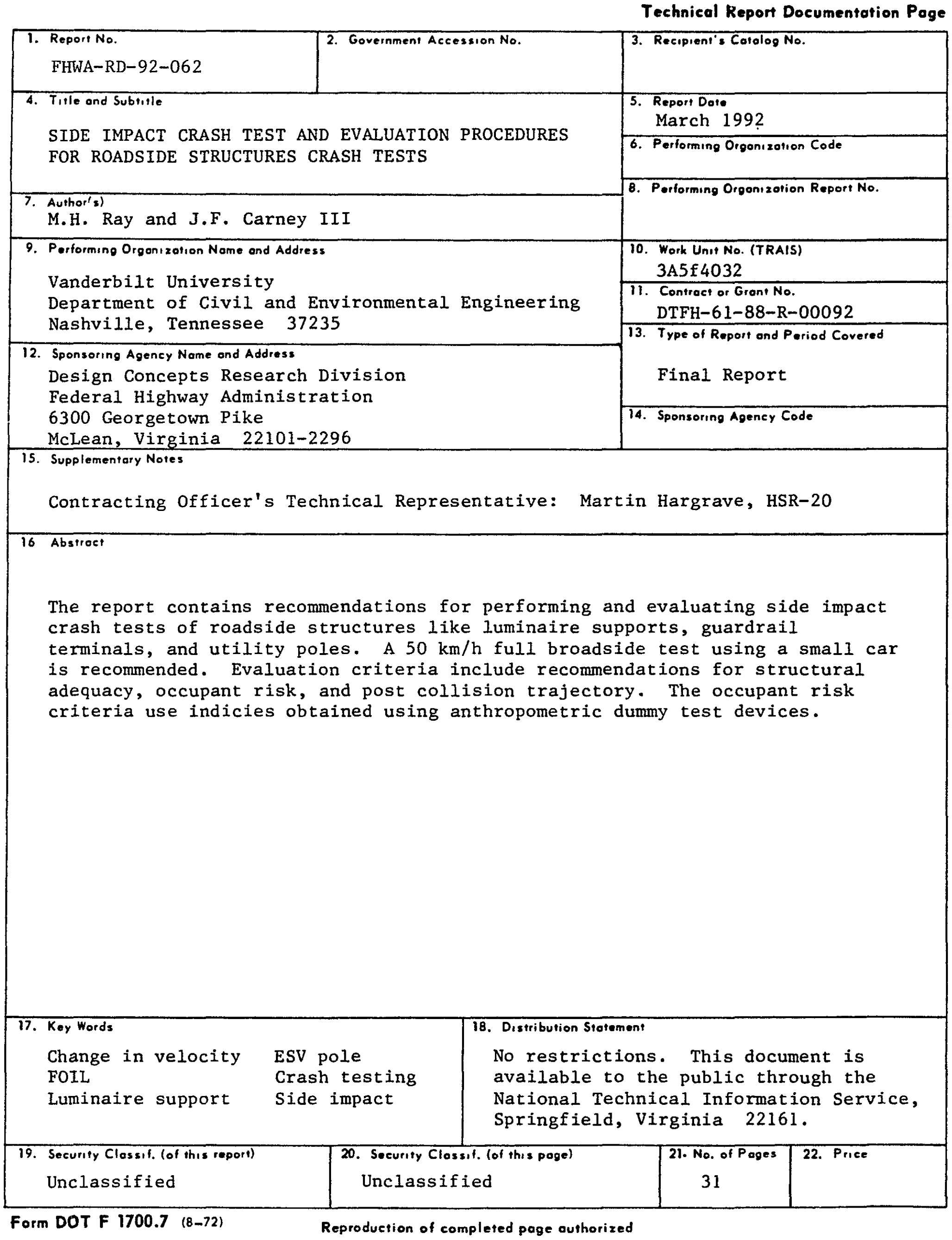
5. Find ASI as the maximum of the series of the ^{k}ASI.

APPENDIX G

SIDE IMPACT TEST AND EVALUATION PROCEDURES FOR ROADSIDE STRUCTURE CRASH TESTS

Concurrent with the effort to prepare an update to *NCHRP Report 230,* an FHWA-sponsored study was being conducted to examine the side impact problem and to develop tentative side impact test procedures for sign and luminaire support structures. This appendix contains a report, copied verbatim, from that study describing recommended test and evaluation procedures for side impact testing. It is included for information purposes only because no recommendations are made relative to the impact performance requirements of a safety feature for side impacts (see discussion in Section 3.5). Until these or other guidelines are nationally accepted, developers of safety features for side impact capabilities may use guidelines presented therein.

Note that references cited in the appendix refer to the list of references given at the end of the Appendix G.



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16 Abstract

The report contains recommendations for performing and evaluating side impact crash tests of roadside structures like luminaire supports, guardrail terminals, and utility poles. A 50 km/h full broadside test using a small car is recommended. Evaluation criteria include recommendations for structural adequacy, occupant risk, and post collision trajectory. The occupant risk criteria use indicies obtained using anthropometric dummy test devices.

17. Key Words

Change in velocity	ESV pole
FOIL	Crash testing
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SI* (MODERN METRIC) CONVERSION FACTORS

APPROXIMATE CONVERSIONS TO SI UNITS

LENGTH

Symbol	When You Know	Multiply By	To Find	Symbol
in	inches	25.4	millimetres	mm
ft	feet	0.305	metres	m
yd	yards	0.914	metres	m
mi	miles	1.61	kilometres	km

AREA

Symbol	When You Know	Multiply By	To Find	Symbol
in^2	square inches	645.2	millimetres squared	mm^2
ft^2	square feet	0.093	metres squared	m^2
yd^2	square yards	0.836	metres squared	m^2
ac	acres	0.405	hectares	ha
mi^2	square miles	2.59	kilometres squared	km^2

VOLUME

Symbol	When You Know	Multiply By	To Find	Symbol
fl oz	fluid ounces	29.57	millilitres	mL
gal	gallons	3.785	litres	L
ft^3	cubic feet	0.028	metres cubed	m^3
yd^3	cubic yards	0.765	metres cubed	m^3

NOTE: Volumes greater than 1000 L shall be shown in m^3.

MASS

Symbol	When You Know	Multiply By	To Find	Symbol
oz	ounces	28.35	grams	g
lb	pounds	0.454	kilograms	kg
T	short tons (2000 lb)	0.907	megagrams	Mg

TEMPERATURE (exact)

Symbol	When You Know	Multiply By	To Find	Symbol
°F	Fahrenheit temperature	5(F-32)/9	Celcius temperature	°C

APPROXIMATE CONVERSIONS FROM SI UNITS

LENGTH

Symbol	When You Know	Multiply By	To Find	Symbol
mm	millimetres	0.039	inches	in
m	metres	3.28	feet	ft
m	metres	1.09	yards	yd
km	kilometres	0.621	miles	mi

AREA

Symbol	When You Know	Multiply By	To Find	Symbol
mm^2	millimetres squared	0.0016	square inches	in^2
m^2	metres squared	10.764	square feet	ft^2
ha	hectares	2.47	acres	ac
km^2	kilometres squared	0.386	square miles	mi^2

VOLUME

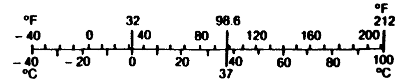
Symbol	When You Know	Multiply By	To Find	Symbol
mL	millilitres	0.034	fluid ounces	fl oz
L	litres	0.264	gallons	gal
m^3	metres cubed	35.315	cubic feet	ft^3
m^3	metres cubed	1.308	cubic yards	yd^3

MASS

Symbol	When You Know	Multiply By	To Find	Symbol
g	grams	0.035	ounces	oz
kg	kilograms	2.205	pounds	lb
Mg	megagrams	1.102	short tons (2000 lb)	T

TEMPERATURE (exact)

Symbol	When You Know	Multiply By	To Find	Symbol
°C	Celcius temperature	1.8C + 32	Fahrenheit temperature	°F



* SI is the symbol for the International System of Measurement

(Revised April 1989)

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1 Introduction

Side impact collisions involving fixed roadside objects like utility poles, trees and luminaire supports account for about 1600 fatalities and 60,000 injured vehicle occupants each year in the United States [28] [31]. This type of collision appears to cause a disproportionate number of fatalities and serious injuries [16][28] [31].

Many of the fixed objects struck on the roadside are intentionally placed there to provide lighting, power transmission, or to convey information. While usually serving a benign purpose, these objects can become a hazard if they are not designed to breakaway, collapse, or fracture in an impact with an errant vehicle. This report recommends testing conditions and evaluation criteria for side impact full-scale crash tests of structures placed on the roadside.

These recommendations supplement guidelines published by the National Cooperative Highway Research Program (NCHRP) [19] [29]. Where a particular guideline is not specifically addressed herein, the more general NCHRP guidelines should be applied. For example, test documentation is not specifically addressed so all documentation items recommended in the NCHRP guidelines should be included in a complete side impact test report. These recommendations are organized much like the latest NCHRP guidelines [29].

This document is based largely on research described in several other reports and papers [32] [27] [28] [31] [14]. The Fatal Accident Reporting System (FARS) and National Accident Sampling System (NASS) were investigated to learn about the characteristics of side impacts with fixed roadside objects [32] [28]. A number of side impact crash tests of luminaire supports and guardrail terminals were performed by the Federal Highway Administration (FHWA) during the past decade [27] [12] [10]. More information on particular aspects of these recommendations can be found in these other documents.

The National Highway Traffic and Safety Administration (NHTSA) has published rules on performing vehicle-to-vehicle side impact crash tests to evaluate the crashworthiness of production automobiles and light trucks [23] [21] [22]. Although there are important differences between the objectives of these NHTSA tests and the tests addressed by this report, the proposed NHTSA rules were used as a guide wherever possible in formulating these recommendations [24]. Any linkage that can be forged between NHTSA and FHWA side impact crash tests would be beneficial in the future as more is learned by both research communities about side impact collisions.

2 Test Parameters

2.1 Test Facilities

Side impact crash tests are significantly more difficult to perform than typical safety appurtenance crash tests. Accelerating the vehicle laterally requires test facilities that are not commonly found in the roadside research community. Side impact crash tests have been performed using:

- A differentially braked vehicle on a slick pavement [9].
- A cable-guided cable-towed carriage with the vehicle mounted sideways [2].
- A cable-towed wooden pallet with the vehicle resting sideways [33].
- A monorail and outrigger assembly with the vehicle resting on casters [11].

The Federal Outdoor Impact Laboratory (FOIL), shown in figure 1, was used for most of the side impact crash tests performed to assess the performance of roadside structures. The vehicle is transported on a monorail. A stabilizing outrigger rail runs parallel to the monorail. Rolling carriages are mounted on the underside of the vehicle body. The monorail and outrigger rail end approximately 2 m from the test device. The vehicle is brought up to the desired test speed using a drop-weight accelerator. The vehicle, with the attached roller carriages, drops off the rails and slides the remaining distance to the test device. Information about the construction, capabilities, and operation of the FOIL can be found elsewhere [11].

Test devices should be mounted in as realistic a manner as possible. Some objects like foundation mounted luminaire supports may be rigidly connected to a universal foundation if they are normally supported on a rigid foundation in the field. Soil mounted structures like guardrail terminals, utility poles and signs should be mounted in a soil representative of the soil type typically found in the field.

The vehicle should slide laterally at least two vehicle track widths to allow the vehicle to stabilize after dropping off the monorail. The sliding should occur on pavement or wood since soft earth may trip a vehicle sliding broadside over a large distance. Wetting down the approach area with water just prior to the test will help reduce friction between the surface and the vehicle tires. Accidents certainly occur on dry non-paved surfaces but experimental difficulties with the stability of the vehicle and repeatability of the test make reducing the sliding friction advisable.

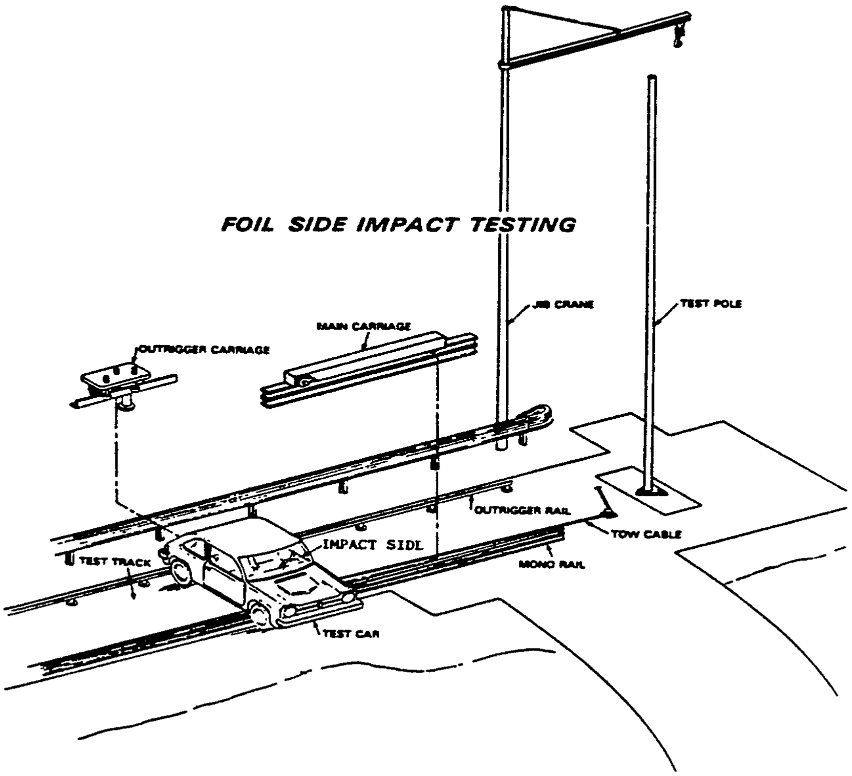

FOIL SIDE IMPACT TESTING
JIB CRANE
TEST POLE
MAIN CARRIAGE
OUTRIGGER CARRIAGE
OUTRIGGER RAIL
TOW CABLE
IMPACT SIDE
TEST TRACK
MONO RAIL
TEST CAR

Figure 1: Federal Outdoor Impact Laboratory [11].

2.2 Test Articles

An investigation of the Fatal Accident Reporting System (FARS) and National Accident Sampling System (NASS) showed that narrow objects accounted for 60 percent of the accidents but 80 percent of the fatalities in side impact accidents involving fixed roadside objects. Narrow objects subject the side of a vehicle to highly concentrated loadings that are difficult to resist without extensive vehicle deformation.

Highway safety appurtenances in the narrow-object category (table 1) include luminaire supports, utility poles, sign supports, guardrail terminals and narrow crash cushions. These recommendations should be used during an assessment program after all the applicable frontal tests have been performed. The side impact test is much more demanding than the frontal tests. After successful performance is observed in the frontal tests, the side impact test should be performed.

Table 1. Test devices appropriate for side impact crash testing.

Luminaire Supports
Large and Small Sign Supports
Guardrail Terminals Narrow Crash Cushions
Breakaway Utility Poles

2.3 Vehicle

A two-door 820-kg small vehicle should be used in side impact crash testing of roadside hardware. This vehicle is identical to the 820C vehicle recommended in the NCHRP guidelines with the exception that only two-door models should be used [29]. All the requirements for mass tolerances, vehicle age and condition recommended in the NCHRP guidelines should also be satisfied for side impact crash tests as should recommended vehicle dimensions.

Examination of the **FARS** accident data has shown that the fatality rate in smaller vehicles is no different than for larger vehicles in side-impact fixed-object accidents [28]. Partyka has shown that, in general, the fatality rate is not a function of weight in single-vehicle accidents where rollover does not occur [25]. The choice of a lighter test vehicle, therefore, cannot be justified on the grounds that the occupant is more at risk.

Instead, the lighter vehicle was chosen in order to (1) minimize the kinetic energy available for device activation and (2) maximize the probability of vehicle instability during the post collision trajectory. Most of the devices targeted by these recommended procedures function by breaking away, collapsing, yielding or fracturing. The 820C vehicle provides a reasonable minimum amount of kinetic energy in an impact.

While vehicle stability was not normally a problem in side impact crash tests with fixed objects, smaller vehicles tend to be less stable than larger vehicles because of narrower track widths, smaller masses, and the position of their centers of gravity. Stability problems like rolling over are more easily identified when smaller vehicles are used so their use is recommended.

The door on a two-door vehicle spans a larger distance than in comparable four-door models. This larger span on two-door models makes the door inherently weaker than the four-door model. The two-door small car minimizes the side impact resistance of the vehicle.

Side impact crash tests of narrow fixed objects sponsored by the FHWA have been performed using the Honda Civic Si, the Dodge Colt, the Plymouth Champ and the Volkswagen Rabbit [27]. All the vehicles used were two-door models manufactured between 1978 and 1986 and conform to the requirements shown in table 2.

Table 2. Side impact test vehicle.

Vehicle Type	NCHRP 820C	
Test Inertial - kg	820	± 25
Dummy - kg	75	
Max. Ballast - kg	50	
Gross Static - kg	915	± 25
Engine Location	Front	
Drive Axle Location	Front	
Number of Doors	Two	

3 Test Conditions

3.1 General

Side impact tests will be supplementary to the usual frontal crash tests specified in *NCHRP Report 230.* The standard frontal matrix of tests includes a 30 km/h test of the breakaway mechanism. Since this frontal test examines the breakaway mechanism at the 30 kJ level, there should be no need to retest in a side impact configuration. In frontal tests, the amount of kinetic energy transformed to vehicle deformation is usually a relatively small percentage of the total energy; most of the energy can be used to activate the device. In side impacts, vehicle deformation accounts for a much larger proportion of the initial kinetic energy. It cannot be assumed a *priori,* then, that a device that activates in a low speed frontal collision will activate in a side impact collision. While testing at a lower speed would produce a more demanding test in terms of device activation, it may not by satisfactory for evaluating the risk to vehicle occupants. The side impact test recommended herein will focus on the response of the hypothetical occupant.

Impact conditions for full-scale crash tests have generally been designed to represent the practical worse-case impact scenario [19]. With this perspective in mind, accident data from the Fatal Accident Reporting System (FARS) and the National Accident Sampling System (NASS) Continuous Sampling System (CSS) were investigated to examine the characteristics of side impact fixed object accidents [32] [28] [31].

The following recommended test conditions for side-impact fixed object collisions are a compromise between the most realistic conditions and those easiest to obtain in controlled experiments. The accident data provides indications of the speeds and orientations common in side impact collisions. Because of the shortcomings of accident data the estimates of impact conditions must be viewed as tentative. The accident data does, however, provide the only view of the real-world accident problem. It appears that most side impacts occur at relatively low lateral speeds and high angles.

Table 3 shows the recommended impact conditions for side impact testing of roadside structures. A collision between a fixed roadside object and the center of the driver's side door is recommended. The lateral impact speed for test SI-should be 50 km/h. Test SI-2 is an optional higher velocity test that can be included when the performance of a device is expected to degrade at higher velocities. While real accidents involve longitudinal and angular velocity components, experimental limitations often preclude testing with these additional velocity components. These impact conditions are suggested as a reasonable set of experimentally achievable test conditions for exploring the performance of roadside hardware in side impact collisions.

Table 3. Side impact test conditions.

	SI-1	SI-2 (optional)
Lateral Velocity	50 km/h	60 km/h
Longitudinal Velocity	0 km/h	0 km/h
Yaw Angle	90 degrees	90 degrees
Yaw Rate	0 degrees/sec	0 degrees/sec
Impact Point	Center of Door	Center of Door

3.2 Vehicle Orientation

Side impacts have been shown to be associated primarily with narrow fixed objects such as utility poles and luminaire supports [28] [31]. The impact angle is usually defined in a crash test as the angle between the longitudinal axis of a device and the approach path of the vehicle. Since many narrow objects have no longitudinal axis the impact angle is technically undefined. The vehicle orientation and yaw angle can be defined instead in

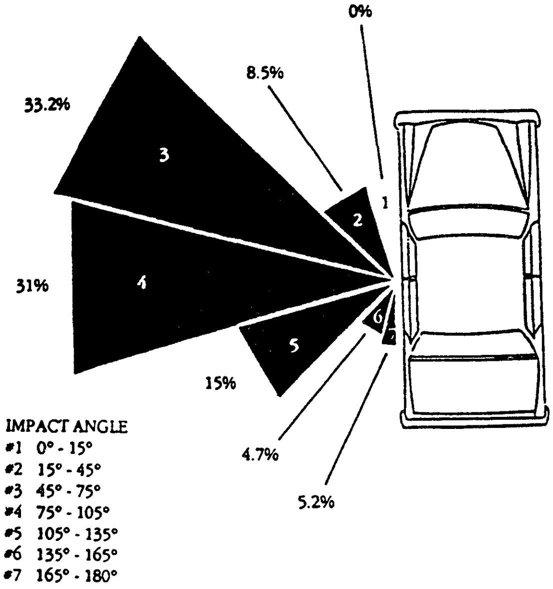


Figure 2. Directions of force in side impact collisions with fixed roadside objects involving the passenger compartment (1982 - 1985 NASS) [32].

terms of the orientation of the traveled way as shown in figure 3. This ensures that the basic accident scenario being investigated is the same even if the devices are different.

There is no direct measure of the yaw angle given in the NASS-CSS accident data. An estimate of these quantities can be made, however, using the direction-of-force variable and the longitudinal and lateral changes in velocity. The yaw angle is the angle between the longitudinal direction of the vehicle and the direction of the velocity vector.

The direction-of-force (DOF variable is an estimate of the orientation of the resultant force during the collision. A DOF of 0 indicates that the force interaction was parallel with the center line of the vehicle whereas a DOF of 90 would indicate a perpendicular force. Figure 2 shows the direction of force distribution for side impacts with fixed objects from the 1982 through 1985 NASS-CSS data [32]. The mean direction of force was found to be 56 degrees and the median value was 60 degrees. The most frequently observed direction of force was 90 degrees, a full broadside collision. Figure 2 also shows that the vehicle had a forward component of velocity in more than 80 percent of the impacts. Angles between 45 and 105 degrees accounted for almost 50 percent of the side impact yaw impact angles.

Another rough estimate of the yaw angle can be obtained by calculating the arctangent of the longitudinal and lateral change in velocity. These estimates are very approximate because of the uncertainties in calculating the two velocity values [30]. The mean yaw

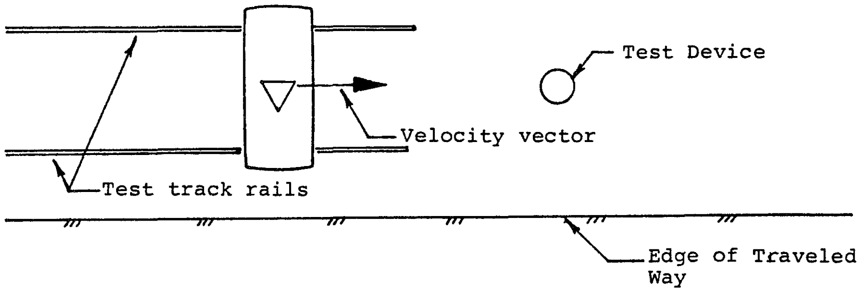


Figure 3. Vehicle orientation for side impact crash tests.

angle was found to be 57 degrees with a standard deviation of 19 degrees [32]. Both estimates of the yaw angle, then, indicate that the mean yaw angle is approximately 60 degrees. The final rule on side impact testing published by NHTSA specifies a test with a crabbed impactor bogie. The principal direction of force on the test vehicle is 63 degrees. This choice of a crabbed vehicle was based on NHTSA's analysis of vehicle-to-vehicle side impact collisions.

Performing tests with a yaw angle of 60 degrees was considered but abandoned in favor of a 90-degree orientation for several reasons:

- Mounting a vehicle on a carriage with a 60 degree yaw angles causes some experimental problems in balancing the vehicle on the guidance rails and mounting the carriages under the vehicle. In NHTSA tests the impacted vehicle is stationary so it does not have to balanced and accelerated. The impactor bogie, though yawed, is a tracking vehicle since the wheel are attached at an angle.

- When a yawed vehicle releases from the carriage it will tend to roll ahead making the vehicle difficult to control. Obtaining repeatable impact locations would be impossible with a yawing vehicle.
- A full broadside orientation constitutes a reasonable worst-case (possible unsurvivable) scenario in terms of the side-door strength of the vehicle. The side of the vehicle is relatively weak and the perpendicular orientation maximizes the loading in the weak direction.

Side impact crash tests of roadside structures should be performed with the vehicle perpendicular to the traveled way with the front of the vehicle facing the traveled way.

This orientation, shown in figure 3, represents the common accident scenario of leaving the road on the wrong side, partially recovering and striking a fixed object nearly broadside.

3.3 Velocity

The velocity change values reported in the 1982 through 1985 NASS-CSS data for side impact fixed objects accidents centered on the occupant compartment were used to estimate the impact velocity [32]. The change in velocity can be assumed to be close to the impact velocity if it is assumed that the vehicle was brought to rest as a result of the collision. The most frequently struck objects in the NASS-CSS data are trees, utility poles and other narrow objects. While some of these objects, like sign supports and delineator posts, do break away or yield the majority of collisions are with objects that do not break away such as trees and utility poles. When the fixed object does not break away, the vehicle must come to rest. The assumption that the changes in lateral velocity can be used to represent lateral impact velocities, then, seems to be a reasonable first approximation for side impacts with fixed roadside structures.

The mean lateral change in velocity, as shown in table 4, was 24 km/h. The maximum lateral velocity observed in this sample was 63 km/h. If the distribution of lateral velocities is assumed to be exponential, 85 percent of the cases would occur at less than 45 km/h and 70 percent of the cases would occur at less than 30 km/h. An exponential distribution is a reasonable assumption since the most common lateral velocity should be zero and large lateral velocities should be quite rare. An exponentially distributed lateral velocity distribution is suggested by the NASS data [27].

Table 4. Change in velocity statistics for side impact accidents with fixed objects (1982-1985 NASS) [32].

Velocity Direction	Maximum Velocity (km/h)	Mean Velocity (km/h)	Standard Deviation (km/h)
Lateral	63	24	16
Longitudinal	38	11	10
Total	67	29	18

Table 5 illustrates the increasing severity of injury with increasing total velocity change. More than 60 percent of all minor injuries occurred in accidents where the lateral change in velocity was less than 10 km/h. In contrast 75 percent of the severe and fatal injuries occurred in accidents where the lateral change in velocity was greater than 31 km/h. Clearly, the amount of energy dissipated is related to the severity of injury experienced by the vehicle occupants. It has been suggested that injury can be defined as exposure to energy [4]; more energy should be correlated with a higher proportion of severe injuries. The proportion of severe and moderately injured occupants increases as the lateral change in velocity increases.

Table 5. Injury as a function of lateral change in velocity for side ipacts centered on the passenger compartment [32].

ΔV_{total}	Minor		Moderate		Severe		Unknown	Total	
	$0 \leq AIS < 2$		$2 \leq AIS < 3$		$AIS \geq 4$				
(km/h)	(no.)	(%)	(no.)	(%)	(no.)	(%)	(no.)	(no.)	(%)
0-10	34	65	7	33	2	25	2	44	54
11-20	4	8	1	5	0	0	0	5	6
21-30	9	17	0	0	0	0	0	9	11
31-40	2	4	6	29	1	12	0	9	11
41-50	3	6	1	5	1	13	0	5	6
51-60	0	0	5	23	3	38	0	8	10
60 >	0	0	1	5	1	12	0	2	2
Total	52	100	21	100	8	100	2	83	
Missing								23	

Severe injuries ($AIS > 3$) can be observed across the range of impact speeds but 75 percent occur at velocities greater than 30 .km/h. The mean velocity for occupants who received $AIS > 3$ injuries was approximately 40 km/h. Impacts occurring in the 30 to 60 km/h range resulted in 1 chance in 18 of sustaining an $AIS > 3$ injury. A test velocity of 50 km/h was selected since successful performance at this speed would imply protection for nearly 90 percent of the vehicle occupants in this sample. Specifying a higher test velocity would probably exceed the point of diminishing returns.

3.4 Impact Point

The impact point for side impact crash tests of roadside structures should be at the center of the driver's side door on a small 2-door passenger vehicle. This location is near the longitudinal center of gravity of the 820C vehicle and about 250 mm in front of the dummy shoulder. The door is weakest at the center so the maximum amount of intrusion should be observed when the impact is located at this point.

One of the CRASH3 data items collected in the NASS data is the distance from the vehicle center of gravity to the centroid of the damaged area. Nearly 60 percent of the side impacts in the study sample occurred between the A and B pillar [32]. Impacts that occur between the A and B pillars will be located on the front door, very close to the front seat occupant.

Earlier tests [12] have used an impact point centered on the front seat occupant. While this orientation represents a practical worst case scenario, recent testing has indicated that obtaining useful anthropometric dummy responses is very difficult when the dummy directly contacts the intruding object [27]. Accidents where the occupant's head directly contacted an intruding pole are not difficult to find in the literature or in the accident data.

When this occurs the occupant is nearly always severely injured, even when the lateral impact velocity is very low. The absence of any protection between the head and the window makes protection of the occupant in this situation nearly impossible.

The recommended impact location is slightly in front of the dummy shoulder. This location, while not the worst, subjects the dummy to large impact loadings but is far enough removed to yield more repeatable and meaningful dummy responses.

3.5 Anthropometric Dummy Position

NHTSA vehicle-to-vehicle side impact tests require the use of an instrumented Part 572 Subpart F side impact dummy (SID) [24]. The side impact tests performed at the FOIL since 1985 have all used this type of dummy. The long-term objective of this research is to specify criteria that will allow side impact crash tests to be evaluated without dummies. There are several reasons for not including dummies in crash tests of safety appurtenances. A recent FHWA staff studies found that in most typical appurtenance crash tests, the data obtained from the anthropometric dummy was rarely used and the responses were often subcritical [17].

The use of dummies in the early stages of side impact research, however, is inescapable. Judging the performance of a test article ultimately involves judging the risk of serious injury to vehicle occupants in real accidents. The anthropometric dummy is the best available device that, at least in principal, links the performance of the device on the test pad to the performance in the real world. Instrumented side impact dummies should, therefore, be used in the side impact crash tests of roadside objects for the foreseeable future.

The Part 572 dummy is recommended primarily because all previous side impact crash tests of roadside structures have used this device, and it is unlikely that newer side impact anthropometric devices like the EuroSID or BioSID will be made available to roadside appurtenance researchers in the near future.

The seat should be positioned as far to the rear as the normal seat adjustment will allow in order to fit both the anthropometric test device and the displacement transducer in the occupant compartment (see section 4.3). The displacement transducer includes a string that stretches from the impact-side door to the non-impact-side door. If the dummy is not position far enough back the string could interfere with the dummy response.

The dummy was placed in the driver's position in all the side impact tests performed at the Federal Outdoor Impact Laboratory. The dummy should be placed in the front seat position on the impact side on the vehicle. Although many injuries in real accidents result from an unrestrained non-impact-side occupant flailing across the passenger compartment, the impact-side occupant is always at greater risk of injury.

Table 6. Anthropometric side impact test dummy.

Type	Part 572 Subpart F (SID)
Seating Position	Impact Side (Driver-side preferable)
Seat Adjustment	Maximum rearward position
Seat Back Position	Normal upright position
Restraint	Available restraints

The seat back should be positioned in the "normal," unadjusted position. This is usually the most vertical orientation. Some recent research has suggested that ensuring the head is relatively level is an important factor in obtaining repeatable HIC values in side impacts [24]. While explicit leveling of the head is not necessary, the upright position of the seat back will result in a more level head form.

Traditionally, restraints have not been used in full-scale crash tests of safety hardware. Seat belts are not effective in side impacts when the impact is on the same side as the occupant. The lap belt restrains the pelvis but not the upper body, allowing the head and thorax to contact the side structure of the vehicle. Seat belts do, however, help keep the dummy in-position as the vehicle is being transported down the test track. Prior research has indicated that all occupant response measures are extremely sensitive to the position of the dummy at impact. The seat belts help keep the dummy from bouncing out of position during the acceleration and sliding phases prior to impact. All available restraints should be used in side-impact fixed object crash tests.

4 Data Acquisition

4.1 General

Side impact crash tests require the same types of vehicle data as more traditional full-scale crash tests [29]. The vehicle should be instrumented with accelerometers to measure all six degrees of freedom of the vehicle. Photographic coverage should conform to the usual practice in appurtenance tests. Film analysis of the vehicle motions should be performed as well as analyses of the vehicle accelerometer outputs. An on-board high-speed camera is useful for understanding the response of the vehicle occupants and possible sources of injury and is essential for determining the pre-impact position of the dummy.

Relating observable results of crash tests and the risk of severe injury in such collisions is the long-term goal of side impact crash testing. Table 7 shows the data elements that should be calculated, collected, recorded and reported in side impact crash tests. To date there are relatively few side impact crash tests available for analysis. These data elements

Table 7. Data elements required in side impact crash tests.

Parameter	Symbol	Acquisition Device
Impact Velocity	V_i	Film
Actual Dummy Impact Velocity	V_{occ}	Calculations
Maximum static vehicle crush (exterior)	c_e	NHTSA 6-point sketch
Area of crush (exterior)	C_{area}	NHTSA 6-point sketch
Maximum dynamic intrusion (interior)	c_i	Displacement Transducer
Average intrusion rate (interior)	$\dot{c}_i$	Displacement Transducer
Maximum 10-msec vehicle acceleration prior to dummy contact with interior.	a_{pc}	Vehicle accelerometers
Maximum 10-msec vehicle acceleration during dummy contact with interior.	a_{dc}	Vehicle accelerometers
Maximum 10-msec vehicle acceleration after dummy contact with interior.	a_{rd}	Vehicle accelerometers
Thoracic Trauma Index	TTI	Anthropometric Dummy
Head Injury Criteria	HIC	Anthropometric Dummy
Maximum pelvis acceleration	a_p	Anthropometric Dummy
Longitudinal distance from the center of the dummy head to impact point	r	Onboard Film
Lateral distance from the left side of the dummy head to edge of passenger compartment	s	Onboard Film
Distance from impact point to front axle	D	Post Test Measurement

are thought to be important characteristics of the collision that might be useful in building models that predict the risk of severe occupant injury. Collecting these data will allow a data base of important side impact parameters to be assembled. Table 7 also shows the data acquisition device needed to obtained each parameter. Most involve typical acquisition methods like vehicle accelerometers and film analysis. In addition to the usual data acquisition methods, a fully instrumented anthropometric dummy and a displacement transducer to measure the deformation of the door should be used.

4.2 Anthropometric Dummy

While anthropometric dummies were routinely used in the past in full-scale safety appurtenance crash tests, they did not provide much useful information [17]. The forces in most longitudinal barrier tests are well below the level necessary to result in significant dummy responses. In recent years anthropometric dummies have been included in most crash tests only to represent the occupant's inertia and to enhance the on-board photographic record.

The severity of the crash loading in side impacts, however, places the dummy in a much more extreme environment and meaningful dummy responses can be obtained. An instrumented SID should be used in side impact crash tests with roadside structures.

The dummy should be instrumented so that the Thoracic Trauma Index (TTI), the Head Injury Criteria (HIC) and the maximum pelvis accelerations (a_p) can be calculated. Calculation of the TTI requires accelerometers located on the impact-side ribs and T12 segment of the spine. The HIC is calculated based on the resultant of a triaxial accelerometer mounted in the head form of the Part 572 dummy [24].

4.3 Displacement Transducer

The intrusion of the door into the passenger compartment is one of the most hazardous characteristics of side impact accidents. The occupant strikes the intruding door structure in a typical side impact event. Penetration of the passenger compartment has long been recognized as a very hazardous event in roadside collisions. Any significant penetration or deformation of the passenger compartment is disallowed in all other types of full-scale appurtenance crash tests. The severity of side impact collisions, however, makes this an unreasonable and unobtainable restriction. In order to determine the effect of the intrusion and more particularly the intrusion rate, the use of a displacement transducer is recommended.

A Celesco PT510 string pot transducer has been used successfully in several side impact crash tests of guardrail terminals [27]. This device can accurately record distances between 0 and 2 m. Devices of this type are readily available, inexpensive and very robust. Figure 4 shows how the transducer was mounted in several recent tests. The transducer unit was attached to the inner window sill of the non-impact side door. The end of the string was stretched across the passenger compartment and screwed into the structure of the impact-side door. The string should be perpendicular to the door. The transducer measures the instantaneous width of the passenger compartment and the slope of this line represents the velocity of the inner surface of the intruding door. Figure 5 shows the output of a string pot transducer in a side impact test of a guardrail terminal.

5 Evaluation Criteria

5.1 General

Using standard testing conditions ensures that different tests performed by different testing agencies can be compared directly. Standardizing test conditions does not indicate how well a device performs; evaluation criteria do. Evaluation criteria are a set of quantifiable

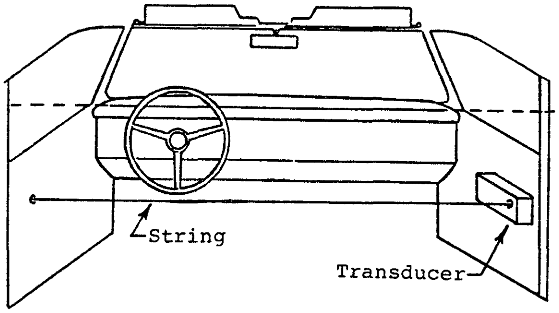
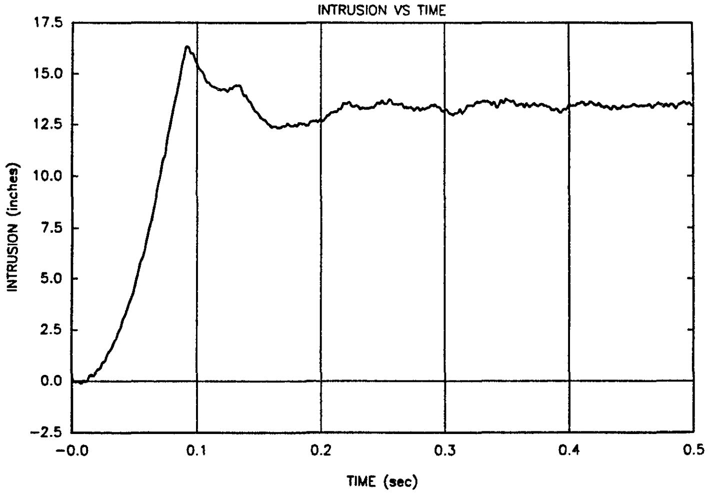


Figure 4. Displacement transducer.



1 in = 25.4 mm

Figure 5. Example Displacement Transducer Output.

limits that, taken together, suggest how well a roadside structure can be expected to perform in real-world impacts.

The NCHRP guidelines recommend three separate criteria for evaluating crash tests: structural adequacy, occupant risk, and vehicle trajectory. These three criteria have evolved over the years to ensure that hardware performs as designed (structural adequacy criteria), it does so without undue risk to vehicle occupants (occupant risk criteria), and the probability of subsequent accident events is minimized (vehicle trajectory). Although past criteria have not addressed side impacts specifically, the three general criteria are as applicable to the side impact scenario as other, more typical accident scenarios. The three general NCHRP evaluation criteria are used as a framework for developing side impact crash test evaluation procedures.

5.2 Structural Adequacy Criteria

The structural adequacy criteria requires that a roadside structure be structurally capable of accomplishing its primary purpose. For longitudinal barriers, this primary structural purpose is preventing the vehicle from crossing the barrier line; for breakaway hardware the primary purpose is to yield or breakaway without penetrating the passenger compartment or scattering debris onto the roadway.

In side impacts, roadside structures should be expected to breakaway, fracture, collapse or yield allowing the vehicle to either pass by or stop. The suggested structural adequacy criteria for side impacts are shown in table 8. Italicized text represents an addition to NCHRP criterion B

5.3 Occupant Risk Criteria

The occupant risk criteria have evolved into the most important single evaluation criteria in testing roadside hardware. The ultimate objective of all safety hardware is to prevent or minimize the potential for injury to occupants of vehicles that leave the traveled way.
Unfortunately, establishing a linkage between parameters measured in crash tests and real occupants of vehicles in accidents has been an extraordinarily difficult task.

Report 230 introduced the concept of the flail space occupant risk criteria. The flail space method calculates the hypothetical impact velocity of an occupant head with the interior of the vehicle. The impact velocity between the occupant and the vehicle interior did not prove to be a good predictor of dummy response even when the flail space method was modified to account for the intrusion rate of the door.

Relating the forces experienced by anthropometric test devices to the potential for

Table 8. Side impact crash test evaluation criteria.

Structural Adequacy	NCHRP-B.	The test article should readily activate in a predictable manner by *collapsing*, breaking away, fracturing or yielding.
Occupant Risk	NCHRP-F	The vehicle should remain upright during and after the collision although moderate roll, pitching and yawing are acceptable.
	SI-H	The Head Injury Criteria (HIC) measured using a side impact dummy (part 572 subpart F) should be less than 1000. If the dummy was not in the normal seating position at the time of impact, the HIC may be normalized using the following expression: $HIC_{norm} = HIC_{obs}\ 0.9925^{r}\ 0.9883^{s}$ where $r =$ Longitudinal distance from dummy shoulder to impact point (mm). $s =$ Lateral distance from the left side of the dummy head to door window (mm).
	SI-T	The Thoracic Trauma Index (TTI) measured using a side impact dummy (part 572 subpart F) should be less than 90. If the dummy was not in the normal seating position at the time of impact, the TTI may be normalized using the following expression: $TTI_{norm} = TTI_{obs}\ 0.9960^{r}\ 0.9975^{s}$ where $r =$ Longitudinal distance from dummy shoulder to impact point (mm). $s =$ Lateral distance from the left side of the dummy head to door window (mm).
	SI-P	The pelvic acceleration measured using a side impact dummy must be less than 130 g's.
Vehicle Trajectory	SI-V.	After collision the vehicle trajectory should not intrude into adjacent traffic lanes.

serious injury is a challenging area of research that has been pursued by NHTSA, the military and the automotive design communities for decades. The measures of injury promoted by NHTSA are recommended since that agency has the most expertise and ability in the area of biomechanics and human tolerance. Conforming to the NHTSA recommendations will allow the roadside safety community to take advantage of a wealth of biomechanics experience while also facilitating the exchange of information between these two agencies in the future. While the HIC and TTI could certainly be improved, they have a better linkage to real human trauma than the flail space for side impacts.

The recommended occupant risk criteria, discussed below, is composed of four subcriterion: a vehicle stability criterion, a thoracic trauma criterion, a head injury criterion and a pelvis acceleration criterion.

5.3.1 Vehicle Stability Criterion

Roll over of the vehicle has long been recognized has a very hazardous event in single vehicle accidents [25]. Roadside structures should breakaway, collapse or yield in an impact without causing the vehicle to rollover or completely loose contact with the ground.

5.3.2 Head Injury Criterion

The Head Injury Criteria (HIC) evolved from several earlier techniques for measuring the resultant accelerations experienced by the head form of the Part 572 dummy [18]. The HIC has been used for many years in frontal barrier crash tests by NHTSA as well as by the roadside design community. A HIC of 1000 has generally been considered the threshold for severe injury.

The purpose of any occupant response measure is to estimate the risk to occupants in real accidents. A cumulative probability density function relating the probability of sustaining an $AIS > 3$ injury based on the observed HIC has been developed from the results of cadaver testing [26]. According to this curve, a $HIC = 1000$ implies a risk of $AIS > 3$ injury of 0.18: 18 percent of occupants with a $HIC = 1000$ will be severely injured.

Unfortunately, the HIC was not developed to measure head injury potential in side impacts. The differences between longitudinal and lateral head impact tolerance and the degree to which the Part 572 head form predicts human injury have been debated but no consensus has been reached [18]. It is widely agreed, however, that the head is probably *less* tolerant in lateral impacts than in frontal impacts so the HIC should certainly be no greater than 1000. There is a great need for the biomechanics research community to address the issue of an appropriate lateral HIC limit or, more generally, head injury criteria for the side of the head. A *HIC* = 1000 has been used in a recent study of head form impacts with upper vehicle-interior structures like the A-pillar, roof rails and B-pillar [6]. A limit of 1000 appears to be the best available link between the dynamics of an impacting head and the potential for serious injury.

Anthropometric dummies should be used in side impact crash tests of roadside structures as long as the possibility of serious damage to the dummy is minimal. The *HIC* should be evaluated in the same manner typically used for frontal collisions. Details on computationally efficient *HIC* algorithms can be found in a variety of papers in the literature [18] [3] [13].

The exact location of the dummy at the time of impact has been a problem in performing side impact crash tests of roadside structures. Dummies in vehicle-to-vehicle crash tests do not move prior to the impact because the struck vehicle is stationary so correct dummy position can be guaranteed. In roadside structure crash tests the vehicle and dummy must be accelerated to the desired impact velocity since the structure is fixed and the vehicle is accelerated. Ideally, the dummy should be in the "normal" seating position. This would correspond to a location about 250 mm behind the impact point and 165 mm from the head to the side window.

When the dummy is not in the correct position at impact, the HIC can be normalized using the following expression:

$$HIC_{norm} = HIC_{obs}\ 0.9925^{r}\ 0.9883^{s} \qquad (1)$$

where
r = Longitudinal distance from dummy shoulder to impact point in mm.
s = Lateral distance from dummy shoulder to impact point in mm.

This expression was derived from a regression analysis of 15 side impact crash tests of poles. The HIC appears to decay exponentially as the distance between the head and the impact point increases. The worst-case impact location is one that is centered on the occupant's head when the occupant is in contact with the door window (r = 0, *s* = 0). Figure 6 shows the definition of the coordinates for equation 1.

It is very important to normalize the HIC when different tests are being compared since a large HIC may be due to the head impact being too close to the intruding object rather than a difference in performance between one device and another.

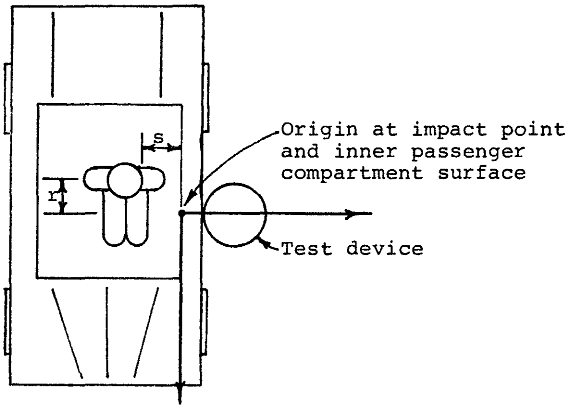

s
r
Origin at impact point and inner passenger compartment surface
Test device

Figure 6. Coordinates for occupant position in side impact crash tests.

5.3.3 Thoracic Trauma Criterion

The thoracic trauma index (TTI) was developed by NHTSA to measure the chance of severely injuring the human thorax during a collision [20] [5] [8] [7]. The formulation of TTI has gone through several revisions, the most recent being found in the 1990 final amendment for Federal Motor Vehicle Safety Standard 214 [24]. The TTI is the average acceleration experienced by accelerometers located on the left upper rib (LURY) and the twelfth spinal segment (T12P) as shown in equation 2. The accelerations should be filtered using the FIR100 finite impulse response filter as specified in FMVSS No. 214 [24].

$$TTI = \frac{G_R + G_{LS}}{2} \qquad (2)$$

where
G_R = The greater peak acceleration of either the upper or lower rib in g's, and
G_{LR} = The peak acceleration of the lower spine (T12) [24].

The TTI is not the only possible measure of thoracic trauma. Researchers at General Motors Research Laboratory, for example, developed a competing injury scale, the viscous criteria (VC) [34] [15]. Unfortunately, the data required to calculate VC are only obtainable using BioSID dummies which were not available in any of the side impact crash tests performed to date at the FOIL. In contrast, the TTI can be calculated using the more common Part 572 SID. Since most of the FHWA tests and all of the NHTSA tests contained data that could be used to calculate the TTI, the TTI was preferred as a measure of thoracic occupant trauma.

Instrumented anthropometric dummies should be used in side impact crash tests of roadside structures. The TTI for the in-position dummy should be less than 90 g's. NHTSA, in its 1990 final rules on side impact, requires the TTI be less than 90 g's in tests of 2-door passenger cars. The recommended criteria therefore conform to the NHTSA design limits. The TTI has been related to the probability of various levels of injury using the Abbreviated Injury Score (AIS) [1]. The cumulative density function of TTI was found to be a Weibull extreme value distribution [20]. A $TTI = 90$ corresponds to a 0.16 probability of an $AIS > 3$ injury. This level of risk is roughly the same used for the HIC so these criteria represent an internally consistent risk of trauma for evaluating side impact tests.

Maintaining correct dummy position, as discussed earlier, is often very difficult in a crash test where the vehicle and dummy must be accelerated up to a target test speed. If the dummy is out of position (i.e. *r* „ 250 mm and *s* „ 165 mm) the TTI should be normalized to the hypothetical in-position response using equation 3. The coordinate system for occupant motions is shown in figure 6. This expression is based on the empirical observation that the TTI, like HIC, seems to decay exponentially as the distance from the occupant increases.

$$TTI_{norm} = TTI_{obs}\ 0.9960^{r}\ 0.9975^{s} \qquad (3)$$

where
r = Longitudinal distance from dummy shoulder to impact point in mm
s = Lateral distance from dummy shoulder to impact point in mm.

5.3.4 Pelvis Acceleration Criterion

Although no side impact crash tests of roadside hardware have collected the pelvic acceleration, it is a component of the NHTSA final rules on FMVSS-214 [24]. The pelvis accelerations should be filtered using the FIR100 finite impulse response filter as specified in FMVSS No. 214. This rule specifies that the pelvic must not experience an acceleration greater than 130 g's during the test.

$$a_p \leq 130$$

Where a_p, is the maximum acceleration of the pelvis of the side impact dummy. The probability of experiencing a fatal fracture is relatively low at this level. Including the pelvis acceleration also helps to ensure that improvements in the TTI and HIC do not come at the expense of shifting the load path through lower parts of the vehicle.

5.4 Vehicle Trajectory Criterion

The purpose of the vehicle trajectory criteria is to reduce the chance of a subsequent harmful event after the appurtenance collision. Since the vehicle is sliding sideways in a side impact it will usually loose speed rapidly and come to rest near the first impact point. Sometimes, after the collision, the laterally sliding tires may begin to rotate causing the vehicle to roll forward. Since the front of the vehicle is pointing toward the traveled way in the standard orientation (figure 3) there is a danger that the vehicle can reenter the roadway or even travel completely across it. Reentry of the vehicle into the roadway, especially at the high angles resulting from a side impact, is not acceptable.

Criteria SI-V, shown in table 8, is very similar to NCHRP evaluation criteria K [29] except more restrictive language is used. This slightly more stringent criteria is recommended because, after a side impact, a vehicle could reenter the roadway at a high angle, perhaps even perpendicular to the roadway. After the vehicle comes to rest it will probably require towing since vehicle damage is usually extensive in a side impact.

6 Estimating Dummy Responses

6.1 General

Ultimately anthropometric dummies should be eliminated from full scale tests. The environment in many full scale crash tests of roadside safety appurtenances is so severe it is often not advisable to place dummies in the vehicle. Good dummy results require careful and frequent calibration which has traditionally been a problem for roadside safety applications. Since the response of the vehicle can be easily measured in a full-scale test, vehicle-based evaluation parameters that estimate the response of hypothetical humans are preferable to the use of fully instrumented anthropometric dummies.

As discussed earlier, there are three primary injury mechanisms that are active in side impact collisions: thoracic trauma, head injury and pelvic fracture. Data from 15 previous crash tests were analyzed to determine if there were any relationships between the observed vehicle-based parameters and the vales of TTI and HIC. Pelvic acceleration was not modeled since there has been no data collected as yet. The same type of modeling activity could be performed to estimate the accelerations of the pelvis based on vehicle-based parameters once sufficient data has been collected. The 15 tests used represent all the tests that used instrumented SID dummies. The following sections summarize the findings of these investigations [27].

6.2 Thoracic Trauma Index

A multiple linear regression analysis was performed on 15 tests of small cars side impacting a variety of poles. Values for all the parameters listed in table 7 were collected and entered into a data base of values. A variety of regression models were evaluated. Models that included the effect of occupant position (parameters *r* and s) and impact velocity were required. Beyond these three basic parameters the model with the fewest predictors and highest R^2 were preferred. The best five parameter model was:

$$TTI = 0.5(10)^{-3}\,\{0.9960^{r}0.9975^{s}\}\,V_i^{2.5}\left\{\frac{c_i^{1.25}}{\sqrt{\dot{c}_i}}\right\} \tag{5}$$

$R^2 = 0.90$

where r = Longitudinal distance from occupant head to impact point in mm.
s = Lateral distance from occupant head to the impact point in mm.
V_i = Vehicle impact velocity in m/sec.
c_i = Maximum static passenger compartment crush in mm.
c_i = Average passenger compartment intrusion rate in m/sec.

The coefficient of regression squared *(R^2)* for this model was quite good for this type of experimental data. The components of this model seem reasonable: severity should increase as the occupant gets closer to the impact point *(r* and *s)* and severity should be a function of the impact velocity (V_i) since this is a measure of the total amount of kinetic energy at the start of the impact event. Passenger compartment crush and crush rate were also thought to be directly related to the TTI since thoracic injuries are caused by contact with the side door panels.

In a test with no dummy, three of these parameters (r, *s,* and V_i) are specified. Only crush and crush rate are measurable results of the test. The desirable, in-position location of an occupant is at r = 250 mm, s = 165 mm and the standard impact speed is 50 km/h (14 m/sec). The maximum allowable TTI from table 8 is 90 g's. These values can be substituted into equation 5 and solved for the quantity $\left\{\frac{c_i^{1.25}}{\sqrt{\dot{c}_i}}\right\}$. Doing so results in a criterion for allowable thoracic trauma.

$$1000 \geq \frac{c_i^{1.25}}{\sqrt{\dot{c}_i}}$$

If the crush and crush rate result in a value less than 1000, the probability of observing a TTI greater than 90 is relatively small.

This expression was developed using the results of 15 side impact tests of slip-base and ESV poles. The degree to which this expression will predict TTI scores for other types of devices is not known. The range of crush and crush rate in these tests was between 200 and 900 mm and 1 and 10 m/sec, respectively. These expressions might not yield appropriate estimates for tests where the crush or crush rate was substantially more than the tested range. These expressions should be used as a guide when direct measures of the TTI are not available.

6.3 Head Injury Criteria

The same type of stepwise regression analysis was performed to find models for the HIC. The results of this analysis were not as attractive as the TTI model described in the previous section. It is presented here to serve as an approximate guide for tests where no dummies were included.

Most of the 15 tests were conducted with the dummy head aligned with the impacting pole. This caused exceptionally high HIC values since there was often direct contact between the pole and the head. This extreme test condition may be more demanding than the SID dummy capabilities. For this reason, a longitudinal impact point (r) of 250 mm is recommended for future tests. The severity of the loading caused problems in developing a model for HIC. The $r = 0$ position appears to represent a singularity in the response of the dummy. Future research with dummies at positions other than r = 0 should help to refine the model presented herein. Equation 7 represents the model with the best R^2 value which included terms for occupant position *(r* and *s).*

$R^2 = 0.56$

where r = Longitudinal distance from occupant head to impact point in mm.
s = Lateral distance from occupant head to the impact point in mm.
V_{occ}= Occupant impact velocity with intruding vehicle interior in m/sec
c_i = Maximum static passenger compartment crush in mm.
c_i = Average passenger compartment intrusion rate in m/sec.

When there is no dummy in the test vehicle, the above expression can be solved for limiting values of the quantity $\cdot\left\{\frac{c^{1.64}}{\dot{c}\,{V_{occ}}^{0.15}}\right\}$. Substituting HIC=1000, r=250 , and s=165, yields a value of 165.An approximate criteria that would predict acceptable HIC values could be stated as :

$$165 \geq \left\{\frac{c^{1.64}}{\dot{c}\,{V_{occ}}^{0.15}}\right\} \tag{8}$$

As with the TTI model described above, this model may not be appropriate for devices that are not breakaway poles and for impacts outside the range of typical values used in building the regression models. Equation 7 is presented as a guide for tests where it is not possible to use a dummy in the test vehicle. This model, due to the underlying data. should only be used in tests where there is a possibility of direct contact between the head and the intruding object. In tests of guardrail terminals, for example, there is no possible contact between the head and the terminal so the HIC should not be evaluated. The HIC should always be evaluated for tall, narrow objects like luminaires, utility poles and signs.

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APPENDIX H

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APPENDIX I

GLOSSARY

AASHTO American Association of State Highway and Transportation Officials.ACI American Concrete Institute.

AISC American Institute of Steel Construction.

AISI American Iron and Steel Institute.

ASI Acceleration Severity Index.

ASTM American Society for Testing and Materials.

Ballast Mass added to vehicle, other than simulated occupant(s) and instrumentation, to simulate cargo and/or to achieve desired test inertial mass.

Bogie A device used as a surrogate for a production model test vehicle. Existing bogies are four-wheeled devices that are towedinto the test article. They are typically designed to replicate the dynamic response of a vehicle for specific tests, e.g., tests of breakaway features. Bogies typically can be used for both lowand high-speed tests.

Center of Mass (c.m.) Point within test vehicle at which its total mass can be assumed to be concentrated.

Clear Zone The total roadside border area, starting at the edge of the traveled way, available for safe use by errant vehicles. This area may consist of a shoulder, a recoverable slope, a nonrecoverable slope, and/or a clear run-out area. The desired width is dependent on the traffic volumes and speeds and on the roadside geometry.

Crash Cushion A device designed primarily to safely stop a vehicle within a relatively short distance. A *redirective crash cushion* is designed to contain and redirect a vehicle impacting downstream from the nose of the cushion. A *nonredirective crash cushion* is designed to contain and capture a vehicle impacting downstream from the nose of the cushion.

Crash Test A test in which a production model test vehicle or a surrogate test vehicle impacts or traverses a highway feature.

Critical Impact Angle (CIA) For a given test and the attendant range of vehicular impact angles, the CIA is the angle within this range judged to have the greatest potential for causing a failure when the test is assessed by the recommended evaluation criteria. For most tests, impact angles can range from 0 up to 25 degrees.

Critical Impact Point (CIP) For a given test, the CIP is the initial point(s) of vehicular contact along the longitudinal dimension of a feature judged to have the greatest potential for causing a failure when the test is assessed by the recommended evaluation criteria.

Curb Mass Mass of test vehicle with standard equipment, maximum capacity of engine fuel, oil and coolant and, if so equipped, air conditioning and additional optional mass engine. It does not include occupants or cargo.

Device Refers to a design or a specific part thereof, such as a breakaway device. Note that the terms "device" and "feature"are often synonymous.

Evaluation Criteria Criteria used to assess the results of a crash test or to assess the in-service performance of a feature.

Feature Refers to a specific element of a highway. It may be a hardware item and its associated foundation, such as a sign orbarrier installation, or it may be a geometric element, such as a side slope or a ditch cross section.

FHWA Federal Highway Administration.

Flail Space Hypothetical space in which a hypothetical occupant is permitted to move during impact.

Gating Device (Feature) A device designed to allow controlled penetration of a vehicle when impacted upstream of the beginning of the length of need (LON). Note there is some distance between the end of a gating device and the beginning of the LON of the device.

Geometric Feature A roadside cross section element such as a ditch section, an embankment, a driveway or a median crossover, or a curb. It also includes drainage structures such as inlets and culvert ends and devices such as grates used to enhance safety of these features.

Gross Static Mass Sum of test inertial mass and mass of surrogate occupant(s).

HVOSM Highway-Vehicle-Object-Simulation-Model computer program.

Hybrid III Dummy An anthropomorphic dummy, representing the 50th percentile male, the specifications of which are contained in part 572, Subpart E, Title 49 of the Code of Federal Regulations, Chapter V-(10-1-88 Edition).

Impact Angle (Θ) Angle between normal direction of traffic and approach path of test vehicle into the test article. The test article should be oriented as it would typically be in service with respect to the normal direction of traffic.

Impact Point The initial point on a test article contacted by the impacting test vehicle.

Impact Severity (IS) A measure of the impact severity of a vehicle of mass M, impacting at a speed V, at an impact angle Θ. It is defined as follows: $IS = 1/2M(V \sin \Theta)^2$

Length of Need (LON) That part of a longitudinal barrier or terminal designed to contain and redirect an errant vehicle.

Longitudinal Barrier A device whose primary functions are to prevent vehicular penetration and to safely redirect an errant vehicle away from a roadside or median hazard. The three types of longitudinal barriers are roadside barriers, median barriers, and bridge rails.

Nongating Device A device with redirectional capabilities along its entire length. Note that the end of a nongating device is the beginning of the length of need for the device.

Occupant Impact Velocity Velocity at which a hypothetical "point mass" occupant impacts a surface of a hypothetical occupant compartment.

Pendulum A device used as a surrogate for a production model test vehicle. A mass is attached to cables, which are in turn suspended from a fixed point. The mass is raised to a selected height and released, allowing gravity to accelerate the mass as it swings into the test article. The structure of the mass can be designed to replicate the dynamic crush properties of a production model test vehicle. It is basically a low-speed test device.

Permanent Feature (Device) A feature with an anticipated long duration of service, as opposed to those used in a work or construction zone having a relatively short duration of service.

Pocketing If, on impact, a redirective device undergoes relatively large lateral displacements within a relatively short longitudinal distance, pocketing is said to have occurred. Depending on the degree, pocketing can cause large and unacceptable vehicular decelerations.

PHD Post-Impact Head Deceleration.

Production Model Test Vehicle A commercially available vehicle with properties matching those required in a given test. Ridedown Acceleration Acceleration experienced by a hypothetical "point mass" occupant subsequent to impact with a hypothetical occupant compartment.

SAE Society of Automotive Engineers.

Sprung Mass All mass that is supported by a vehicle's suspension system, including portions of the mass of the suspensionmembers.

Snagging When a portion of a test vehicle, such as a wheel,engages a vertical element in a redirective device, such as apost, snagging is said to have occurred. The degree of snagging depends on the degree of engagement. Snagging may cause large and unacceptable vehicular decelerations.

Support Structure A system used to support a sign panel, chevronpanel, luminaire, utility lines, mailbox, or emergency call box.The system includes the post(s), pole(s), structural elements, foundation, breakaway mechanism if used, and accompanying hardware used to support the given feature.

SI International System of Units.

Surrogate Occupant A dummy, set of sand bags, or other artifact used to simulate the effects and/or to study the dynamic responseof an occupant in a vehicle.

Surrogate Test Vehicle A bogie, pendulum device, or other substitute device designed to replicate the dynamic response of aproduction model vehicle when in collision with a roadside feature.

Temporary Feature (Device) A feature used in a work, construction, or maintenance zone. Its duration of use is normally relatively short, usually one year or less.

Terminal A device designed to treat the end of a longitudinal barrier. A terminal may function by (a) decelerating a vehicle to a safe stop within a relatively short distance, (b) permitting controlled penetration of the vehicle behind the device, (c) containing and redirecting the vehicle, or (d) a combination of a, b, and c.

Test Article (Test Feature) All components of a system, including the foundation as relevant, being evaluated in a crash test. Notethat the system may be a geometric feature such as a ditch or driveway slope.

Test Inertial Mass Mass of test vehicle and all items rigidly attached to vehicle's structure, including ballast and instrumentation. Mass of surrogate occupant(s), if used, is not included in test inertial mass.

Test Level (TL) A set of conditions, defined in terms of vehiculartype and mass, vehicular impact speed, and vehicular impact angle, that quantifies the impact severity of a matrix of tests.

Test Vehicle A commercially available, production model vehicle or an approved surrogate vehicle used in a crash test to evaluate the impact performance of a test article.

THIV Theoretical Head Impact Velocity.

Track Width Center-of-tire-to-center-of-tire distance for a givenaxle of a vehicle.

Transition That part of a longitudinal barrier system between and connecting sections of differing lateral stiffness and/or sectionsof differing design or geometry.

Truck-Mounted Attenuator (TMA) An energy-absorbing deviceattached to the rear of a truck or utility vehicle. A TMA isdesigned to provide a controlled stop of a vehicle impacting the rear of the truck.

Unsprung Mass All mass which is not carried by the suspensionsystem but is supported directly by the tire or wheel and considered to move with it.

Utility Pole A support structure used to support power transmission lines or communication lines.

Work Zone Traffic Control Device A device used in a work zone to regulate, warn, and guide road users and advise them totraverse a section of highway or street in the proper manner. Work zone traffic control devices of interest herein include signs, plastic drums, and lights that may be used thereon; cones, barricades, chevron panels, and their support system; and any other such device(s) commonly exposed to traffic that may pose a hazard to occupants of a vehicle and/or to work zone personnel.

APPENDIX J

SI CONVERSIONS

To convert from	To	Multiply by[1]
	ACCELERATION	
Meter per second squared (m/s^2)	ft/s^2	3.280 840 E+00
	AREA	
Square meter (m^2)	ft^2	1.076 391 E+01
	ENERGY	
Joule (J)	ft-lbf	7.375 621 E-01
	FORCE	
Newton (N)	pound-force (lbf)	2.248 089 E-01
	LENGTH	
Meter (m)	ft	3.280 840 E+00
Meter (m)	in	3.937 008 E+01
Centimeter (cm)	in	3.937 008 E-01
	MASS	
Kilogram (kg)	pound (lb avoirdupois)	2.204 623 E+00
	PRESSURE OR STRESS	
Pascal (Pa)	psi	1.450 377 E-04
	VELOCITY	
Kilometer per hour (km/h)	miles per hour (mi/h)	6.213 712 E-01
Kilometer per hour (km/h)	ft/s	9.113 444 E-01
Meter per second (m/s)	ft/s	3.280 840 E+00

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[1] Numbers to the left of "E" are multiplied by 10 raised to an exponent equal to the number, and accompanying sign, to the right of "E."

THE TRANSPORTATION RESEARCH BOARD is a unit of the National Research Council, which serves the National Academy of Sciences and the National Academy of Engineering. It evolved in 1974 from the Highway Research Board which was established in 1920. The TRB incorporates all former HRB activities and also performs additional functions under a broader scope involving all modes of transportation and the interactions of transportation with society. The Board's purpose is to stimulate research concerning the nature and performance of transportation systems, to disseminate information that the research produces, and to encourage the application of appropriate research findings. The Board's program is carried out by more than 270 committees, task forces, and panels composed of more than 3,300 administrators, engineers, social scientists, attorneys, educators, and others concerned with transportation; they serve without compensation. The program is supported by state transportation and highway departments, the modal administrations of the U.S. Department of Transportation, the Association of American Railroads, the National Highway Traffic Safety Administration, and other organizations and individuals interested in the development of transportation.

The National Academy of Sciences is a private, nonprofit, self-perpetuating society of distinguished scholars engaged in scientific and engineering research, dedicated to the furtherance of science and technology and to their use for the general welfare. Upon the authority of the charter granted to it by the Congress in 1863, the Academy has a mandate that requires it to advise the federal government on scientific and technical matters. Dr. Frank Press is president of the National Academy of Sciences.

The National Academy of Engineering was established in 1964, under the charter of the National Academy of Sciences, as a parallel organization of outstanding engineers. It is autonomous in its administration and in the selection of its members, sharing with the National Academy of Sciences the responsibility for advising the federal government. The National Academy of Engineering also sponsors engineering programs aimed at meeting national needs, encourages education and research and recognizes the superior achievements of engineers. Dr. Robert M. White is president of the National Academy of Engineering.

The Institute of Medicine was established in 1970 by the National Academy of Sciences to secure the services of eminent members of appropriate professions in the examination of policy matters pertaining to the health of the public. The Institute acts under the responsibility given to the National Academy of Sciences by its congressional charter to be an adviser to the federal government and, upon its own initiative, to identify issues of medical care, research, and education. Dr. Kenneth I. Shine is president of the Institute of Medicine.

The National Research Council was organized by the National Academy of Sciences in 1916 to associate the broad community of science and technology with the Academy's purpose of furthering knowledge and advising the federal government. Functioning in accordance with general policies determined by the Academy, the Council has become the principal operating agency of both the National Academy of Sciences and the National Academy of Engineering in providing services to the government, the public, and the scientific and engineering communities. The Council is administered jointly by both Academies and the Institute of Medicine. Dr. Frank Press and Dr. Robert M. White are chairman and vice chairman, respectively, of the National Research Council.

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ADDRESS CORRECTION REQUESTED